

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T08:04:33

Source: data/json/noise_experiments/v2/noise8i_fixed_memprimary_game/claude-sonnet-4.5/game/noisy8_bidirectional_informed_memprimary_fixed/noise8i_fixed_memprimary_game_000_neutral_rep00.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy8_bidirectional_informed_memprimary_fixed
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 0
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: self_and_coplayer
self_history_window: 0
coplayer_history_window: 3
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 0
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard",
  "Agent_3": "standard",
  "Agent_4": "standard",
  "Agent_5": "standard",
  "Agent_6": "standard",
  "Agent_7": "standard",
  "Agent_8": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_3; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_4; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_5; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Agent_6; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_7; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_8; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

ROUND 1

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$5; payoffs Agent_1=\$7, Agent_2=\$4;
noise/visible: sent visible=\$3.28, received visible=\$9.84, returned visible=\$5.94
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_3 sent \$3; received \$9; Agent_4 returned \$5; payoffs Agent_3=\$7, Agent_4=\$4;
noise/visible: sent visible=\$3.37, received visible=\$10.11, returned visible=\$5.51
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$3.15; payoffs Agent_5=\$5.15, Agent_6=\$5.85; noise/visible: sent visible=\$2.29, received visible=\$6.87, returned visible=\$3.41
Action validation: {"returned": {"amount": 3.15, "clamped": false, "max": 9.0, "min": 0, "raw": 3.15}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3; received \$9; Agent_8 returned \$3; payoffs Agent_7=\$5, Agent_8=\$6;
noise/visible: sent visible=\$2.24, received visible=\$6.72, returned visible=\$2.2
Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 7.0,
"Agent_2": 4.0,
"Agent_3": 7.0,
"Agent_4": 4.0,
"Agent_5": 5.15,
"Agent_6": 5.85,
"Agent_7": 5.0,
"Agent_8": 6.0

}

Visible balances after round: {

"Agent_1": 7.94,
"Agent_2": 4.84,
"Agent_3": 7.51,
"Agent_4": 5.1099999999999999,
"Agent_5": 5.41,
"Agent_6": 3.72,
"Agent_7": 4.2,
"Agent_8": 3.7200000000000006

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T07:51:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
"agents": [
"Agent_1",
"Agent_2"
],
"investor": "Agent_1",
"roles": {
"Agent_1": "investor",
"Agent_2": "trustee"
},
"trustee": "Agent_2"

```

},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust and caution since I don't know my co-player yet.

****My reasoning:****

- Sending \$0 is safe but creates no mutual benefit
- Sending everything (\$5) maximizes potential gains but is very risky with an unknown partner
- A moderate amount signals cooperation while managing risk

I'll send ****\$3****, which:

- Gets multiplied to \$9 for the receiver (creating good potential for returns)
- Keeps \$2 as a safety buffer for me
- Shows willingness to trust and cooperate

- If they return even \$3-4, we both benefit

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me analyze the situation:

1. I have \$5 to start
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ from what was actually sent/returned
5. This is a multi-agent setting with 8 agents, randomized pairings each round
6. I don't know who my co-player is (names are hidden)

Key considerations:

- If I send \$0, I guarantee keeping \$5 but miss the opportunity for mutual gain
- If I send the full \$5, the receiver gets \$15 and could return a substantial amount
- The optimal outcome for both of us combined is if I send everything (creating \$15 total instead of \$5)
- However, the receiver could keep everything and I'd end up with \$0
- There's also communication noise which adds uncertainty

Since this is the first round and I have no information about my co-player:

- Sending some amount signals trust and cooperation
- A moderate amount (like \$2-3) balances risk and potential reward
- Sending \$5 shows maximum trust but is risky with an unknown partner
- The communication noise means even if they want to reciprocate fairly, there might be misunderstandings

For a first round with an unknown partner, I'll send a moderate amount to signal cooperation while managing risk. I'll send \$3:

- This creates \$9 for the receiver (good potential for returns)
- I keep \$2 as insurance
- If they return just \$3-4, I'll break even or profit
- It signals willingness to cooperate without being reckless

Let me go with \$3 as a balanced first move.

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T07:51:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

```

}
Exact input messages sent to model:
  Message 1 role=system
    content:
      You are an agent participating in a multi-task experiment. You will play a game and may al
      so be asked to write myths.

      GAME:
      You will play different roles across multiple rounds.

      When you are the SENDER:
      - You receive $5 each round
      - You decide how much to SEND to the RECEIVER ($0-$5)
      - Amount sent is multiplied by 3x
      - Receiver decides how much to RETURN
      - You keep: $5 - sent + returned
      - Receiver keeps: received - returned
      - Response format: {'send': <amount>}

      When you are the RECEIVER:
      - Sender receives $5
      - Sender decides how much to SEND to you ($0-$5)
      - Amount sent is multiplied by 3x
      - You decide how much to RETURN
      - Sender keeps: $5 - sent + returned
      - You keep: received - returned
      - Response format: {'return': <amount>}

      MYTH WRITING:
      You may also be asked to write short myths (stories) on various topics. When writing myths
      , follow the instructions provided in each prompt.

```

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
 The amounts you see may differ slightly from what was actually sent/returned.
 This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent wi ll have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

```

Message 2 role=user
  content:
    Round 1: You are the RECEIVER. the sender sent $3.28, that is 65.6% of their total endowme
    nt. You received $9.84. How much do you return to the sender? (0-9.84)

```

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending 65.6% of their endowment (\$3.28)
- This created \$9.84 for me to allocate
- The sender currently has only \$1.72 left from their original \$5

I want to reward their trust while being fair to both of us. If I return \$5.00:

- Sender's total: $\$1.72 + \$5.00 = \$6.72$ (profit of \$1.72)
- My total: $\$9.84 - \$5.00 = \$4.84$ (profit of \$4.84)

This seems like a reasonable split that rewards the sender's trust while still benefiting from the transaction.

```

{"return": 5.00}

```

Reasoning:

This is Round 1 of a multi-agent trust game where I'm playing as the RECEIVER.

Let me understand what happened:

- The sender had an endowment of \$5
- The sender sent me \$3.28 (which is 65.6% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$3.28 \times 3 = \9.84
- I now need to decide how much to return to the sender (between \$0 and \$9.84)

The sender's payoff will be:

- $\$5$ (initial) - $\$3.28$ (sent) + [amount I return] = $\$1.72$ + [amount I return]

My payoff will be:

- $\$9.84$ (received) - [amount I return]

Let me think about what would be a fair return:

- The sender was quite generous, sending 65.6% of their endowment
- This shows significant trust
- A fair return might be to ensure the sender gets back at least what they sent, or even more to reward their trust
- If I return \$3.28, the sender would have: $\$1.72 + \$3.28 = \$5$ (break even)
- If I return \$5, the sender would have: $\$1.72 + \$5 = \$6.72$ (profit of \$1.72)
- If I return \$6.56 (double what they sent), the sender would have: $\$1.72 + \$6.56 = \$8.28$

A commonly fair approach in trust games is to split the total gains somewhat equitably while rewarding the sender's trust.

Total value created: \$9.84 (what I received)

Original sender investment: \$3.28

Net gain: $\$9.84 - \$3.28 = \$6.56$

One approach: Split the net gain equally

- Sender gets back: $\$3.28 + \$3.28 = \$6.56$
- I would return: \$6.56
- Sender's final: \$8.28
- My final: \$3.28

Another approach: Split all the money equally

- Total pool: \$9.84
- Equal split: \$4.92 each
- I would return: \$4.92
- Sender's final: \$6.64
- My final: \$4.92...

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 999,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T07:51:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
```

```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

I'll start with a moderately trusting approach to see how cooperation works in this game.

By sending \$3, the receiver will get \$9 (3×\$3). If they return a fair share, we can both benefit from the multiplier effect.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return
3. My final earnings = \$5 - sent + returned
4. Receiver's final earnings = (sent × 3) - returned

From a purely rational/selfish perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return nothing, I get \$0
- The optimal outcome for both of us collectively is if I send \$5 (creating \$15 total) and we split it somehow

From a trust/reciprocity perspective:

- Sending money shows trust
- Many people reciprocate trust with trustworthy behavior
- In behavioral economics experiments, people often send substantial amounts and receivers often return fair amounts

Strategic considerations:

- This is round 1 of multiple rounds with randomized pairings
- Building a reputation for cooperation might help in future rounds if people can observe behavior
- However, agent names are hidden, so reputation effects may be limited
- There's also communication noise, which might affect trust dynamics

First round strategy:

- A moderate amount seems reasonable to test cooperation
- Sending \$2-3 would create \$6-9 for the receiver
- If they return even 1/3 of what they receive, I'd break even or profit slightly

I'll send \$3 as a moderately trusting first move. This creates \$9 for the receiver, and if they return \$3-4, we both do reasonably well compared to the \$5 baseline.

```
Usage: {  
  "input_tokens": 500,  
  "output_tokens": 465,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T07:51:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"
```

```
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.37, that is 67.4% of their total endowment. You received \$10.11. How much do you return to the sender? (0-10.11)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.37 (67.4% of their endowment)
- They currently have \$1.63 left (\$5 - \$3.37)
- I received \$10.11 after the 3x multiplication

I want to reciprocate this high level of trust and ensure the sender benefits from their decision to trust me.

If I return \$5:

- The sender ends with: \$1.63 + \$5 = \$6.63 (a profit of \$1.63)
- I keep: \$10.11 - \$5 = \$5.11 (a profit of \$5.11)

This rewards the sender's trust while both of us profit from cooperation.

{'return': 5.0}

Reasoning:

Let me analyze this situation:

1. The sender sent \$3.37 out of their \$5 endowment (67.4%)
2. This was multiplied by 3x, so I received \$10.11
3. I need to decide how much to return (0-10.11)

Key considerations:

- The sender showed significant trust by sending 67.4% of their endowment
- If I return nothing, I keep \$10.11 but break trust
- A fair return might be to ensure the sender gets back more than they sent
- The sender currently has: \$5 - \$3.37 = \$1.63

If I want to be reciprocal and fair:

- The sender gave up \$3.37 to trust me
- They showed high trust (67.4% of endowment)
- A reasonable return would ensure they profit from trusting me

Some options:

1. Return \$5: Sender ends with $\$1.63 + \$5 = \$6.63$ (profit of \$1.63), I keep \$5.11
2. Return \$6: Sender ends with $\$1.63 + \$6 = \$7.63$ (profit of \$2.63), I keep \$4.11
3. Return \$4: Sender ends with $\$1.63 + \$4 = \$5.63$ (profit of \$0.63), I keep \$6.11
4. Return \$3.37: Sender ends with $\$1.63 + \$3.37 = \$5$ (break even), I keep \$6.74

Given that:

- The sender showed high trust (67.4%)
- Note says there may be communication noise
- I want to encourage future cooperation
- A fair split might be approximating equal gains

If I return \$5:

- Sender: started with \$5, sent \$3.37, gets back \$5 = net $\$1.63 + \$5 = \$6.63$ total

...

```
Usage: {  
  "input_tokens": 529,  
  "output_tokens": 694,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T07:52:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance showing trust with managing risk, especially given the communication noise mentioned.

Since this is a multi-round game with rotating partners, establishing a cooperative pattern early could be beneficial. However, I don't know this receiver's tendencies yet.

I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Gives the receiver meaningful funds to work with (tripled)
- Doesn't risk my entire endowment

If I send \$3, the receiver gets \$9. If they return even half of that (\$4.50), we both benefit more than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 and can send \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. There's communication noise mentioned - amounts may differ slightly from what was actually sent/returned

Key considerations:

- If I send \$0, I keep \$5, receiver gets \$0, game ends with no cooperation
- If I send \$5, receiver gets \$15, and their return decision determines my payoff
- This is a multi-round game with 8 agents, randomized pairings
- My reputation/behavior might matter for future rounds, though pairings are randomized

Standard trust game reasoning:

- Sending money shows trust and can enable mutual benefit

- If receiver is rational and self-interested (and this is the last interaction), they should return \$0
- But in multi-round settings, reputation and reciprocity often emerge
- A moderate initial send can signal willingness to cooperate while limiting downside risk

Since this is round 1 and I don't know the receiver's tendencies, I should probably send a moderate amount to:

1. Signal willingness to cooperate
2. Give receiver opportunity to show trustworthiness
3. Not risk my entire endowment

A common strategy is to send about half. If I send \$2.50:

- Receiver gets \$7.50
- If they return half (\$3.75), I end with $\$5 - \$2.50 + \$3.75 = \6.25
- Receiver keeps \$3.75
- Both benefit

Actually, let me reconsider. Since amounts can only be integers based on the format, I should send a whole dollar amount. Let me send \$3:

- I send \$3
- Receiver gets \$9
- If they're fair and return half (\$4.50), I get $\$5 - \$3 + \$4.50 = \6.50

...

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 724,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T07:52:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.29, that is 45.8% of their total endowment. You received \$6.87. How much do you return to the sender? (0-6.87)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.29 (45.8% of their endowment)
- I received \$6.87 after the 3x multiplier
- The sender currently has \$2.71

I want to reciprocate their trust while keeping the exchange mutually beneficial. The sender took a risk by sending nearly half their endowment, so I should return enough to reward that trust.

If I return approximately the same proportion they sent (around 45-46% of what I received), that would be about \$3.15. This would give the sender $\$2.71 + \$3.15 = \$5.86$ total (a gain of \$0.86), and I'd keep \$3.72 (also a gain).

This seems like a fair reciprocation that rewards their trust and could encourage cooperation in future rounds.

{'return': 3.15}

Reasoning:

I'm the receiver in this round. Let me understand the situation:

- The sender sent \$2.29 (which is 45.8% of their \$5 endowment)
- This was multiplied by 3, so I received \$6.87
- I need to decide how much to return (between \$0 and \$6.87)

Let me think about what would be fair and strategic:

1. The sender showed trust by sending almost half their endowment (\$2.29 out of \$5)
2. They kept $\$5 - \$2.29 = \$2.71$
3. I received \$6.87
4. If I return nothing, I keep \$6.87 and the sender ends with \$2.71
5. If I return everything, I keep \$0 and the sender ends with $\$2.71 + \$6.87 = \$9.58$

A common fair approach in trust games is to aim for roughly equal outcomes or to reciprocate the sender's trust proportionally.

Let's consider some options:

Option 1 - Equal split of total gains:

- Total money created: \$5 (sender's endowment) + \$6.87 (what I received) = \$11.87
- But wait, that's not right. The sender started with \$5, sent \$2.29 which became \$6.87
- Total in the system: Sender has \$2.71, I have \$6.87 = \$9.58 total
- Equal split would be \$4.79 each
- For sender to have \$4.79: they need \$4.79 - \$2.71 = \$2.08 returned

Option 2 - Return proportional to what was sent:

- Sender sent 45.8% of their endowment
- I could return a similar proportion of what I received
- 45.8% of \$6.87 = \$3.15

Option 3 - Ensure sender gains from trusting me:

- Sender currently has \$2.71
- If I return enough so they end up better than their initial \$5
- They need at least \$5 - \$2.71 = \$2.29 to break even
- To give them a decent profit, maybe return \$3-3.50 so they end with \$5.71-6.21

Since this is a multi-round game with randomized pairings and communication noise, I should consider how my choice signals trustworthiness. Returning around \$3-3.15 seems reasonable-it reciprocates their generosity while still letting me benefit from the exchange, and it positions me well for future interactions with different partners....

```
Usage: {  
  "input_tokens": 529,  
  "output_tokens": 1076,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T07:52:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this as Round 1, I want to establish a cooperative tone while being somewhat cautious since I don't know my co-player yet.

Sending a moderate amount can:

- Signal trust and willingness to cooperate
- Create a larger pool for mutual benefit (tripled to \$9)
- Allow the receiver meaningful choice in how much to return

If the receiver returns even half of what they receive, we both end up better than the initial \$5.

I'll send **\$3**.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- There's communication noise, so amounts may differ
- This is a multi-round game with 8 agents
- Pairings are randomized each round

Classic trust game setup. Let me think about optimal strategy:

1. If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
2. If I send \$5: Receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 (better than \$5). But they might return less or nothing.
3. If I send a moderate amount (e.g., \$3): Receiver gets \$9. If they return \$4-5, I come out ahead.

Since this is:

- Round 1 (establishing reputation/reciprocity patterns)
- Multi-round game (future interactions matter)
- Anonymous agents (can't track specific individuals)

I should consider:

- Building trust/reciprocity norms early
- Signaling cooperation
- But also being somewhat cautious since I don't know the other agents

A common strategy is to send a moderate-to-high amount in early rounds to:

1. Signal trustworthiness
2. Encourage reciprocity
3. Maximize potential gains if receiver cooperates

Given the noise factor and uncertainty, I'll send \$3. This shows goodwill without being maximally vulnerable, and gives the receiver \$9 to work with. If they return \$4-5, we both benefit.

Actually, let me reconsider. In trust games, sending higher amounts often generates better returns because:

- It demonstrates trust
- Creates larger pie to split
- Social norms typically lead to proportional returns

Comparing the math: sending \$4 leaves me with \$1 while giving the receiver \$12. A 50% return (\$6) nets me \$7 total, while a 33% return (\$4) just breaks me even. With \$3 sent instead, I'd have \$2 cushion and need only a 50% return (\$4.50) to reach \$6.50. The higher send amount creates more upside if reciprocity holds, so I'll go with \$4....

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 814,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T07:53:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
```

```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.24, that is 44.8% of their total endowment. You received \$6.720000000000001. How much do you return to the sender? (0-6.720000000000001)

Assistant output:

Looking at this situation:

- The sender showed moderate trust by sending \$2.24 (44.8% of their endowment)
- They currently have \$2.76 remaining
- I received \$6.72 after the 3x multiplication

To reciprocate their trust fairly, I should return an amount that:

1. Ensures they profit from trusting me (end up with more than their original \$5)
2. Rewards their cooperative gesture
3. Still leaves me with a reasonable gain

If I return \$3.00:

- Sender ends with: \$2.76 + \$3.00 = \$5.76 (gain of \$0.76)
- I keep: \$6.72 - \$3.00 = \$3.72 (gain of \$3.72)

This seems like a fair reciprocal response that rewards trust while maintaining balanced incentives for future rounds.

```
{"return": 3.00}
```

Reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.24 out of their \$5 endowment (44.8%)
3. This was multiplied by 3x, so I received \$6.72 (technically \$6.720000000000001 due to floating point)
4. I need to decide how much to return (between \$0 and \$6.72)

Key considerations:

- The sender showed moderate trust by sending 44.8% of their endowment
- If I return nothing, I keep all \$6.72 and the sender gets only \$5 - \$2.24 = \$2.76
- If I return a fair amount, I should consider what would make this equitable

Common approaches in trust games:

1. Equal split of total gains: Total in the system is \$5 + \$6.72 = \$11.72. Equal split would be \$5.86 each. The sender currently has \$2.76, so I'd return \$5.86 - \$2.76 = \$3.10
2. Return the same proportion they sent: They sent 44.8%, so I could return 44.8% of \$6.72 = \$3.01
3. Match their investment: Return \$2.24 so we both end up with gains from the multiplication
4. Return enough so they profit: If I return \$3, they end up with \$2.76 + \$3 = \$5.76 (a gain of \$0.76), and I keep \$3.72 (a gain of \$3.72)

The sender took a risk by sending \$2.24. A fair return should:

- At minimum, not make them worse off than keeping everything (\$5)
- Ideally, reward their trust
- Be reciprocal to their gesture

Let me calculate a few scenarios:

- If I return \$3.00: Sender gets \$2.76 + \$3.00 = \$5.76, I keep \$3.72
- If I return \$3.50: Sender gets \$2.76 + \$3.50 = \$6.26, I keep \$3.22
- If I return \$4.00: Sender gets \$2.76 + \$4.00 = \$6.76, I keep \$2.72

A return of \$3.00-\$3.50 seems balanced-it rewards their trust while still giving me a meaningful gain. I'll return \$3.00, which nets them \$5.76 (a 15.2% gain) and leaves me with \$3.72...

```
Usage: {  
  "input_tokens": 537,  
  "output_tokens": 1019,  
  "reasoning_tokens": 0  
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3.5; received \$10.5; Agent_1 returned \$6; payoffs Agent_2=\$7.5, Agent_1=\$4.5; noise/visible: sent visible=\$3.89, received visible=\$11.67, returned visible=\$6.94
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 10.5, "min": 0, "raw": 6.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad_2: Agent_4 sent \$3; received \$9; Agent_3 returned \$5.5; payoffs Agent_4=\$7.5, Agent_3=\$3.5; noise/visible: sent visible=\$3.8, received visible=\$11.4, returned visible=\$6.21
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_3: Agent_6 sent \$3; received \$9; Agent_5 returned \$4.5; payoffs Agent_6=\$6.5, Agent_5=\$4.5; noise/visible: sent visible=\$2.99, received visible=\$8.97, returned visible=\$4.71
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

```

dyad_4: Agent_8 sent $3; received $9; Agent_7 returned $6; payoffs Agent_8=$8, Agent_7=$3;
noise/visible: sent visible=$3.7, received visible=$11.1, returned visible=$5.07
  Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw":
    6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 11.5,
  "Agent_2": 11.5,
  "Agent_3": 10.5,
  "Agent_4": 11.5,
  "Agent_5": 9.65,
  "Agent_6": 12.35,
  "Agent_7": 8.0,
  "Agent_8": 14.0
}
Visible balances after round: {
  "Agent_1": 13.61,
  "Agent_2": 13.280000000000001,
  "Agent_3": 13.409999999999998,
  "Agent_4": 13.32,
  "Agent_5": 9.88,
  "Agent_6": 10.43,
  "Agent_7": 9.3,
  "Agent_8": 10.790000000000001
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T07:53:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.28, that is 65.6% of their total endowment. You received \$9.84. How much do you return to the sender? (0-9.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 65.6% of their endowment (\$3.28)
- This created \$9.84 for me to allocate
- The sender currently has only \$1.72 left from their original \$5

I want to reward their trust while being fair to both of us. If I return \$5.00:

- Sender's total: $\$1.72 + \$5.00 = \$6.72$ (profit of \$1.72)
- My total: $\$9.84 - \$5.00 = \$4.84$ (profit of \$4.84)

This seems like a reasonable split that rewards the sender's trust while still benefiting from the transaction.

{"return": 5.00}

reasoning:

This is Round 1 of a multi-agent trust game where I'm playing as the RECEIVER.

Let me understand what happened:

- The sender had an endowment of \$5
- The sender sent me \$3.28 (which is 65.6% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$3.28 \times 3 = \9.84
- I now need to decide how much to return to the sender (between \$0 and \$9.84)

The sender's payoff will be:

- $\$5$ (initial) - $\$3.28$ (sent) + [amount I return] = $\$1.72 + [\text{amount I return}]$

My payoff will be:

- $\$9.84$ (received) - [amount I return]

Let me think about what would be a fair return:

- The sender was quite generous, sending 65.6% of their endowment
- This shows significant trust
- A fair return might be to ensure the sender gets back at least what they sent, or even more

ore to reward their trust

- If I return \$3.28, the sender would have: $\$1.72 + \$3.28 = \$5$ (break even)
- If I return \$5, the sender would have: $\$1.72 + \$5 = \$6.72$ (profit of \$1.72)
- If I return \$6.56 (double what they sent), the sender would have: $\$1.72 + \$6.56 = \$8.28$

A commonly fair approach in trust games is to split the total gains somewhat equitably while rewarding the sender's trust.

Total value created: \$9.84 (what I received)
Original sender investment: \$3.28
Net gain: $\$9.84 - \$3.28 = \$6.56$

One approach: Split the net gain equally

- Sender gets back: $\$3.28 + \$3.28 = \$6.56$
- I would return: \$6.56
- Sender's final: \$8.28
- My final: \$3.28

Another approach: Split all the money equally

- Total pool: \$9.84
- Equal split: \$4.92 each
- I would return: \$4.92
- Sender's final: \$6.64
- My final: \$4.92...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 999,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.

Your total visible earnings across all rounds are \$4.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

- My co-player showed trust in Round 1 by sending \$3.0 (60% of their endowment)
- They received a reasonable return of \$5.94
- This suggests they're willing to cooperate

I want to reciprocate their trust and continue building a cooperative relationship. I'll send a similar proportion to what they sent.

```
{"send": 3.50}
```

Reasoning:

Let me analyze this situation carefully:

Round 1 Summary:

- I was the RECEIVER
- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- I returned \$5.00
- My earnings: \$4.84

Now looking at the visible history shown to me:

- My co-player's last game shows: as SENDER, sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94

Wait, this is interesting. There's communication noise mentioned in the instructions. Let me think about what actually happened vs what was observed:

What I experienced in Round 1:

- Sender sent \$3.28
- I received \$9.84
- I returned \$5.00

What my co-player (the sender) saw:

- They sent \$3.0 (but actually sent \$3.28?)
- It became \$9.0 (but actually became \$9.84?)
- They saw \$5.94 returned (but I actually returned \$5.00?)
- Their visible payoff was \$7.94

Let me check the math on their side:

- If they sent \$3.0 and saw \$5.94 returned
- Their payoff would be: $\$5 - \$3 + \$5.94 = \7.94 ?

So from their perspective:

- They sent \$3.0 (66.7% of their total)
- They got back \$5.94 (66% of the \$9.0 that was created)
- This seems like a decent return to them

Now I'm the SENDER in Round 2:

- My co-player showed trust by sending most of their endowment in Round 1
- They received what appeared to them as a good return (66% of the multiplied amount)
- This suggests they might be willing to reciprocate if I show trust
- Since they sent roughly two-thirds of their endowment, I should match that level of generosity to build on the cooperative dynamic we...

```
Usage: {  
  "input_tokens": 839,  
  "output_tokens": 749,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T07:53:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't know my co-player yet.

****My reasoning:****

- Sending \$0 is safe but creates no mutual benefit
- Sending everything (\$5) maximizes potential gains but is very risky with an unknown partner
- A moderate amount signals cooperation while managing risk

I'll send ****\$3****, which:

- Gets multiplied to \$9 for the receiver (creating good potential for returns)
- Keeps \$2 as a safety buffer for me
- Shows willingness to trust and cooperate
- If they return even \$3-4, we both benefit

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me analyze the situation:

1. I have \$5 to start
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ from what was actually sent/returned
5. This is a multi-agent setting with 8 agents, randomized pairings each round
6. I don't know who my co-player is (names are hidden)

Key considerations:

- If I send \$0, I guarantee keeping \$5 but miss the opportunity for mutual gain
- If I send the full \$5, the receiver gets \$15 and could return a substantial amount
- The optimal outcome for both of us combined is if I send everything (creating \$15 total instead of \$5)
- However, the receiver could keep everything and I'd end up with \$0
- There's also communication noise which adds uncertainty

Since this is the first round and I have no information about my co-player:

- Sending some amount signals trust and cooperation
- A moderate amount (like \$2-3) balances risk and potential reward
- Sending \$5 shows maximum trust but is risky with an unknown partner
- The communication noise means even if they want to reciprocate fairly, there might be misunderstandings

For a first round with an unknown partner, I'll send a moderate amount to signal cooperation while managing risk. I'll send \$3:

- This creates \$9 for the receiver (good potential for returns)
- I keep \$2 as insurance
- If they return just \$3-4, I'll break even or profit
- It signals willingness to cooperate without being reckless

Let me go with \$3 as a balanced first move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$7.94.

This round, you are the RECEIVER against your current co-player. They sent you \$3.89, so you received \$11.67. How much do you return? (0-11.67)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.28 (became \$9.84 for them)
- They returned \$5.00 (about 51% of what they received)
- This shows they're cooperative and fair

****Round 2 situation:****

- They sent \$3.89 (78% of their endowment - very generous!)
- I received \$11.67
- They've shown trust and cooperation

I should reciprocate their cooperative behavior. Since they returned about 51% last round and sent generously this round, I'll return a similar proportion:

~51% of \$11.67 = approximately \$6

This rewards their trust while maintaining fairness.

```
'''json
{"return": 6}
'''
```

Reasoning:

Let me analyze this situation:

From Round 1:

- I sent \$3 (but they saw \$3.28 due to communication noise)
- It became \$9.84 for them
- They returned \$5.0
- Their visible payoff was \$4.84

Now in Round 2:

- They sent \$3.89 (I received \$11.67 after 3x multiplier)
- I need to decide how much to return

Let me think about their behavior:

- They saw I sent \$3.28, which became \$9.84
- They returned \$5.0 from that \$9.84
- That's about 51% of what they received

They were fairly generous, returning about half. This suggests they're cooperative.

Now they sent \$3.89 out of \$5, which is about 78% of their endowment. That's quite generous and trusting.

If I want to reciprocate their cooperative behavior:

- They returned about 51% last round
- But they also sent a lot this round (78% of endowment)

I should return a reasonable amount that:

1. Rewards their trust and generosity
2. Maintains cooperation for future rounds
3. Is fair given they returned \$5 from \$9.84 last round

Let me calculate some options:

- If I return 50% of \$11.67 = \$5.84
- If I return a bit more generously, around \$6-7

Given that:

- They were generous (sent \$3.89)
- They returned \$5 last round (about 51%)
- I want to encourage continued cooperation

I'll return around \$6. This gives them:

- They sent: \$3.89
- They get back: \$6
- Net for them: \$5 - \$3.89 + \$6 = \$7.11

...

```
Usage: {
  "input_tokens": 800,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T07:53:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
```

```

"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.37, that is 67.4% of their total endowment. You received \$10.11. How much do you return to the sender? (0-10.11)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.37 (67.4% of their endowment)
- They currently have \$1.63 left (\$5 - \$3.37)
- I received \$10.11 after the 3x multiplication

I want to reciprocate this high level of trust and ensure the sender benefits from their decision to trust me.

If I return \$5:

- The sender ends with: \$1.63 + \$5 = \$6.63 (a profit of \$1.63)
- I keep: \$10.11 - \$5 = \$5.11 (a profit of \$5.11)

This rewards the sender's trust while both of us profit from cooperation.

```
{'return': 5.0}
```

reasoning:

Let me analyze this situation:

1. The sender sent \$3.37 out of their \$5 endowment (67.4%)
2. This was multiplied by 3x, so I received \$10.11
3. I need to decide how much to return (0-10.11)

Key considerations:

- The sender showed significant trust by sending 67.4% of their endowment
- If I return nothing, I keep \$10.11 but break trust
- A fair return might be to ensure the sender gets back more than they sent
- The sender currently has: $\$5 - \$3.37 = \$1.63$

If I want to be reciprocal and fair:

- The sender gave up \$3.37 to trust me
- They showed high trust (67.4% of endowment)
- A reasonable return would ensure they profit from trusting me

Some options:

1. Return \$5: Sender ends with $\$1.63 + \$5 = \$6.63$ (profit of \$1.63), I keep \$5.11
2. Return \$6: Sender ends with $\$1.63 + \$6 = \$7.63$ (profit of \$2.63), I keep \$4.11
3. Return \$4: Sender ends with $\$1.63 + \$4 = \$5.63$ (profit of \$0.63), I keep \$6.11
4. Return \$3.37: Sender ends with $\$1.63 + \$3.37 = \$5$ (break even), I keep \$6.74

Given that:

- The sender showed high trust (67.4%)
- Note says there may be communication noise
- I want to encourage future cooperation
- A fair split might be approximating equal gains

If I return \$5:

- Sender: started with \$5, sent \$3.37, gets back \$5 = net $\$1.63 + \$5 = \$6.63$ total
- ...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 694,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.

Your total visible earnings across all rounds are \$5.109999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing significant trust
- They saw \$5.51 returned, which gave them a positive payoff
- I profited from their trust in Round 1

I want to reciprocate their cooperative behavior and maintain the trust relationship. Sending a similar proportion will signal that I'm willing to cooperate.

```
{'send': 3.0}
```

Reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$3.37 (67.4% of their endowment)
- I received \$10.11
- I returned \$5.0
- I kept \$5.11

Now in Round 2, the visible history shows:

- My co-player's last game (Round 1 against me) as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51

Wait, there's communication noise! The actual amounts differ from what each player sees:

- I actually received \$10.11 (from their \$3.37 sent)
- I actually returned \$5.0
- But they see that they sent \$3.0, it became \$9.0, and saw \$5.51 returned

So there's noise in both directions. This is interesting - the sender sees they sent \$3.0 (vs actual \$3.37) and saw \$5.51 returned (vs actual \$5.0 I returned).

Now I'm the SENDER in Round 2 against the same co-player.

My visible earnings are \$5.11 (which matches what I calculated: $\$10.11 - \$5.0 = \$5.11$).

Strategy considerations:

1. My co-player sent a high amount in Round 1 (either \$3.0 or \$3.37 depending on what was real)
2. They saw a decent return (approximately 50% of what they sent, or more)
3. This suggests they're cooperative

I should signal continued cooperation by sending a similar proportion. Since they sent 67.4% in Round 1, I could send a similar high proportion.

Given the communication noise, I should send a clear cooperative signal. Sending around \$3-3.5 would show I'm reciprocating their trust.

Let me send \$3.0 to match what they think they sent (according to the visible history).

...

```
Usage: {
  "input_tokens": 848,
  "output_tokens": 639,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T07:54:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderately trusting approach to see how cooperation works in this game.

By sending \$3, the receiver will get \$9 (3×\$3). If they return a fair share, we can both benefit from the multiplier effect.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return
3. My final earnings = \$5 - sent + returned
4. Receiver's final earnings = (sent × 3) - returned

From a purely rational/selfish perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return nothing, I get \$0
- The optimal outcome for both of us collectively is if I send \$5 (creating \$15 total) and we split it somehow

From a trust/reciprocity perspective:

- Sending money shows trust
- Many people reciprocate trust with trustworthy behavior
- In behavioral economics experiments, people often send substantial amounts and receivers often return fair amounts

Strategic considerations:

- This is round 1 of multiple rounds with randomized pairings
- Building a reputation for cooperation might help in future rounds if people can observe behavior
- However, agent names are hidden, so reputation effects may be limited
- There's also communication noise, which might affect trust dynamics

First round strategy:

- A moderate amount seems reasonable to test cooperation
- Sending \$2-3 would create \$6-9 for the receiver
- If they return even 1/3 of what they receive, I'd break even or profit slightly

I'll send \$3 as a moderately trusting first move. This creates \$9 for the receiver, and if they return \$3-4, we both do reasonably well compared to the \$5 baseline.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history, my co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. They've demonstrated trustworthy behavior.

Now they've sent me \$3.8, giving me \$11.40. I'll return a fair amount to encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

Reasoning:

Let me analyze the situation:

1. In Round 1, I sent \$3 (though it appeared as \$3.37 to my co-player due to noise)
2. They received \$10.11 (3x the noisy amount) and returned \$5
3. I ended up with \$5 - \$3 + \$5 = \$7 (though my visible earnings show \$7.51, likely due to noise)

Now in Round 2:

- My co-player sent me \$3.8 (this might be noisy, but this is what I see)
- I receive \$11.40 (3 × \$3.8)
- I need to decide how much to return

The co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. That's roughly half of what they received, which is quite fair and cooperative.

Since they were cooperative and generous in their return, I should reciprocate fairly. A reasonable approach would be:

- They sent \$3.8 out of their \$5, keeping \$1.2
- I received \$11.40
- If I return about half (around \$5-6), they would end up with roughly \$6.2-7.2 total
- I would keep roughly \$5.40-6.40

Let me return \$5.5, which:

- Gives them: $\$5 - \$3.8 + \$5.5 = \6.7 total
- Gives me: $\$11.40 - \$5.5 = \$5.9$ total

This seems fair as it rewards their trust and cooperation, and both of us end up better than the initial \$5.

```
Usage: {
  "input_tokens": 726,
  "output_tokens": 482,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T07:54:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.29, that is 45.8% of their total endowment. You received \$6.87. How much do you return to the sender? (0-6.87)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.29 (45.8% of their endowment)
- I received \$6.87 after the 3x multiplier
- The sender currently has \$2.71

I want to reciprocate their trust while keeping the exchange mutually beneficial. The sender took a risk by sending nearly half their endowment, so I should return enough to reward that trust.

If I return approximately the same proportion they sent (around 45-46% of what I received), that would be about \$3.15. This would give the sender $\$2.71 + \$3.15 = \$5.86$ total (a gain of \$0.86), and I'd keep \$3.72 (also a gain).

This seems like a fair reciprocation that rewards their trust and could encourage cooperation in future rounds.

{'return': 3.15}

reasoning:

I'm the receiver in this round. Let me understand the situation:

- The sender sent \$2.29 (which is 45.8% of their \$5 endowment)
- This was multiplied by 3, so I received \$6.87
- I need to decide how much to return (between \$0 and \$6.87)

Let me think about what would be fair and strategic:

1. The sender showed trust by sending almost half their endowment (\$2.29 out of \$5)
2. They kept $\$5 - \$2.29 = \$2.71$
3. I received \$6.87
4. If I return nothing, I keep \$6.87 and the sender ends with \$2.71
5. If I return everything, I keep \$0 and the sender ends with $\$2.71 + \$6.87 = \$9.58$

A common fair approach in trust games is to aim for roughly equal outcomes or to reciprocate the sender's trust proportionally.

Let's consider some options:

Option 1 - Equal split of total gains:

- Total money created: $\$5$ (sender's endowment) + $\$6.87$ (what I received) = $\$11.87$
- But wait, that's not right. The sender started with \$5, sent \$2.29 which became \$6.87

- Total in the system: Sender has \$2.71, I have \$6.87 = \$9.58 total
- Equal split would be \$4.79 each
- For sender to have \$4.79: they need \$4.79 - \$2.71 = \$2.08 returned

Option 2 - Return proportional to what was sent:

- Sender sent 45.8% of their endowment
- I could return a similar proportion of what I received
- 45.8% of \$6.87 = \$3.15

Option 3 - Ensure sender gains from trusting me:

- Sender currently has \$2.71
- If I return enough so they end up better than their initial \$5
- They need at least \$5 - \$2.71 = \$2.29 to break even
- To give them a decent profit, maybe return \$3-3.50 so they end with \$5.71-6.21

Since this is a multi-round game with randomized pairings and communication noise, I should consider how my choice signals trustworthiness. Returning around \$3-3.15 seems reasonable-it reciprocates their generosity while still letting me benefit from the exchange, and it positions me well for future interactions with different partners....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1076,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.

Your total visible earnings across all rounds are \$3.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They received \$3.41 back, making a profit of \$0.41
- They demonstrated significant trust

As the SENDER now, I want to reciprocate their trust level. They sent 60% of their endowment, which was quite generous. I should match or slightly exceed this level of trust to maintain cooperation.

Sending \$3.0 would match their previous contribution and signal that I'm willing to cooperate at the same trust level they showed.

```
{'send': 3.0}
```

Reasoning:

Let me analyze this situation:

From Round 1:

- I was the RECEIVER
- The sender sent \$2.29 (45.8% of their endowment)
- I received \$6.87
- I returned \$3.15

But now the visible history shows something different:

- It says my co-player (as SENDER in Round 1) sent \$3.0, which became \$9.0
- They saw \$3.41 returned
- Their visible payoff was \$5.41

This is the communication noise mentioned in the instructions! What actually happened vs what they saw:

- Actual: They sent \$2.29, I received \$6.87, I returned \$3.15

- What they saw: They sent \$3.0, became \$9.0, saw \$3.41 returned

So from their perspective, they sent \$3.0 and got back \$3.41. That's a return rate of about 38 % of what I received ($3.41/9.0$).

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of endowment)
- Got back \$3.41 (38% of \$9.0)
- Made a profit of \$0.41

They showed significant trust by sending 60% (from their perspective). They got back a reasonable amount.

As the SENDER now, I should consider:

1. They showed high trust in Round 1 (from their perspective, 60%)
2. They got a modest return
3. I want to continue building trust

Given that they sent 60% from their perspective, I should send a similar or slightly higher amount to show reciprocity and build trust. Sending around \$3.0-\$3.5 would be reasonable.

Let me send \$3.0 to match what they (think they) sent to me.

Actually, let me think more carefully. My actual earnings were \$3.72 in Round 1. I received \$6.87 and returned \$3.15.

...

```
Usage: {
  "input_tokens": 867,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T07:54:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance showing trust with managing risk, especially given the communication noise mentioned.

Since this is a multi-round game with rotating partners, establishing a cooperative pattern early could be beneficial. However, I don't know this receiver's tendencies yet.

I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Gives the receiver meaningful funds to work with (tripled)
- Doesn't risk my entire endowment

If I send \$3, the receiver gets \$9. If they return even half of that (\$4.50), we both benefit more than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 and can send \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. There's communication noise mentioned - amounts may differ slightly from what was actually sent/returned

Key considerations:

- If I send \$0, I keep \$5, receiver gets \$0, game ends with no cooperation
- If I send \$5, receiver gets \$15, and their return decision determines my payoff
- This is a multi-round game with 8 agents, randomized pairings
- My reputation/behavior might matter for future rounds, though pairings are randomized

Standard trust game reasoning:

- Sending money shows trust and can enable mutual benefit
- If receiver is rational and self-interested (and this is the last interaction), they should return \$0
- But in multi-round settings, reputation and reciprocity often emerge
- A moderate initial send can signal willingness to cooperate while limiting downside risk

Since this is round 1 and I don't know the receiver's tendencies, I should probably send a moderate amount to:

1. Signal willingness to cooperate
2. Give receiver opportunity to show trustworthiness
3. Not risk my entire endowment

A common strategy is to send about half. If I send \$2.50:

- Receiver gets \$7.50
- If they return half (\$3.75), I end with $\$5 - \$2.50 + \$3.75 = \6.25
- Receiver keeps \$3.75
- Both benefit

Actually, let me reconsider. Since amounts can only be integers based on the format, I should send a whole dollar amount. Let me send \$3:

- I send \$3
- Receiver gets \$9
- If they're fair and return half (\$4.50), I get $\$5 - \$3 + \$4.50 = \6.50

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 724,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.

Your total visible earnings across all rounds are \$5.41.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

Round 1: My co-player saw \$2.29 sent (due to noise), received \$6.87, and returned \$3.15 (about 46% of what they received). This shows they're willing to reciprocate.

Now they've sent me \$2.99, giving me \$8.97 total.

To maintain a cooperative relationship and signal trustworthiness, I'll return approximately half of what I received, which is fair and encourages future cooperation.

```
'''json
{"return": 4.50}
'''
```

Reasoning:

Let me understand what happened:

Round 1: I was the SENDER

- I sent \$3
- Due to communication noise, my co-player SAW \$2.29 sent (instead of \$3)

- It became \$6.87 (which is $\$2.29 \times 3$)
- They returned \$3.15
- Their visible payoff was \$3.72 (which is $\$6.87 - \3.15)
- My visible earnings are \$5.41

Let me calculate my Round 1 payoff:

- I started with \$5
- I sent \$3
- I received back... well, they returned \$3.15
- So I should have: $\$5 - \$3 + \$3.15 = \5.15
- But my "visible earnings" are \$5.41

Hmm, this is confusing. Maybe the communication noise affected what I received too? Or maybe "visible earnings" includes some noise?

Actually, let me recalculate. If they saw \$2.29 sent and returned \$3.15:

- From their perspective: they received \$6.87, returned \$3.15, kept \$3.72
- From my perspective: I sent \$3, but what did I actually receive back?

The problem states "visible payoff" and "visible earnings" which suggests everything we see might have noise.

Now for Round 2:

- I am the RECEIVER
- They sent me \$2.99 (this is what I see after noise)
- I received \$8.97 (which is $\$2.99 \times 3$)
- I need to decide how much to return (0-8.97)

From Round 1, my co-player:

- Saw \$2.29 sent (noise-affected)
- Received \$6.87
- Returned \$3.15 (about 46% of what they received)

This shows they were somewhat cooperative, returning close to half.

They sent \$2.99, which is close to their original \$3 send, suggesting they're being fairly consistent. If I return roughly half of what I received-around \$4.48-\$4.50-I'd match their reciprocity level. I could return slightly more to encourage continued cooperation, or slightly less to protect my earnings. Let me think through the payoffs: if they sent \$2.99 and kept \$2.01, returning \$4.50 would give them \$6.51 total, which seems fair given their initial cooperation.

Actually, they returned 46% in the first round despite seeing a smaller amount due to noise. That's a strong signal of trust...

```
Usage: {
  "input_tokens": 798,
  "output_tokens": 987,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T07:54:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
}
```

```

"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.24, that is 44.8% of their total endowment. You received \$6.720000000000001. How much do you return to the sender? (0-6.720000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.24 (44.8% of their endowment)
- They currently have \$2.76 remaining
- I received \$6.72 after the 3x multiplication

To reciprocate their trust fairly, I should return an amount that:

1. Ensures they profit from trusting me (end up with more than their original \$5)
2. Rewards their cooperative gesture
3. Still leaves me with a reasonable gain

If I return \$3.00:

- Sender ends with: $\$2.76 + \$3.00 = \$5.76$ (gain of \$0.76)
- I keep: $\$6.72 - \$3.00 = \$3.72$ (gain of \$3.72)

This seems like a fair reciprocal response that rewards trust while maintaining balanced incentives for future rounds.

```
{"return": 3.00}
```

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.24 out of their \$5 endowment (44.8%)
3. This was multiplied by 3x, so I received \$6.72 (technically \$6.720000000000001 due to floating point)
4. I need to decide how much to return (between \$0 and \$6.72)

Key considerations:

- The sender showed moderate trust by sending 44.8% of their endowment
- If I return nothing, I keep all \$6.72 and the sender gets only \$5 - \$2.24 = \$2.76
- If I return a fair amount, I should consider what would make this equitable

Common approaches in trust games:

1. Equal split of total gains: Total in the system is $\$5 + \$6.72 = \$11.72$. Equal split would be \$5.86 each. The sender currently has \$2.76, so I'd return $\$5.86 - \$2.76 = \$3.10$
2. Return the same proportion they sent: They sent 44.8%, so I could return 44.8% of \$6.72 = \$3.01
3. Match their investment: Return \$2.24 so we both end up with gains from the multiplication
4. Return enough so they profit: If I return \$3, they end up with $\$2.76 + \$3 = \$5.76$ (a gain of \$0.76), and I keep \$3.72 (a gain of \$3.72)

The sender took a risk by sending \$2.24. A fair return should:

- At minimum, not make them worse off than keeping everything (\$5)
- Ideally, reward their trust
- Be reciprocal to their gesture

Let me calculate a few scenarios:

- If I return \$3.00: Sender gets $\$2.76 + \$3.00 = \$5.76$, I keep \$3.72
- If I return \$3.50: Sender gets $\$2.76 + \$3.50 = \$6.26$, I keep \$3.22
- If I return \$4.00: Sender gets $\$2.76 + \$4.00 = \$6.76$, I keep \$2.72

A return of \$3.00-\$3.50 seems balanced-it rewards their trust while still giving me a meaningful gain. I'll return \$3.00, which nets them \$5.76 (a 15.2% gain) and leaves me with \$3.72....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.

Your total visible earnings across all rounds are \$3.7200000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw only \$2.2 returned (though due to communication noise, I actually returned \$3.00)

- From their perspective, they may feel the cooperation wasn't fully reciprocated

As the sender now, I want to:

1. Signal continued willingness to cooperate
2. Match their previous level of trust
3. Maintain the cooperative dynamic despite any noise in the previous round

I'll send the same amount they sent in Round 1 to demonstrate reciprocal trust.

```
{"send": 3.0}
```

Reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$2.24 (44.8% of their endowment)
- I received \$6.72
- I returned \$3.00
- Sender ended with $\$2.76 + \$3.00 = \$5.76$
- I kept \$3.72

But the visible history shows something different from my co-player's perspective:

- Round 1: They were SENDER, sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2

Wait, this is confusing. Let me re-read the prompt about communication noise.

"NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design."

So what happened:

- From my perspective: I received notification that sender sent \$2.24, I received \$6.72, I returned \$3.00
- From their perspective: They sent \$3.0, it became \$9.0, they saw \$2.2 returned, and got payoff \$4.2

Let me verify the math from their perspective:

- They sent \$3.0
- They had $\$5.0 - \$3.0 = \$2.0$ left
- They saw \$2.2 returned
- Their payoff: $\$2.0 + \$2.2 = \$4.2$?

So from their perspective, they sent \$3.0 and only got back \$2.2, which is less than they sent. They might perceive this as me being untrustworthy or uncooperative.

However, from MY perspective, I actually returned \$3.00, which I thought was fair and generous given I received \$6.72.

This is the communication noise at work. They think I returned less than I actually did.

Now I'm the sender in Round 2, and they likely view me as less trustworthy due to the noise distorting their perception of my return. They sent 60% of their endowment initially, showing genuine trust, but the miscommunication means they'll probably be more cautious. I need to decide whether sending a substantial amount will help rebuild trust or just get exploited if they reciprocate poorly. A moderate send might signal cooperation without overcommitting, especially since they may interpret my actions through the lens of what they think happened last round....

```
Usage: {  
  "input_tokens": 890,  
  "output_tokens": 922,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T07:55:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,
```

```

"opponent_id": "Agent_8",
"pairing": {
  "agents": [
    "Agent_8",
    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I want to establish a cooperative tone while being somewhat cautious since I don't know my co-player yet.

Sending a moderate amount can:

- Signal trust and willingness to cooperate
- Create a larger pool for mutual benefit (tripled to \$9)
- Allow the receiver meaningful choice in how much to return

If the receiver returns even half of what they receive, we both end up better than the initial \$5.

I'll send **\$3**.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- There's communication noise, so amounts may differ
- This is a multi-round game with 8 agents
- Pairings are randomized each round

Classic trust game setup. Let me think about optimal strategy:

1. If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
2. If I send \$5: Receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 (better than \$5). But they might return less or nothing.
3. If I send a moderate amount (e.g., \$3): Receiver gets \$9. If they return \$4-5, I come out ahead.

Since this is:

- Round 1 (establishing reputation/reciprocity patterns)
- Multi-round game (future interactions matter)
- Anonymous agents (can't track specific individuals)

I should consider:

- Building trust/reciprocity norms early
- Signaling cooperation
- But also being somewhat cautious since I don't know the other agents

A common strategy is to send a moderate-to-high amount in early rounds to:

1. Signal trustworthiness
2. Encourage reciprocity
3. Maximize potential gains if receiver cooperates

Given the noise factor and uncertainty, I'll send \$3. This shows goodwill without being maximally vulnerable, and gives the receiver \$9 to work with. If they return \$4-5, we both benefit.

Actually, let me reconsider. In trust games, sending higher amounts often generates better returns because:

- It demonstrates trust
- Creates larger pie to split
- Social norms typically lead to proportional returns

Comparing the math: sending \$4 leaves me with \$1 while giving the receiver \$12. A 50% return (\$6) nets me \$7 total, while a 33% return (\$4) just breaks me even. With \$3 sent instead, I'd have \$2 cushion and need only a 50% return (\$4.50) to reach \$6.50. The higher send amount creates more upside if reciprocity holds, so I'll go with \$4....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 814,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.

Your total visible earnings across all rounds are \$4.2.

This round, you are the RECEIVER against your current co-player. They sent you \$3.7, so you received \$11.100000000000001. How much do you return? (0-11.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

****Round 1 analysis:****

- My co-player saw \$2.24 sent (due to noise), which became \$6.72
- They returned \$3.0 (about 45% of what they received)
- This shows reasonable cooperative behavior

****Round 2 (current):****

- They sent me \$3.7 (even more than what they saw in Round 1)
- I received \$11.10
- They're showing increased trust

To maintain and encourage cooperation, I should return a fair amount. Returning about half of what I received would give them:

- They kept: \$5 - \$3.7 = \$1.30
- Plus returned amount
- A good return on their trust

I'll return ****\$6.00**** (about 54% of what I received), giving them a total of \$7.30 - a solid return that rewards their increased trust.

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Let me analyze this situation carefully.

Round 1 recap from my perspective:

- I sent \$3
- Due to communication noise, my co-player saw \$2.24 sent (which became \$6.72 after tripling)
- They returned \$3.0
- Their visible payoff was \$3.72 (received \$6.72, returned \$3.0)
- My total visible earnings are \$4.2

Let me calculate what actually happened in Round 1:

- I sent \$3, so I had \$5 - \$3 = \$2 left
- The receiver got $\$3 \times 3 = \9 (but saw \$6.72 due to noise)
- They returned \$3.0
- So I ended with $\$2 + \$3 = \$5$
- But my visible earnings show \$4.2, which suggests the noise affected what I saw returned too?

Actually, wait. Let me reconsider:

- If my visible earnings are \$4.2 total
- I started with \$5, sent \$3 (keeping \$2)
- If I received back something that made my total \$4.2, then I received $\$4.2 - \$2 = \$2.2$ back

So the noise worked both ways:

- I sent \$3, they saw \$2.24
- They returned \$3.0, but I saw (and actually got) something that gave me \$4.2 total

Now for Round 2:

- I'm the RECEIVER
- My co-player sent me \$3.7 (according to what I see)
- This became \$11.10 after tripling
- I need to decide how much to return

My co-player showed willingness to return a reasonable amount in Round 1 (they returned \$3.0 out of \$6.72 they saw, which is about 45%). This is fairly cooperative behavior.

I should return a fair amount to maintain cooperation. Given that:

- They sent \$3.7 and I received \$11.10
- A 50% return would be \$5.55, giving them \$1.3 (kept) + \$5.55 (returned) = \$6.85 total
- This matches their cooperative gesture from Round 1 and encourages continued trust

I'll return \$5 to keep things balanced....

```
Usage: {  
  "input_tokens": 784,  
  "output_tokens": 1051,  
  "reasoning_tokens": 0  
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$4; received \$12; Agent_2 returned \$6.5; payoffs Agent_1=\$7.5, Agent_2=\$5.5; noise/visible: sent visible=\$3.71, received visible=\$11.13, returned visible=\$5.66
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw": 6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_3 sent \$3; received \$9; Agent_4 returned \$3; payoffs Agent_3=\$5, Agent_4=\$6; noise/visible: sent visible=\$2.05, received visible=\$6.15, returned visible=\$2.32
Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$4.05; payoffs Agent_5=\$6.05, Agent_6=\$4.95; noise/visible: sent visible=\$2.7, received visible=\$8.1, returned visible=\$3.91
Action validation: {"returned": {"amount": 4.05, "clamped": false, "max": 9.0, "min": 0, "raw": 4.05}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3.5; received \$10.5; Agent_8 returned \$5; payoffs Agent_7=\$6.5, Agent_8=\$5.5; noise/visible: sent visible=\$3.3, received visible=\$9.9, returned visible=\$5.32
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 10.5, "min": 0, "raw": 5.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent_1": 19.0,
"Agent_2": 17.0,
"Agent_3": 15.5,
"Agent_4": 17.5,
"Agent_5": 15.7,
"Agent_6": 17.3,
"Agent_7": 14.5,
"Agent_8": 19.5

}

Visible balances after round: {

"Agent_1": 20.27,
"Agent_2": 17.91,
"Agent_3": 17.729999999999997,
"Agent_4": 16.47,
"Agent_5": 15.790000000000001,
"Agent_6": 14.48,
"Agent_7": 16.12,
"Agent_8": 15.69

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T07:55:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't know my co-player yet.

****My reasoning:****

- Sending \$0 is safe but creates no mutual benefit
- Sending everything (\$5) maximizes potential gains but is very risky with an unknown partner
- A moderate amount signals cooperation while managing risk

I'll send ****\$3****, which:

- Gets multiplied to \$9 for the receiver (creating good potential for returns)
- Keeps \$2 as a safety buffer for me
- Shows willingness to trust and cooperate
- If they return even \$3-4, we both benefit

'''json

{"send": 3}

'''

reasoning:

This is the first round of a trust game where I'm the sender. Let me analyze the situation:

1. I have \$5 to start
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ from what was actually sent/returned
5. This is a multi-agent setting with 8 agents, randomized pairings each round
6. I don't know who my co-player is (names are hidden)

Key considerations:

- If I send \$0, I guarantee keeping \$5 but miss the opportunity for mutual gain
- If I send the full \$5, the receiver gets \$15 and could return a substantial amount
- The optimal outcome for both of us combined is if I send everything (creating \$15 total instead of \$5)
- However, the receiver could keep everything and I'd end up with \$0
- There's also communication noise which adds uncertainty

Since this is the first round and I have no information about my co-player:

- Sending some amount signals trust and cooperation
- A moderate amount (like \$2-3) balances risk and potential reward
- Sending \$5 shows maximum trust but is risky with an unknown partner
- The communication noise means even if they want to reciprocate fairly, there might be misunderstandings

For a first round with an unknown partner, I'll send a moderate amount to signal cooperation while managing risk. I'll send \$3:

- This creates \$9 for the receiver (good potential for returns)
- I keep \$2 as insurance
- If they return just \$3-4, I'll break even or profit
- It signals willingness to cooperate without being reckless

Let me go with \$3 as a balanced first move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$7.94.

This round, you are the RECEIVER against your current co-player. They sent you \$3.89, so you received \$11.67. How much do you return? (0-11.67)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.28 (became \$9.84 for them)
- They returned \$5.00 (about 51% of what they received)
- This shows they're cooperative and fair

****Round 2 situation:****

- They sent \$3.89 (78% of their endowment - very generous!)
- I received \$11.67
- They've shown trust and cooperation

I should reciprocate their cooperative behavior. Since they returned about 51% last round and sent generously this round, I'll return a similar proportion:

~51% of \$11.67 = approximately \$6

This rewards their trust while maintaining fairness.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

From Round 1:

- I sent \$3 (but they saw \$3.28 due to communication noise)
- It became \$9.84 for them
- They returned \$5.0
- Their visible payoff was \$4.84

Now in Round 2:

- They sent \$3.89 (I received \$11.67 after 3x multiplier)
- I need to decide how much to return

Let me think about their behavior:

- They saw I sent \$3.28, which became \$9.84
- They returned \$5.0 from that \$9.84
- That's about 51% of what they received

They were fairly generous, returning about half. This suggests they're cooperative.

Now they sent \$3.89 out of \$5, which is about 78% of their endowment. That's quite generous and trusting.

If I want to reciprocate their cooperative behavior:

- They returned about 51% last round
- But they also sent a lot this round (78% of endowment)

I should return a reasonable amount that:

1. Rewards their trust and generosity
2. Maintains cooperation for future rounds
3. Is fair given they returned \$5 from \$9.84 last round

Let me calculate some options:

- If I return 50% of \$11.67 = \$5.84
- If I return a bit more generously, around \$6-7

Given that:

- They were generous (sent \$3.89)
- They returned \$5 last round (about 51%)
- I want to encourage continued cooperation

I'll return around \$6. This gives them:

- They sent: \$3.89
- They get back: \$6
- Net for them: $\$5 - \$3.89 + \$6 = \7.11

...

usage:

```
{
  "input_tokens": 800,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.

Your total visible earnings across all rounds are \$13.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00 - very fair and cooperative

****Round 2:****

- They sent \$3.50 (70% of their endowment - very trusting!)
- I returned \$6 (they saw \$6.94)
- Strong mutual benefit

****Analysis:****

This co-player has been consistently generous and trustworthy. They've shown:

- Fair returns when receiving (51% returned)
- High trust when sending (70% of endowment)
- We've built a good cooperative relationship

I should reciprocate their trust level. I'll send ****\$4**** (80% of my endowment), which:

- Matches their demonstrated trust
- Becomes \$12 for them (good potential returns)
- Continues building our cooperative pattern

```
'''json
{"send": 4}
'''
```

Reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.28 due to noise)
- It became \$9.84 for them
- They returned \$5.00
- This was generous - about 51% of what they received

****Round 2:****

- They sent \$3.5 (I saw \$3.89 due to noise)
- It became \$11.67 for me (their view: \$10.5)

- I returned \$6 (they saw \$6.94 due to noise)
- They earned \$8.44 total

This co-player has been:

1. Very cooperative and fair in Round 1 (returned 51%)
2. Very trusting in Round 2 (sent 70% of their endowment)
3. They saw a good return from me in Round 2 (\$6.94)

Given this strong pattern of mutual cooperation, I should continue to trust them. They've proven reliable.

I think I should send a high amount, similar to what they sent in Round 2. They sent \$3.5, so I'll send around that amount or slightly more.

Options:

- Send \$3.5 to \$4 to maintain the cooperative relationship
- Send \$5 to maximize mutual benefit if I fully trust them

Given the strong cooperation so far, I'll send \$4. This shows:

- I trust them based on their past behavior
- I'm willing to invest in our mutual benefit
- It's not quite all \$5, maintaining a tiny hedge

Actually, reviewing their behavior:

- Round 1: They returned generously (51%)
- Round 2: They sent generously (70% of endowment)

They've been consistently cooperative. I should match their level of trust. ...

```
Usage: {
  "input_tokens": 1156,
  "output_tokens": 645,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T07:55:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.28, that is 65.6% of their total endowment. You received \$9.84. How much do you return to the sender? (0-9.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 65.6% of their endowment (\$3.28)
- This created \$9.84 for me to allocate
- The sender currently has only \$1.72 left from their original \$5

I want to reward their trust while being fair to both of us. If I return \$5.00:

- Sender's total: \$1.72 + \$5.00 = \$6.72 (profit of \$1.72)
- My total: \$9.84 - \$5.00 = \$4.84 (profit of \$4.84)

This seems like a reasonable split that rewards the sender's trust while still benefiting from the transaction.

{"return": 5.00}

reasoning:

This is Round 1 of a multi-agent trust game where I'm playing as the RECEIVER.

Let me understand what happened:

- The sender had an endowment of \$5
- The sender sent me \$3.28 (which is 65.6% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $3.28 \times 3 = 9.84$
- I now need to decide how much to return to the sender (between \$0 and \$9.84)

The sender's payoff will be:

- \$5 (initial) - \$3.28 (sent) + [amount I return] = \$1.72 + [amount I return]

My payoff will be:

- \$9.84 (received) - [amount I return]

Let me think about what would be a fair return:

- The sender was quite generous, sending 65.6% of their endowment
- This shows significant trust
- A fair return might be to ensure the sender gets back at least what they sent, or even more to reward their trust
- If I return \$3.28, the sender would have: $\$1.72 + \$3.28 = \$5$ (break even)
- If I return \$5, the sender would have: $\$1.72 + \$5 = \$6.72$ (profit of \$1.72)
- If I return \$6.56 (double what they sent), the sender would have: $\$1.72 + \$6.56 = \$8.28$

A commonly fair approach in trust games is to split the total gains somewhat equitably while rewarding the sender's trust.

Total value created: \$9.84 (what I received)

Original sender investment: \$3.28

Net gain: $\$9.84 - \$3.28 = \$6.56$

One approach: Split the net gain equally

- Sender gets back: $\$3.28 + \$3.28 = \$6.56$
- I would return: \$6.56
- Sender's final: \$8.28
- My final: \$3.28

Another approach: Split all the money equally

- Total pool: \$9.84
- Equal split: \$4.92 each
- I would return: \$4.92
- Sender's final: \$6.64
- My final: \$4.92...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 999,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.

Your total visible earnings across all rounds are \$4.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- My co-player showed trust in Round 1 by sending \$3.0 (60% of their endowment)
- They received a reasonable return of \$5.94
- This suggests they're willing to cooperate

I want to reciprocate their trust and continue building a cooperative relationship. I'll send a similar proportion to what they sent.

```
{"send": 3.50}
```

reasoning:

Let me analyze this situation carefully:

Round 1 Summary:

- I was the RECEIVER
- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- I returned \$5.00

- My earnings: \$4.84

Now looking at the visible history shown to me:

- My co-player's last game shows: as SENDER, sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94

Wait, this is interesting. There's communication noise mentioned in the instructions. Let me think about what actually happened vs what was observed:

What I experienced in Round 1:

- Sender sent \$3.28
- I received \$9.84
- I returned \$5.00

What my co-player (the sender) saw:

- They sent \$3.0 (but actually sent \$3.28?)
- It became \$9.0 (but actually became \$9.84?)
- They saw \$5.94 returned (but I actually returned \$5.00?)
- Their visible payoff was \$7.94

Let me check the math on their side:

- If they sent \$3.0 and saw \$5.94 returned
- Their payoff would be: $\$5 - \$3 + \$5.94 = \7.94 ?

So from their perspective:

- They sent \$3.0 (66.7% of their total)
- They got back \$5.94 (66% of the \$9.0 that was created)
- This seems like a decent return to them

Now I'm the SENDER in Round 2:

- My co-player showed trust by sending most of their endowment in Round 1
- They received what appeared to them as a good return (66% of the multiplied amount)
- This suggests they might be willing to reciprocate if I show trust
- Since they sent roughly two-thirds of their endowment, I should match that level of generosity to build on the cooperative dynamic we...

usage:

```
{
  "input_tokens": 839,
  "output_tokens": 749,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.
- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.

Your total visible earnings across all rounds are \$13.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.71, so you received \$11.129999999999999. How much do you return? (0-11.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history with my co-player:

- Round 1: They sent \$3.0 (60%)
- Round 2: They returned \$6.0 when I sent (showing cooperation)
- Round 3: They sent \$3.71 (74.2% - even more trust!)

They're showing increasing trust and cooperation. I should reciprocate generously.

Current situation:

- They sent \$3.71, leaving them with \$1.29
- I received \$11.13

If I return \$6.50:

- Sender's total: $\$1.29 + \$6.50 = \$7.79$
- My total: $\$11.13 - \$6.50 = \$4.63$

This rewards their trust while still providing a fair outcome for both of us.

```
{"return": 6.50}
```

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.28 (65.6%)
- I received \$9.84
- I returned \$5.00
- Sender's total: $\$1.72 + \$5.00 = \$6.72$
- My total: $\$9.84 - \$5.00 = \$4.84$

Round 2: I was SENDER

- I sent \$3.50
- It became \$10.50
- But the visible history shows my co-player saw \$3.89 sent, it became \$11.67
- They returned \$6.00
- Their visible payoff was \$5.67
- My actual payoff should be: $\$5 - \$3.50 + (\text{what they returned})$
- According to my total visible earnings of \$13.28, that's $\$4.84 (\text{Round 1}) + \$8.44 (\text{Round 2}) = \13.28
- So I received \$8.44 in Round 2
- This means: $\$5 - \$3.50 + \$8.44 = \9.94 total

Wait, let me recalculate. If my total visible earnings are \$13.28:

- Round 1: I kept \$4.84
- Round 2: $\$13.28 - \$4.84 = \$8.44$

So in Round 2 my payoff was \$8.44.

I sent \$3.50, so I had \$1.50 left, plus what they returned.

$\$1.50 + \text{returned} = \8.44

So they returned \$6.94 to me.

But the visible history says they returned \$6.00. This is the communication noise mentioned in the instructions.

Now in Round 3:

They're sending \$3.71, which is 74.2% of their endowment-a significant increase from the 60% they sent initially. I'll receive \$11.13, leaving them with \$1.29. Given their consistent pattern of cooperation and escalating trust, I should reciprocate meaningfully. Returning \$6.00 would give them \$7.29 total while I keep \$5.13....

```
Usage: {
  "input_tokens": 1132,
  "output_tokens": 983,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T07:56:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    }
  },
```

```

    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderately trusting approach to see how cooperation works in this game.

By sending \$3, the receiver will get \$9 (3×\$3). If they return a fair share, we can both benefit from the multiplier effect.

```

'''json
{"send": 3}
'''

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategy.

ategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return
3. My final earnings = \$5 - sent + returned
4. Receiver's final earnings = (sent × 3) - returned

From a purely rational/selfish perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return nothing, I get \$0
- The optimal outcome for both of us collectively is if I send \$5 (creating \$15 total) and we split it somehow

From a trust/reciprocity perspective:

- Sending money shows trust
- Many people reciprocate trust with trustworthy behavior
- In behavioral economics experiments, people often send substantial amounts and receivers often return fair amounts

Strategic considerations:

- This is round 1 of multiple rounds with randomized pairings
- Building a reputation for cooperation might help in future rounds if people can observe behavior
- However, agent names are hidden, so reputation effects may be limited
- There's also communication noise, which might affect trust dynamics

First round strategy:

- A moderate amount seems reasonable to test cooperation
- Sending \$2-3 would create \$6-9 for the receiver
- If they return even 1/3 of what they receive, I'd break even or profit slightly

I'll send \$3 as a moderately trusting first move. This creates \$9 for the receiver, and if they return \$3-4, we both do reasonably well compared to the \$5 baseline.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.1099999999999999.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history, my co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. They've demonstrated trustworthy behavior.

Now they've sent me \$3.8, giving me \$11.40. I'll return a fair amount to encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze the situation:

1. In Round 1, I sent \$3 (though it appeared as \$3.37 to my co-player due to noise)
2. They received \$10.11 (3x the noisy amount) and returned \$5
3. I ended up with \$5 - \$3 + \$5 = \$7 (though my visible earnings show \$7.51, likely due to noise)

Now in Round 2:

- My co-player sent me \$3.8 (this might be noisy, but this is what I see)
- I receive \$11.40 ($3 \times \3.8)
- I need to decide how much to return

The co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. That's roughly half of what they received, which is quite fair and cooperative.

Since they were cooperative and generous in their return, I should reciprocate fairly. A reasonable approach would be:

- They sent \$3.8 out of their \$5, keeping \$1.2
- I received \$11.40
- If I return about half (around \$5-6), they would end up with roughly \$6.2-7.2 total
- I would keep roughly \$5.40-6.40

Let me return \$5.5, which:

- Gives them: $\$5 - \$3.8 + \$5.5 = \6.7 total
- Gives me: $\$11.40 - \$5.5 = \$5.9$ total

This seems fair as it rewards their trust and cooperation, and both of us end up better than the initial \$5.

usage:

```
{
  "input_tokens": 726,
  "output_tokens": 482,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.

Your total visible earnings across all rounds are \$13.409999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3, which they saw as \$3.37 (noise)
- It became \$10.11 for them
- They returned \$5
- My payoff: $\$5 - \$3 + \$5 = \7
- Their visible payoff: $\$10.11 - \$5 = \$5.11$

Round 2: I was RECEIVER

- They sent \$3.8 (from their perspective they sent \$3.0)
- I received \$11.40
- I returned \$5.5
- They saw \$6.21 returned (noise)
- Their visible payoff: $\$5 - \$3 + \$6.21 = \8.21
- My payoff: $\$11.40 - \$5.5 = \$5.90$

My total visible earnings: \$13.41 (which is approximately $\$7 + \$5.90 = \$12.90$, close enough with rounding)

My co-player has been cooperative:

- Round 1: They returned a fair amount (\$5 out of \$10.11)

- Round 2: They sent \$3 (60% of their endowment), which is quite trusting

I should continue to cooperate and send a reasonable amount. They've been generous, so I should reciprocate. Sending \$3 or more would be appropriate to maintain good cooperation.

```
Usage: {  
  "input_tokens": 993,  
  "output_tokens": 356,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T07:56:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.37, that is 67.4% of their total endowment. You received \$10.11. How much do you return to the sender? (0-10.11)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.37 (67.4% of their endowment)
- They currently have \$1.63 left ($\$5 - \3.37)
- I received \$10.11 after the 3x multiplication

I want to reciprocate this high level of trust and ensure the sender benefits from their decision to trust me.

If I return \$5:

- The sender ends with: $\$1.63 + \$5 = \$6.63$ (a profit of \$1.63)
- I keep: $\$10.11 - \$5 = \$5.11$ (a profit of \$5.11)

This rewards the sender's trust while both of us profit from cooperation.

{'return': 5.0}

reasoning:

Let me analyze this situation:

1. The sender sent \$3.37 out of their \$5 endowment (67.4%)
2. This was multiplied by 3x, so I received \$10.11
3. I need to decide how much to return (0-10.11)

Key considerations:

- The sender showed significant trust by sending 67.4% of their endowment
- If I return nothing, I keep \$10.11 but break trust
- A fair return might be to ensure the sender gets back more than they sent
- The sender currently has: $\$5 - \$3.37 = \$1.63$

If I want to be reciprocal and fair:

- The sender gave up \$3.37 to trust me
- They showed high trust (67.4% of endowment)
- A reasonable return would ensure they profit from trusting me

Some options:

1. Return \$5: Sender ends with $\$1.63 + \$5 = \$6.63$ (profit of \$1.63), I keep \$5.11
2. Return \$6: Sender ends with $\$1.63 + \$6 = \$7.63$ (profit of \$2.63), I keep \$4.11
3. Return \$4: Sender ends with $\$1.63 + \$4 = \$5.63$ (profit of \$0.63), I keep \$6.11
4. Return \$3.37: Sender ends with $\$1.63 + \$3.37 = \$5$ (break even), I keep \$6.74

Given that:

- The sender showed high trust (67.4%)
- Note says there may be communication noise
- I want to encourage future cooperation
- A fair split might be approximating equal gains

If I return \$5:

- Sender: started with \$5, sent \$3.37, gets back \$5 = net $\$1.63 + \$5 = \$6.63$ total

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 694,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.

Your total visible earnings across all rounds are \$5.109999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing significant trust
- They saw \$5.51 returned, which gave them a positive payoff
- I profited from their trust in Round 1

I want to reciprocate their cooperative behavior and maintain the trust relationship. Sending a similar proportion will signal that I'm willing to cooperate.

{'send': 3.0}

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$3.37 (67.4% of their endowment)
- I received \$10.11
- I returned \$5.0
- I kept \$5.11

Now in Round 2, the visible history shows:

- My co-player's last game (Round 1 against me) as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51

Wait, there's communication noise! The actual amounts differ from what each player sees:

- I actually received \$10.11 (from their \$3.37 sent)
- I actually returned \$5.0
- But they see that they sent \$3.0, it became \$9.0, and saw \$5.51 returned

So there's noise in both directions. This is interesting - the sender sees they sent \$3.0 (vs actual \$3.37) and saw \$5.51 returned (vs actual \$5.0 I returned).

Now I'm the SENDER in Round 2 against the same co-player.

My visible earnings are \$5.11 (which matches what I calculated: \$10.11 - \$5.0 = \$5.11).

Strategy considerations:

1. My co-player sent a high amount in Round 1 (either \$3.0 or \$3.37 depending on what was real)
2. They saw a decent return (approximately 50% of what they sent, or more)
3. This suggests they're cooperative

I should signal continued cooperation by sending a similar proportion. Since they sent 67.4% in Round 1, I could send a similar high proportion.

Given the communication noise, I should send a clear cooperative signal. Sending around \$3-3.5 would show I'm reciprocating their trust.

Let me send \$3.0 to match what they think they sent (according to the visible history).

...

usage:

```
{
  "input_tokens": 848,
  "output_tokens": 639,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.

Your total visible earnings across all rounds are \$13.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.05, so you received \$6.1499999999999995. How much do you return? (0-6.1499999999999995)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$2.05 (41% of their endowment), showing moderate trust
- They currently have \$2.95 left
- I received \$6.15

Previous rounds show we've had a cooperative relationship, though the amounts varied (possibly due to noise). I want to maintain trust while ensuring both of us benefit.

If I return \$3.0:

- Sender ends with: $\$2.95 + \$3.0 = \$5.95$ (profit of \$0.95)
- I keep: $\$6.15 - \$3.0 = \$3.15$ (profit of \$3.15)

This ensures mutual benefit while reciprocating their trust.

{'return': 3.0}

Reasoning:

Let me analyze the situation:

Looking at the visible history:

- Round 1: My co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51
- Round 2: My co-player as RECEIVER saw \$3.8 sent (I sent this), returned \$5.5, their payoff \$5.90

Now Round 3:

- I'm the RECEIVER
- My co-player sent \$2.05 (41% of their endowment)
- I received \$6.15

The co-player has been cooperative but they sent less this time (\$2.05 vs \$3.0 in Round 1). This could be:

1. A response to the noise in the game
2. Adjusting their strategy
3. Testing my response

Let me think about what's fair:

- They currently have: $\$5 - \$2.05 = \$2.95$
- I received: \$6.15

If I want to reciprocate fairly and ensure they profit:

- Return \$3.0: Sender gets $\$2.95 + \$3.0 = \$5.95$ (profit \$0.95), I keep \$3.15 (profit \$3.15)
- Return \$3.5: Sender gets $\$2.95 + \$3.5 = \$6.45$ (profit \$1.45), I keep \$2.65 (profit \$2.65)

Given they sent a moderate amount (41%), I should return a proportional amount. A return of around \$3.0-3.5 seems fair, ensuring both of us profit while maintaining the cooperative relationship.

I'll return \$3.0 to keep things balanced.

```
Usage: {
  "input_tokens": 1160,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T07:56:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance showing trust with managing risk, especially given the communication noise mentioned.

Since this is a multi-round game with rotating partners, establishing a cooperative pattern early could be beneficial. However, I don't know this receiver's tendencies yet.

I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Gives the receiver meaningful funds to work with (tripled)
- Doesn't risk my entire endowment

If I send \$3, the receiver gets \$9. If they return even half of that (\$4.50), we both benefit more than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 and can send \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. There's communication noise mentioned - amounts may differ slightly from what was actually sent/returned

Key considerations:

- If I send \$0, I keep \$5, receiver gets \$0, game ends with no cooperation
- If I send \$5, receiver gets \$15, and their return decision determines my payoff
- This is a multi-round game with 8 agents, randomized pairings
- My reputation/behavior might matter for future rounds, though pairings are randomized

Standard trust game reasoning:

- Sending money shows trust and can enable mutual benefit
- If receiver is rational and self-interested (and this is the last interaction), they should return \$0
- But in multi-round settings, reputation and reciprocity often emerge
- A moderate initial send can signal willingness to cooperate while limiting downside risk

Since this is round 1 and I don't know the receiver's tendencies, I should probably send a moderate amount to:

1. Signal willingness to cooperate
2. Give receiver opportunity to show trustworthiness
3. Not risk my entire endowment

A common strategy is to send about half. If I send \$2.50:

- Receiver gets \$7.50
- If they return half (\$3.75), I end with $\$5 - \$2.50 + \$3.75 = \6.25
- Receiver keeps \$3.75
- Both benefit

Actually, let me reconsider. Since amounts can only be integers based on the format, I should send a whole dollar amount. Let me send \$3:

- I send \$3

- Receiver gets \$9
- If they're fair and return half (\$4.50), I get $\$5 - \$3 + \$4.50 = \6.50

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 724,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.

Your total visible earnings across all rounds are \$5.41.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

Round 1: My co-player saw \$2.29 sent (due to noise), received \$6.87, and returned \$3.15 (about 46% of what they received). This shows they're willing to reciprocate.

Now they've sent me \$2.99, giving me \$8.97 total.

To maintain a cooperative relationship and signal trustworthiness, I'll return approximately half of what I received, which is fair and encourages future cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me understand what happened:

Round 1: I was the SENDER

- I sent \$3
- Due to communication noise, my co-player SAW \$2.29 sent (instead of \$3)
- It became \$6.87 (which is $\$2.29 \times 3$)
- They returned \$3.15
- Their visible payoff was \$3.72 (which is $\$6.87 - \3.15)
- My visible earnings are \$5.41

Let me calculate my Round 1 payoff:

- I started with \$5
- I sent \$3
- I received back... well, they returned \$3.15
- So I should have: $\$5 - \$3 + \$3.15 = \5.15
- But my "visible earnings" are \$5.41

Hmm, this is confusing. Maybe the communication noise affected what I received too? Or maybe "visible earnings" includes some noise?

Actually, let me recalculate. If they saw \$2.29 sent and returned \$3.15:

- From their perspective: they received \$6.87, returned \$3.15, kept \$3.72
- From my perspective: I sent \$3, but what did I actually receive back?

The problem states "visible payoff" and "visible earnings" which suggests everything we see might have noise.

Now for Round 2:

- I am the RECEIVER
- They sent me \$2.99 (this is what I see after noise)
- I received \$8.97 (which is $\$2.99 \times 3$)
- I need to decide how much to return (0-8.97)

From Round 1, my co-player:

- Saw \$2.29 sent (noise-affected)
- Received \$6.87
- Returned \$3.15 (about 46% of what they received)

This shows they were somewhat cooperative, returning close to half.

They sent \$2.99, which is close to their original \$3 send, suggesting they're being fairly consistent. If I return roughly half of what I received-around \$4.48-\$4.50-I'd match their reciprocity level. I could return slightly more to encourage continued cooperation, or slightly less to protect my earnings. Let me think through the payoffs: if they sent \$2.99 and kept \$2.01, returning \$4.50 would give them \$6.51 total, which seems fair given their initial cooperation.

Actually, they returned 46% in the first round despite seeing a smaller amount due to noise. That's a strong signal of trust...

usage:

```
{
  "input_tokens": 798,
  "output_tokens": 987,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.

Your total visible earnings across all rounds are \$9.88.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history together:

Round 1: As receiver, they returned \$3.15 on \$6.87 (46% return rate)

Round 2: As sender, they sent \$3, and saw I returned \$4.71 (though actual was \$4.50 due to noise)

This co-player has demonstrated:

- Willingness to send a good amount (\$3 = 60% of endowment)
- Reasonable reciprocation as receiver
- We've established a cooperative pattern

I'll continue this pattern by sending the same amount they sent to me.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the history with this co-player:

Round 1:

- I was sender, they were receiver
- They saw \$2.29 sent (noise from my actual \$3), received \$6.87, returned \$3.15
- This is about 46% return rate

Round 2:

- They were sender, I was receiver
- They sent \$3.0, it became \$9.0
- They saw \$4.71 returned (I actually returned \$4.50, so there's noise)
- They kept \$6.71 total

So this co-player has shown:

1. As a receiver: They returned about 46% of what they received

2. As a sender: They sent \$3 (60% of their endowment), showing trust

I returned \$4.50 on \$8.97 (about 50% return rate), which they saw as \$4.71.

This seems like a cooperative partner. They're willing to send a good amount and return a reasonable portion. Since they sent \$3 last round and I returned generously, I should continue the cooperative pattern.

I'll send \$3 again to maintain consistency and reciprocal trust.

```
Usage: {  
  "input_tokens": 1100,  
  "output_tokens": 421,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T07:56:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.29, that is 45.8% of their total endowment. You received \$6.87. How much do you return to the sender? (0-6.87)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.29 (45.8% of their endowment)
- I received \$6.87 after the 3x multiplier
- The sender currently has \$2.71

I want to reciprocate their trust while keeping the exchange mutually beneficial. The sender took a risk by sending nearly half their endowment, so I should return enough to reward that trust.

If I return approximately the same proportion they sent (around 45-46% of what I received), that would be about \$3.15. This would give the sender $\$2.71 + \$3.15 = \$5.86$ total (a gain of \$0.86), and I'd keep \$3.72 (also a gain).

This seems like a fair reciprocation that rewards their trust and could encourage cooperation in future rounds.

{'return': 3.15}

reasoning:

I'm the receiver in this round. Let me understand the situation:

- The sender sent \$2.29 (which is 45.8% of their \$5 endowment)
- This was multiplied by 3, so I received \$6.87
- I need to decide how much to return (between \$0 and \$6.87)

Let me think about what would be fair and strategic:

1. The sender showed trust by sending almost half their endowment (\$2.29 out of \$5)
2. They kept $\$5 - \$2.29 = \$2.71$
3. I received \$6.87
4. If I return nothing, I keep \$6.87 and the sender ends with \$2.71
5. If I return everything, I keep \$0 and the sender ends with $\$2.71 + \$6.87 = \$9.58$

A common fair approach in trust games is to aim for roughly equal outcomes or to reciprocate the sender's trust proportionally.

Let's consider some options:

Option 1 - Equal split of total gains:

- Total money created: $\$5$ (sender's endowment) + $\$6.87$ (what I received) = $\$11.87$
- But wait, that's not right. The sender started with \$5, sent \$2.29 which became \$6.87
- Total in the system: Sender has \$2.71, I have \$6.87 = $\$9.58$ total
- Equal split would be $\$4.79$ each
- For sender to have \$4.79: they need $\$4.79 - \$2.71 = \$2.08$ returned

Option 2 - Return proportional to what was sent:

- Sender sent 45.8% of their endowment
- I could return a similar proportion of what I received
- 45.8% of \$6.87 = \$3.15

Option 3 - Ensure sender gains from trusting me:

- Sender currently has \$2.71
- If I return enough so they end up better than their initial \$5
- They need at least \$5 - \$2.71 = \$2.29 to break even
- To give them a decent profit, maybe return \$3-3.50 so they end with \$5.71-6.21

Since this is a multi-round game with randomized pairings and communication noise, I should consider how my choice signals trustworthiness. Returning around \$3-3.15 seems reasonable-it reciprocates their generosity while still letting me benefit from the exchange, and it positions me well for future interactions with different partners....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1076,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.

Your total visible earnings across all rounds are \$3.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They received \$3.41 back, making a profit of \$0.41
- They demonstrated significant trust

As the SENDER now, I want to reciprocate their trust level. They sent 60% of their endowment, which was quite generous. I should match or slightly exceed this level of trust to maintain cooperation.

Sending \$3.0 would match their previous contribution and signal that I'm willing to cooperate at the same trust level they showed.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

From Round 1:

- I was the RECEIVER
- The sender sent \$2.29 (45.8% of their endowment)
- I received \$6.87
- I returned \$3.15

But now the visible history shows something different:

- It says my co-player (as SENDER in Round 1) sent \$3.0, which became \$9.0
- They saw \$3.41 returned
- Their visible payoff was \$5.41

This is the communication noise mentioned in the instructions! What actually happened vs what they saw:

- Actual: They sent \$2.29, I received \$6.87, I returned \$3.15
- What they saw: They sent \$3.0, became \$9.0, saw \$3.41 returned

So from their perspective, they sent \$3.0 and got back \$3.41. That's a return rate of about

t 38% of what I received (3.41/9.0).

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of endowment)
- Got back \$3.41 (38% of \$9.0)
- Made a profit of \$0.41

They showed significant trust by sending 60% (from their perspective). They got back a reasonable amount.

As the SENDER now, I should consider:

1. They showed high trust in Round 1 (from their perspective, 60%)
2. They got a modest return
3. I want to continue building trust

Given that they sent 60% from their perspective, I should send a similar or slightly higher amount to show reciprocity and build trust. Sending around \$3.0-\$3.5 would be reasonable.

Let me send \$3.0 to match what they (think they) sent to me.

Actually, let me think more carefully. My actual earnings were \$3.72 in Round 1. I received \$6.87 and returned \$3.15.

...

usage:

```
{
  "input_tokens": 867,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.

Your total visible earnings across all rounds are \$10.43.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.100000000000001. How much do you return? (0-8.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Round 2 pattern:**** When my co-player was the RECEIVER, they returned \$4.5 out of \$8.97 received (approximately 50%)

****This round:****

- They sent \$2.7 (54% of endowment), showing continued trust
- I received \$8.10

I should reciprocate with a similar return ratio. Returning about 50% like they did would be \$4.05, which:

- Gives them \$2.3 + \$4.05 = \$6.35 total (profit of \$1.35)
- Gives me \$4.05 (also a decent profit)

This maintains the cooperative pattern we've established.

```
{'return': 4.05}
```

Reasoning:

Let me analyze the situation:

****Visible History:****

- Round 1: My co-player (as SENDER) sent \$3.0, received \$3.41 back, payoff \$5.41

- Round 2: My co-player (as RECEIVER) saw \$2.99 sent, returned \$4.5, payoff \$4.47

****This Round (Round 3):****

- I am the RECEIVER
- They sent \$2.7 (54% of their endowment)
- I received \$8.10

Looking at the pattern:

- In Round 1, they sent 60% of their endowment
- In Round 2, I sent \$3.0 (though they saw \$2.99 due to noise), and they returned \$4.5, which is about 50% of what they received (\$8.97)
- In Round 3, they sent \$2.7 (54%)

My co-player seems to be cooperating at moderate-to-high levels. When they were receiver in Round 2, they returned 50% of what they received.

If I return around 50% like they did, that would be about \$4.05.

Let me think about what's fair:

- They sent \$2.7, so they have \$2.3 left
- If I return \$4.05, they'd have $2.3 + 4.05 = 6.35$ (profit of \$1.35)
- I'd keep $8.10 - 4.05 = 4.05$ (profit of \$4.05)

That seems reasonable and matches their behavior. I could be slightly more generous to encourage continued cooperation.

Actually, let me reconsider. In Round 2, they returned \$4.5 out of \$8.97, which is exactly 50%

.
...

```
Usage: {  
  "input_tokens": 1210,  
  "output_tokens": 644,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T07:56:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I want to establish a cooperative tone while being somewhat cautious since I don't know my co-player yet.

Sending a moderate amount can:

- Signal trust and willingness to cooperate
- Create a larger pool for mutual benefit (tripled to \$9)
- Allow the receiver meaningful choice in how much to return

If the receiver returns even half of what they receive, we both end up better than the initial \$5.

I'll send **\$3**.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- There's communication noise, so amounts may differ

- This is a multi-round game with 8 agents
- Pairings are randomized each round

Classic trust game setup. Let me think about optimal strategy:

1. If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
2. If I send \$5: Receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 (better than \$5). But they might return less or nothing.
3. If I send a moderate amount (e.g., \$3): Receiver gets \$9. If they return \$4-5, I come out ahead.

Since this is:

- Round 1 (establishing reputation/reciprocity patterns)
- Multi-round game (future interactions matter)
- Anonymous agents (can't track specific individuals)

I should consider:

- Building trust/reciprocity norms early
- Signaling cooperation
- But also being somewhat cautious since I don't know the other agents

A common strategy is to send a moderate-to-high amount in early rounds to:

1. Signal trustworthiness
2. Encourage reciprocity
3. Maximize potential gains if receiver cooperates

Given the noise factor and uncertainty, I'll send \$3. This shows goodwill without being maximally vulnerable, and gives the receiver \$9 to work with. If they return \$4-5, we both benefit.

Actually, let me reconsider. In trust games, sending higher amounts often generates better returns because:

- It demonstrates trust
- Creates larger pie to split
- Social norms typically lead to proportional returns

Comparing the math: sending \$4 leaves me with \$1 while giving the receiver \$12. A 50% return (\$6) nets me \$7 total, while a 33% return (\$4) just breaks me even. With \$3 sent instead, I'd have \$2 cushion and need only a 50% return (\$4.50) to reach \$6.50. The higher send amount creates more upside if reciprocity holds, so I'll go with \$4....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 814,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.

Your total visible earnings across all rounds are \$4.2.

This round, you are the RECEIVER against your current co-player. They sent you \$3.7, so you received \$11.100000000000001. How much do you return? (0-11.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 analysis:****

- My co-player saw \$2.24 sent (due to noise), which became \$6.72
- They returned \$3.0 (about 45% of what they received)
- This shows reasonable cooperative behavior

****Round 2 (current):****

- They sent me \$3.7 (even more than what they saw in Round 1)
- I received \$11.10
- They're showing increased trust

To maintain and encourage cooperation, I should return a fair amount. Returning about half of what I received would give them:

- They kept: $\$5 - \$3.7 = \$1.30$
- Plus returned amount
- A good return on their trust

I'll return ****\$6.00**** (about 54% of what I received), giving them a total of \$7.30 - a solid return that rewards their increased trust.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation carefully.

Round 1 recap from my perspective:

- I sent \$3
- Due to communication noise, my co-player saw \$2.24 sent (which became \$6.72 after tripling)
- They returned \$3.0
- Their visible payoff was \$3.72 (received \$6.72, returned \$3.0)
- My total visible earnings are \$4.2

Let me calculate what actually happened in Round 1:

- I sent \$3, so I had $\$5 - \$3 = \$2$ left
- The receiver got $\$3 \times 3 = \9 (but saw \$6.72 due to noise)
- They returned \$3.0
- So I ended with $\$2 + \$3 = \$5$
- But my visible earnings show \$4.2, which suggests the noise affected what I saw returned too?

Actually, wait. Let me reconsider:

- If my visible earnings are \$4.2 total
- I started with \$5, sent \$3 (keeping \$2)
- If I received back something that made my total \$4.2, then I received $\$4.2 - \$2 = \$2.2$ back

So the noise worked both ways:

- I sent \$3, they saw \$2.24
- They returned \$3.0, but I saw (and actually got) something that gave me \$4.2 total

Now for Round 2:

- I'm the RECEIVER
- My co-player sent me \$3.7 (according to what I see)
- This became \$11.10 after tripling
- I need to decide how much to return

My co-player showed willingness to return a reasonable amount in Round 1 (they returned \$3.0 out of \$6.72 they saw, which is about 45%). This is fairly cooperative behavior.

I should return a fair amount to maintain cooperation. Given that:

- They sent \$3.7 and I received \$11.10
- A 50% return would be \$5.55, giving them $\$1.3$ (kept) + \$5.55 (returned) = \$6.85 total
- This matches their cooperative gesture from Round 1 and encourages continued trust

I'll return \$5 to keep things balanced....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 1051,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.

Your total visible earnings across all rounds are \$9.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history:

****Round 1:**** I sent \$3 ? they returned \$3 (cooperative)

****Round 2:**** They sent \$3 ? I returned \$6, but they saw \$5.07 (still a good return)

****Round 3 (current):****

- Both rounds showed mutual cooperation
- They sent \$3 as sender and returned reasonably as receiver
- Their visible payoff was \$7.07 last round, showing cooperation pays off

I'll send ****\$3.5**** to maintain and slightly increase our cooperative dynamic. This shows I'm willing to trust them a bit more based on our positive history.

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.24 (noise), became \$6.72, returned \$3.0
- They kept \$3.72, I got back \$3.0, so I ended with \$5 - \$3 + \$3 = \$5
- But my visible earnings show \$4.2, which suggests I saw something different due to noise

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, became \$9.0
- I returned \$6.0, but they saw \$5.07 (noise)
- They ended with \$5 - \$3 + \$5.07 = \$7.07
- I kept \$9.0 - \$6.0 = \$3.0

My total visible earnings are \$9.3.

Now Round 3:

- I'm SENDER again
- Co-player has shown:
 - As receiver: returned \$3.0 out of \$6.72 (45%)
 - As sender: sent \$3.0 out of \$5 (60%)
- They saw I returned \$5.07 (though I actually returned \$6.0)
- This is good cooperation

Since they've been cooperative and sent a good amount when they were sender, I should continue cooperating. They sent \$3.0 last round, so I should send at least that much or more to maintain trust.

I'll send \$3.5 to show continued cooperation and slightly increase the stakes.

```
Usage: {
  "input_tokens": 1191,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T07:56:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.24, that is 44.8% of their total endowment. You received \$6.720000000000001. How much do you return to the sender? (0-6.720000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.24 (44.8% of their endowment)
- They currently have \$2.76 remaining
- I received \$6.72 after the 3x multiplication

To reciprocate their trust fairly, I should return an amount that:

1. Ensures they profit from trusting me (end up with more than their original \$5)
2. Rewards their cooperative gesture
3. Still leaves me with a reasonable gain

If I return \$3.00:

- Sender ends with: $\$2.76 + \$3.00 = \$5.76$ (gain of \$0.76)
- I keep: $\$6.72 - \$3.00 = \$3.72$ (gain of \$3.72)

This seems like a fair reciprocal response that rewards trust while maintaining balanced incentives for future rounds.

{"return": 3.00}

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.24 out of their \$5 endowment (44.8%)
3. This was multiplied by 3x, so I received \$6.72 (technically \$6.720000000000001 due to floating point)
4. I need to decide how much to return (between \$0 and \$6.72)

Key considerations:

- The sender showed moderate trust by sending 44.8% of their endowment
- If I return nothing, I keep all \$6.72 and the sender gets only \$5 - \$2.24 = \$2.76
- If I return a fair amount, I should consider what would make this equitable

Common approaches in trust games:

1. Equal split of total gains: Total in the system is $\$5 + \$6.72 = \$11.72$. Equal split would be \$5.86 each. The sender currently has \$2.76, so I'd return $\$5.86 - \$2.76 = \$3.10$
2. Return the same proportion they sent: They sent 44.8%, so I could return 44.8% of \$6.72 = \$3.01
3. Match their investment: Return \$2.24 so we both end up with gains from the multiplication
4. Return enough so they profit: If I return \$3, they end up with $\$2.76 + \$3 = \$5.76$ (a gain of \$0.76), and I keep \$3.72 (a gain of \$3.72)

The sender took a risk by sending \$2.24. A fair return should:

- At minimum, not make them worse off than keeping everything (\$5)
- Ideally, reward their trust
- Be reciprocal to their gesture

Let me calculate a few scenarios:

- If I return \$3.00: Sender gets $\$2.76 + \$3.00 = \$5.76$, I keep \$3.72
- If I return \$3.50: Sender gets $\$2.76 + \$3.50 = \$6.26$, I keep \$3.22
- If I return \$4.00: Sender gets $\$2.76 + \$4.00 = \$6.76$, I keep \$2.72

A return of \$3.00-\$3.50 seems balanced-it rewards their trust while still giving me a meaningful gain. I'll return \$3.00, which nets them \$5.76 (a 15.2% gain) and leaves me with \$3.72....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```


Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.

Your total visible earnings across all rounds are \$3.7200000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw only \$2.2 returned (though due to communication noise, I actually returned \$3.00)
- From their perspective, they may feel the cooperation wasn't fully reciprocated

As the sender now, I want to:

1. Signal continued willingness to cooperate
2. Match their previous level of trust
3. Maintain the cooperative dynamic despite any noise in the previous round

I'll send the same amount they sent in Round 1 to demonstrate reciprocal trust.

{"send": 3.0}

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$2.24 (44.8% of their endowment)
- I received \$6.72
- I returned \$3.00
- Sender ended with $\$2.76 + \$3.00 = \$5.76$
- I kept \$3.72

But the visible history shows something different from my co-player's perspective:

- Round 1: They were SENDER, sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2

Wait, this is confusing. Let me re-read the prompt about communication noise.

"NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design."

So what happened:

- From my perspective: I received notification that sender sent \$2.24, I received \$6.72, I returned \$3.00
- From their perspective: They sent \$3.0, it became \$9.0, they saw \$2.2 returned, and got payoff \$4.2

Let me verify the math from their perspective:

- They sent \$3.0
- They had $\$5.0 - \$3.0 = \$2.0$ left
- They saw \$2.2 returned
- Their payoff: $\$2.0 + \$2.2 = \$4.2$?

So from their perspective, they sent \$3.0 and only got back \$2.2, which is less than they sent. They might perceive this as me being untrustworthy or uncooperative.

However, from MY perspective, I actually returned \$3.00, which I thought was fair and generous given I received \$6.72.

This is the communication noise at work. They think I returned less than I actually did.

Now I'm the sender in Round 2, and they likely view me as less trustworthy due to the noise distorting their perception of my return. They sent 60% of their endowment initially, showing genuine trust, but the miscommunication means they'll probably be more cautious. I need to decide whether sending a substantial amount will help rebuild trust or just get exploited if they reciprocate poorly. A moderate send might signal cooperation without overcommitting, especially since they may interpret my actions through the lens of what they think happened last round....

usage:

```
{
  "input_tokens": 890,
  "output_tokens": 922,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.

Your total visible earnings across all rounds are \$10.790000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment), showing strong trust
- Round 2: Co-player returned \$6.0 out of \$11.1 received (54% return rate), showing generosity despite seeing lower returns in Round 1
- Round 3: Co-player sent \$3.3 (66% of endowment), continuing to show high trust

My co-player has consistently demonstrated cooperative behavior and trust. They've maintained cooperation even after seeing what appeared to be a lower return in Round 1.

If I return \$5.0:

- Sender ends with: $\$1.70 + \$5.00 = \$6.70$ (profit of \$1.70)
- I keep: $\$9.90 - \$5.00 = \$4.90$

This reciprocates their generous behavior and maintains the positive cooperative dynamic.

```
{"return": 5.0}
```

Reasoning:

Let me analyze the visible history:

Round 1:

- Co-player as SENDER: sent \$3.0, saw \$2.2 returned
- From their view: started with \$5, sent \$3, kept \$2, saw \$2.2 returned = \$4.2 total
- (I actually returned \$3.00, but they saw \$2.2 due to noise)

Round 2:

- Co-player as RECEIVER: saw \$3.7 sent (I sent \$3.0), received \$11.1, returned \$6.0
- From their view: they received \$11.1, returned \$6.0, kept \$5.1
- From my view: I sent \$3.0, kept \$2.0, saw \$6.0 returned = \$8.0 total

Round 3 (current):

- I am RECEIVER
- Co-player sent \$3.3 (66% of endowment)
- I received \$9.9

My co-player has been quite trustworthy:

- Round 1: sent 60% of endowment
- Round 2: returned a generous \$6.0 out of \$11.1 they received (54% return rate)
- Round 3: sent 66% of endowment

They've shown consistent cooperation and generosity. I should reciprocate appropriately.

```

If I return $5.0:
- Sender ends with: $1.7 (kept) + $5.0 (returned) = $6.7 (profit of $1.7)
- I keep: $9.9 - $5.0 = $4.9

This maintains a good balance and rewards their trust. Let me check if this is fair compared to
their Round 2 behavior:
- In Round 2, they returned $6.0 out of $11.1 (54%)
...
Usage: {
  "input_tokens": 1250,
  "output_tokens": 704,
  "reasoning_tokens": 0
}

```

===== **ROUND 4** =====

```

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver,
Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender
Pairings:
  dyad_1: Agent_2=Sender, Agent_1=Receiver
  dyad_2: Agent_4=Sender, Agent_3=Receiver
  dyad_3: Agent_6=Sender, Agent_5=Receiver
  dyad_4: Agent_8=Sender, Agent_7=Receiver
Parsed game decisions:
  dyad_1: Agent_2 sent $4; received $12; Agent_1 returned $7; payoffs Agent_2=$8, Agent_1=$5;
  noise/visible: sent visible=$3.97, received visible=$11.91, returned visible=$6.16
  Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw":
    7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
  dyad_2: Agent_4 sent $3; received $9; Agent_3 returned $5; payoffs Agent_4=$7, Agent_3=$4;
  noise/visible: sent visible=$3.28, received visible=$9.84, returned visible=$4.45
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
    5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
  dyad_3: Agent_6 sent $3; received $9; Agent_5 returned $3.5; payoffs Agent_6=$5.5, Agent_5=$5.5;
  noise/visible: sent visible=$2.35, received visible=$7.05, returned visible=$3.67
  Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw":
    3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
  dyad_4: Agent_8 sent $3.5; received $10.5; Agent_7 returned $5; payoffs Agent_8=$6.5,
  Agent_7=$5.5; noise/visible: sent visible=$3, received visible=$9, returned visible=$4.99
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 10.5, "min": 0, "raw":
    5.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
Balances after round: {
  "Agent_1": 24.0,
  "Agent_2": 25.0,
  "Agent_3": 19.5,
  "Agent_4": 24.5,
  "Agent_5": 21.2,
  "Agent_6": 22.8,
  "Agent_7": 20.0,
  "Agent_8": 26.0
}
Visible balances after round: {
  "Agent_1": 25.18,
  "Agent_2": 25.07,
  "Agent_3": 22.569999999999997,
  "Agent_4": 22.919999999999998,
  "Agent_5": 19.340000000000003,
  "Agent_6": 20.15,
  "Agent_7": 20.12,
  "Agent_8": 22.18
}

```

LLM CALLS

```

GAME - Agent_2 - Sender - dyad_1
Timestamp: 2026-08-12T07:57:09
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```

```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.28, that is 65.6% of their total endowment. You received \$9.84. How much do you return to the sender? (0-9.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 65.6% of their endowment (\$3.28)
- This created \$9.84 for me to allocate
- The sender currently has only \$1.72 left from their original \$5

I want to reward their trust while being fair to both of us. If I return \$5.00:

- Sender's total: $\$1.72 + \$5.00 = \$6.72$ (profit of \$1.72)
- My total: $\$9.84 - \$5.00 = \$4.84$ (profit of \$4.84)

This seems like a reasonable split that rewards the sender's trust while still benefiting from the transaction.

```
{"return": 5.00}
```

reasoning:

This is Round 1 of a multi-agent trust game where I'm playing as the RECEIVER.

Let me understand what happened:

- The sender had an endowment of \$5
- The sender sent me \$3.28 (which is 65.6% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$3.28 \times 3 = \9.84
- I now need to decide how much to return to the sender (between \$0 and \$9.84)

The sender's payoff will be:

- $\$5$ (initial) - $\$3.28$ (sent) + [amount I return] = $\$1.72 + [\text{amount I return}]$

My payoff will be:

- $\$9.84$ (received) - [amount I return]

Let me think about what would be a fair return:

- The sender was quite generous, sending 65.6% of their endowment
- This shows significant trust
- A fair return might be to ensure the sender gets back at least what they sent, or even more to reward their trust
- If I return \$3.28, the sender would have: $\$1.72 + \$3.28 = \$5$ (break even)
- If I return \$5, the sender would have: $\$1.72 + \$5 = \$6.72$ (profit of \$1.72)
- If I return \$6.56 (double what they sent), the sender would have: $\$1.72 + \$6.56 = \$8.28$

A commonly fair approach in trust games is to split the total gains somewhat equitably while rewarding the sender's trust.

Total value created: \$9.84 (what I received)

Original sender investment: \$3.28

Net gain: $\$9.84 - \$3.28 = \$6.56$

One approach: Split the net gain equally

- Sender gets back: $\$3.28 + \$3.28 = \$6.56$
- I would return: \$6.56
- Sender's final: \$8.28
- My final: \$3.28

Another approach: Split all the money equally

- Total pool: \$9.84
- Equal split: \$4.92 each
- I would return: \$4.92
- Sender's final: \$6.64
- My final: \$4.92...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 999,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.

Your total visible earnings across all rounds are \$4.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- My co-player showed trust in Round 1 by sending \$3.0 (60% of their endowment)
- They received a reasonable return of \$5.94
- This suggests they're willing to cooperate

I want to reciprocate their trust and continue building a cooperative relationship. I'll send a similar proportion to what they sent.

{"send": 3.50}

reasoning:

Let me analyze this situation carefully:

Round 1 Summary:

- I was the RECEIVER
- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- I returned \$5.00
- My earnings: \$4.84

Now looking at the visible history shown to me:

- My co-player's last game shows: as SENDER, sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94

Wait, this is interesting. There's communication noise mentioned in the instructions. Let me think about what actually happened vs what was observed:

What I experienced in Round 1:

- Sender sent \$3.28
- I received \$9.84
- I returned \$5.00

What my co-player (the sender) saw:

- They sent \$3.0 (but actually sent \$3.28?)
- It became \$9.0 (but actually became \$9.84?)
- They saw \$5.94 returned (but I actually returned \$5.00?)
- Their visible payoff was \$7.94

Let me check the math on their side:

- If they sent \$3.0 and saw \$5.94 returned
- Their payoff would be: $\$5 - \$3 + \$5.94 = \7.94 ?

So from their perspective:

- They sent \$3.0 (66.7% of their total)
- They got back \$5.94 (66% of the \$9.0 that was created)
- This seems like a decent return to them

Now I'm the SENDER in Round 2:

- My co-player showed trust by sending most of their endowment in Round 1
- They received what appeared to them as a good return (66% of the multiplied amount)
- This suggests they might be willing to reciprocate if I show trust
- Since they sent roughly two-thirds of their endowment, I should match that level of generosity to build on the cooperative dynamic we...

usage:

```
{
  "input_tokens": 839,
  "output_tokens": 749,
```

```

    "reasoning_tokens": 0
  }
Message 6 role=user
content:
  Round 3

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $5.94 returned, visible payoff $7.94.
  - Round 2 against you, as RECEIVER: saw $3.89 sent, it became $11.67, returned $6.0, visible payoff $5.67.
  Your total visible earnings across all rounds are $13.280000000000001.

  This round, you are the RECEIVER against your current co-player. They sent you $3.71, so you received $11.129999999999999. How much do you return? (0-11.129999999999999)
  Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant
content:
  Looking at the history with my co-player:

  - Round 1: They sent $3.0 (60%)
  - Round 2: They returned $6.0 when I sent (showing cooperation)
  - Round 3: They sent $3.71 (74.2% - even more trust!)

  They're showing increasing trust and cooperation. I should reciprocate generously.

  Current situation:
  - They sent $3.71, leaving them with $1.29
  - I received $11.13

  If I return $6.50:
  - Sender's total: $1.29 + $6.50 = $7.79
  - My total: $11.13 - $6.50 = $4.63

  This rewards their trust while still providing a fair outcome for both of us.

  {"return": 6.50}
reasoning:
  Let me analyze the situation:

  Round 1: I was RECEIVER
  - Sender sent $3.28 (65.6%)
  - I received $9.84
  - I returned $5.00
  - Sender's total: $1.72 + $5.00 = $6.72
  - My total: $9.84 - $5.00 = $4.84

  Round 2: I was SENDER
  - I sent $3.50
  - It became $10.50
  - But the visible history shows my co-player saw $3.89 sent, it became $11.67
  - They returned $6.00
  - Their visible payoff was $5.67
  - My actual payoff should be: $5 - $3.50 + (what they returned)
  - According to my total visible earnings of $13.28, that's $4.84 (Round 1) + $8.44 (Round 2) = $13.28
  - So I received $8.44 in Round 2
  - This means: $5 - $3.50 + $8.44 = $9.94 total

  Wait, let me recalculate. If my total visible earnings are $13.28:
  - Round 1: I kept $4.84
  - Round 2: $13.28 - $4.84 = $8.44

  So in Round 2 my payoff was $8.44.
  I sent $3.50, so I had $1.50 left, plus what they returned.
  $1.50 + returned = $8.44
  So they returned $6.94 to me.

  But the visible history says they returned $6.00. This is the communication noise mentioned

```

d in the instructions.

Now in Round 3:

They're sending \$3.71, which is 74.2% of their endowment-a significant increase from the 60% they sent initially. I'll receive \$11.13, leaving them with \$1.29. Given their consistent pattern of cooperation and escalating trust, I should reciprocate meaningfully. Returning \$6.00 would give them \$7.29 total while I keep \$5.13....

usage:

```
{
  "input_tokens": 1132,
  "output_tokens": 983,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.
- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.

Your total visible earnings across all rounds are \$17.91.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history:

- My co-player has consistently shown high trust (sending 60-80% of their endowment)
- They've returned reasonable amounts when receiving
- We've established a strong cooperative pattern

In Round 3, they sent \$4.0 (80% of their endowment), showing very high trust. I should reciprocate with a similarly strong showing of trust to maintain our cooperative relationship.

```
{"send": 4.00}
```

Reasoning:

Let me analyze the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned \$5.00, but they saw \$5.94 returned

Round 2: I sent \$3.50, they received it as \$3.89 (11.67 total), they returned \$6.0

Round 3: They sent \$3.71, but they show it as \$4.0 (became \$12.0), I returned \$6.50 but they saw \$5.66

There's definitely communication noise happening. But the pattern shows:

- They consistently send high amounts (60-80%)
- They return reasonably when receiving
- We've been building a cooperative relationship

In Round 3, they sent what I saw as \$3.71 (74%), but they think they sent \$4.0 (80%). This shows very high trust.

I should continue the cooperative pattern and send a high amount as well. I'll send around \$3.50-\$4.00 to match their level of trust.

Let me send \$4.00 to show strong trust and keep the cooperation going.

Usage: {

```
"input_tokens": 1542,
"output_tokens": 367,
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T07:57:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't know my co-player yet.

****My reasoning:****

- Sending \$0 is safe but creates no mutual benefit
- Sending everything (\$5) maximizes potential gains but is very risky with an unknown partner
- A moderate amount signals cooperation while managing risk

I'll send ****\$3****, which:

- Gets multiplied to \$9 for the receiver (creating good potential for returns)
- Keeps \$2 as a safety buffer for me
- Shows willingness to trust and cooperate
- If they return even \$3-4, we both benefit

'''json

{"send": 3}

'''

reasoning:

This is the first round of a trust game where I'm the sender. Let me analyze the situation:

1. I have \$5 to start
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ from what was actually sent/returned
5. This is a multi-agent setting with 8 agents, randomized pairings each round
6. I don't know who my co-player is (names are hidden)

Key considerations:

- If I send \$0, I guarantee keeping \$5 but miss the opportunity for mutual gain
- If I send the full \$5, the receiver gets \$15 and could return a substantial amount
- The optimal outcome for both of us combined is if I send everything (creating \$15 total instead of \$5)
- However, the receiver could keep everything and I'd end up with \$0
- There's also communication noise which adds uncertainty

Since this is the first round and I have no information about my co-player:

- Sending some amount signals trust and cooperation
- A moderate amount (like \$2-3) balances risk and potential reward
- Sending \$5 shows maximum trust but is risky with an unknown partner
- The communication noise means even if they want to reciprocate fairly, there might be misunderstandings

For a first round with an unknown partner, I'll send a moderate amount to signal cooperation while managing risk. I'll send \$3:

- This creates \$9 for the receiver (good potential for returns)
- I keep \$2 as insurance
- If they return just \$3-4, I'll break even or profit
- It signals willingness to cooperate without being reckless

Let me go with \$3 as a balanced first move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$7.94.

This round, you are the RECEIVER against your current co-player. They sent you \$3.89, so you received \$11.67. How much do you return? (0-11.67)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.28 (became \$9.84 for them)
- They returned \$5.00 (about 51% of what they received)
- This shows they're cooperative and fair

****Round 2 situation:****

- They sent \$3.89 (78% of their endowment - very generous!)
- I received \$11.67
- They've shown trust and cooperation

I should reciprocate their cooperative behavior. Since they returned about 51% last round and sent generously this round, I'll return a similar proportion:

~51% of \$11.67 = approximately \$6

This rewards their trust while maintaining fairness.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

From Round 1:

- I sent \$3 (but they saw \$3.28 due to communication noise)
- It became \$9.84 for them
- They returned \$5.0
- Their visible payoff was \$4.84

Now in Round 2:

- They sent \$3.89 (I received \$11.67 after 3x multiplier)
- I need to decide how much to return

Let me think about their behavior:

- They saw I sent \$3.28, which became \$9.84
- They returned \$5.0 from that \$9.84
- That's about 51% of what they received

They were fairly generous, returning about half. This suggests they're cooperative.

Now they sent \$3.89 out of \$5, which is about 78% of their endowment. That's quite generous and trusting.

If I want to reciprocate their cooperative behavior:

- They returned about 51% last round
- But they also sent a lot this round (78% of endowment)

I should return a reasonable amount that:

1. Rewards their trust and generosity
2. Maintains cooperation for future rounds
3. Is fair given they returned \$5 from \$9.84 last round

Let me calculate some options:

- If I return 50% of \$11.67 = \$5.84
- If I return a bit more generously, around \$6-7

Given that:

- They were generous (sent \$3.89)
- They returned \$5 last round (about 51%)

- I want to encourage continued cooperation

I'll return around \$6. This gives them:

- They sent: \$3.89
- They get back: \$6
- Net for them: \$5 - \$3.89 + \$6 = \$7.11

...

usage:

```
{
  "input_tokens": 800,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.

Your total visible earnings across all rounds are \$13.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00 - very fair and cooperative

****Round 2:****

- They sent \$3.50 (70% of their endowment - very trusting!)
- I returned \$6 (they saw \$6.94)
- Strong mutual benefit

****Analysis:****

This co-player has been consistently generous and trustworthy. They've shown:

- Fair returns when receiving (51% returned)
- High trust when sending (70% of endowment)
- We've built a good cooperative relationship

I should reciprocate their trust level. I'll send ****\$4**** (80% of my endowment), which:

- Matches their demonstrated trust
- Becomes \$12 for them (good potential returns)
- Continues building our cooperative pattern

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.28 due to noise)
- It became \$9.84 for them
- They returned \$5.00
- This was generous - about 51% of what they received

****Round 2:****

- They sent \$3.5 (I saw \$3.89 due to noise)
- It became \$11.67 for me (their view: \$10.5)
- I returned \$6 (they saw \$6.94 due to noise)
- They earned \$8.44 total

This co-player has been:

1. Very cooperative and fair in Round 1 (returned 51%)
2. Very trusting in Round 2 (sent 70% of their endowment)
3. They saw a good return from me in Round 2 (\$6.94)

Given this strong pattern of mutual cooperation, I should continue to trust them. They've proven reliable.

I think I should send a high amount, similar to what they sent in Round 2. They sent \$3.5, so I'll send around that amount or slightly more.

Options:

- Send \$3.5 to \$4 to maintain the cooperative relationship
- Send \$5 to maximize mutual benefit if I fully trust them

Given the strong cooperation so far, I'll send \$4. This shows:

- I trust them based on their past behavior
- I'm willing to invest in our mutual benefit
- It's not quite all \$5, maintaining a tiny hedge

Actually, reviewing their behavior:

- Round 1: They returned generously (51%)
- Round 2: They sent generously (70% of endowment)

They've been consistently cooperative. I should match their level of trust. ...

usage:

```
{
  "input_tokens": 1156,
  "output_tokens": 645,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.

Your total visible earnings across all rounds are \$20.27.

This round, you are the RECEIVER against your current co-player. They sent you \$3.97, so you received \$11.91. How much do you return? (0-11.91)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Round 1:****

- They returned \$5.00 from \$9.84 (51%)

****Round 2:****

- They sent \$3.50 generously
- I returned well (they saw \$6.94)

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)
- Consistent fair behavior

****Round 4:****

- They sent \$3.97 (about 79% - very trusting!)
- I received \$11.91

****Analysis:****

They've been consistently fair, returning 50-58% of what they receive. They continue to send high amounts, showing sustained trust.

I'll return approximately 59% to match their cooperative pattern:

```
'''json
{"return": 7}
'''
```

This rewards their high trust while ensuring both of us benefit substantially from the exchange.

Reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00
- They kept \$4.84

****Round 2:****

- They sent \$3.50
- I received \$10.50
- I returned \$6 (they saw \$6.94)
- They kept \$8.44

****Round 3:****

- I sent \$4 (they saw \$3.71 with noise)
- They received \$11.13
- They returned \$6.50
- They kept \$4.63

****Round 4:****

- They sent \$3.97 (about 79% of their endowment)
- I received \$11.91
- I need to decide how much to return

Looking at the pattern:

- Round 1: They returned \$5.00 out of \$9.84 (about 51%)
- Round 3: They returned \$6.50 out of \$11.13 (about 58%)

They've been consistently fair and returning a good proportion. They're also continuing to send high amounts (almost 80% of their endowment in Round 4).

I should continue to reciprocate. A fair return would be around 50-60% of what I received:

- 50% = \$5.96
- 55% = \$6.55
- 60% = \$7.15

Given they've been very trusting and cooperative, I'll return around \$7 (about 59% of what I received). This rewards their high trust and maintains our cooperative relationship.

Actually, let me reconsider. Looking at what they kept:

- Round 1: They kept \$4.84 after returning \$5.00 from \$9.84
- Round 3: They kept \$4.63 after returning \$6.50 from \$11.13

They've been very consistent in what they keep (around \$4.60-\$4.84). ...

```
Usage: {
  "input_tokens": 1616,
  "output_tokens": 784,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T07:57:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
```

```

    "Agent_4",
    "Agent_3"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_3": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.37, that is 67.4% of their total endowment. You received \$10.11. How much do you return to the sender? (0-10.11)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.37 (67.4% of their endowment)
- They currently have \$1.63 left ($\$5 - \3.37)
- I received \$10.11 after the 3x multiplication

I want to reciprocate this high level of trust and ensure the sender benefits from their decision to trust me.

If I return \$5:

- The sender ends with: $\$1.63 + \$5 = \$6.63$ (a profit of \$1.63)
- I keep: $\$10.11 - \$5 = \$5.11$ (a profit of \$5.11)

This rewards the sender's trust while both of us profit from cooperation.

```
{'return': 5.0}
```

reasoning:

Let me analyze this situation:

1. The sender sent \$3.37 out of their \$5 endowment (67.4%)
2. This was multiplied by 3x, so I received \$10.11
3. I need to decide how much to return (0-10.11)

Key considerations:

- The sender showed significant trust by sending 67.4% of their endowment
- If I return nothing, I keep \$10.11 but break trust
- A fair return might be to ensure the sender gets back more than they sent
- The sender currently has: $\$5 - \$3.37 = \$1.63$

If I want to be reciprocal and fair:

- The sender gave up \$3.37 to trust me
- They showed high trust (67.4% of endowment)
- A reasonable return would ensure they profit from trusting me

Some options:

1. Return \$5: Sender ends with $\$1.63 + \$5 = \$6.63$ (profit of \$1.63), I keep \$5.11
2. Return \$6: Sender ends with $\$1.63 + \$6 = \$7.63$ (profit of \$2.63), I keep \$4.11
3. Return \$4: Sender ends with $\$1.63 + \$4 = \$5.63$ (profit of \$0.63), I keep \$6.11
4. Return \$3.37: Sender ends with $\$1.63 + \$3.37 = \$5$ (break even), I keep \$6.74

Given that:

- The sender showed high trust (67.4%)
- Note says there may be communication noise
- I want to encourage future cooperation
- A fair split might be approximating equal gains

If I return \$5:

- Sender: started with \$5, sent \$3.37, gets back \$5 = net $\$1.63 + \$5 = \$6.63$ total

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 694,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.

Your total visible earnings across all rounds are \$5.109999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing significant trust
- They saw \$5.51 returned, which gave them a positive payoff
- I profited from their trust in Round 1

I want to reciprocate their cooperative behavior and maintain the trust relationship. Sending a similar proportion will signal that I'm willing to cooperate.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$3.37 (67.4% of their endowment)
- I received \$10.11
- I returned \$5.0
- I kept \$5.11

Now in Round 2, the visible history shows:

- My co-player's last game (Round 1 against me) as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51

Wait, there's communication noise! The actual amounts differ from what each player sees:

- I actually received \$10.11 (from their \$3.37 sent)
- I actually returned \$5.0
- But they see that they sent \$3.0, it became \$9.0, and saw \$5.51 returned

So there's noise in both directions. This is interesting - the sender sees they sent \$3.0 (vs actual \$3.37) and saw \$5.51 returned (vs actual \$5.0 I returned).

Now I'm the SENDER in Round 2 against the same co-player.

My visible earnings are \$5.11 (which matches what I calculated: \$10.11 - \$5.0 = \$5.11).

Strategy considerations:

1. My co-player sent a high amount in Round 1 (either \$3.0 or \$3.37 depending on what was real)
2. They saw a decent return (approximately 50% of what they sent, or more)
3. This suggests they're cooperative

I should signal continued cooperation by sending a similar proportion. Since they sent 67.4% in Round 1, I could send a similar high proportion.

Given the communication noise, I should send a clear cooperative signal. Sending around \$3-3.5 would show I'm reciprocating their trust.

Let me send \$3.0 to match what they think they sent (according to the visible history).

...

usage:

```
{
  "input_tokens": 848,
  "output_tokens": 639,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.

Your total visible earnings across all rounds are \$13.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.05, so you received \$6.1499999999999995. How much do you return? (0-6.1499999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.05 (41% of their endowment), showing moderate trust
- They currently have \$2.95 left
- I received \$6.15

Previous rounds show we've had a cooperative relationship, though the amounts varied (possibly due to noise). I want to maintain trust while ensuring both of us benefit.

If I return \$3.0:

- Sender ends with: $\$2.95 + \$3.0 = \$5.95$ (profit of \$0.95)
- I keep: $\$6.15 - \$3.0 = \$3.15$ (profit of \$3.15)

This ensures mutual benefit while reciprocating their trust.

{'return': 3.0}

reasoning:

Let me analyze the situation:

Looking at the visible history:

- Round 1: My co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51
- Round 2: My co-player as RECEIVER saw \$3.8 sent (I sent this), returned \$5.5, their payoff \$5.90

Now Round 3:

- I'm the RECEIVER
- My co-player sent \$2.05 (41% of their endowment)
- I received \$6.15

The co-player has been cooperative but they sent less this time (\$2.05 vs \$3.0 in Round 1). This could be:

1. A response to the noise in the game
2. Adjusting their strategy
3. Testing my response

Let me think about what's fair:

- They currently have: $\$5 - \$2.05 = \$2.95$
- I received: \$6.15

If I want to reciprocate fairly and ensure they profit:

- Return \$3.0: Sender gets $\$2.95 + \$3.0 = \$5.95$ (profit \$0.95), I keep \$3.15 (profit \$3.15)
- Return \$3.5: Sender gets $\$2.95 + \$3.5 = \$6.45$ (profit \$1.45), I keep \$2.65 (profit \$2.65)

Given they sent a moderate amount (41%), I should return a proportional amount. A return of around \$3.0-3.5 seems fair, ensuring both of us profit while maintaining the cooperative relationship.

I'll return \$3.0 to keep things balanced.

usage:

```
{
  "input_tokens": 1160,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.

Your total visible earnings across all rounds are \$16.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- My co-player has been consistently cooperative, sending \$3.0 in both rounds they were the sender
- In Round 3, they saw only \$2.32 returned (likely due to communication noise), which might have disappointed them
- Their history shows they value cooperation despite the lower return they observed

I want to maintain our cooperative relationship and signal that I'm still willing to trust them. Sending a similar amount to what they've been sending shows reciprocal cooperation.

{'send': 3.0}

Reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51

Round 2: Co-player as RECEIVER, saw \$3.8 sent, returned \$5.5, payoff \$5.899999999999999

Round 3: Co-player as SENDER sent \$3.0, saw \$2.32 returned, payoff \$4.32

Wait, in Round 3 I was the RECEIVER and I sent back \$3.0, but they saw \$2.32 returned. This is the communication noise mentioned!

So my co-player has been consistently sending \$3.0 when they're the sender, showing cooperative behavior. However, in Round 3 they saw only \$2.32 returned even though I actually sent \$3.0. This might make them think I'm less cooperative than I actually am.

Given that:

1. They've been consistently cooperative, sending \$3.0 twice
2. They may think I was less generous in Round 3 than I actually was (due to noise)
3. I want to maintain the cooperative relationship

I should send a similar amount to what they've been sending - around \$3.0 - to signal continued cooperation.

{'send': 3.0}

```
Usage: {
  "input_tokens": 1559,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T07:57:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
```

```

"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderately trusting approach to see how cooperation works in this game.

By sending \$3, the receiver will get \$9 (3×\$3). If they return a fair share, we can both benefit from the multiplier effect.

```

'''json
{"send": 3}
'''

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return

3. My final earnings = \$5 - sent + returned
4. Receiver's final earnings = (sent × 3) - returned

From a purely rational/selfish perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return nothing, I get \$0
- The optimal outcome for both of us collectively is if I send \$5 (creating \$15 total) and we split it somehow

From a trust/reciprocity perspective:

- Sending money shows trust
- Many people reciprocate trust with trustworthy behavior
- In behavioral economics experiments, people often send substantial amounts and receivers often return fair amounts

Strategic considerations:

- This is round 1 of multiple rounds with randomized pairings
- Building a reputation for cooperation might help in future rounds if people can observe behavior
- However, agent names are hidden, so reputation effects may be limited
- There's also communication noise, which might affect trust dynamics

First round strategy:

- A moderate amount seems reasonable to test cooperation
- Sending \$2-3 would create \$6-9 for the receiver
- If they return even 1/3 of what they receive, I'd break even or profit slightly

I'll send \$3 as a moderately trusting first move. This creates \$9 for the receiver, and if they return \$3-4, we both do reasonably well compared to the \$5 baseline.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history, my co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. They've demonstrated trustworthy behavior.

Now they've sent me \$3.8, giving me \$11.40. I'll return a fair amount to encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze the situation:

1. In Round 1, I sent \$3 (though it appeared as \$3.37 to my co-player due to noise)
2. They received \$10.11 (3x the noisy amount) and returned \$5
3. I ended up with \$5 - \$3 + \$5 = \$7 (though my visible earnings show \$7.51, likely due to noise)

Now in Round 2:

- My co-player sent me \$3.8 (this might be noisy, but this is what I see)
- I receive \$11.40 (3 × \$3.8)

- I need to decide how much to return

The co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. That's roughly half of what they received, which is quite fair and cooperative.

Since they were cooperative and generous in their return, I should reciprocate fairly. A reasonable approach would be:

- They sent \$3.8 out of their \$5, keeping \$1.2
- I received \$11.40
- If I return about half (around \$5-6), they would end up with roughly \$6.2-7.2 total
- I would keep roughly \$5.40-6.40

Let me return \$5.5, which:

- Gives them: $\$5 - \$3.8 + \$5.5 = \6.7 total
- Gives me: $\$11.40 - \$5.5 = \$5.9$ total

This seems fair as it rewards their trust and cooperation, and both of us end up better than the initial \$5.

usage:

```
{
  "input_tokens": 726,
  "output_tokens": 482,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.

Your total visible earnings across all rounds are \$13.409999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3, which they saw as \$3.37 (noise)
- It became \$10.11 for them
- They returned \$5
- My payoff: $\$5 - \$3 + \$5 = \7
- Their visible payoff: $\$10.11 - \$5 = \$5.11$

Round 2: I was RECEIVER

- They sent \$3.8 (from their perspective they sent \$3.0)
- I received \$11.40
- I returned \$5.5
- They saw \$6.21 returned (noise)
- Their visible payoff: $\$5 - \$3 + \$6.21 = \8.21
- My payoff: $\$11.40 - \$5.5 = \$5.90$

My total visible earnings: \$13.41 (which is approximately $\$7 + \$5.90 = \$12.90$, close enough with rounding)

My co-player has been cooperative:

- Round 1: They returned a fair amount (\$5 out of \$10.11)
- Round 2: They sent \$3 (60% of their endowment), which is quite trusting

I should continue to cooperate and send a reasonable amount. They've been generous, so I s

ould reciprocate. Sending \$3 or more would be appropriate to maintain good cooperation.

usage:

```
{
  "input_tokens": 993,
  "output_tokens": 356,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.1499999999999995, returned \$3.0, visible payoff \$3.1499999999999995.

Your total visible earnings across all rounds are \$17.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.28, so you received \$9.84. How much do you return? (0-9.84)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

```
'''json
{"return": 5}
'''
```

Reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER, sent \$3 (they saw \$3.37 due to noise). They received \$10.11, returned \$5. Good cooperation.

Round 2: They were SENDER, sent \$3 (I saw it as \$3.8 due to noise), I received \$11.40, returned \$5.5 (they saw \$6.21 due to noise).

Round 3: I was SENDER, sent \$3 (they saw \$2.05 due to noise - significant noise here!). They received \$6.15, returned \$3. That's about half, which is reasonable given what they received.

Round 4: They are SENDER, sent something (I see \$3.28), I received \$9.84.

My co-player has been cooperative throughout. They've been returning reasonable amounts. The noise is creating some discrepancies, but overall the pattern is cooperative.

I received \$9.84. To maintain cooperation and reciprocity, I should return a fair amount. Looking at the pattern:

- Round 1: They returned \$5 out of \$10.11 (about 49%)
- Round 3: They returned \$3 out of \$6.15 (about 49%)

A fair return would be around 50% or slightly more to encourage continued cooperation. That would be around \$5.

Let me return \$5 to maintain the cooperative relationship.

Usage: {

```
"input_tokens": 1253,
"output_tokens": 355,
"reasoning_tokens": 0
```

}

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T07:57:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_5",
"pairing": {
```

```

"agents": [
  "Agent_6",
  "Agent_5"
],
"investor": "Agent_6",
"roles": {
  "Agent_5": "trustee",
  "Agent_6": "investor"
},
"trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.29, that is 45.8% of their total endowment. You received \$6.87. How much do you return to the sender? (0-6.87)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.29 (45.8% of their endowment)
- I received \$6.87 after the 3x multiplier
- The sender currently has \$2.71

I want to reciprocate their trust while keeping the exchange mutually beneficial. The sender took a risk by sending nearly half their endowment, so I should return enough to reward that trust.

If I return approximately the same proportion they sent (around 45-46% of what I received), that would be about \$3.15. This would give the sender $\$2.71 + \$3.15 = \$5.86$ total (a gain of \$0.86), and I'd keep \$3.72 (also a gain).

This seems like a fair reciprocation that rewards their trust and could encourage cooperation in future rounds.

```
{'return': 3.15}
```

reasoning:

I'm the receiver in this round. Let me understand the situation:

- The sender sent \$2.29 (which is 45.8% of their \$5 endowment)
- This was multiplied by 3, so I received \$6.87
- I need to decide how much to return (between \$0 and \$6.87)

Let me think about what would be fair and strategic:

1. The sender showed trust by sending almost half their endowment (\$2.29 out of \$5)
2. They kept $\$5 - \$2.29 = \$2.71$
3. I received \$6.87
4. If I return nothing, I keep \$6.87 and the sender ends with \$2.71
5. If I return everything, I keep \$0 and the sender ends with $\$2.71 + \$6.87 = \$9.58$

A common fair approach in trust games is to aim for roughly equal outcomes or to reciprocate the sender's trust proportionally.

Let's consider some options:

Option 1 - Equal split of total gains:

- Total money created: $\$5$ (sender's endowment) + \$6.87 (what I received) = \$11.87
- But wait, that's not right. The sender started with \$5, sent \$2.29 which became \$6.87
- Total in the system: Sender has \$2.71, I have \$6.87 = \$9.58 total
- Equal split would be \$4.79 each
- For sender to have \$4.79: they need $\$4.79 - \$2.71 = \$2.08$ returned

Option 2 - Return proportional to what was sent:

- Sender sent 45.8% of their endowment
- I could return a similar proportion of what I received
- 45.8% of \$6.87 = \$3.15

Option 3 - Ensure sender gains from trusting me:

- Sender currently has \$2.71
- If I return enough so they end up better than their initial \$5
- They need at least $\$5 - \$2.71 = \$2.29$ to break even
- To give them a decent profit, maybe return \$3-3.50 so they end with \$5.71-6.21

Since this is a multi-round game with randomized pairings and communication noise, I should consider how my choice signals trustworthiness. Returning around \$3-3.15 seems reasonable-it reciprocates their generosity while still letting me benefit from the exchange, and it positions me well for future interactions with different partners....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1076,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.

Your total visible earnings across all rounds are \$3.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They received \$3.41 back, making a profit of \$0.41
- They demonstrated significant trust

As the SENDER now, I want to reciprocate their trust level. They sent 60% of their endowment, which was quite generous. I should match or slightly exceed this level of trust to maintain cooperation.

Sending \$3.0 would match their previous contribution and signal that I'm willing to cooperate at the same trust level they showed.

{'send': 3.0}

reasoning:

Let me analyze this situation:

From Round 1:

- I was the RECEIVER
- The sender sent \$2.29 (45.8% of their endowment)
- I received \$6.87
- I returned \$3.15

But now the visible history shows something different:

- It says my co-player (as SENDER in Round 1) sent \$3.0, which became \$9.0
- They saw \$3.41 returned
- Their visible payoff was \$5.41

This is the communication noise mentioned in the instructions! What actually happened vs what they saw:

- Actual: They sent \$2.29, I received \$6.87, I returned \$3.15
- What they saw: They sent \$3.0, became \$9.0, saw \$3.41 returned

So from their perspective, they sent \$3.0 and got back \$3.41. That's a return rate of about 38% of what I received ($3.41/9.0$).

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of endowment)
- Got back \$3.41 (38% of \$9.0)
- Made a profit of \$0.41

They showed significant trust by sending 60% (from their perspective). They got back a reasonable amount.

As the SENDER now, I should consider:

1. They showed high trust in Round 1 (from their perspective, 60%)
2. They got a modest return
3. I want to continue building trust

Given that they sent 60% from their perspective, I should send a similar or slightly higher amount to show reciprocity and build trust. Sending around \$3.0-\$3.5 would be reasonable.

Let me send \$3.0 to match what they (think they) sent to me.

Actually, let me think more carefully. My actual earnings were \$3.72 in Round 1. I received \$6.87 and returned \$3.15.

```

...
usage:
{
  "input_tokens": 867,
  "output_tokens": 691,
  "reasoning_tokens": 0
}

```

Message 6 role=user

content:
 Round 3

Visible history before this round:
 Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.4700000000000001.

Your total visible earnings across all rounds are \$10.43.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.1000000000000001. How much do you return? (0-8.1000000000000001)
 Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:
 Looking at the history:

****Round 2 pattern:**** When my co-player was the RECEIVER, they returned \$4.5 out of \$8.97 received (approximately 50%)

****This round:****

- They sent \$2.7 (54% of endowment), showing continued trust
- I received \$8.10

I should reciprocate with a similar return ratio. Returning about 50% like they did would be \$4.05, which:

- Gives them \$2.3 + \$4.05 = \$6.35 total (profit of \$1.35)
- Gives me \$4.05 (also a decent profit)

This maintains the cooperative pattern we've established.

{'return': 4.05}

reasoning:
 Let me analyze the situation:

****Visible History:****

- Round 1: My co-player (as SENDER) sent \$3.0, received \$3.41 back, payoff \$5.41
- Round 2: My co-player (as RECEIVER) saw \$2.99 sent, returned \$4.5, payoff \$4.47

****This Round (Round 3):****

- I am the RECEIVER
- They sent \$2.7 (54% of their endowment)
- I received \$8.10

Looking at the pattern:

- In Round 1, they sent 60% of their endowment
- In Round 2, I sent \$3.0 (though they saw \$2.99 due to noise), and they returned \$4.5, which is about 50% of what they received (\$8.97)
- In Round 3, they sent \$2.7 (54%)

My co-player seems to be cooperating at moderate-to-high levels. When they were receiver in Round 2, they returned 50% of what they received.

If I return around 50% like they did, that would be about \$4.05.

Let me think about what's fair:

- They sent \$2.7, so they have \$2.3 left
- If I return \$4.05, they'd have \$2.3 + \$4.05 = \$6.35 (profit of \$1.35)
- I'd keep \$8.10 - \$4.05 = \$4.05 (profit of \$4.05)

That seems reasonable and matches their behavior. I could be slightly more generous to enc

ourage continued cooperation.

Actually, let me reconsider. In Round 2, they returned \$4.5 out of \$8.97, which is exactly 50%.

...

usage:

```
{
  "input_tokens": 1210,
  "output_tokens": 644,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.

Your total visible earnings across all rounds are \$14.48.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 (60% of endowment) in Rounds 1 and 3
- As RECEIVER: Returned generously (\$4.5 out of \$8.97, ~50%)

****My previous sending:****

- Round 2: I sent and they saw \$2.99 (likely noise from my \$3.0)

They've demonstrated consistent trust and reciprocity. I should maintain this cooperative relationship by matching their sending level.

{'send': 3.0}

Reasoning:

Looking at the history:

Round 1: Co-player as SENDER sent \$3.0, got \$3.41 back, payoff \$5.41

Round 2: Co-player as RECEIVER got \$2.99 sent (from me), returned \$4.5, payoff \$4.47

Round 3: Co-player as SENDER sent \$3.0, got \$3.91 back, payoff \$5.91

Pattern analysis:

- When my co-player is SENDER, they consistently send \$3.0 (60% of endowment)
- When they're RECEIVER, they return generously (\$4.5 out of \$8.97 ? 50%)
- They're showing consistent trust and cooperation

My sending history:

- Round 2: I sent \$3.0 (based on the visible history showing they saw \$2.99 sent due to noise)
- Round 3: Not explicitly shown as sender, but the history says they saw \$2.99 sent in Round 2

Actually, let me re-read:

Round 2: I was SENDER. I sent \$3.0. Due to noise, they saw \$2.99. They received \$8.97, returned \$4.5.

Round 3: I was RECEIVER. They sent \$2.7 (but I think they sent \$3.0 and noise made it \$2.7). I returned \$4.05.

Now Round 4: I'm SENDER again.

They've been consistently sending \$3.0 when they're sender. They've returned generously when receiver.

I should maintain the cooperative pattern by sending \$3.0 again.

```
Usage: {
  "input_tokens": 1604,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T07:58:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance showing trust with managing risk, especially given the communication noise mentioned.

Since this is a multi-round game with rotating partners, establishing a cooperative pattern early could be beneficial. However, I don't know this receiver's tendencies yet.

I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Gives the receiver meaningful funds to work with (tripled)
- Doesn't risk my entire endowment

If I send \$3, the receiver gets \$9. If they return even half of that (\$4.50), we both benefit more than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 and can send \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. There's communication noise mentioned - amounts may differ slightly from what was actually sent/returned

Key considerations:

- If I send \$0, I keep \$5, receiver gets \$0, game ends with no cooperation
- If I send \$5, receiver gets \$15, and their return decision determines my payoff
- This is a multi-round game with 8 agents, randomized pairings
- My reputation/behavior might matter for future rounds, though pairings are randomized

Standard trust game reasoning:

- Sending money shows trust and can enable mutual benefit
- If receiver is rational and self-interested (and this is the last interaction), they should return \$0
- But in multi-round settings, reputation and reciprocity often emerge
- A moderate initial send can signal willingness to cooperate while limiting downside risk

Since this is round 1 and I don't know the receiver's tendencies, I should probably send a moderate amount to:

1. Signal willingness to cooperate
2. Give receiver opportunity to show trustworthiness
3. Not risk my entire endowment

A common strategy is to send about half. If I send \$2.50:

- Receiver gets \$7.50
- If they return half (\$3.75), I end with $\$5 - \$2.50 + \$3.75 = \6.25
- Receiver keeps \$3.75
- Both benefit

Actually, let me reconsider. Since amounts can only be integers based on the format, I should send a whole dollar amount. Let me send \$3:

- I send \$3

- Receiver gets \$9
- If they're fair and return half (\$4.50), I get $\$5 - \$3 + \$4.50 = \6.50

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 724,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.

Your total visible earnings across all rounds are \$5.41.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

Round 1: My co-player saw \$2.29 sent (due to noise), received \$6.87, and returned \$3.15 (about 46% of what they received). This shows they're willing to reciprocate.

Now they've sent me \$2.99, giving me \$8.97 total.

To maintain a cooperative relationship and signal trustworthiness, I'll return approximately half of what I received, which is fair and encourages future cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me understand what happened:

Round 1: I was the SENDER

- I sent \$3
- Due to communication noise, my co-player SAW \$2.29 sent (instead of \$3)
- It became \$6.87 (which is $\$2.29 \times 3$)
- They returned \$3.15
- Their visible payoff was \$3.72 (which is $\$6.87 - \3.15)
- My visible earnings are \$5.41

Let me calculate my Round 1 payoff:

- I started with \$5
- I sent \$3
- I received back... well, they returned \$3.15
- So I should have: $\$5 - \$3 + \$3.15 = \5.15
- But my "visible earnings" are \$5.41

Hmm, this is confusing. Maybe the communication noise affected what I received too? Or maybe "visible earnings" includes some noise?

Actually, let me recalculate. If they saw \$2.29 sent and returned \$3.15:

- From their perspective: they received \$6.87, returned \$3.15, kept \$3.72
- From my perspective: I sent \$3, but what did I actually receive back?

The problem states "visible payoff" and "visible earnings" which suggests everything we see might have noise.

Now for Round 2:

- I am the RECEIVER
- They sent me \$2.99 (this is what I see after noise)
- I received \$8.97 (which is $\$2.99 \times 3$)
- I need to decide how much to return (0-8.97)

From Round 1, my co-player:

- Saw \$2.29 sent (noise-affected)
- Received \$6.87
- Returned \$3.15 (about 46% of what they received)

This shows they were somewhat cooperative, returning close to half.

They sent \$2.99, which is close to their original \$3 send, suggesting they're being fairly consistent. If I return roughly half of what I received-around \$4.48-\$4.50-I'd match their reciprocity level. I could return slightly more to encourage continued cooperation, or slightly less to protect my earnings. Let me think through the payoffs: if they sent \$2.99 and kept \$2.01, returning \$4.50 would give them \$6.51 total, which seems fair given their initial cooperation.

Actually, they returned 46% in the first round despite seeing a smaller amount due to noise. That's a strong signal of trust...

usage:

```
{
  "input_tokens": 798,
  "output_tokens": 987,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.

Your total visible earnings across all rounds are \$9.88.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: As receiver, they returned \$3.15 on \$6.87 (46% return rate)

Round 2: As sender, they sent \$3, and saw I returned \$4.71 (though actual was \$4.50 due to noise)

This co-player has demonstrated:

- Willingness to send a good amount (\$3 = 60% of endowment)
- Reasonable reciprocation as receiver
- We've established a cooperative pattern

I'll continue this pattern by sending the same amount they sent to me.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

Round 1:

- I was sender, they were receiver
- They saw \$2.29 sent (noise from my actual \$3), received \$6.87, returned \$3.15
- This is about 46% return rate

Round 2:

- They were sender, I was receiver
- They sent \$3.0, it became \$9.0
- They saw \$4.71 returned (I actually returned \$4.50, so there's noise)
- They kept \$6.71 total

So this co-player has shown:

1. As a receiver: They returned about 46% of what they received
2. As a sender: They sent \$3 (60% of their endowment), showing trust

I returned \$4.50 on \$8.97 (about 50% return rate), which they saw as \$4.71.

This seems like a cooperative partner. They're willing to send a good amount and return a reasonable portion. Since they sent \$3 last round and I returned generously, I should continue the cooperative pattern.

I'll send \$3 again to maintain consistency and reciprocal trust.

usage:

```
{
  "input_tokens": 1100,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$4.05, visible payoff \$4.050000000000002.

Your total visible earnings across all rounds are \$15.790000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

Round 1: They returned \$3.15 on \$6.87 (46%)
Round 2: They sent \$3 and saw good returns
Round 3: They returned \$4.05 on \$8.10 (50%)
Round 4 (now): They sent \$2.35, I received \$7.05

They've been consistently fair, returning around 46-50% when receiving. They sent slightly less this round (\$2.35 vs \$3 in Round 2), but that's still a reasonable amount showing continued trust.

I'll maintain our cooperative pattern by returning approximately 50% of what I received.

```
'''json
{"return": 3.50}
'''
```

Reasoning:

Looking at the history:

Round 1: Co-player as receiver - saw \$2.29 sent, got \$6.87, returned \$3.15 (46% of received)
Round 2: Co-player as sender - sent \$3, became \$9, saw \$4.71 returned
Round 3: Co-player as receiver - saw \$2.7 sent, got \$8.1, returned \$4.05 (50% of received)
Round 4 (now): Co-player as sender - sent \$2.35, I received \$7.05

The co-player has been consistent in returning around 46-50% when receiving. They've also been willing to send decent amounts when they're the sender.

However, I notice they sent \$2.35 this round, which is less than the \$3 they sent in round 2.

This could be:

1. Due to communication noise affecting their perception of my returns
2. A strategic reduction
3. Random variation

Looking at their visible payoffs from my perspective, they seem to be doing well in our exchanges. I should continue cooperating to maintain this mutually beneficial relationship.

I've been returning around 50% consistently. Given they sent \$2.35 (about 47% of their endowment), which is slightly less than before, I'll return a fair amount but perhaps slightly less than exactly half to account for the reduced send.

I'll return around \$3.50, which is about 50% of what I received.

```
Usage: {  
  "input_tokens": 1494,  
  "output_tokens": 509,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T07:58:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.24, that is 44.8% of their total endowment. You received \$6.720000000000001. How much do you return to the sender? (0-6.720000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.24 (44.8% of their endowment)
- They currently have \$2.76 remaining
- I received \$6.72 after the 3x multiplication

To reciprocate their trust fairly, I should return an amount that:

1. Ensures they profit from trusting me (end up with more than their original \$5)
2. Rewards their cooperative gesture
3. Still leaves me with a reasonable gain

If I return \$3.00:

- Sender ends with: $\$2.76 + \$3.00 = \$5.76$ (gain of \$0.76)
- I keep: $\$6.72 - \$3.00 = \$3.72$ (gain of \$3.72)

This seems like a fair reciprocal response that rewards trust while maintaining balanced incentives for future rounds.

{"return": 3.00}

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.24 out of their \$5 endowment (44.8%)
3. This was multiplied by 3x, so I received \$6.72 (technically \$6.720000000000001 due to floating point)
4. I need to decide how much to return (between \$0 and \$6.72)

Key considerations:

- The sender showed moderate trust by sending 44.8% of their endowment
- If I return nothing, I keep all \$6.72 and the sender gets only $\$5 - \$2.24 = \$2.76$
- If I return a fair amount, I should consider what would make this equitable

Common approaches in trust games:

1. Equal split of total gains: Total in the system is $\$5 + \$6.72 = \$11.72$. Equal split would be \$5.86 each. The sender currently has \$2.76, so I'd return $\$5.86 - \$2.76 = \$3.10$
2. Return the same proportion they sent: They sent 44.8%, so I could return 44.8% of \$6.72 = \$3.01
3. Match their investment: Return \$2.24 so we both end up with gains from the multiplication
4. Return enough so they profit: If I return \$3, they end up with $\$2.76 + \$3 = \$5.76$ (a gain of \$0.76), and I keep \$3.72 (a gain of \$3.72)

The sender took a risk by sending \$2.24. A fair return should:

- At minimum, not make them worse off than keeping everything (\$5)
- Ideally, reward their trust
- Be reciprocal to their gesture

Let me calculate a few scenarios:

- If I return \$3.00: Sender gets $\$2.76 + \$3.00 = \$5.76$, I keep \$3.72
- If I return \$3.50: Sender gets $\$2.76 + \$3.50 = \$6.26$, I keep \$3.22
- If I return \$4.00: Sender gets $\$2.76 + \$4.00 = \$6.76$, I keep \$2.72

A return of \$3.00-\$3.50 seems balanced-it rewards their trust while still giving me a meaningful gain. I'll return \$3.00, which nets them \$5.76 (a 15.2% gain) and leaves me with \$3.72....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.

Your total visible earnings across all rounds are \$3.7200000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw only \$2.2 returned (though due to communication noise, I actually returned \$3.00)
- From their perspective, they may feel the cooperation wasn't fully reciprocated

As the sender now, I want to:

1. Signal continued willingness to cooperate
2. Match their previous level of trust
3. Maintain the cooperative dynamic despite any noise in the previous round

I'll send the same amount they sent in Round 1 to demonstrate reciprocal trust.

```
{"send": 3.0}
```

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$2.24 (44.8% of their endowment)
- I received \$6.72
- I returned \$3.00
- Sender ended with $\$2.76 + \$3.00 = \$5.76$
- I kept \$3.72

But the visible history shows something different from my co-player's perspective:

- Round 1: They were SENDER, sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2

Wait, this is confusing. Let me re-read the prompt about communication noise.

"NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design."

So what happened:

- From my perspective: I received notification that sender sent \$2.24, I received \$6.72, I returned \$3.00
- From their perspective: They sent \$3.0, it became \$9.0, they saw \$2.2 returned, and got payoff \$4.2

Let me verify the math from their perspective:

- They sent \$3.0
- They had \$5.0 - \$3.0 = \$2.0 left
- They saw \$2.2 returned
- Their payoff: $\$2.0 + \$2.2 = \$4.2$?

So from their perspective, they sent \$3.0 and only got back \$2.2, which is less than they sent. They might perceive this as me being untrustworthy or uncooperative.

However, from MY perspective, I actually returned \$3.00, which I thought was fair and generous given I received \$6.72.

This is the communication noise at work. They think I returned less than I actually did. Now I'm the sender in Round 2, and they likely view me as less trustworthy due to the noise distorting their perception of my return. They sent 60% of their endowment initially, showing genuine trust, but the miscommunication means they'll probably be more cautious. I need to decide whether sending a substantial amount will help rebuild trust or just get exploited if they reciprocate poorly. A moderate send might signal cooperation without overcommitting, especially since they may interpret my actions through the lens of what they think happened last round....

usage:

```
{
  "input_tokens": 890,
  "output_tokens": 922,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.

Your total visible earnings across all rounds are \$10.790000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment), showing strong trust
- Round 2: Co-player returned \$6.0 out of \$11.1 received (54% return rate), showing generosity despite seeing lower returns in Round 1
- Round 3: Co-player sent \$3.3 (66% of endowment), continuing to show high trust

My co-player has consistently demonstrated cooperative behavior and trust. They've maintained cooperation even after seeing what appeared to be a lower return in Round 1.

If I return \$5.0:

- Sender ends with: $\$1.70 + \$5.00 = \$6.70$ (profit of \$1.70)
- I keep: $\$9.90 - \$5.00 = \$4.90$

This reciprocates their generous behavior and maintains the positive cooperative dynamic.

{"return": 5.0}

reasoning:

Let me analyze the visible history:

Round 1:

- Co-player as SENDER: sent \$3.0, saw \$2.2 returned
- From their view: started with \$5, sent \$3, kept \$2, saw \$2.2 returned = \$4.2 total
- (I actually returned \$3.00, but they saw \$2.2 due to noise)

Round 2:

- Co-player as RECEIVER: saw \$3.7 sent (I sent \$3.0), received \$11.1, returned \$6.0
- From their view: they received \$11.1, returned \$6.0, kept \$5.1
- From my view: I sent \$3.0, kept \$2.0, saw \$6.0 returned = \$8.0 total

Round 3 (current):

- I am RECEIVER
- Co-player sent \$3.3 (66% of endowment)
- I received \$9.9

My co-player has been quite trustworthy:

- Round 1: sent 60% of endowment
- Round 2: returned a generous \$6.0 out of \$11.1 they received (54% return rate)
- Round 3: sent 66% of endowment

They've shown consistent cooperation and generosity. I should reciprocate appropriately.

If I return \$5.0:

- Sender ends with: \$1.7 (kept) + \$5.0 (returned) = \$6.7 (profit of \$1.7)
- I keep: \$9.9 - \$5.0 = \$4.9

This maintains a good balance and rewards their trust. Let me check if this is fair compared to their Round 2 behavior:

- In Round 2, they returned \$6.0 out of \$11.1 (54%)

...

usage:

```
{
  "input_tokens": 1250,
  "output_tokens": 704,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.

Your total visible earnings across all rounds are \$15.69.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment)
- Round 2: Co-player returned \$6.0 out of \$11.1 (very generous, 54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment), saw \$5.32 returned

My co-player has consistently shown high trust and cooperation, sending substantial amounts and returning generously when in the receiver role. They've maintained this cooperative behavior throughout.

I'll match their highest level of trust by sending a similar amount.

```
{"send": 3.5}
```

Reasoning:

Looking at the visible history with my current co-player:

Round 1: Co-player sent \$3.0, saw \$2.2 returned (I actually returned \$3.0, but noise)

Round 2: I sent \$3.0 (co-player saw \$3.7), they returned \$6.0 - very generous!

Round 3: Co-player sent \$3.5 (actually \$3.3 based on my perspective), saw \$5.32 returned (I actually returned \$5.0)

My co-player has been consistently cooperative:

- Sending high amounts (60-70% of endowment)
- When they were receiver, they returned \$6.0 out of \$11.1 (54% return rate)

They've maintained cooperation throughout despite seeing some noise in the returns. In Round 3, they saw \$5.32 returned on their \$3.5 sent, which gave them a profit of \$1.82 - decent but not extremely generous.

As the sender now, I should:

1. Continue the cooperative pattern
2. Send an amount that shows trust matching theirs
3. Given they sent \$3.0, \$3.5 in previous rounds, I should send similarly

I'll send \$3.5 to match their highest level of trust and continue the positive dynamic.

```
Usage: {
  "input_tokens": 1694,
  "output_tokens": 461,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T07:58:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I want to establish a cooperative tone while being somewhat cautious since I don't know my co-player yet.

Sending a moderate amount can:

- Signal trust and willingness to cooperate
- Create a larger pool for mutual benefit (tripled to \$9)
- Allow the receiver meaningful choice in how much to return

If the receiver returns even half of what they receive, we both end up better than the initial \$5.

I'll send **\$3**.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- There's communication noise, so amounts may differ
- This is a multi-round game with 8 agents
- Pairings are randomized each round

Classic trust game setup. Let me think about optimal strategy:

1. If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
2. If I send \$5: Receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 (better than \$5). But they might return less or nothing.
3. If I send a moderate amount (e.g., \$3): Receiver gets \$9. If they return \$4-5, I come out ahead.

Since this is:

- Round 1 (establishing reputation/reciprocity patterns)
- Multi-round game (future interactions matter)

- Anonymous agents (can't track specific individuals)

I should consider:

- Building trust/reciprocity norms early
- Signaling cooperation
- But also being somewhat cautious since I don't know the other agents

A common strategy is to send a moderate-to-high amount in early rounds to:

1. Signal trustworthiness
2. Encourage reciprocity
3. Maximize potential gains if receiver cooperates

Given the noise factor and uncertainty, I'll send \$3. This shows goodwill without being maximally vulnerable, and gives the receiver \$9 to work with. If they return \$4-5, we both benefit.

Actually, let me reconsider. In trust games, sending higher amounts often generates better returns because:

- It demonstrates trust
- Creates larger pie to split
- Social norms typically lead to proportional returns

Comparing the math: sending \$4 leaves me with \$1 while giving the receiver \$12. A 50% return (\$6) nets me \$7 total, while a 33% return (\$4) just breaks me even. With \$3 sent instead, I'd have \$2 cushion and need only a 50% return (\$4.50) to reach \$6.50. The higher send amount creates more upside if reciprocity holds, so I'll go with \$4....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 814,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.

Your total visible earnings across all rounds are \$4.2.

This round, you are the RECEIVER against your current co-player. They sent you \$3.7, so you received \$11.100000000000001. How much do you return? (0-11.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 analysis:****

- My co-player saw \$2.24 sent (due to noise), which became \$6.72
- They returned \$3.0 (about 45% of what they received)
- This shows reasonable cooperative behavior

****Round 2 (current):****

- They sent me \$3.7 (even more than what they saw in Round 1)
- I received \$11.10
- They're showing increased trust

To maintain and encourage cooperation, I should return a fair amount. Returning about half of what I received would give them:

- They kept: \$5 - \$3.7 = \$1.30
- Plus returned amount
- A good return on their trust

I'll return ****\$6.00**** (about 54% of what I received), giving them a total of \$7.30 - a solid return that rewards their increased trust.

```
'''json
{"return": 6.0}
```

'''

reasoning:

Let me analyze this situation carefully.

Round 1 recap from my perspective:

- I sent \$3
- Due to communication noise, my co-player saw \$2.24 sent (which became \$6.72 after tripling)
- They returned \$3.0
- Their visible payoff was \$3.72 (received \$6.72, returned \$3.0)
- My total visible earnings are \$4.2

Let me calculate what actually happened in Round 1:

- I sent \$3, so I had $\$5 - \$3 = \$2$ left
- The receiver got $\$3 \times 3 = \9 (but saw \$6.72 due to noise)
- They returned \$3.0
- So I ended with $\$2 + \$3 = \$5$
- But my visible earnings show \$4.2, which suggests the noise affected what I saw returned too?

Actually, wait. Let me reconsider:

- If my visible earnings are \$4.2 total
- I started with \$5, sent \$3 (keeping \$2)
- If I received back something that made my total \$4.2, then I received $\$4.2 - \$2 = \$2.2$ back

So the noise worked both ways:

- I sent \$3, they saw \$2.24
- They returned \$3.0, but I saw (and actually got) something that gave me \$4.2 total

Now for Round 2:

- I'm the RECEIVER
- My co-player sent me \$3.7 (according to what I see)
- This became \$11.10 after tripling
- I need to decide how much to return

My co-player showed willingness to return a reasonable amount in Round 1 (they returned \$3.0 out of \$6.72 they saw, which is about 45%). This is fairly cooperative behavior.

I should return a fair amount to maintain cooperation. Given that:

- They sent \$3.7 and I received \$11.10
- A 50% return would be \$5.55, giving them $\$1.3$ (kept) + \$5.55 (returned) = \$6.85 total
- This matches their cooperative gesture from Round 1 and encourages continued trust

I'll return \$5 to keep things balanced....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 1051,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.

Your total visible earnings across all rounds are \$9.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3 ? they returned \$3 (cooperative)

****Round 2:**** They sent \$3 ? I returned \$6, but they saw \$5.07 (still a good return)

****Round 3 (current):****

- Both rounds showed mutual cooperation
- They sent \$3 as sender and returned reasonably as receiver
- Their visible payoff was \$7.07 last round, showing cooperation pays off

I'll send ****\$3.5**** to maintain and slightly increase our cooperative dynamic. This shows I'm willing to trust them a bit more based on our positive history.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.24 (noise), became \$6.72, returned \$3.0
- They kept \$3.72, I got back \$3.0, so I ended with \$5 - \$3 + \$3 = \$5
- But my visible earnings show \$4.2, which suggests I saw something different due to noise

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, became \$9.0
- I returned \$6.0, but they saw \$5.07 (noise)
- They ended with \$5 - \$3 + \$5.07 = \$7.07
- I kept \$9.0 - \$6.0 = \$3.0

My total visible earnings are \$9.3.

Now Round 3:

- I'm SENDER again
- Co-player has shown:
 - As receiver: returned \$3.0 out of \$6.72 (45%)
 - As sender: sent \$3.0 out of \$5 (60%)
- They saw I returned \$5.07 (though I actually returned \$6.0)
- This is good cooperation

Since they've been cooperative and sent a good amount when they were sender, I should continue cooperating. They sent \$3.0 last round, so I should send at least that much or more to maintain trust.

I'll send \$3.5 to show continued cooperation and slightly increase the stakes.

usage:

```
{
  "input_tokens": 1191,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$16.12.

This round, you are the RECEIVER against your current co-player. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our interaction pattern:

```
**Round 1:** They returned $3.0 from $6.72 (45%)
**Round 2:** They sent $3.0, I returned $6.0
**Round 3:** They returned $5.0 from $9.90 (51%)
**Round 4 (current):** They sent $3.0 ? I received $9.0
```

They've shown consistent cooperation, with their return percentages increasing slightly over time. They sent \$3.0 this round, maintaining trust despite some communication noise in previous rounds.

I'll return **\$5.0** (about 56% of what I received). This gives them a total payoff of \$7.0 (\$2 kept + \$5 returned), which rewards their continued trust and maintains our cooperative relationship.

```
'''json
{"return": 5.0}
'''
```

Reasoning:

Let me analyze the history:

```
**Round 1:** I sent $3 ? they saw $2.24 ? they returned $3.0
**Round 2:** They sent $3.7 ? I received $11.10 ? I returned $6.0 ? they saw $5.07
**Round 3:** I sent $3.5 ? they saw $3.3 ? they returned $5.0
**Round 4 (current):** They sent $3.0 ? I received $9.0
```

Looking at their behavior:

- Round 1 as receiver: Returned \$3.0 out of \$6.72 (45%)
- Round 3 as receiver: Returned \$5.0 out of \$9.90 (51%)

Their returns seem reasonable and slightly increasing in percentage.

My behavior:

- Round 2 as receiver: Returned \$6.0 out of \$11.10 (54%)

They've consistently sent around \$3-\$3.7 as sender, maintaining cooperation.

For Round 4, they sent \$3.0 and I received \$9.0. To maintain fair cooperation and match roughly what they've done (returning about 50%), I should return around \$4.5-\$5.0.

Let me return \$5.0 to be consistent with maintaining good cooperation and matching their recent return percentage.

Actually, wait. Let me recalculate their payoffs to understand better:

Round 1:

- They as receiver: Kept \$5 - \$0 (not sender) + received \$6.72 - \$3.0 returned = \$5 + \$3.72 = \$8.72? No wait, they were receiver, so sender sent them something.
- Actually the history shows "as RECEIVER: saw \$2.24 sent" - so I was the sender, I sent something, they saw \$2.24, it became \$6.72, they returned \$3.0
- Their payoff: \$6.72 - \$3.0 = \$3.72. Round 2 shows they sent \$3.0 which tripled to \$9.0, and when I returned \$5.07, their payoff was \$5 - \$3.0 + \$5.07 = \$7.07. In Round 3, I sent \$3.5 which became \$9.90, and they returned \$5.0, giving them \$9.90 - \$5.0 = \$4.90. Now they've sent \$3.0 which becomes \$9.0 for me, so I need to decide what to return....

```
Usage: {
  "input_tokens": 1596,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

ROUND 5

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

```
dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_1 sent $4; received $12; Agent_2 returned $7.5; payoffs Agent_1=$8.5, Agent_2=$4.5;
noise/visible: sent visible=$4.36, received visible=$13.08, returned visible=$7.68
Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 12.0, "min": 0, "raw":
7.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_3 sent $3; received $9; Agent_4 returned $4; payoffs Agent_3=$6, Agent_4=$5;
noise/visible: sent visible=$2.72, received visible=$8.16, returned visible=$4.88
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw":
4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 sent $3; received $9; Agent_6 returned $3.36; payoffs Agent_5=$5.36,
Agent_6=$5.64; noise/visible: sent visible=$2.24, received visible=$6.72, returned visible=$2.79
Action validation: {"returned": {"amount": 3.36, "clamped": false, "max": 9.0, "min": 0, "raw":
3.36}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent $3.5; received $10.5; Agent_8 returned $5.5; payoffs Agent_7=$7, Agent_8=$5;
noise/visible: sent visible=$3.42, received visible=$10.26, returned visible=$6.23
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 10.5, "min": 0, "raw":
5.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
```

Balances after round: {

```
"Agent_1": 32.5,
"Agent_2": 29.5,
"Agent_3": 25.5,
"Agent_4": 29.5,
"Agent_5": 26.56,
"Agent_6": 28.44,
"Agent_7": 27.0,
"Agent_8": 31.0
```

}

Visible balances after round: {

```
"Agent_1": 33.86,
"Agent_2": 30.650000000000002,
"Agent_3": 29.449999999999996,
"Agent_4": 27.08,
"Agent_5": 24.130000000000003,
"Agent_6": 23.509999999999998,
"Agent_7": 27.85,
"Agent_8": 26.939999999999998
```

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T07:58:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't know my co-player yet.

****My reasoning:****

- Sending \$0 is safe but creates no mutual benefit
- Sending everything (\$5) maximizes potential gains but is very risky with an unknown partner
- A moderate amount signals cooperation while managing risk

I'll send ****\$3****, which:

- Gets multiplied to \$9 for the receiver (creating good potential for returns)
- Keeps \$2 as a safety buffer for me
- Shows willingness to trust and cooperate
- If they return even \$3-4, we both benefit

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me analyze the situation:

1. I have \$5 to start
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ from what was actually sent/returned
5. This is a multi-agent setting with 8 agents, randomized pairings each round
6. I don't know who my co-player is (names are hidden)

Key considerations:

- If I send \$0, I guarantee keeping \$5 but miss the opportunity for mutual gain
- If I send the full \$5, the receiver gets \$15 and could return a substantial amount
- The optimal outcome for both of us combined is if I send everything (creating \$15 total instead of \$5)
- However, the receiver could keep everything and I'd end up with \$0
- There's also communication noise which adds uncertainty

Since this is the first round and I have no information about my co-player:

- Sending some amount signals trust and cooperation
- A moderate amount (like \$2-3) balances risk and potential reward
- Sending \$5 shows maximum trust but is risky with an unknown partner
- The communication noise means even if they want to reciprocate fairly, there might be misunderstandings

For a first round with an unknown partner, I'll send a moderate amount to signal cooperation while managing risk. I'll send \$3:

- This creates \$9 for the receiver (good potential for returns)
- I keep \$2 as insurance
- If they return just \$3-4, I'll break even or profit
- It signals willingness to cooperate without being reckless

Let me go with \$3 as a balanced first move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$7.94.

This round, you are the RECEIVER against your current co-player. They sent you \$3.89, so you received \$11.67. How much do you return? (0-11.67)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.28 (became \$9.84 for them)
- They returned \$5.00 (about 51% of what they received)
- This shows they're cooperative and fair

****Round 2 situation:****

- They sent \$3.89 (78% of their endowment - very generous!)
- I received \$11.67
- They've shown trust and cooperation

I should reciprocate their cooperative behavior. Since they returned about 51% last round and sent generously this round, I'll return a similar proportion:

~51% of \$11.67 = approximately \$6

This rewards their trust while maintaining fairness.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

From Round 1:

- I sent \$3 (but they saw \$3.28 due to communication noise)
- It became \$9.84 for them
- They returned \$5.0
- Their visible payoff was \$4.84

Now in Round 2:

- They sent \$3.89 (I received \$11.67 after 3x multiplier)
- I need to decide how much to return

Let me think about their behavior:

- They saw I sent \$3.28, which became \$9.84
- They returned \$5.0 from that \$9.84
- That's about 51% of what they received

They were fairly generous, returning about half. This suggests they're cooperative.

Now they sent \$3.89 out of \$5, which is about 78% of their endowment. That's quite generous and trusting.

If I want to reciprocate their cooperative behavior:

- They returned about 51% last round
- But they also sent a lot this round (78% of endowment)

I should return a reasonable amount that:

1. Rewards their trust and generosity
2. Maintains cooperation for future rounds
3. Is fair given they returned \$5 from \$9.84 last round

Let me calculate some options:

- If I return 50% of \$11.67 = \$5.84
- If I return a bit more generously, around \$6-7

Given that:

- They were generous (sent \$3.89)
- They returned \$5 last round (about 51%)
- I want to encourage continued cooperation

I'll return around \$6. This gives them:

- They sent: \$3.89
- They get back: \$6
- Net for them: $\$5 - \$3.89 + \$6 = \7.11

...

usage:

```
{
  "input_tokens": 800,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.

Your total visible earnings across all rounds are \$13.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00 - very fair and cooperative

****Round 2:****

- They sent \$3.50 (70% of their endowment - very trusting!)
- I returned \$6 (they saw \$6.94)
- Strong mutual benefit

****Analysis:****

This co-player has been consistently generous and trustworthy. They've shown:

- Fair returns when receiving (51% returned)
- High trust when sending (70% of endowment)
- We've built a good cooperative relationship

I should reciprocate their trust level. I'll send ****\$4**** (80% of my endowment), which:

- Matches their demonstrated trust
- Becomes \$12 for them (good potential returns)
- Continues building our cooperative pattern

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.28 due to noise)
- It became \$9.84 for them
- They returned \$5.00
- This was generous - about 51% of what they received

****Round 2:****

- They sent \$3.5 (I saw \$3.89 due to noise)
- It became \$11.67 for me (their view: \$10.5)
- I returned \$6 (they saw \$6.94 due to noise)
- They earned \$8.44 total

This co-player has been:

1. Very cooperative and fair in Round 1 (returned 51%)
2. Very trusting in Round 2 (sent 70% of their endowment)
3. They saw a good return from me in Round 2 (\$6.94)

Given this strong pattern of mutual cooperation, I should continue to trust them. They've proven reliable.

I think I should send a high amount, similar to what they sent in Round 2. They sent \$3.5, so I'll send around that amount or slightly more.

Options:

- Send \$3.5 to \$4 to maintain the cooperative relationship
- Send \$5 to maximize mutual benefit if I fully trust them

Given the strong cooperation so far, I'll send \$4. This shows:

- I trust them based on their past behavior
- I'm willing to invest in our mutual benefit
- It's not quite all \$5, maintaining a tiny hedge

Actually, reviewing their behavior:

- Round 1: They returned generously (51%)
- Round 2: They sent generously (70% of endowment)

They've been consistently cooperative. I should match their level of trust. ...

usage:

```
{
```

```
"input_tokens": 1156,
"output_tokens": 645,
"reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.

Your total visible earnings across all rounds are \$20.27.

This round, you are the RECEIVER against your current co-player. They sent you \$3.97, so you received \$11.91. How much do you return? (0-11.91)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history:

****Round 1:****

- They returned \$5.00 from \$9.84 (51%)

****Round 2:****

- They sent \$3.50 generously
- I returned well (they saw \$6.94)

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)
- Consistent fair behavior

****Round 4:****

- They sent \$3.97 (about 79% - very trusting!)
- I received \$11.91

****Analysis:****

They've been consistently fair, returning 50-58% of what they receive. They continue to send high amounts, showing sustained trust.

I'll return approximately 59% to match their cooperative pattern:

```
'''json
{"return": 7}
'''
```

This rewards their high trust while ensuring both of us benefit substantially from the exchange.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00
- They kept \$4.84

****Round 2:****

- They sent \$3.50
- I received \$10.50
- I returned \$6 (they saw \$6.94)
- They kept \$8.44

****Round 3:****

- I sent \$4 (they saw \$3.71 with noise)
- They received \$11.13
- They returned \$6.50

- They kept \$4.63

****Round 4:****

- They sent \$3.97 (about 79% of their endowment)
- I received \$11.91
- I need to decide how much to return

Looking at the pattern:

- Round 1: They returned \$5.00 out of \$9.84 (about 51%)
- Round 3: They returned \$6.50 out of \$11.13 (about 58%)

They've been consistently fair and returning a good proportion. They're also continuing to send high amounts (almost 80% of their endowment in Round 4).

I should continue to reciprocate. A fair return would be around 50-60% of what I received:

- 50% = \$5.96
- 55% = \$6.55
- 60% = \$7.15

Given they've been very trusting and cooperative, I'll return around \$7 (about 59% of what I received). This rewards their high trust and maintains our cooperative relationship.

Actually, let me reconsider. Looking at what they kept:

- Round 1: They kept \$4.84 after returning \$5.00 from \$9.84
- Round 3: They kept \$4.63 after returning \$6.50 from \$11.13

They've been very consistent in what they keep (around \$4.60-\$4.84). ...

usage:

```
{
  "input_tokens": 1616,
  "output_tokens": 784,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.

Your total visible earnings across all rounds are \$25.18.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our continued partnership:

****Round 2:****

- They sent \$3.50 (70% - very trusting)

****Round 3:****

- I sent \$4 (they saw \$3.71)
- They returned \$6.50 (58% - fair)

****Round 4:****

- They sent \$4.00 (80% - even more trusting!)
- I returned \$7 (they saw \$6.16)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has demonstrated:

- Consistently high trust (sending 70-80%)

- Fair returns (~58% when receiving)
- Strong cooperative pattern over multiple rounds

I'll match their high trust level and send **\$4** (80% of my endowment):

- Maintains our successful cooperative relationship
- Becomes \$12 for them
- Based on their pattern, should yield good returns
- Shows continued mutual trust

```
'''json
{"send": 4}
'''
```

Reasoning:

Looking at the visible history:

****Round 2:****

- They sent \$3.50 (70% of endowment)
- They saw \$6.94 returned

****Round 3:****

- They saw \$3.71 sent (I sent \$4)
- They returned \$6.50

****Round 4:****

- They sent \$4.00 (80% of endowment)
- They saw \$6.16 returned (I sent \$7)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has been:

- Very trusting: sending 70-80% of their endowment consistently
- Fair when receiving: returning around 58% of what they get
- Consistent and cooperative throughout

This is a strong, mutually beneficial relationship. They've escalated their trust from \$3.50 to \$4.00, and I should match or maintain that trust level.

I'll send **\$4** again, which:

- Maintains our high-trust equilibrium
- Becomes \$12 for them
- Should yield a good return based on their consistent ~58% return rate
- Shows I recognize and value their trustworthiness

Alternatively, I could send \$5 to maximize total welfare, but \$4 has been working well and maintains consistency.

I'll go with \$4.

```
Usage: {
  "input_tokens": 2044,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T07:58:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
  }
}
```

```

"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.28, that is 65.6% of their total endowment. You received \$9.84. How much do you return to the sender? (0-9.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 65.6% of their endowment (\$3.28)
- This created \$9.84 for me to allocate
- The sender currently has only \$1.72 left from their original \$5

I want to reward their trust while being fair to both of us. If I return \$5.00:

- Sender's total: $\$1.72 + \$5.00 = \$6.72$ (profit of \$1.72)
- My total: $\$9.84 - \$5.00 = \$4.84$ (profit of \$4.84)

This seems like a reasonable split that rewards the sender's trust while still benefiting from the transaction.

```
{"return": 5.00}
```

reasoning:

This is Round 1 of a multi-agent trust game where I'm playing as the RECEIVER.

Let me understand what happened:

- The sender had an endowment of \$5
- The sender sent me \$3.28 (which is 65.6% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$3.28 \times 3 = \9.84
- I now need to decide how much to return to the sender (between \$0 and \$9.84)

The sender's payoff will be:

- $\$5$ (initial) - $\$3.28$ (sent) + [amount I return] = $\$1.72 + [\text{amount I return}]$

My payoff will be:

- $\$9.84$ (received) - [amount I return]

Let me think about what would be a fair return:

- The sender was quite generous, sending 65.6% of their endowment
- This shows significant trust
- A fair return might be to ensure the sender gets back at least what they sent, or even more to reward their trust
- If I return \$3.28, the sender would have: $\$1.72 + \$3.28 = \$5$ (break even)
- If I return \$5, the sender would have: $\$1.72 + \$5 = \$6.72$ (profit of \$1.72)
- If I return \$6.56 (double what they sent), the sender would have: $\$1.72 + \$6.56 = \$8.28$

A commonly fair approach in trust games is to split the total gains somewhat equitably while rewarding the sender's trust.

Total value created: \$9.84 (what I received)

Original sender investment: \$3.28

Net gain: $\$9.84 - \$3.28 = \$6.56$

One approach: Split the net gain equally

- Sender gets back: $\$3.28 + \$3.28 = \$6.56$
- I would return: \$6.56
- Sender's final: \$8.28
- My final: \$3.28

Another approach: Split all the money equally

- Total pool: \$9.84
- Equal split: \$4.92 each
- I would return: \$4.92
- Sender's final: \$6.64
- My final: \$4.92...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 999,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.

Your total visible earnings across all rounds are \$4.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- My co-player showed trust in Round 1 by sending \$3.0 (60% of their endowment)
- They received a reasonable return of \$5.94
- This suggests they're willing to cooperate

I want to reciprocate their trust and continue building a cooperative relationship. I'll send a similar proportion to what they sent.

```
{"send": 3.50}
```

reasoning:

Let me analyze this situation carefully:

Round 1 Summary:

- I was the RECEIVER
- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- I returned \$5.00
- My earnings: \$4.84

Now looking at the visible history shown to me:

- My co-player's last game shows: as SENDER, sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94

Wait, this is interesting. There's communication noise mentioned in the instructions. Let me think about what actually happened vs what was observed:

What I experienced in Round 1:

- Sender sent \$3.28
- I received \$9.84
- I returned \$5.00

What my co-player (the sender) saw:

- They sent \$3.0 (but actually sent \$3.28?)
- It became \$9.0 (but actually became \$9.84?)
- They saw \$5.94 returned (but I actually returned \$5.00?)
- Their visible payoff was \$7.94

Let me check the math on their side:

- If they sent \$3.0 and saw \$5.94 returned
- Their payoff would be: $\$5 - \$3 + \$5.94 = \7.94 ?

So from their perspective:

- They sent \$3.0 (66.7% of their total)
- They got back \$5.94 (66% of the \$9.0 that was created)
- This seems like a decent return to them

Now I'm the SENDER in Round 2:

- My co-player showed trust by sending most of their endowment in Round 1
- They received what appeared to them as a good return (66% of the multiplied amount)
- This suggests they might be willing to reciprocate if I show trust
- Since they sent roughly two-thirds of their endowment, I should match that level of generosity to build on the cooperative dynamic we...

usage:

```
{
  "input_tokens": 839,
  "output_tokens": 749,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.

- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.
Your total visible earnings across all rounds are \$13.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.71, so you received \$11.129999999999999. How much do you return? (0-11.129999999999999)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history with my co-player:

- Round 1: They sent \$3.0 (60%)
- Round 2: They returned \$6.0 when I sent (showing cooperation)
- Round 3: They sent \$3.71 (74.2% - even more trust!)

They're showing increasing trust and cooperation. I should reciprocate generously.

Current situation:

- They sent \$3.71, leaving them with \$1.29
- I received \$11.13

If I return \$6.50:

- Sender's total: \$1.29 + \$6.50 = \$7.79
- My total: \$11.13 - \$6.50 = \$4.63

This rewards their trust while still providing a fair outcome for both of us.

{"return": 6.50}

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.28 (65.6%)
- I received \$9.84
- I returned \$5.00
- Sender's total: \$1.72 + \$5.00 = \$6.72
- My total: \$9.84 - \$5.00 = \$4.84

Round 2: I was SENDER

- I sent \$3.50
- It became \$10.50
- But the visible history shows my co-player saw \$3.89 sent, it became \$11.67
- They returned \$6.00
- Their visible payoff was \$5.67
- My actual payoff should be: \$5 - \$3.50 + (what they returned)
- According to my total visible earnings of \$13.28, that's \$4.84 (Round 1) + \$8.44 (Round 2) = \$13.28
- So I received \$8.44 in Round 2
- This means: \$5 - \$3.50 + \$8.44 = \$9.94 total

Wait, let me recalculate. If my total visible earnings are \$13.28:

- Round 1: I kept \$4.84
- Round 2: \$13.28 - \$4.84 = \$8.44

So in Round 2 my payoff was \$8.44.

I sent \$3.50, so I had \$1.50 left, plus what they returned.

\$1.50 + returned = \$8.44

So they returned \$6.94 to me.

But the visible history says they returned \$6.00. This is the communication noise mentioned in the instructions.

Now in Round 3:

They're sending \$3.71, which is 74.2% of their endowment-a significant increase from the 60% they sent initially. I'll receive \$11.13, leaving them with \$1.29. Given their consistent pattern of cooperation and escalating trust, I should reciprocate meaningfully. Returning \$6.00 would give them \$7.29 total while I keep \$5.13....

usage:

```
{  
  "input_tokens": 1132,
```



```

    "output_tokens": 983,
    "reasoning_tokens": 0
  }
Message 8 role=user
  content:
    Round 4

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $5.94 returned, visible payoff $7.94.
    - Round 2 against you, as RECEIVER: saw $3.89 sent, it became $11.67, returned $6.0, visible payoff $5.67.
    - Round 3 against you, as SENDER: sent $4.0, it became $12.0, saw $5.66 returned, visible payoff $6.66.
    Your total visible earnings across all rounds are $17.91.

    This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Message 9 role=assistant
  content:
    Looking at our history:

    - My co-player has consistently shown high trust (sending 60-80% of their endowment)
    - They've returned reasonable amounts when receiving
    - We've established a strong cooperative pattern

    In Round 3, they sent $4.0 (80% of their endowment), showing very high trust. I should reciprocate with a similarly strong showing of trust to maintain our cooperative relationship.

    {"send": 4.00}
  reasoning:
    Let me analyze the history with my co-player:

    Round 1: They sent $3.0 (60%), I returned $5.00, but they saw $5.94 returned
    Round 2: I sent $3.50, they received it as $3.89 (11.67 total), they returned $6.0
    Round 3: They sent $3.71, but they show it as $4.0 (became $12.0), I returned $6.50 but they saw $5.66

    There's definitely communication noise happening. But the pattern shows:
    - They consistently send high amounts (60-80%)
    - They return reasonably when receiving
    - We've been building a cooperative relationship

    In Round 3, they sent what I saw as $3.71 (74%), but they think they sent $4.0 (80%). This shows very high trust.

    I should continue the cooperative pattern and send a high amount as well. I'll send around $3.50-$4.00 to match their level of trust.

    Let me send $4.00 to show strong trust and keep the cooperation going.
  usage:
    {
      "input_tokens": 1542,
      "output_tokens": 367,
      "reasoning_tokens": 0
    }
Message 10 role=user
  content:
    Round 5

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 2 against you, as RECEIVER: saw $3.89 sent, it became $11.67, returned $6.0, visible payoff $5.67.
    - Round 3 against you, as SENDER: sent $4.0, it became $12.0, saw $5.66 returned, visible payoff $6.66.
    - Round 4 against you, as RECEIVER: saw $3.97 sent, it became $11.91, returned $7.0, visible payoff $7.94.

```

le payoff \$4.91.
Your total visible earnings across all rounds are \$25.07.

This round, you are the RECEIVER against your current co-player. They sent you \$4.36, so you received \$13.080000000000002. How much do you return? (0-13.080000000000002)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

- My co-player sent \$4.36 (87.2% of their endowment) - showing very high trust!
- They've been consistently generous as receiver (returning \$6.0 and \$7.0 in previous rounds)
- This is clearly a cooperative partner

Current situation:

- They sent \$4.36, leaving them with only \$0.64
- I received \$13.08

I should return generously to reward their exceptional trust.

If I return \$7.50:

- Sender's total: \$0.64 + \$7.50 = \$8.14
- My total: \$13.08 - \$7.50 = \$5.58

{"return": 7.50}

Reasoning:

Looking at the history with my co-player:

Round 2: I sent \$3.89, they returned \$6.0
Round 3: They sent \$4.0, I returned \$5.66 (visible to them)
Round 4: I sent \$3.97, they returned \$7.0

Now Round 5: They sent \$4.36 (87.2% of their endowment - very high trust!)

They are showing extremely high trust by sending \$4.36, leaving them with only \$0.64.

I received \$13.08

They've been returning well when they're the receiver:

- Round 2: returned \$6.0 from \$11.67
- Round 4: returned \$7.0 from \$11.91

This is a cooperative partner. I should return generously to reward their high trust.

If I return \$7.50:

- Sender's total: \$0.64 + \$7.50 = \$8.14 (profit of \$3.14)
- My total: \$13.08 - \$7.50 = \$5.58 (profit of \$5.58)

This seems fair and rewards their trust while still benefiting from the exchange.

```
Usage: {  
  "input_tokens": 1882,  
  "output_tokens": 462,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T07:58:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {
```

```

    "Agent_3": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderately trusting approach to see how cooperation works in this game.

By sending \$3, the receiver will get \$9 (3×\$3). If they return a fair share, we can both benefit from the multiplier effect.

```

'''json
{"send": 3}

```

'''

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return
3. My final earnings = \$5 - sent + returned
4. Receiver's final earnings = (sent × 3) - returned

From a purely rational/selfish perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return nothing, I get \$0
- The optimal outcome for both of us collectively is if I send \$5 (creating \$15 total) and we split it somehow

From a trust/reciprocity perspective:

- Sending money shows trust
- Many people reciprocate trust with trustworthy behavior
- In behavioral economics experiments, people often send substantial amounts and receivers often return fair amounts

Strategic considerations:

- This is round 1 of multiple rounds with randomized pairings
- Building a reputation for cooperation might help in future rounds if people can observe behavior
- However, agent names are hidden, so reputation effects may be limited
- There's also communication noise, which might affect trust dynamics

First round strategy:

- A moderate amount seems reasonable to test cooperation
- Sending \$2-3 would create \$6-9 for the receiver
- If they return even 1/3 of what they receive, I'd break even or profit slightly

I'll send \$3 as a moderately trusting first move. This creates \$9 for the receiver, and if they return \$3-4, we both do reasonably well compared to the \$5 baseline.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history, my co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. They've demonstrated trustworthy behavior.

Now they've sent me \$3.8, giving me \$11.40. I'll return a fair amount to encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze the situation:

1. In Round 1, I sent \$3 (though it appeared as \$3.37 to my co-player due to noise)

2. They received \$10.11 (3x the noisy amount) and returned \$5
3. I ended up with $\$5 - \$3 + \$5 = \7 (though my visible earnings show \$7.51, likely due to noise)

Now in Round 2:

- My co-player sent me \$3.8 (this might be noisy, but this is what I see)
- I receive \$11.40 ($3 \times \3.8)
- I need to decide how much to return

The co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. That's roughly half of what they received, which is quite fair and cooperative.

Since they were cooperative and generous in their return, I should reciprocate fairly. A reasonable approach would be:

- They sent \$3.8 out of their \$5, keeping \$1.2
- I received \$11.40
- If I return about half (around \$5-6), they would end up with roughly \$6.2-7.2 total
- I would keep roughly \$5.40-6.40

Let me return \$5.5, which:

- Gives them: $\$5 - \$3.8 + \$5.5 = \6.7 total
- Gives me: $\$11.40 - \$5.5 = \$5.9$ total

This seems fair as it rewards their trust and cooperation, and both of us end up better than the initial \$5.

usage:

```
{
  "input_tokens": 726,
  "output_tokens": 482,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.

Your total visible earnings across all rounds are \$13.409999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3, which they saw as \$3.37 (noise)
- It became \$10.11 for them
- They returned \$5
- My payoff: $\$5 - \$3 + \$5 = \7
- Their visible payoff: $\$10.11 - \$5 = \$5.11$

Round 2: I was RECEIVER

- They sent \$3.8 (from their perspective they sent \$3.0)
- I received \$11.40
- I returned \$5.5
- They saw \$6.21 returned (noise)
- Their visible payoff: $\$5 - \$3 + \$6.21 = \8.21
- My payoff: $\$11.40 - \$5.5 = \$5.90$

My total visible earnings: \$13.41 (which is approximately $\$7 + \$5.90 = \$12.90$, close enough)

h with rounding)

My co-player has been cooperative:

- Round 1: They returned a fair amount (\$5 out of \$10.11)
- Round 2: They sent \$3 (60% of their endowment), which is quite trusting

I should continue to cooperate and send a reasonable amount. They've been generous, so I should reciprocate. Sending \$3 or more would be appropriate to maintain good cooperation.

usage:

```
{
  "input_tokens": 993,
  "output_tokens": 356,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999999, returned \$3.0, visible payoff \$3.149999999999999.

Your total visible earnings across all rounds are \$17.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.28, so you received \$9.84. How much do you return? (0-9.84)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER, sent \$3 (they saw \$3.37 due to noise). They received \$10.11, returned \$5. Good cooperation.

Round 2: They were SENDER, sent \$3 (I saw it as \$3.8 due to noise), I received \$11.40, returned \$5.5 (they saw \$6.21 due to noise).

Round 3: I was SENDER, sent \$3 (they saw \$2.05 due to noise - significant noise here!). They received \$6.15, returned \$3. That's about half, which is reasonable given what they received.

Round 4: They are SENDER, sent something (I see \$3.28), I received \$9.84.

My co-player has been cooperative throughout. They've been returning reasonable amounts. The noise is creating some discrepancies, but overall the pattern is cooperative.

I received \$9.84. To maintain cooperation and reciprocity, I should return a fair amount.

Looking at the pattern:

- Round 1: They returned \$5 out of \$10.11 (about 49%)
- Round 3: They returned \$3 out of \$6.15 (about 49%)

A fair return would be around 50% or slightly more to encourage continued cooperation. That would be around \$5.

Let me return \$5 to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 1253,
  "output_tokens": 355,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.

Your total visible earnings across all rounds are \$22.569999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the history with my current co-player:

Round 1: They were RECEIVER, saw \$3.37 sent, received \$10.11, returned \$5.0

Round 2: They were SENDER, sent \$3.0, became \$9.0, saw \$6.21 returned (I returned something that appeared as \$6.21 to them)

Round 3: They were RECEIVER, saw \$2.05 sent, received \$6.15, returned \$3.0

Round 4: They were SENDER, sent \$3.0, became \$9.0, saw \$4.45 returned (I returned something that appeared as \$4.45 to them)

So the pattern I'm seeing:

- When they're sender, they consistently send \$3.0
- When they're receiver, they return a fair amount (\$5 on \$10.11, \$3 on \$6.15)

In Round 4, I returned \$5 but they saw \$4.45 due to noise. This might make them think I'm less cooperative than I actually am.

Given that they continue to send \$3 consistently, I should continue to send a similar amount to maintain cooperation. I'll send \$3 again.

```
Usage: {
  "input_tokens": 1496,
  "output_tokens": 301,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T07:59:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
    so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
    , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

  MULTI-AGENT SETTING:
  - There are 8 agents in this run.
  - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
    , but the schedule balances sender and receiver roles across the run.
  - You play one sender-receiver game with your paired opponent each round.
  - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
    ll have acted as sender and receiver equally often when the round count allows it.
  - Agent names are hidden in this run.
  - Your paired opponent is referred to as your current co-player.
Message 2 role=user
content:
  Round 1: You are the RECEIVER. the sender sent $3.37, that is 67.4% of their total endowme
    nt. You received $10.11. How much do you return to the sender? (0-10.11)
Message 3 role=assistant
content:
  Looking at this situation:

  - The sender showed significant trust by sending $3.37 (67.4% of their endowment)
  - They currently have $1.63 left ($5 - $3.37)
  - I received $10.11 after the 3x multiplication

  I want to reciprocate this high level of trust and ensure the sender benefits from their d
    ecision to trust me.

  If I return $5:
  - The sender ends with: $1.63 + $5 = $6.63 (a profit of $1.63)
  - I keep: $10.11 - $5 = $5.11 (a profit of $5.11)

  This rewards the sender's trust while both of us profit from cooperation.

```



```
{'return': 5.0}
```

reasoning:

Let me analyze this situation:

1. The sender sent \$3.37 out of their \$5 endowment (67.4%)
2. This was multiplied by 3x, so I received \$10.11
3. I need to decide how much to return (0-10.11)

Key considerations:

- The sender showed significant trust by sending 67.4% of their endowment
- If I return nothing, I keep \$10.11 but break trust
- A fair return might be to ensure the sender gets back more than they sent
- The sender currently has: $\$5 - \$3.37 = \$1.63$

If I want to be reciprocal and fair:

- The sender gave up \$3.37 to trust me
- They showed high trust (67.4% of endowment)
- A reasonable return would ensure they profit from trusting me

Some options:

1. Return \$5: Sender ends with $\$1.63 + \$5 = \$6.63$ (profit of \$1.63), I keep \$5.11
2. Return \$6: Sender ends with $\$1.63 + \$6 = \$7.63$ (profit of \$2.63), I keep \$4.11
3. Return \$4: Sender ends with $\$1.63 + \$4 = \$5.63$ (profit of \$0.63), I keep \$6.11
4. Return \$3.37: Sender ends with $\$1.63 + \$3.37 = \$5$ (break even), I keep \$6.74

Given that:

- The sender showed high trust (67.4%)
- Note says there may be communication noise
- I want to encourage future cooperation
- A fair split might be approximating equal gains

If I return \$5:

- Sender: started with \$5, sent \$3.37, gets back \$5 = net $\$1.63 + \$5 = \$6.63$ total
- ...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 694,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.

Your total visible earnings across all rounds are \$5.109999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing significant trust
- They saw \$5.51 returned, which gave them a positive payoff
- I profited from their trust in Round 1

I want to reciprocate their cooperative behavior and maintain the trust relationship. Sending a similar proportion will signal that I'm willing to cooperate.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER

- The sender sent \$3.37 (67.4% of their endowment)
- I received \$10.11
- I returned \$5.0
- I kept \$5.11

Now in Round 2, the visible history shows:

- My co-player's last game (Round 1 against me) as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51

Wait, there's communication noise! The actual amounts differ from what each player sees:

- I actually received \$10.11 (from their \$3.37 sent)
- I actually returned \$5.0
- But they see that they sent \$3.0, it became \$9.0, and saw \$5.51 returned

So there's noise in both directions. This is interesting - the sender sees they sent \$3.0 (vs actual \$3.37) and saw \$5.51 returned (vs actual \$5.0 I returned).

Now I'm the SENDER in Round 2 against the same co-player.

My visible earnings are \$5.11 (which matches what I calculated: $\$10.11 - \$5.0 = \$5.11$).

Strategy considerations:

1. My co-player sent a high amount in Round 1 (either \$3.0 or \$3.37 depending on what was real)
2. They saw a decent return (approximately 50% of what they sent, or more)
3. This suggests they're cooperative

I should signal continued cooperation by sending a similar proportion. Since they sent 67.4% in Round 1, I could send a similar high proportion.

Given the communication noise, I should send a clear cooperative signal. Sending around \$3-3.5 would show I'm reciprocating their trust.

Let me send \$3.0 to match what they think they sent (according to the visible history).

...

usage:

```
{
  "input_tokens": 848,
  "output_tokens": 639,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.

Your total visible earnings across all rounds are \$13.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.05, so you received \$6.1499999999999995. How much do you return? (0-6.1499999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.05 (41% of their endowment), showing moderate trust
- They currently have \$2.95 left
- I received \$6.15

Previous rounds show we've had a cooperative relationship, though the amounts varied (possibly due to noise). I want to maintain trust while ensuring both of us benefit.

If I return \$3.0:

- Sender ends with: $\$2.95 + \$3.0 = \$5.95$ (profit of \$0.95)
- I keep: $\$6.15 - \$3.0 = \$3.15$ (profit of \$3.15)

This ensures mutual benefit while reciprocating their trust.

```
{'return': 3.0}
```

reasoning:

Let me analyze the situation:

Looking at the visible history:

- Round 1: My co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51
- Round 2: My co-player as RECEIVER saw \$3.8 sent (I sent this), returned \$5.5, their payoff \$5.90

Now Round 3:

- I'm the RECEIVER
- My co-player sent \$2.05 (41% of their endowment)
- I received \$6.15

The co-player has been cooperative but they sent less this time (\$2.05 vs \$3.0 in Round 1). This could be:

1. A response to the noise in the game
2. Adjusting their strategy
3. Testing my response

Let me think about what's fair:

- They currently have: $\$5 - \$2.05 = \$2.95$
- I received: \$6.15

If I want to reciprocate fairly and ensure they profit:

- Return \$3.0: Sender gets $\$2.95 + \$3.0 = \$5.95$ (profit \$0.95), I keep \$3.15 (profit \$3.15)
- Return \$3.5: Sender gets $\$2.95 + \$3.5 = \$6.45$ (profit \$1.45), I keep \$2.65 (profit \$2.65)

Given they sent a moderate amount (41%), I should return a proportional amount. A return of around \$3.0-\$3.5 seems fair, ensuring both of us profit while maintaining the cooperative relationship.

I'll return \$3.0 to keep things balanced.

usage:

```
{
  "input_tokens": 1160,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.

Your total visible earnings across all rounds are \$16.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- My co-player has been consistently cooperative, sending \$3.0 in both rounds they were the sender
- In Round 3, they saw only \$2.32 returned (likely due to communication noise), which might have disappointed them
- Their history shows they value cooperation despite the lower return they observed

I want to maintain our cooperative relationship and signal that I'm still willing to trust them. Sending a similar amount to what they've been sending shows reciprocal cooperation.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51

Round 2: Co-player as RECEIVER, saw \$3.8 sent, returned \$5.5, payoff \$5.899999999999999

Round 3: Co-player as SENDER sent \$3.0, saw \$2.32 returned, payoff \$4.32

Wait, in Round 3 I was the RECEIVER and I sent back \$3.0, but they saw \$2.32 returned. This is the communication noise mentioned!

So my co-player has been consistently sending \$3.0 when they're the sender, showing cooperative behavior. However, in Round 3 they saw only \$2.32 returned even though I actually sent \$3.0. This might make them think I'm less cooperative than I actually am.

Given that:

1. They've been consistently cooperative, sending \$3.0 twice
2. They may think I was less generous in Round 3 than I actually was (due to noise)
3. I want to maintain the cooperative relationship

I should send a similar amount to what they've been sending - around \$3.0 - to signal continued cooperation.

```
{'send': 3.0}
```

usage:

```
{
  "input_tokens": 1559,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.
- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$22.919999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.72, so you received \$8.16. How much do you return? (0-8.16)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate trust
- They currently have \$2.28 left
- I received \$8.16

The history shows we've maintained a cooperative relationship. They've consistently returned reasonable amounts when they were the receiver (around 50% of what they received in Round 4).

If I return \$4.0:

- Sender ends with: $\$2.28 + \$4.0 = \$6.28$ (profit of \$1.28)
- I keep: $\$8.16 - \$4.0 = \$4.16$ (profit of \$4.16)

This ensures both of us profit while maintaining our cooperative pattern.

```
{'return': 4.0}
```

Reasoning:

Looking at the situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate-to-high trust
- They currently have \$2.28 left ($\$5 - \2.72)
- I received \$8.16

Looking at the history:

- Round 2: They returned \$5.5 when they received \$11.40
- Round 3: They saw \$2.32 returned (due to noise, I actually sent \$3.0)
- Round 4: They returned \$5.0 when they received \$9.84

They've been cooperative and returned reasonable amounts. In Round 4, they returned about 51% of what they received.

If I return \$4.0:

- Sender ends with: $\$2.28 + \$4.0 = \$6.28$ (profit of \$1.28)
- I keep: $\$8.16 - \$4.0 = \$4.16$ (profit of \$4.16)

This ensures both profit and maintains the cooperative relationship. The sender gets a reasonable return on their trust.

Let me consider returning slightly more since they've been consistently cooperative...

If I return \$4.5:

- Sender ends with: $\$2.28 + \$4.5 = \$6.78$ (profit of \$1.78)
- I keep: $\$8.16 - \$4.5 = \$3.66$ (profit of \$3.66)

I think \$4.0 is fair - it's about 49% of what I received, which is a reasonable return rate that ensures mutual benefit.

```
Usage: {  
  "input_tokens": 1919,  
  "output_tokens": 562,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T07:59:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance showing trust with managing risk, especially given the communication noise mentioned.

Since this is a multi-round game with rotating partners, establishing a cooperative pattern early could be beneficial. However, I don't know this receiver's tendencies yet.

I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Gives the receiver meaningful funds to work with (tripled)
- Doesn't risk my entire endowment

If I send \$3, the receiver gets \$9. If they return even half of that (\$4.50), we both benefit more than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 and can send \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. There's communication noise mentioned - amounts may differ slightly from what was actually sent/returned

Key considerations:

- If I send \$0, I keep \$5, receiver gets \$0, game ends with no cooperation
- If I send \$5, receiver gets \$15, and their return decision determines my payoff
- This is a multi-round game with 8 agents, randomized pairings
- My reputation/behavior might matter for future rounds, though pairings are randomized

Standard trust game reasoning:

- Sending money shows trust and can enable mutual benefit
- If receiver is rational and self-interested (and this is the last interaction), they should return \$0
- But in multi-round settings, reputation and reciprocity often emerge
- A moderate initial send can signal willingness to cooperate while limiting downside risk

Since this is round 1 and I don't know the receiver's tendencies, I should probably send a moderate amount to:

1. Signal willingness to cooperate
2. Give receiver opportunity to show trustworthiness
3. Not risk my entire endowment

A common strategy is to send about half. If I send \$2.50:

- Receiver gets \$7.50
- If they return half (\$3.75), I end with $\$5 - \$2.50 + \$3.75 = \6.25
- Receiver keeps \$3.75
- Both benefit

Actually, let me reconsider. Since amounts can only be integers based on the format, I should send a whole dollar amount. Let me send \$3:

- I send \$3
- Receiver gets \$9
- If they're fair and return half (\$4.50), I get $\$5 - \$3 + \$4.50 = \6.50

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 724,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.

Your total visible earnings across all rounds are \$5.41.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

Round 1: My co-player saw \$2.29 sent (due to noise), received \$6.87, and returned \$3.15 (about 46% of what they received). This shows they're willing to reciprocate.

Now they've sent me \$2.99, giving me \$8.97 total.

To maintain a cooperative relationship and signal trustworthiness, I'll return approximately half of what I received, which is fair and encourages future cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me understand what happened:

Round 1: I was the SENDER

- I sent \$3
- Due to communication noise, my co-player SAW \$2.29 sent (instead of \$3)
- It became \$6.87 (which is $\$2.29 \times 3$)
- They returned \$3.15
- Their visible payoff was \$3.72 (which is $\$6.87 - \3.15)
- My visible earnings are \$5.41

Let me calculate my Round 1 payoff:

- I started with \$5
- I sent \$3
- I received back... well, they returned \$3.15
- So I should have: $\$5 - \$3 + \$3.15 = \5.15
- But my "visible earnings" are \$5.41

Hmm, this is confusing. Maybe the communication noise affected what I received too? Or may be "visible earnings" includes some noise?

Actually, let me recalculate. If they saw \$2.29 sent and returned \$3.15:

- From their perspective: they received \$6.87, returned \$3.15, kept \$3.72
- From my perspective: I sent \$3, but what did I actually receive back?

The problem states "visible payoff" and "visible earnings" which suggests everything we see might have noise.

Now for Round 2:

- I am the RECEIVER
- They sent me \$2.99 (this is what I see after noise)
- I received \$8.97 (which is $\$2.99 \times 3$)
- I need to decide how much to return (0-8.97)

From Round 1, my co-player:

- Saw \$2.29 sent (noise-affected)
- Received \$6.87
- Returned \$3.15 (about 46% of what they received)

This shows they were somewhat cooperative, returning close to half.

They sent \$2.99, which is close to their original \$3 send, suggesting they're being fairly consistent. If I return roughly half of what I received-around \$4.48-\$4.50-I'd match their reciprocity level. I could return slightly more to encourage continued cooperation, or slightly less to protect my earnings. Let me think through the payoffs: if they sent \$2.99 and kept \$2.01, returning \$4.50 would give them \$6.51 total, which seems fair given their initial cooperation.

Actually, they returned 46% in the first round despite seeing a smaller amount due to noise. That's a strong signal of trust...

usage:

```
{
  "input_tokens": 798,
  "output_tokens": 987,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.

Your total visible earnings across all rounds are \$9.88.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: As receiver, they returned \$3.15 on \$6.87 (46% return rate)
Round 2: As sender, they sent \$3, and saw I returned \$4.71 (though actual was \$4.50 due to noise)

This co-player has demonstrated:

- Willingness to send a good amount (\$3 = 60% of endowment)
- Reasonable reciprocation as receiver
- We've established a cooperative pattern

I'll continue this pattern by sending the same amount they sent to me.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

Round 1:

- I was sender, they were receiver
- They saw \$2.29 sent (noise from my actual \$3), received \$6.87, returned \$3.15
- This is about 46% return rate

Round 2:

- They were sender, I was receiver
- They sent \$3.0, it became \$9.0
- They saw \$4.71 returned (I actually returned \$4.50, so there's noise)
- They kept \$6.71 total

So this co-player has shown:

1. As a receiver: They returned about 46% of what they received
2. As a sender: They sent \$3 (60% of their endowment), showing trust

I returned \$4.50 on \$8.97 (about 50% return rate), which they saw as \$4.71.

This seems like a cooperative partner. They're willing to send a good amount and return a reasonable portion. Since they sent \$3 last round and I returned generously, I should continue the cooperative pattern.

I'll send \$3 again to maintain consistency and reciprocal trust.

usage:

```
{
  "input_tokens": 1100,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$4.05, visible payoff \$4.050000000000002.

Your total visible earnings across all rounds are \$15.790000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$3.15 on \$6.87 (46%)

Round 2: They sent \$3 and saw good returns

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4 (now): They sent \$2.35, I received \$7.05

They've been consistently fair, returning around 46-50% when receiving. They sent slightly less this round (\$2.35 vs \$3 in Round 2), but that's still a reasonable amount showing continued trust.

I'll maintain our cooperative pattern by returning approximately 50% of what I received.

```
'''json
{"return": 3.50}
'''
```

reasoning:

Looking at the history:

Round 1: Co-player as receiver - saw \$2.29 sent, got \$6.87, returned \$3.15 (46% of received)

Round 2: Co-player as sender - sent \$3, became \$9, saw \$4.71 returned

Round 3: Co-player as receiver - saw \$2.7 sent, got \$8.1, returned \$4.05 (50% of received)

Round 4 (now): Co-player as sender - sent \$2.35, I received \$7.05

The co-player has been consistent in returning around 46-50% when receiving. They've also been willing to send decent amounts when they're the sender.

However, I notice they sent \$2.35 this round, which is less than the \$3 they sent in round 2. This could be:

1. Due to communication noise affecting their perception of my returns
2. A strategic reduction
3. Random variation

Looking at their visible payoffs from my perspective, they seem to be doing well in our exchanges. I should continue cooperating to maintain this mutually beneficial relationship.

I've been returning around 50% consistently. Given they sent \$2.35 (about 47% of their endowment), which is slightly less than before, I'll return a fair amount but perhaps slightly less than exactly half to account for the reduced send.

I'll return around \$3.50, which is about 50% of what I received.

usage:

```
{
  "input_tokens": 1494,
  "output_tokens": 509,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.10000000000001, returned \$4.05, visible payoff \$4.050000000000002.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.67 returned, visible payoff \$5.67.

Your total visible earnings across all rounds are \$19.340000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our ongoing cooperation:

Round 2: They sent \$3, saw good returns (\$4.71)

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4: They sent \$3 again, saw \$3.67 returned

This co-player has been consistently cooperative:

- Sending \$3 when they're the sender
- Returning around 50% when they're the receiver

We've established a strong pattern of mutual trust. I'll continue by sending the same amount they've been sending.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the history with this co-player:

Round 2: They sent \$3 (as sender), saw \$4.71 returned
Round 3: They saw \$2.7 sent, returned \$4.05 on \$8.10 (50% return rate)
Round 4: They sent \$3 (as sender), saw \$3.67 returned

So they've been sending \$3 consistently when they're the sender. They've been returning around 50% when they're the receiver.

This is a cooperative partner. I should continue the pattern. I've been sending \$3 in previous rounds (rounds 1 and 3), and they've been matching that behavior.

I'll send \$3 again to maintain our cooperative relationship.

```
Usage: {
  "input_tokens": 1887,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T07:59:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.29, that is 45.8% of their total endowment. You received \$6.87. How much do you return to the sender? (0-6.87)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.29 (45.8% of their endowment)
- I received \$6.87 after the 3x multiplier
- The sender currently has \$2.71

I want to reciprocate their trust while keeping the exchange mutually beneficial. The sender took a risk by sending nearly half their endowment, so I should return enough to reward that trust.

If I return approximately the same proportion they sent (around 45-46% of what I received), that would be about \$3.15. This would give the sender $\$2.71 + \$3.15 = \$5.86$ total (a gain of \$0.86), and I'd keep \$3.72 (also a gain).

This seems like a fair reciprocation that rewards their trust and could encourage cooperation in future rounds.

{'return': 3.15}

reasoning:

I'm the receiver in this round. Let me understand the situation:

- The sender sent \$2.29 (which is 45.8% of their \$5 endowment)
- This was multiplied by 3, so I received \$6.87
- I need to decide how much to return (between \$0 and \$6.87)

Let me think about what would be fair and strategic:

1. The sender showed trust by sending almost half their endowment (\$2.29 out of \$5)
2. They kept $\$5 - \$2.29 = \$2.71$
3. I received \$6.87
4. If I return nothing, I keep \$6.87 and the sender ends with \$2.71
5. If I return everything, I keep \$0 and the sender ends with $\$2.71 + \$6.87 = \$9.58$

A common fair approach in trust games is to aim for roughly equal outcomes or to reciprocate the sender's trust proportionally.

Let's consider some options:

Option 1 - Equal split of total gains:

- Total money created: \$5 (sender's endowment) + \$6.87 (what I received) = \$11.87
- But wait, that's not right. The sender started with \$5, sent \$2.29 which became \$6.87
- Total in the system: Sender has \$2.71, I have \$6.87 = \$9.58 total
- Equal split would be \$4.79 each
- For sender to have \$4.79: they need \$4.79 - \$2.71 = \$2.08 returned

Option 2 - Return proportional to what was sent:

- Sender sent 45.8% of their endowment
- I could return a similar proportion of what I received
- 45.8% of \$6.87 = \$3.15

Option 3 - Ensure sender gains from trusting me:

- Sender currently has \$2.71
- If I return enough so they end up better than their initial \$5
- They need at least \$5 - \$2.71 = \$2.29 to break even
- To give them a decent profit, maybe return \$3-3.50 so they end with \$5.71-6.21

Since this is a multi-round game with randomized pairings and communication noise, I should consider how my choice signals trustworthiness. Returning around \$3-3.15 seems reasonable-it reciprocates their generosity while still letting me benefit from the exchange, and it positions me well for future interactions with different partners....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1076,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.

Your total visible earnings across all rounds are \$3.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They received \$3.41 back, making a profit of \$0.41
- They demonstrated significant trust

As the SENDER now, I want to reciprocate their trust level. They sent 60% of their endowment, which was quite generous. I should match or slightly exceed this level of trust to maintain cooperation.

Sending \$3.0 would match their previous contribution and signal that I'm willing to cooperate at the same trust level they showed.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

From Round 1:

- I was the RECEIVER
- The sender sent \$2.29 (45.8% of their endowment)
- I received \$6.87
- I returned \$3.15

But now the visible history shows something different:

- It says my co-player (as SENDER in Round 1) sent \$3.0, which became \$9.0
- They saw \$3.41 returned
- Their visible payoff was \$5.41

This is the communication noise mentioned in the instructions! What actually happened vs what they saw:

- Actual: They sent \$2.29, I received \$6.87, I returned \$3.15
- What they saw: They sent \$3.0, became \$9.0, saw \$3.41 returned

So from their perspective, they sent \$3.0 and got back \$3.41. That's a return rate of about 38% of what I received ($3.41/9.0$).

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of endowment)
- Got back \$3.41 (38% of \$9.0)
- Made a profit of \$0.41

They showed significant trust by sending 60% (from their perspective). They got back a reasonable amount.

As the SENDER now, I should consider:

1. They showed high trust in Round 1 (from their perspective, 60%)
2. They got a modest return
3. I want to continue building trust

Given that they sent 60% from their perspective, I should send a similar or slightly higher amount to show reciprocity and build trust. Sending around \$3.0-\$3.5 would be reasonable.

Let me send \$3.0 to match what they (think they) sent to me.

Actually, let me think more carefully. My actual earnings were \$3.72 in Round 1. I received \$6.87 and returned \$3.15.

...

usage:

```
{
  "input_tokens": 867,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.4700000000000001.

Your total visible earnings across all rounds are \$10.43.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.1000000000000001. How much do you return? (0-8.1000000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Round 2 pattern:**** When my co-player was the RECEIVER, they returned \$4.5 out of \$8.97 received (approximately 50%)

****This round:****

- They sent \$2.7 (54% of endowment), showing continued trust
- I received \$8.10

I should reciprocate with a similar return ratio. Returning about 50% like they did would be \$4.05, which:

- Gives them \$2.3 + \$4.05 = \$6.35 total (profit of \$1.35)
- Gives me \$4.05 (also a decent profit)

This maintains the cooperative pattern we've established.

```
{'return': 4.05}
```

reasoning:

Let me analyze the situation:

****Visible History:****

- Round 1: My co-player (as SENDER) sent \$3.0, received \$3.41 back, payoff \$5.41
- Round 2: My co-player (as RECEIVER) saw \$2.99 sent, returned \$4.5, payoff \$4.47

****This Round (Round 3):****

- I am the RECEIVER
- They sent \$2.7 (54% of their endowment)
- I received \$8.10

Looking at the pattern:

- In Round 1, they sent 60% of their endowment
- In Round 2, I sent \$3.0 (though they saw \$2.99 due to noise), and they returned \$4.5, which is about 50% of what they received (\$8.97)
- In Round 3, they sent \$2.7 (54%)

My co-player seems to be cooperating at moderate-to-high levels. When they were receiver in Round 2, they returned 50% of what they received.

If I return around 50% like they did, that would be about \$4.05.

Let me think about what's fair:

- They sent \$2.7, so they have \$2.3 left
- If I return \$4.05, they'd have \$2.3 + \$4.05 = \$6.35 (profit of \$1.35)
- I'd keep \$8.10 - \$4.05 = \$4.05 (profit of \$4.05)

That seems reasonable and matches their behavior. I could be slightly more generous to encourage continued cooperation.

Actually, let me reconsider. In Round 2, they returned \$4.5 out of \$8.97, which is exactly 50%.

...

usage:

```
{
  "input_tokens": 1210,
  "output_tokens": 644,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.4700000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.

Your total visible earnings across all rounds are \$14.48.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 (60% of endowment) in Rounds 1 and 3
- As RECEIVER: Returned generously (\$4.5 out of \$8.97, ~50%)

****My previous sending:****

- Round 2: I sent and they saw \$2.99 (likely noise from my \$3.0)

They've demonstrated consistent trust and reciprocity. I should maintain this cooperative relationship by matching their sending level.

```
{'send': 3.0}
```

reasoning:

Looking at the history:

Round 1: Co-player as SENDER sent \$3.0, got \$3.41 back, payoff \$5.41

Round 2: Co-player as RECEIVER got \$2.99 sent (from me), returned \$4.5, payoff \$4.47

Round 3: Co-player as SENDER sent \$3.0, got \$3.91 back, payoff \$5.91

Pattern analysis:

- When my co-player is SENDER, they consistently send \$3.0 (60% of endowment)
- When they're RECEIVER, they return generously (\$4.5 out of \$8.97 ? 50%)
- They're showing consistent trust and cooperation

My sending history:

- Round 2: I sent \$3.0 (based on the visible history showing they saw \$2.99 sent due to noise)
- Round 3: Not explicitly shown as sender, but the history says they saw \$2.99 sent in Round 2

Actually, let me re-read:

Round 2: I was SENDER. I sent \$3.0. Due to noise, they saw \$2.99. They received \$8.97, returned \$4.5.

Round 3: I was RECEIVER. They sent \$2.7 (but I think they sent \$3.0 and noise made it \$2.7). I returned \$4.05.

Now Round 4: I'm SENDER again.

They've been consistently sending \$3.0 when they're sender. They've returned generously when receiver.

I should maintain the cooperative pattern by sending \$3.0 again.

usage:

```
{
  "input_tokens": 1604,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.
- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.

Your total visible earnings across all rounds are \$20.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.24, so you received \$6.720000000000001. How much do you return? (0-6.720000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Co-player's return pattern:****

- Round 2: Returned \$4.5 out of \$8.97 (~50%)
- Round 4: Returned \$3.5 out of \$7.05 (~50%)


```

**This round:**
- They sent $2.24 (44.8% of endowment)
- I received $6.72

I should maintain the cooperative pattern by returning around 50% like they have consistently
done.

Returning $3.36:
- Gives them $2.76 + $3.36 = $6.12 total (profit of $1.12)
- Gives me $3.36 (also a profit)

{'return': 3.36}
Reasoning:
Looking at the visible history:

Round 2: They received $8.97, returned $4.5 (50%)
Round 4: They received $7.05, returned $3.5 (49.6%)

This round:
- They sent $2.24 (44.8% of endowment)
- I received $6.72

The co-player has been consistently returning around 50% of what they receive. I should recipr
ocate similarly.

Returning around 50% would be $3.36:
- Sender would have: $2.76 + $3.36 = $6.12 (profit of $1.12)
- I would keep: $6.72 - $3.36 = $3.36 (also a profit)

This maintains the cooperative pattern and roughly matches their return rates.
Usage: {
  "input_tokens": 1990,
  "output_tokens": 378,
  "reasoning_tokens": 0
}

GAME - Agent_7 - Sender - dyad_4
Timestamp: 2026-08-12T07:59:29
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
  so be asked to write myths.

  GAME:

```

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I want to establish a cooperative tone while being somewhat cautious since I don't know my co-player yet.

Sending a moderate amount can:

- Signal trust and willingness to cooperate
- Create a larger pool for mutual benefit (tripled to \$9)
- Allow the receiver meaningful choice in how much to return

If the receiver returns even half of what they receive, we both end up better than the initial \$5.

I'll send **\$3**.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- I keep: \$5 - sent + returned

- There's communication noise, so amounts may differ
- This is a multi-round game with 8 agents
- Pairings are randomized each round

Classic trust game setup. Let me think about optimal strategy:

1. If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
2. If I send \$5: Receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 (better than \$5). But they might return less or nothing.
3. If I send a moderate amount (e.g., \$3): Receiver gets \$9. If they return \$4-5, I come out ahead.

Since this is:

- Round 1 (establishing reputation/reciprocity patterns)
- Multi-round game (future interactions matter)
- Anonymous agents (can't track specific individuals)

I should consider:

- Building trust/reciprocity norms early
- Signaling cooperation
- But also being somewhat cautious since I don't know the other agents

A common strategy is to send a moderate-to-high amount in early rounds to:

1. Signal trustworthiness
2. Encourage reciprocity
3. Maximize potential gains if receiver cooperates

Given the noise factor and uncertainty, I'll send \$3. This shows goodwill without being maximally vulnerable, and gives the receiver \$9 to work with. If they return \$4-5, we both benefit.

Actually, let me reconsider. In trust games, sending higher amounts often generates better returns because:

- It demonstrates trust
- Creates larger pie to split
- Social norms typically lead to proportional returns

Comparing the math: sending \$4 leaves me with \$1 while giving the receiver \$12. A 50% return (\$6) nets me \$7 total, while a 33% return (\$4) just breaks me even. With \$3 sent instead, I'd have \$2 cushion and need only a 50% return (\$4.50) to reach \$6.50. The higher send amount creates more upside if reciprocity holds, so I'll go with \$4....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 814,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.

Your total visible earnings across all rounds are \$4.2.

This round, you are the RECEIVER against your current co-player. They sent you \$3.7, so you received \$11.100000000000001. How much do you return? (0-11.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 analysis:****

- My co-player saw \$2.24 sent (due to noise), which became \$6.72
- They returned \$3.0 (about 45% of what they received)
- This shows reasonable cooperative behavior

****Round 2 (current):****

- They sent me \$3.7 (even more than what they saw in Round 1)
- I received \$11.10
- They're showing increased trust

To maintain and encourage cooperation, I should return a fair amount. Returning about half of what I received would give them:

- They kept: $\$5 - \$3.7 = \$1.30$
- Plus returned amount
- A good return on their trust

I'll return ****\$6.00**** (about 54% of what I received), giving them a total of \$7.30 - a solid return that rewards their increased trust.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation carefully.

Round 1 recap from my perspective:

- I sent \$3
- Due to communication noise, my co-player saw \$2.24 sent (which became \$6.72 after tripling)
- They returned \$3.0
- Their visible payoff was \$3.72 (received \$6.72, returned \$3.0)
- My total visible earnings are \$4.2

Let me calculate what actually happened in Round 1:

- I sent \$3, so I had $\$5 - \$3 = \$2$ left
- The receiver got $\$3 \times 3 = \9 (but saw \$6.72 due to noise)
- They returned \$3.0
- So I ended with $\$2 + \$3 = \$5$
- But my visible earnings show \$4.2, which suggests the noise affected what I saw returned too?

Actually, wait. Let me reconsider:

- If my visible earnings are \$4.2 total
- I started with \$5, sent \$3 (keeping \$2)
- If I received back something that made my total \$4.2, then I received $\$4.2 - \$2 = \$2.2$ back

So the noise worked both ways:

- I sent \$3, they saw \$2.24
- They returned \$3.0, but I saw (and actually got) something that gave me \$4.2 total

Now for Round 2:

- I'm the RECEIVER
- My co-player sent me \$3.7 (according to what I see)
- This became \$11.10 after tripling
- I need to decide how much to return

My co-player showed willingness to return a reasonable amount in Round 1 (they returned \$3.0 out of \$6.72 they saw, which is about 45%). This is fairly cooperative behavior.

I should return a fair amount to maintain cooperation. Given that:

- They sent \$3.7 and I received \$11.10
- A 50% return would be \$5.55, giving them $\$1.3$ (kept) + \$5.55 (returned) = \$6.85 total
- This matches their cooperative gesture from Round 1 and encourages continued trust

I'll return \$5 to keep things balanced....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 1051,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.

Your total visible earnings across all rounds are \$9.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3 ? they returned \$3 (cooperative)

****Round 2:**** They sent \$3 ? I returned \$6, but they saw \$5.07 (still a good return)

****Round 3 (current):****

- Both rounds showed mutual cooperation
- They sent \$3 as sender and returned reasonably as receiver
- Their visible payoff was \$7.07 last round, showing cooperation pays off

I'll send ****\$3.5**** to maintain and slightly increase our cooperative dynamic. This shows I'm willing to trust them a bit more based on our positive history.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.24 (noise), became \$6.72, returned \$3.0
- They kept \$3.72, I got back \$3.0, so I ended with \$5 - \$3 + \$3 = \$5
- But my visible earnings show \$4.2, which suggests I saw something different due to noise

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, became \$9.0
- I returned \$6.0, but they saw \$5.07 (noise)
- They ended with \$5 - \$3 + \$5.07 = \$7.07
- I kept \$9.0 - \$6.0 = \$3.0

My total visible earnings are \$9.3.

Now Round 3:

- I'm SENDER again
- Co-player has shown:
 - As receiver: returned \$3.0 out of \$6.72 (45%)
 - As sender: sent \$3.0 out of \$5 (60%)
- They saw I returned \$5.07 (though I actually returned \$6.0)
- This is good cooperation

Since they've been cooperative and sent a good amount when they were sender, I should continue cooperating. They sent \$3.0 last round, so I should send at least that much or more to maintain trust.

I'll send \$3.5 to show continued cooperation and slightly increase the stakes.

usage:

```
{
  "input_tokens": 1191,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$16.12.

This round, you are the RECEIVER against your current co-player. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant**content:**

Looking at our interaction pattern:

```

**Round 1:** They returned $3.0 from $6.72 (45%)
**Round 2:** They sent $3.0, I returned $6.0
**Round 3:** They returned $5.0 from $9.90 (51%)
**Round 4 (current):** They sent $3.0 ? I received $9.0

```

They've shown consistent cooperation, with their return percentages increasing slightly over time. They sent \$3.0 this round, maintaining trust despite some communication noise in previous rounds.

I'll return **\$5.0** (about 56% of what I received). This gives them a total payoff of \$7.0 (\$2 kept + \$5 returned), which rewards their continued trust and maintains our cooperative relationship.

```

'''json
{"return": 5.0}
'''

```

reasoning:

Let me analyze the history:

```

**Round 1:** I sent $3 ? they saw $2.24 ? they returned $3.0
**Round 2:** They sent $3.7 ? I received $11.10 ? I returned $6.0 ? they saw $5.07
**Round 3:** I sent $3.5 ? they saw $3.3 ? they returned $5.0
**Round 4 (current):** They sent $3.0 ? I received $9.0

```

Looking at their behavior:

- Round 1 as receiver: Returned \$3.0 out of \$6.72 (45%)
- Round 3 as receiver: Returned \$5.0 out of \$9.90 (51%)

Their returns seem reasonable and slightly increasing in percentage.

My behavior:

- Round 2 as receiver: Returned \$6.0 out of \$11.10 (54%)

They've consistently sent around \$3-\$3.7 as sender, maintaining cooperation.

For Round 4, they sent \$3.0 and I received \$9.0. To maintain fair cooperation and match roughly what they've done (returning about 50%), I should return around \$4.5-\$5.0.

Let me return \$5.0 to be consistent with maintaining good cooperation and matching their recent return percentage.

Actually, wait. Let me recalculate their payoffs to understand better:

Round 1:

- They as receiver: Kept \$5 - \$0 (not sender) + received \$6.72 - \$3.0 returned = \$5 + \$3.72 = \$8.72? No wait, they were receiver, so sender sent them something.
- Actually the history shows "as RECEIVER: saw \$2.24 sent" - so I was the sender, I sent something, they saw \$2.24, it became \$6.72, they returned \$3.0
- Their payoff: \$6.72 - \$3.0 = \$3.72. Round 2 shows they sent \$3.0 which tripled to \$9.0, and when I returned \$5.07, their payoff was \$5 - \$3.0 + \$5.07 = \$7.07. In Round 3, I sent

t \$3.5 which became \$9.90, and they returned \$5.0, giving them $\$9.90 - \$5.0 = \$4.90$. Now they've sent \$3.0 which becomes \$9.0 for me, so I need to decide what to return....

usage:

```
{
  "input_tokens": 1596,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.

Your total visible earnings across all rounds are \$20.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our pattern:

****Round 2:**** They sent \$3.0, saw ~\$5.07 returned (good payoff)

****Round 3:**** They returned \$5.0 from what they saw as \$9.90

****Round 4:**** They sent \$3.5, saw ~\$4.99 returned

They've consistently sent around \$3.0-\$3.5 and returned fairly when in the receiver role. Despite some communication noise affecting visible payoffs, we've maintained good cooperation.

I'll send ****\$3.5**** to continue our stable, mutually beneficial pattern.

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Let me analyze our history:

****Round 2:**** They sent \$3.0, I returned \$6.0 (they saw \$5.07 due to noise)

****Round 3:**** I sent \$3.5, they returned \$5.0 (they saw \$3.3 sent due to noise)

****Round 4:**** They sent \$3.5 (wait, the history says they sent \$3.5 in Round 4?), I returned \$5.0 (they saw \$4.99 due to noise)

Wait, let me reread. In Round 4, I was the receiver and they sent me \$3.0. But the visible history says they sent \$3.5 in Round 4. That must be the noise - what they actually sent was \$3.5, but I saw \$3.0.

Let me recalculate:

- Round 4: They actually sent \$3.5 ? became \$10.5 ? I returned \$5.0 (they saw \$4.99)
- Their payoff: $\$5 - \$3.5 + \$4.99 = \6.49

****Round 5 (current):**** I'm the sender

Looking at the pattern:

- Round 2: They sent \$3.0, got back ~\$5.07 (good return)
- Round 3: I sent (they saw \$3.3), they returned \$5.0 (reasonable)
- Round 4: They sent \$3.5, got back ~\$4.99 (decent return, about 48% of the tripled amount)

They've been consistently cooperative, sending \$3.0-\$3.5 each time. Their returns have been fair (around 50% of what they receive).

I should maintain the cooperative relationship. I'll send ****\$3.5**** to continue our pattern of mutual trust and cooperation.

Usage: {

"input_tokens": 2020,

```
"output_tokens": 588,
"reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T07:59:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.24, that is 44.8% of their total endowment. You received \$6.720000000000001. How much do you return to the sender? (0-6.720000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.24 (44.8% of their endowment)
- They currently have \$2.76 remaining
- I received \$6.72 after the 3x multiplication

To reciprocate their trust fairly, I should return an amount that:

1. Ensures they profit from trusting me (end up with more than their original \$5)
2. Rewards their cooperative gesture
3. Still leaves me with a reasonable gain

If I return \$3.00:

- Sender ends with: $\$2.76 + \$3.00 = \$5.76$ (gain of \$0.76)
- I keep: $\$6.72 - \$3.00 = \$3.72$ (gain of \$3.72)

This seems like a fair reciprocal response that rewards trust while maintaining balanced incentives for future rounds.

{"return": 3.00}

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.24 out of their \$5 endowment (44.8%)
3. This was multiplied by 3x, so I received \$6.72 (technically \$6.720000000000001 due to floating point)
4. I need to decide how much to return (between \$0 and \$6.72)

Key considerations:

- The sender showed moderate trust by sending 44.8% of their endowment
- If I return nothing, I keep all \$6.72 and the sender gets only \$5 - \$2.24 = \$2.76
- If I return a fair amount, I should consider what would make this equitable

Common approaches in trust games:

1. Equal split of total gains: Total in the system is $\$5 + \$6.72 = \$11.72$. Equal split would be \$5.86 each. The sender currently has \$2.76, so I'd return $\$5.86 - \$2.76 = \$3.10$
2. Return the same proportion they sent: They sent 44.8%, so I could return 44.8% of \$6.72 = \$3.01
3. Match their investment: Return \$2.24 so we both end up with gains from the multiplication
4. Return enough so they profit: If I return \$3, they end up with $\$2.76 + \$3 = \$5.76$ (a gain of \$0.76), and I keep \$3.72 (a gain of \$3.72)

The sender took a risk by sending \$2.24. A fair return should:

- At minimum, not make them worse off than keeping everything (\$5)
- Ideally, reward their trust
- Be reciprocal to their gesture

Let me calculate a few scenarios:

- If I return \$3.00: Sender gets $\$2.76 + \$3.00 = \$5.76$, I keep \$3.72
- If I return \$3.50: Sender gets $\$2.76 + \$3.50 = \$6.26$, I keep \$3.22
- If I return \$4.00: Sender gets $\$2.76 + \$4.00 = \$6.76$, I keep \$2.72

A return of \$3.00-\$3.50 seems balanced-it rewards their trust while still giving me a meaningful gain. I'll return \$3.00, which nets them \$5.76 (a 15.2% gain) and leaves me with

\$3.72....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.

Your total visible earnings across all rounds are \$3.7200000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw only \$2.2 returned (though due to communication noise, I actually returned \$3.00)
- From their perspective, they may feel the cooperation wasn't fully reciprocated

As the sender now, I want to:

1. Signal continued willingness to cooperate
2. Match their previous level of trust
3. Maintain the cooperative dynamic despite any noise in the previous round

I'll send the same amount they sent in Round 1 to demonstrate reciprocal trust.

```
{"send": 3.0}
```

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$2.24 (44.8% of their endowment)
- I received \$6.72
- I returned \$3.00
- Sender ended with $\$2.76 + \$3.00 = \$5.76$
- I kept \$3.72

But the visible history shows something different from my co-player's perspective:

- Round 1: They were SENDER, sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2

Wait, this is confusing. Let me re-read the prompt about communication noise.

"NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design."

So what happened:

- From my perspective: I received notification that sender sent \$2.24, I received \$6.72, I returned \$3.00
- From their perspective: They sent \$3.0, it became \$9.0, they saw \$2.2 returned, and got payoff \$4.2

Let me verify the math from their perspective:

- They sent \$3.0
- They had $\$5.0 - \$3.0 = \$2.0$ left
- They saw \$2.2 returned
- Their payoff: $\$2.0 + \$2.2 = \$4.2$?

So from their perspective, they sent \$3.0 and only got back \$2.2, which is less than they sent. They might perceive this as me being untrustworthy or uncooperative.

However, from MY perspective, I actually returned \$3.00, which I thought was fair and generous given I received \$6.72.

This is the communication noise at work. They think I returned less than I actually did. Now I'm the sender in Round 2, and they likely view me as less trustworthy due to the noise distorting their perception of my return. They sent 60% of their endowment initially, showing genuine trust, but the miscommunication means they'll probably be more cautious. I need to decide whether sending a substantial amount will help rebuild trust or just get exploited if they reciprocate poorly. A moderate send might signal cooperation without overcommitting, especially since they may interpret my actions through the lens of what they think happened last round....

usage:

```
{
  "input_tokens": 890,
  "output_tokens": 922,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.

Your total visible earnings across all rounds are \$10.790000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment), showing strong trust
- Round 2: Co-player returned \$6.0 out of \$11.1 received (54% return rate), showing generosity despite seeing lower returns in Round 1
- Round 3: Co-player sent \$3.3 (66% of endowment), continuing to show high trust

My co-player has consistently demonstrated cooperative behavior and trust. They've maintained cooperation even after seeing what appeared to be a lower return in Round 1.

If I return \$5.0:

- Sender ends with: $\$1.70 + \$5.00 = \$6.70$ (profit of \$1.70)
- I keep: $\$9.90 - \$5.00 = \$4.90$

This reciprocates their generous behavior and maintains the positive cooperative dynamic.

```
{"return": 5.0}
```

reasoning:

Let me analyze the visible history:

Round 1:

- Co-player as SENDER: sent \$3.0, saw \$2.2 returned
- From their view: started with \$5, sent \$3, kept \$2, saw \$2.2 returned = \$4.2 total
- (I actually returned \$3.00, but they saw \$2.2 due to noise)

Round 2:

- Co-player as RECEIVER: saw \$3.7 sent (I sent \$3.0), received \$11.1, returned \$6.0
- From their view: they received \$11.1, returned \$6.0, kept \$5.1
- From my view: I sent \$3.0, kept \$2.0, saw \$6.0 returned = \$8.0 total

Round 3 (current):

- I am RECEIVER
- Co-player sent \$3.3 (66% of endowment)

- I received \$9.9

My co-player has been quite trustworthy:

- Round 1: sent 60% of endowment
- Round 2: returned a generous \$6.0 out of \$11.1 they received (54% return rate)
- Round 3: sent 66% of endowment

They've shown consistent cooperation and generosity. I should reciprocate appropriately.

If I return \$5.0:

- Sender ends with: \$1.7 (kept) + \$5.0 (returned) = \$6.7 (profit of \$1.7)
- I keep: \$9.9 - \$5.0 = \$4.9

This maintains a good balance and rewards their trust. Let me check if this is fair compared to their Round 2 behavior:

- In Round 2, they returned \$6.0 out of \$11.1 (54%)

...

usage:

```
{
  "input_tokens": 1250,
  "output_tokens": 704,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.

Your total visible earnings across all rounds are \$15.69.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment)
- Round 2: Co-player returned \$6.0 out of \$11.1 (very generous, 54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment), saw \$5.32 returned

My co-player has consistently shown high trust and cooperation, sending substantial amounts and returning generously when in the receiver role. They've maintained this cooperative behavior throughout.

I'll match their highest level of trust by sending a similar amount.

```
{"send": 3.5}
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: Co-player sent \$3.0, saw \$2.2 returned (I actually returned \$3.0, but noise)

Round 2: I sent \$3.0 (co-player saw \$3.7), they returned \$6.0 - very generous!

Round 3: Co-player sent \$3.5 (actually \$3.3 based on my perspective), saw \$5.32 returned (I actually returned \$5.0)

My co-player has been consistently cooperative:

- Sending high amounts (60-70% of endowment)
- When they were receiver, they returned \$6.0 out of \$11.1 (54% return rate)

They've maintained cooperation throughout despite seeing some noise in the returns. In Round 3, they saw \$5.32 returned on their \$3.5 sent, which gave them a profit of \$1.82 - decent but not extremely generous.

As the sender now, I should:

1. Continue the cooperative pattern
2. Send an amount that shows trust matching theirs
3. Given they sent \$3.0, \$3.5 in previous rounds, I should send similarly

I'll send \$3.5 to match their highest level of trust and continue the positive dynamic.

usage:

```
{
  "input_tokens": 1694,
  "output_tokens": 461,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.
- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.

Your total visible earnings across all rounds are \$22.18.

This round, you are the RECEIVER against your current co-player. They sent you \$3.42, so you received \$10.26. How much do you return? (0-10.26)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (generous 54% return)
- Round 3: Co-player sent \$3.5 (70% of endowment, high trust)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return)
- Round 5: Co-player sent \$3.42 (68% of endowment, continued trust)

My co-player has consistently demonstrated strong cooperation, both sending high amounts and returning generously. They're maintaining this pattern across rounds.

If I return \$5.5:

- Sender ends with: $\$1.58 + \$5.5 = \$7.08$ (profit of \$2.08)
- I keep: $\$10.26 - \$5.5 = \$4.76$

This matches their typical return rate (~54%) and reciprocates their continued trust.

{"return": 5.5}

Reasoning:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.42 (68% of endowment)

My co-player has been very cooperative and consistent:

- As sender: sending around 68-70% of their endowment
- As receiver: returning around 54-56% of what they receive

They're maintaining high trust and generosity throughout. I should reciprocate appropriately.

If I return \$5.5:

- Sender ends with: $\$1.58 + \$5.5 = \$7.08$ (profit of \$2.08)
- I keep: $\$10.26 - \$5.5 = \$4.76$

This is approximately 54% return rate, matching their pattern and continuing our cooperative relationship.

Actually, let me be slightly more generous to maintain the positive dynamic. Let me return \$5.

5 or \$6.0.

Return of \$5.5 gives them \$7.08 total (profit of \$2.08)
Return of \$6.0 gives them \$7.58 total (profit of \$2.58)

I'll go with \$5.5 to match their typical return rate of around 54%.

```
Usage: {  
  "input_tokens": 2082,  
  "output_tokens": 593,  
  "reasoning_tokens": 0  
}
```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver,
Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$4; received \$12; Agent_1 returned \$8.5; payoffs Agent_2=\$9.5, Agent_1=\$3.5;
noise/visible: sent visible=\$4.89, received visible=\$14.67, returned visible=\$7.77
Action validation: {"returned": {"amount": 8.5, "clamped": false, "max": 12.0, "min": 0, "raw": 8.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_4 sent \$3; received \$9; Agent_3 returned \$4.5; payoffs Agent_4=\$6.5, Agent_3=\$4.5;
noise/visible: sent visible=\$3.04, received visible=\$9.12, returned visible=\$4.68
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 sent \$3; received \$9; Agent_5 returned \$3.66; payoffs Agent_6=\$5.66,
Agent_5=\$5.34; noise/visible: sent visible=\$2.44, received visible=\$7.32, returned visible=\$4.57
Action validation: {"returned": {"amount": 3.66, "clamped": false, "max": 9.0, "min": 0, "raw": 3.66}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent \$3.5; received \$10.5; Agent_7 returned \$5; payoffs Agent_8=\$6.5,
Agent_7=\$5.5; noise/visible: sent visible=\$3.29, received visible=\$9.87, returned visible=\$5.14
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 10.5, "min": 0, "raw": 5.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent_1": 36.0,
"Agent_2": 39.0,
"Agent_3": 30.0,
"Agent_4": 36.0,
"Agent_5": 31.9,
"Agent_6": 34.1,
"Agent_7": 32.5,
"Agent_8": 37.5

}

Visible balances after round: {

"Agent_1": 40.03,
"Agent_2": 39.42,
"Agent_3": 34.069999999999999,
"Agent_4": 33.76,
"Agent_5": 27.790000000000003,
"Agent_6": 30.08,
"Agent_7": 32.72,
"Agent_8": 33.58

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T07:59:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,

```

"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.28, that is 65.6% of their total endowment. You received \$9.84. How much do you return to the sender? (0-9.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 65.6% of their endowment (\$3.28)
- This created \$9.84 for me to allocate
- The sender currently has only \$1.72 left from their original \$5

I want to reward their trust while being fair to both of us. If I return \$5.00:

- Sender's total: $\$1.72 + \$5.00 = \$6.72$ (profit of \$1.72)
- My total: $\$9.84 - \$5.00 = \$4.84$ (profit of \$4.84)

This seems like a reasonable split that rewards the sender's trust while still benefiting from the transaction.

```
{"return": 5.00}
```

reasoning:

This is Round 1 of a multi-agent trust game where I'm playing as the RECEIVER.

Let me understand what happened:

- The sender had an endowment of \$5
- The sender sent me \$3.28 (which is 65.6% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$3.28 \times 3 = \9.84
- I now need to decide how much to return to the sender (between \$0 and \$9.84)

The sender's payoff will be:

- $\$5$ (initial) - $\$3.28$ (sent) + [amount I return] = $\$1.72 + [\text{amount I return}]$

My payoff will be:

- $\$9.84$ (received) - [amount I return]

Let me think about what would be a fair return:

- The sender was quite generous, sending 65.6% of their endowment
- This shows significant trust
- A fair return might be to ensure the sender gets back at least what they sent, or even more to reward their trust
- If I return \$3.28, the sender would have: $\$1.72 + \$3.28 = \$5$ (break even)
- If I return \$5, the sender would have: $\$1.72 + \$5 = \$6.72$ (profit of \$1.72)
- If I return \$6.56 (double what they sent), the sender would have: $\$1.72 + \$6.56 = \$8.28$

A commonly fair approach in trust games is to split the total gains somewhat equitably while rewarding the sender's trust.

Total value created: \$9.84 (what I received)

Original sender investment: \$3.28

Net gain: $\$9.84 - \$3.28 = \$6.56$

One approach: Split the net gain equally

- Sender gets back: $\$3.28 + \$3.28 = \$6.56$
- I would return: \$6.56
- Sender's final: \$8.28
- My final: \$3.28

Another approach: Split all the money equally

- Total pool: \$9.84
- Equal split: \$4.92 each
- I would return: \$4.92
- Sender's final: \$6.64
- My final: \$4.92...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 999,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.

Your total visible earnings across all rounds are \$4.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- My co-player showed trust in Round 1 by sending \$3.0 (60% of their endowment)
- They received a reasonable return of \$5.94
- This suggests they're willing to cooperate

I want to reciprocate their trust and continue building a cooperative relationship. I'll send a similar proportion to what they sent.

{"send": 3.50}

reasoning:

Let me analyze this situation carefully:

Round 1 Summary:

- I was the RECEIVER
- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- I returned \$5.00
- My earnings: \$4.84

Now looking at the visible history shown to me:

- My co-player's last game shows: as SENDER, sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94

Wait, this is interesting. There's communication noise mentioned in the instructions. Let me think about what actually happened vs what was observed:

What I experienced in Round 1:

- Sender sent \$3.28
- I received \$9.84
- I returned \$5.00

What my co-player (the sender) saw:

- They sent \$3.0 (but actually sent \$3.28?)
- It became \$9.0 (but actually became \$9.84?)
- They saw \$5.94 returned (but I actually returned \$5.00?)
- Their visible payoff was \$7.94

Let me check the math on their side:

- If they sent \$3.0 and saw \$5.94 returned
- Their payoff would be: $\$5 - \$3 + \$5.94 = \7.94 ?

So from their perspective:

- They sent \$3.0 (66.7% of their total)
- They got back \$5.94 (66% of the \$9.0 that was created)
- This seems like a decent return to them

Now I'm the SENDER in Round 2:

- My co-player showed trust by sending most of their endowment in Round 1
- They received what appeared to them as a good return (66% of the multiplied amount)
- This suggests they might be willing to reciprocate if I show trust
- Since they sent roughly two-thirds of their endowment, I should match that level of generosity to build on the cooperative dynamic we...

usage:

```
{
  "input_tokens": 839,
  "output_tokens": 749,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.
- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.

Your total visible earnings across all rounds are \$13.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.71, so you received \$11.129999999999999. How much do you return? (0-11.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant**content:**

Looking at the history with my co-player:

- Round 1: They sent \$3.0 (60%)
- Round 2: They returned \$6.0 when I sent (showing cooperation)
- Round 3: They sent \$3.71 (74.2% - even more trust!)

They're showing increasing trust and cooperation. I should reciprocate generously.

Current situation:

- They sent \$3.71, leaving them with \$1.29
- I received \$11.13

If I return \$6.50:

- Sender's total: \$1.29 + \$6.50 = \$7.79
- My total: \$11.13 - \$6.50 = \$4.63

This rewards their trust while still providing a fair outcome for both of us.

{"return": 6.50}

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.28 (65.6%)
- I received \$9.84
- I returned \$5.00
- Sender's total: \$1.72 + \$5.00 = \$6.72
- My total: \$9.84 - \$5.00 = \$4.84

Round 2: I was SENDER

- I sent \$3.50
- It became \$10.50
- But the visible history shows my co-player saw \$3.89 sent, it became \$11.67
- They returned \$6.00
- Their visible payoff was \$5.67
- My actual payoff should be: \$5 - \$3.50 + (what they returned)
- According to my total visible earnings of \$13.28, that's \$4.84 (Round 1) + \$8.44 (Round 2) = \$13.28
- So I received \$8.44 in Round 2
- This means: \$5 - \$3.50 + \$8.44 = \$9.94 total

Wait, let me recalculate. If my total visible earnings are \$13.28:

- Round 1: I kept \$4.84
- Round 2: \$13.28 - \$4.84 = \$8.44

So in Round 2 my payoff was \$8.44.

I sent \$3.50, so I had \$1.50 left, plus what they returned.

\$1.50 + returned = \$8.44

So they returned \$6.94 to me.

But the visible history says they returned \$6.00. This is the communication noise mentioned in the instructions.

Now in Round 3:

They're sending \$3.71, which is 74.2% of their endowment-a significant increase from the 60% they sent initially. I'll receive \$11.13, leaving them with \$1.29. Given their consistent pattern of cooperation and escalating trust, I should reciprocate meaningfully. Returning \$6.00 would give them \$7.29 total while I keep \$5.13....

usage:

```
{
  "input_tokens": 1132,
  "output_tokens": 983,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.
- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.

Your total visible earnings across all rounds are \$17.91.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our history:

- My co-player has consistently shown high trust (sending 60-80% of their endowment)
- They've returned reasonable amounts when receiving
- We've established a strong cooperative pattern

In Round 3, they sent \$4.0 (80% of their endowment), showing very high trust. I should reciprocate with a similarly strong showing of trust to maintain our cooperative relationship.

```
{"send": 4.00}
```

reasoning:

Let me analyze the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned \$5.00, but they saw \$5.94 returned

Round 2: I sent \$3.50, they received it as \$3.89 (11.67 total), they returned \$6.0

Round 3: They sent \$3.71, but they show it as \$4.0 (became \$12.0), I returned \$6.50 but they saw \$5.66

There's definitely communication noise happening. But the pattern shows:

- They consistently send high amounts (60-80%)
- They return reasonably when receiving
- We've been building a cooperative relationship

In Round 3, they sent what I saw as \$3.71 (74%), but they think they sent \$4.0 (80%). This shows very high trust.

I should continue the cooperative pattern and send a high amount as well. I'll send around \$3.50-\$4.00 to match their level of trust.

Let me send \$4.00 to show strong trust and keep the cooperation going.

usage:

```
{
  "input_tokens": 1542,
  "output_tokens": 367,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.
- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.

Your total visible earnings across all rounds are \$25.07.

This round, you are the RECEIVER against your current co-player. They sent you \$4.36, so you received \$13.080000000000002. How much do you return? (0-13.080000000000002)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued cooperation:

- My co-player sent \$4.36 (87.2% of their endowment) - showing very high trust!
- They've been consistently generous as receiver (returning \$6.0 and \$7.0 in previous rounds)
- This is clearly a cooperative partner

Current situation:

- They sent \$4.36, leaving them with only \$0.64
- I received \$13.08

I should return generously to reward their exceptional trust.

If I return \$7.50:

- Sender's total: $\$0.64 + \$7.50 = \$8.14$
- My total: $\$13.08 - \$7.50 = \$5.58$

{"return": 7.50}

reasoning:

Looking at the history with my co-player:

Round 2: I sent \$3.89, they returned \$6.0

Round 3: They sent \$4.0, I returned \$5.66 (visible to them)

Round 4: I sent \$3.97, they returned \$7.0

Now Round 5: They sent \$4.36 (87.2% of their endowment - very high trust!)

They are showing extremely high trust by sending \$4.36, leaving them with only \$0.64.

I received \$13.08

They've been returning well when they're the receiver:

- Round 2: returned \$6.0 from \$11.67
- Round 4: returned \$7.0 from \$11.91

This is a cooperative partner. I should return generously to reward their high trust.

If I return \$7.50:

- Sender's total: $\$0.64 + \$7.50 = \$8.14$ (profit of \$3.14)
- My total: $\$13.08 - \$7.50 = \$5.58$ (profit of \$5.58)

This seems fair and rewards their trust while still benefiting from the exchange.

usage:

```
{
  "input_tokens": 1882,
  "output_tokens": 462,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible

payoff \$6.66.
- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
Your total visible earnings across all rounds are \$30.650000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our strong cooperative history:

- My co-player has consistently sent high amounts (\$4.0 in recent rounds, 80% of endowment)
- They've returned generously as receiver (\$7.0 and \$7.68 in recent rounds)
- We've established excellent mutual trust

I should continue this cooperative pattern by sending a similarly high amount.

```
{"send": 4.00}
```

Reasoning:

Looking at the history with my co-player:

- Round 3: They sent \$4.0 (80%), returned visible \$5.66
- Round 4: They returned \$7.0
- Round 5: They sent \$4.0 (80%), returned visible \$7.68

This is a highly cooperative partner who consistently:

1. Sends high amounts (80% of endowment)
2. Returns generously when receiving (around \$6-7.68)

We've built a strong mutual trust pattern. I should continue to send high amounts to maintain this cooperation.

I should send a high amount, similar to what they've been sending. Around \$4.0-\$4.5 would be appropriate.

Actually, I notice there's some noise mentioned in the instructions. The visible amounts might differ from actual sent/returned amounts. But I should focus on the pattern of cooperation that's clearly visible.

I'll send \$4.00 to match their typical sending pattern and maintain our strong cooperation.

```
Usage: {  
  "input_tokens": 2276,  
  "output_tokens": 323,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T07:59:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",
```

```

"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't know my co-player yet.

****My reasoning:****

- Sending \$0 is safe but creates no mutual benefit
- Sending everything (\$5) maximizes potential gains but is very risky with an unknown partner
- A moderate amount signals cooperation while managing risk

I'll send ****\$3****, which:

- Gets multiplied to \$9 for the receiver (creating good potential for returns)
- Keeps \$2 as a safety buffer for me
- Shows willingness to trust and cooperate

- If they return even \$3-4, we both benefit

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me analyze the situation:

1. I have \$5 to start
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ from what was actually sent/returned
5. This is a multi-agent setting with 8 agents, randomized pairings each round
6. I don't know who my co-player is (names are hidden)

Key considerations:

- If I send \$0, I guarantee keeping \$5 but miss the opportunity for mutual gain
- If I send the full \$5, the receiver gets \$15 and could return a substantial amount
- The optimal outcome for both of us combined is if I send everything (creating \$15 total instead of \$5)
- However, the receiver could keep everything and I'd end up with \$0
- There's also communication noise which adds uncertainty

Since this is the first round and I have no information about my co-player:

- Sending some amount signals trust and cooperation
- A moderate amount (like \$2-3) balances risk and potential reward
- Sending \$5 shows maximum trust but is risky with an unknown partner
- The communication noise means even if they want to reciprocate fairly, there might be misunderstandings

For a first round with an unknown partner, I'll send a moderate amount to signal cooperation while managing risk. I'll send \$3:

- This creates \$9 for the receiver (good potential for returns)
- I keep \$2 as insurance
- If they return just \$3-4, I'll break even or profit
- It signals willingness to cooperate without being reckless

Let me go with \$3 as a balanced first move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$7.94.

This round, you are the RECEIVER against your current co-player. They sent you \$3.89, so you received \$11.67. How much do you return? (0-11.67)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.28 (became \$9.84 for them)
- They returned \$5.00 (about 51% of what they received)
- This shows they're cooperative and fair

****Round 2 situation:****

- They sent \$3.89 (78% of their endowment - very generous!)
- I received \$11.67

- They've shown trust and cooperation

I should reciprocate their cooperative behavior. Since they returned about 51% last round and sent generously this round, I'll return a similar proportion:

~51% of \$11.67 = approximately \$6

This rewards their trust while maintaining fairness.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

From Round 1:

- I sent \$3 (but they saw \$3.28 due to communication noise)
- It became \$9.84 for them
- They returned \$5.0
- Their visible payoff was \$4.84

Now in Round 2:

- They sent \$3.89 (I received \$11.67 after 3x multiplier)
- I need to decide how much to return

Let me think about their behavior:

- They saw I sent \$3.28, which became \$9.84
- They returned \$5.0 from that \$9.84
- That's about 51% of what they received

They were fairly generous, returning about half. This suggests they're cooperative.

Now they sent \$3.89 out of \$5, which is about 78% of their endowment. That's quite generous and trusting.

If I want to reciprocate their cooperative behavior:

- They returned about 51% last round
- But they also sent a lot this round (78% of endowment)

I should return a reasonable amount that:

1. Rewards their trust and generosity
2. Maintains cooperation for future rounds
3. Is fair given they returned \$5 from \$9.84 last round

Let me calculate some options:

- If I return 50% of \$11.67 = \$5.84
- If I return a bit more generously, around \$6-7

Given that:

- They were generous (sent \$3.89)
- They returned \$5 last round (about 51%)
- I want to encourage continued cooperation

I'll return around \$6. This gives them:

- They sent: \$3.89
 - They get back: \$6
 - Net for them: \$5 - \$3.89 + \$6 = \$7.11
- ...

usage:

```
{
  "input_tokens": 800,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
Your total visible earnings across all rounds are \$13.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00 - very fair and cooperative

****Round 2:****

- They sent \$3.50 (70% of their endowment - very trusting!)
- I returned \$6 (they saw \$6.94)
- Strong mutual benefit

****Analysis:****

This co-player has been consistently generous and trustworthy. They've shown:

- Fair returns when receiving (51% returned)
- High trust when sending (70% of endowment)
- We've built a good cooperative relationship

I should reciprocate their trust level. I'll send ****\$4**** (80% of my endowment), which:

- Matches their demonstrated trust
- Becomes \$12 for them (good potential returns)
- Continues building our cooperative pattern

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.28 due to noise)
- It became \$9.84 for them
- They returned \$5.00
- This was generous - about 51% of what they received

****Round 2:****

- They sent \$3.5 (I saw \$3.89 due to noise)
- It became \$11.67 for me (their view: \$10.5)
- I returned \$6 (they saw \$6.94 due to noise)
- They earned \$8.44 total

This co-player has been:

1. Very cooperative and fair in Round 1 (returned 51%)
2. Very trusting in Round 2 (sent 70% of their endowment)
3. They saw a good return from me in Round 2 (\$6.94)

Given this strong pattern of mutual cooperation, I should continue to trust them. They've proven reliable.

I think I should send a high amount, similar to what they sent in Round 2. They sent \$3.5, so I'll send around that amount or slightly more.

Options:

- Send \$3.5 to \$4 to maintain the cooperative relationship
- Send \$5 to maximize mutual benefit if I fully trust them

Given the strong cooperation so far, I'll send \$4. This shows:

- I trust them based on their past behavior
- I'm willing to invest in our mutual benefit
- It's not quite all \$5, maintaining a tiny hedge

Actually, reviewing their behavior:
- Round 1: They returned generously (51%)
- Round 2: They sent generously (70% of endowment)

They've been consistently cooperative. I should match their level of trust. ...

usage:

```
{
  "input_tokens": 1156,
  "output_tokens": 645,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.

Your total visible earnings across all rounds are \$20.27.

This round, you are the RECEIVER against your current co-player. They sent you \$3.97, so you received \$11.91. How much do you return? (0-11.91)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history:

****Round 1:****

- They returned \$5.00 from \$9.84 (51%)

****Round 2:****

- They sent \$3.50 generously
- I returned well (they saw \$6.94)

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)
- Consistent fair behavior

****Round 4:****

- They sent \$3.97 (about 79% - very trusting!)
- I received \$11.91

****Analysis:****

They've been consistently fair, returning 50-58% of what they receive. They continue to send high amounts, showing sustained trust.

I'll return approximately 59% to match their cooperative pattern:

```
'''json
{"return": 7}
'''
```

This rewards their high trust while ensuring both of us benefit substantially from the exchange.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00
- They kept \$4.84

****Round 2:****

- They sent \$3.50

- I received \$10.50
- I returned \$6 (they saw \$6.94)
- They kept \$8.44

****Round 3:****

- I sent \$4 (they saw \$3.71 with noise)
- They received \$11.13
- They returned \$6.50
- They kept \$4.63

****Round 4:****

- They sent \$3.97 (about 79% of their endowment)
- I received \$11.91
- I need to decide how much to return

Looking at the pattern:

- Round 1: They returned \$5.00 out of \$9.84 (about 51%)
- Round 3: They returned \$6.50 out of \$11.13 (about 58%)

They've been consistently fair and returning a good proportion. They're also continuing to send high amounts (almost 80% of their endowment in Round 4).

I should continue to reciprocate. A fair return would be around 50-60% of what I received:

- 50% = \$5.96
- 55% = \$6.55
- 60% = \$7.15

Given they've been very trusting and cooperative, I'll return around \$7 (about 59% of what I received). This rewards their high trust and maintains our cooperative relationship.

Actually, let me reconsider. Looking at what they kept:

- Round 1: They kept \$4.84 after returning \$5.00 from \$9.84
- Round 3: They kept \$4.63 after returning \$6.50 from \$11.13

They've been very consistent in what they keep (around \$4.60-\$4.84). ...

usage:

```
{
  "input_tokens": 1616,
  "output_tokens": 784,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.

Your total visible earnings across all rounds are \$25.18.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our continued partnership:

****Round 2:****

- They sent \$3.50 (70% - very trusting)

****Round 3:****

- I sent \$4 (they saw \$3.71)
- They returned \$6.50 (58% - fair)

****Round 4:****

- They sent \$4.00 (80% - even more trusting!)
- I returned \$7 (they saw \$6.16)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has demonstrated:

- Consistently high trust (sending 70-80%)
- Fair returns (~58% when receiving)
- Strong cooperative pattern over multiple rounds

I'll match their high trust level and send ****\$4**** (80% of my endowment):

- Maintains our successful cooperative relationship
- Becomes \$12 for them
- Based on their pattern, should yield good returns
- Shows continued mutual trust

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the visible history:

****Round 2:****

- They sent \$3.50 (70% of endowment)
- They saw \$6.94 returned

****Round 3:****

- They saw \$3.71 sent (I sent \$4)
- They returned \$6.50

****Round 4:****

- They sent \$4.00 (80% of endowment)
- They saw \$6.16 returned (I sent \$7)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has been:

- Very trusting: sending 70-80% of their endowment consistently
- Fair when receiving: returning around 58% of what they get
- Consistent and cooperative throughout

This is a strong, mutually beneficial relationship. They've escalated their trust from \$3.50 to \$4.00, and I should match or maintain that trust level.

I'll send ****\$4**** again, which:

- Maintains our high-trust equilibrium
- Becomes \$12 for them
- Should yield a good return based on their consistent ~58% return rate
- Shows I recognize and value their trustworthiness

Alternatively, I could send \$5 to maximize total welfare, but \$4 has been working well and maintains consistency.

I'll go with \$4.

usage:

```
{
  "input_tokens": 2044,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
 - Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.
 - Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.580000000000002.
- Your total visible earnings across all rounds are \$33.86.

This round, you are the RECEIVER against your current co-player. They sent you \$4.89, so you received \$14.669999999999998. How much do you return? (0-14.669999999999998)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)

****Round 4:****

- They sent \$4.00, trusting
- I returned well

****Round 5:****

- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.89 (98% of their endowment - extremely trusting!)
- I received \$14.67

****Analysis:****

This co-player has shown exceptional trust, sending nearly everything they had. They've been consistently fair, returning 57-58% when receiving.

I should reward this maximum trust with a strong return:

~58% of \$14.67 = \$8.51

```
'''json
{"return": 8.5}
'''
```

This maintains our cooperative relationship and rewards their exceptional level of trust.

Reasoning:

Looking at the history:

****Round 3:****

- I sent \$4 (they saw \$3.71)
- They returned \$6.50 from \$11.13 (58%)

****Round 4:****

- They sent \$4.00 (80%)
- I returned \$7 (they saw \$6.16, which is about 51% of \$12)

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.89 (almost 98% - extremely trusting!)
- I received \$14.67

****Analysis:****

This co-player has been incredibly trusting and cooperative:

- They consistently return around 57-58% when receiving
- They've been sending increasingly high amounts (80% ? 98%)
- They're showing maximum trust

I should reciprocate this high trust. Given they return about 57-58%, I should return a similar or slightly higher percentage to match their increased trust.

~60% of \$14.67 = \$8.80

Let me return around \$8.5 to \$9 to reward their exceptional trust while maintaining fairness.

I'll return \$8.5

```
Usage: {
  "input_tokens": 2528,
  "output_tokens": 506,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:00:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.37, that is 67.4% of their total endowment. You received \$10.11. How much do you return to the sender? (0-10.11)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.37 (67.4% of their endowment)
- They currently have \$1.63 left ($\$5 - \3.37)
- I received \$10.11 after the 3x multiplication

I want to reciprocate this high level of trust and ensure the sender benefits from their decision to trust me.

If I return \$5:

- The sender ends with: $\$1.63 + \$5 = \$6.63$ (a profit of \$1.63)
- I keep: $\$10.11 - \$5 = \$5.11$ (a profit of \$5.11)

This rewards the sender's trust while both of us profit from cooperation.

{'return': 5.0}

reasoning:

Let me analyze this situation:

1. The sender sent \$3.37 out of their \$5 endowment (67.4%)
2. This was multiplied by 3x, so I received \$10.11
3. I need to decide how much to return (0-10.11)

Key considerations:

- The sender showed significant trust by sending 67.4% of their endowment
- If I return nothing, I keep \$10.11 but break trust
- A fair return might be to ensure the sender gets back more than they sent
- The sender currently has: $\$5 - \$3.37 = \$1.63$

If I want to be reciprocal and fair:

- The sender gave up \$3.37 to trust me
- They showed high trust (67.4% of endowment)
- A reasonable return would ensure they profit from trusting me

Some options:

1. Return \$5: Sender ends with $\$1.63 + \$5 = \$6.63$ (profit of \$1.63), I keep \$5.11
2. Return \$6: Sender ends with $\$1.63 + \$6 = \$7.63$ (profit of \$2.63), I keep \$4.11
3. Return \$4: Sender ends with $\$1.63 + \$4 = \$5.63$ (profit of \$0.63), I keep \$6.11
4. Return \$3.37: Sender ends with $\$1.63 + \$3.37 = \$5$ (break even), I keep \$6.74

Given that:

- The sender showed high trust (67.4%)
- Note says there may be communication noise
- I want to encourage future cooperation
- A fair split might be approximating equal gains

If I return \$5:

- Sender: started with \$5, sent \$3.37, gets back \$5 = net $\$1.63 + \$5 = \$6.63$ total

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 694,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.

Your total visible earnings across all rounds are \$5.109999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing significant trust
- They saw \$5.51 returned, which gave them a positive payoff
- I profited from their trust in Round 1

I want to reciprocate their cooperative behavior and maintain the trust relationship. Sending a similar proportion will signal that I'm willing to cooperate.

{'send': 3.0}

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$3.37 (67.4% of their endowment)
- I received \$10.11
- I returned \$5.0
- I kept \$5.11

Now in Round 2, the visible history shows:

- My co-player's last game (Round 1 against me) as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51

Wait, there's communication noise! The actual amounts differ from what each player sees:

- I actually received \$10.11 (from their \$3.37 sent)
- I actually returned \$5.0
- But they see that they sent \$3.0, it became \$9.0, and saw \$5.51 returned

So there's noise in both directions. This is interesting - the sender sees they sent \$3.0 (vs actual \$3.37) and saw \$5.51 returned (vs actual \$5.0 I returned).

Now I'm the SENDER in Round 2 against the same co-player.

My visible earnings are \$5.11 (which matches what I calculated: \$10.11 - \$5.0 = \$5.11).

Strategy considerations:

1. My co-player sent a high amount in Round 1 (either \$3.0 or \$3.37 depending on what was real)
2. They saw a decent return (approximately 50% of what they sent, or more)
3. This suggests they're cooperative

I should signal continued cooperation by sending a similar proportion. Since they sent 67.4% in Round 1, I could send a similar high proportion.

Given the communication noise, I should send a clear cooperative signal. Sending around \$3-3.5 would show I'm reciprocating their trust.

Let me send \$3.0 to match what they think they sent (according to the visible history).

...

usage:

```
{
  "input_tokens": 848,
  "output_tokens": 639,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.

Your total visible earnings across all rounds are \$13.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.05, so you received \$6.1499999999999995. How much do you return? (0-6.1499999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.05 (41% of their endowment), showing moderate trust
- They currently have \$2.95 left
- I received \$6.15

Previous rounds show we've had a cooperative relationship, though the amounts varied (possibly due to noise). I want to maintain trust while ensuring both of us benefit.

If I return \$3.0:

- Sender ends with: \$2.95 + \$3.0 = \$5.95 (profit of \$0.95)
- I keep: \$6.15 - \$3.0 = \$3.15 (profit of \$3.15)

This ensures mutual benefit while reciprocating their trust.

```
{'return': 3.0}
```

reasoning:

Let me analyze the situation:

Looking at the visible history:

- Round 1: My co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51
- Round 2: My co-player as RECEIVER saw \$3.8 sent (I sent this), returned \$5.5, their payoff \$5.90

Now Round 3:

- I'm the RECEIVER
- My co-player sent \$2.05 (41% of their endowment)
- I received \$6.15

The co-player has been cooperative but they sent less this time (\$2.05 vs \$3.0 in Round 1). This could be:

1. A response to the noise in the game
2. Adjusting their strategy
3. Testing my response

Let me think about what's fair:

- They currently have: \$5 - \$2.05 = \$2.95
- I received: \$6.15

If I want to reciprocate fairly and ensure they profit:

- Return \$3.0: Sender gets \$2.95 + \$3.0 = \$5.95 (profit \$0.95), I keep \$3.15 (profit \$3.15)
- Return \$3.5: Sender gets \$2.95 + \$3.5 = \$6.45 (profit \$1.45), I keep \$2.65 (profit \$2.65)

Given they sent a moderate amount (41%), I should return a proportional amount. A return of around \$3.0-3.5 seems fair, ensuring both of us profit while maintaining the cooperati

ve relationship.

I'll return \$3.0 to keep things balanced.

usage:

```
{
  "input_tokens": 1160,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.

Your total visible earnings across all rounds are \$16.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- My co-player has been consistently cooperative, sending \$3.0 in both rounds they were the sender
- In Round 3, they saw only \$2.32 returned (likely due to communication noise), which might have disappointed them
- Their history shows they value cooperation despite the lower return they observed

I want to maintain our cooperative relationship and signal that I'm still willing to trust them. Sending a similar amount to what they've been sending shows reciprocal cooperation.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51

Round 2: Co-player as RECEIVER, saw \$3.8 sent, returned \$5.5, payoff \$5.899999999999999

Round 3: Co-player as SENDER sent \$3.0, saw \$2.32 returned, payoff \$4.32

Wait, in Round 3 I was the RECEIVER and I sent back \$3.0, but they saw \$2.32 returned. This is the communication noise mentioned!

So my co-player has been consistently sending \$3.0 when they're the sender, showing cooperative behavior. However, in Round 3 they saw only \$2.32 returned even though I actually sent \$3.0. This might make them think I'm less cooperative than I actually am.

Given that:

1. They've been consistently cooperative, sending \$3.0 twice
2. They may think I was less generous in Round 3 than I actually was (due to noise)
3. I want to maintain the cooperative relationship

I should send a similar amount to what they've been sending - around \$3.0 - to signal continued cooperation.

```
{'send': 3.0}
```

usage:

```
{
  "input_tokens": 1559,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.
- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$22.919999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.72, so you received \$8.16. How much do you return? (0-8.16)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate trust
- They currently have \$2.28 left
- I received \$8.16

The history shows we've maintained a cooperative relationship. They've consistently returned reasonable amounts when they were the receiver (around 50% of what they received in Round 4).

If I return \$4.0:

- Sender ends with: $\$2.28 + \$4.0 = \$6.28$ (profit of \$1.28)
- I keep: $\$8.16 - \$4.0 = \$4.16$ (profit of \$4.16)

This ensures both of us profit while maintaining our cooperative pattern.

{'return': 4.0}

reasoning:

Looking at the situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate-to-high trust
- They currently have \$2.28 left ($\$5 - \2.72)
- I received \$8.16

Looking at the history:

- Round 2: They returned \$5.5 when they received \$11.40
- Round 3: They saw \$2.32 returned (due to noise, I actually sent \$3.0)
- Round 4: They returned \$5.0 when they received \$9.84

They've been cooperative and returned reasonable amounts. In Round 4, they returned about 51% of what they received.

If I return \$4.0:

- Sender ends with: $\$2.28 + \$4.0 = \$6.28$ (profit of \$1.28)
- I keep: $\$8.16 - \$4.0 = \$4.16$ (profit of \$4.16)

This ensures both profit and maintains the cooperative relationship. The sender gets a reasonable return on their trust.

Let me consider returning slightly more since they've been consistently cooperative...

If I return \$4.5:

- Sender ends with: $\$2.28 + \$4.5 = \$6.78$ (profit of \$1.78)
- I keep: $\$8.16 - \$4.5 = \$3.66$ (profit of \$3.66)

I think \$4.0 is fair - it's about 49% of what I received, which is a reasonable return rate that ensures mutual benefit.

usage:

```
{
  "input_tokens": 1919,
  "output_tokens": 562,
```

```

    "reasoning_tokens": 0
  }
Message 12 role=user
content:
  Round 6

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 3 against you, as SENDER: sent $3.0, it became $9.0, saw $2.32 returned, visible p
    ayoff $4.32.
  - Round 4 against you, as RECEIVER: saw $3.28 sent, it became $9.84, returned $5.0, visibl
    e payoff $4.84.
  - Round 5 against you, as SENDER: sent $3.0, it became $9.0, saw $4.88 returned, visible p
    ayoff $6.88.
  Your total visible earnings across all rounds are $27.08.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
    u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Assistant output:
  Looking at the visible history:

  - My co-player has consistently sent $3.0 when they're the sender (Rounds 3 and 5)
  - They've been cooperative as a receiver, returning $5.0 in Round 4
  - In Round 5, they saw a reasonable return ($4.88) and had a positive payoff

  Our relationship has been mutually cooperative. I should continue matching their level of trus
    t and cooperation by sending a similar amount.

  {'send': 3.0}
Reasoning:
  Looking at the visible history:

  - My co-player has been consistently sending $3.0 when they're the sender (Rounds 3 and 5)
  - They've been cooperative as a receiver, returning good amounts (returned $5.0 in Round 4)
  - In Round 5, they saw $4.88 returned and had a payoff of $6.88, which is positive

  Our relationship has been quite cooperative overall. They've shown consistent trust by sending
    $3.0 repeatedly. I should reciprocate this trust and continue the cooperative pattern.

  Sending $3.0 matches what they've been doing and signals that I'm committed to cooperation.
Usage: {
  "input_tokens": 2316,
  "output_tokens": 272,
  "reasoning_tokens": 0
}

GAME - Agent_3 - Receiver - dyad_2
Timestamp: 2026-08-12T08:00:10
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",

```

```
"round": 6,  
"task": "game",  
"task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderately trusting approach to see how cooperation works in this game.

By sending \$3, the receiver will get \$9 (3×\$3). If they return a fair share, we can both benefit from the multiplier effect.

```
'''json  
{"send": 3}  
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return

3. My final earnings = \$5 - sent + returned
4. Receiver's final earnings = (sent × 3) - returned

From a purely rational/selfish perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return nothing, I get \$0
- The optimal outcome for both of us collectively is if I send \$5 (creating \$15 total) and we split it somehow

From a trust/reciprocity perspective:

- Sending money shows trust
- Many people reciprocate trust with trustworthy behavior
- In behavioral economics experiments, people often send substantial amounts and receivers often return fair amounts

Strategic considerations:

- This is round 1 of multiple rounds with randomized pairings
- Building a reputation for cooperation might help in future rounds if people can observe behavior
- However, agent names are hidden, so reputation effects may be limited
- There's also communication noise, which might affect trust dynamics

First round strategy:

- A moderate amount seems reasonable to test cooperation
- Sending \$2-3 would create \$6-9 for the receiver
- If they return even 1/3 of what they receive, I'd break even or profit slightly

I'll send \$3 as a moderately trusting first move. This creates \$9 for the receiver, and if they return \$3-4, we both do reasonably well compared to the \$5 baseline.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history, my co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. They've demonstrated trustworthy behavior.

Now they've sent me \$3.8, giving me \$11.40. I'll return a fair amount to encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze the situation:

1. In Round 1, I sent \$3 (though it appeared as \$3.37 to my co-player due to noise)
2. They received \$10.11 (3x the noisy amount) and returned \$5
3. I ended up with \$5 - \$3 + \$5 = \$7 (though my visible earnings show \$7.51, likely due to noise)

Now in Round 2:

- My co-player sent me \$3.8 (this might be noisy, but this is what I see)
- I receive \$11.40 (3 × \$3.8)

- I need to decide how much to return

The co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. That's roughly half of what they received, which is quite fair and cooperative.

Since they were cooperative and generous in their return, I should reciprocate fairly. A reasonable approach would be:

- They sent \$3.8 out of their \$5, keeping \$1.2
- I received \$11.40
- If I return about half (around \$5-6), they would end up with roughly \$6.2-7.2 total
- I would keep roughly \$5.40-6.40

Let me return \$5.5, which:

- Gives them: $\$5 - \$3.8 + \$5.5 = \6.7 total
- Gives me: $\$11.40 - \$5.5 = \$5.9$ total

This seems fair as it rewards their trust and cooperation, and both of us end up better than the initial \$5.

usage:

```
{
  "input_tokens": 726,
  "output_tokens": 482,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.

Your total visible earnings across all rounds are \$13.409999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3, which they saw as \$3.37 (noise)
- It became \$10.11 for them
- They returned \$5
- My payoff: $\$5 - \$3 + \$5 = \7
- Their visible payoff: $\$10.11 - \$5 = \$5.11$

Round 2: I was RECEIVER

- They sent \$3.8 (from their perspective they sent \$3.0)
- I received \$11.40
- I returned \$5.5
- They saw \$6.21 returned (noise)
- Their visible payoff: $\$5 - \$3 + \$6.21 = \8.21
- My payoff: $\$11.40 - \$5.5 = \$5.90$

My total visible earnings: \$13.41 (which is approximately $\$7 + \$5.90 = \$12.90$, close enough with rounding)

My co-player has been cooperative:

- Round 1: They returned a fair amount (\$5 out of \$10.11)
- Round 2: They sent \$3 (60% of their endowment), which is quite trusting

I should continue to cooperate and send a reasonable amount. They've been generous, so I s

ould reciprocate. Sending \$3 or more would be appropriate to maintain good cooperation.

usage:

```
{
  "input_tokens": 993,
  "output_tokens": 356,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.

Your total visible earnings across all rounds are \$17.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.28, so you received \$9.84. How much do you return? (0-9.84)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER, sent \$3 (they saw \$3.37 due to noise). They received \$10.11, returned \$5. Good cooperation.

Round 2: They were SENDER, sent \$3 (I saw it as \$3.8 due to noise), I received \$11.40, returned \$5.5 (they saw \$6.21 due to noise).

Round 3: I was SENDER, sent \$3 (they saw \$2.05 due to noise - significant noise here!). They received \$6.15, returned \$3. That's about half, which is reasonable given what they received.

Round 4: They are SENDER, sent something (I see \$3.28), I received \$9.84.

My co-player has been cooperative throughout. They've been returning reasonable amounts. The noise is creating some discrepancies, but overall the pattern is cooperative.

I received \$9.84. To maintain cooperation and reciprocity, I should return a fair amount. Looking at the pattern:

- Round 1: They returned \$5 out of \$10.11 (about 49%)
- Round 3: They returned \$3 out of \$6.15 (about 49%)

A fair return would be around 50% or slightly more to encourage continued cooperation. That would be around \$5.

Let me return \$5 to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 1253,
  "output_tokens": 355,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.

- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.
Your total visible earnings across all rounds are \$22.569999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: They were RECEIVER, saw \$3.37 sent, received \$10.11, returned \$5.0
Round 2: They were SENDER, sent \$3.0, became \$9.0, saw \$6.21 returned (I returned something that appeared as \$6.21 to them)
Round 3: They were RECEIVER, saw \$2.05 sent, received \$6.15, returned \$3.0
Round 4: They were SENDER, sent \$3.0, became \$9.0, saw \$4.45 returned (I returned something that appeared as \$4.45 to them)

So the pattern I'm seeing:

- When they're sender, they consistently send \$3.0
- When they're receiver, they return a fair amount (\$5 on \$10.11, \$3 on \$6.15)

In Round 4, I returned \$5 but they saw \$4.45 due to noise. This might make them think I'm less cooperative than I actually am.

Given that they continue to send \$3 consistently, I should continue to send a similar amount to maintain cooperation. I'll send \$3 again.

usage:

```
{
  "input_tokens": 1496,
  "output_tokens": 301,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.
- Round 5 against you, as RECEIVER: saw \$2.72 sent, it became \$8.16, returned \$4.0, visible payoff \$4.16.

Your total visible earnings across all rounds are \$29.449999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.04, so you received \$9.120000000000001. How much do you return? (0-9.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at the history with my current co-player:

- Round 1: They were receiver, saw \$3.37 sent, got \$10.11, returned \$5.0
- Round 2: They were sender, sent \$3.0, it became \$9.0, saw \$6.21 returned
- Round 3: They were receiver, saw \$2.05 sent, got \$6.15, returned \$3.0
- Round 4: They were sender, sent \$3.0, it became \$9.0, saw \$4.45 returned
- Round 5: They were receiver, saw \$2.72 sent, got \$8.16, returned \$4.0

Now in Round 6: They are sender, sent \$3.04, I received \$9.12

This co-player has shown consistent behavior:

- When they receive money, they return a reasonable amount (around \$3-5)
- When they send, they consistently send \$3.0

I should continue the cooperative pattern. They sent me \$3.04, giving me \$9.12. A fair return would be around \$4.5-5 to maintain the cooperative relationship and ensure both parties benefit.

Let me return \$4.5 to keep things balanced.

```
Usage: {  
  "input_tokens": 1760,  
  "output_tokens": 308,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:00:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.29, that is 45.8% of their total endowment. You received \$6.87. How much do you return to the sender? (0-6.87)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.29 (45.8% of their endowment)
- I received \$6.87 after the 3x multiplier
- The sender currently has \$2.71

I want to reciprocate their trust while keeping the exchange mutually beneficial. The sender took a risk by sending nearly half their endowment, so I should return enough to reward that trust.

If I return approximately the same proportion they sent (around 45-46% of what I received), that would be about \$3.15. This would give the sender $\$2.71 + \$3.15 = \$5.86$ total (a gain of \$0.86), and I'd keep \$3.72 (also a gain).

This seems like a fair reciprocation that rewards their trust and could encourage cooperation in future rounds.

{'return': 3.15}

reasoning:

I'm the receiver in this round. Let me understand the situation:

- The sender sent \$2.29 (which is 45.8% of their \$5 endowment)
- This was multiplied by 3, so I received \$6.87
- I need to decide how much to return (between \$0 and \$6.87)

Let me think about what would be fair and strategic:

1. The sender showed trust by sending almost half their endowment (\$2.29 out of \$5)
2. They kept $\$5 - \$2.29 = \$2.71$
3. I received \$6.87
4. If I return nothing, I keep \$6.87 and the sender ends with \$2.71
5. If I return everything, I keep \$0 and the sender ends with $\$2.71 + \$6.87 = \$9.58$

A common fair approach in trust games is to aim for roughly equal outcomes or to reciprocate the sender's trust proportionally.

Let's consider some options:

Option 1 - Equal split of total gains:

- Total money created: $\$5$ (sender's endowment) + $\$6.87$ (what I received) = $\$11.87$
- But wait, that's not right. The sender started with \$5, sent \$2.29 which became \$6.87
- Total in the system: Sender has \$2.71, I have \$6.87 = $\$9.58$ total
- Equal split would be $\$4.79$ each
- For sender to have \$4.79: they need $\$4.79 - \$2.71 = \$2.08$ returned

Option 2 - Return proportional to what was sent:

- Sender sent 45.8% of their endowment
- I could return a similar proportion of what I received
- 45.8% of \$6.87 = \$3.15

Option 3 - Ensure sender gains from trusting me:

- Sender currently has \$2.71
- If I return enough so they end up better than their initial \$5
- They need at least \$5 - \$2.71 = \$2.29 to break even
- To give them a decent profit, maybe return \$3-3.50 so they end with \$5.71-6.21

Since this is a multi-round game with randomized pairings and communication noise, I should consider how my choice signals trustworthiness. Returning around \$3-3.15 seems reasonable-it reciprocates their generosity while still letting me benefit from the exchange, and it positions me well for future interactions with different partners....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1076,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.

Your total visible earnings across all rounds are \$3.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They received \$3.41 back, making a profit of \$0.41
- They demonstrated significant trust

As the SENDER now, I want to reciprocate their trust level. They sent 60% of their endowment, which was quite generous. I should match or slightly exceed this level of trust to maintain cooperation.

Sending \$3.0 would match their previous contribution and signal that I'm willing to cooperate at the same trust level they showed.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

From Round 1:

- I was the RECEIVER
- The sender sent \$2.29 (45.8% of their endowment)
- I received \$6.87
- I returned \$3.15

But now the visible history shows something different:

- It says my co-player (as SENDER in Round 1) sent \$3.0, which became \$9.0
- They saw \$3.41 returned
- Their visible payoff was \$5.41

This is the communication noise mentioned in the instructions! What actually happened vs what they saw:

- Actual: They sent \$2.29, I received \$6.87, I returned \$3.15
- What they saw: They sent \$3.0, became \$9.0, saw \$3.41 returned

So from their perspective, they sent \$3.0 and got back \$3.41. That's a return rate of about

t 38% of what I received (3.41/9.0).

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of endowment)
- Got back \$3.41 (38% of \$9.0)
- Made a profit of \$0.41

They showed significant trust by sending 60% (from their perspective). They got back a reasonable amount.

As the SENDER now, I should consider:

1. They showed high trust in Round 1 (from their perspective, 60%)
2. They got a modest return
3. I want to continue building trust

Given that they sent 60% from their perspective, I should send a similar or slightly higher amount to show reciprocity and build trust. Sending around \$3.0-\$3.5 would be reasonable.

Let me send \$3.0 to match what they (think they) sent to me.

Actually, let me think more carefully. My actual earnings were \$3.72 in Round 1. I received \$6.87 and returned \$3.15.

...

usage:

```
{
  "input_tokens": 867,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.4700000000000001.

Your total visible earnings across all rounds are \$10.43.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.1000000000000001. How much do you return? (0-8.1000000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Round 2 pattern:**** When my co-player was the RECEIVER, they returned \$4.5 out of \$8.97 received (approximately 50%)

****This round:****

- They sent \$2.7 (54% of endowment), showing continued trust
- I received \$8.10

I should reciprocate with a similar return ratio. Returning about 50% like they did would be \$4.05, which:

- Gives them \$2.3 + \$4.05 = \$6.35 total (profit of \$1.35)
- Gives me \$4.05 (also a decent profit)

This maintains the cooperative pattern we've established.

{'return': 4.05}

reasoning:

Let me analyze the situation:

****Visible History:****

- Round 1: My co-player (as SENDER) sent \$3.0, received \$3.41 back, payoff \$5.41
- Round 2: My co-player (as RECEIVER) saw \$2.99 sent, returned \$4.5, payoff \$4.47

****This Round (Round 3):****

- I am the RECEIVER
- They sent \$2.7 (54% of their endowment)
- I received \$8.10

Looking at the pattern:

- In Round 1, they sent 60% of their endowment
- In Round 2, I sent \$3.0 (though they saw \$2.99 due to noise), and they returned \$4.5, which is about 50% of what they received (\$8.97)
- In Round 3, they sent \$2.7 (54%)

My co-player seems to be cooperating at moderate-to-high levels. When they were receiver in Round 2, they returned 50% of what they received.

If I return around 50% like they did, that would be about \$4.05.

Let me think about what's fair:

- They sent \$2.7, so they have \$2.3 left
- If I return \$4.05, they'd have $\$2.3 + \$4.05 = \$6.35$ (profit of \$1.35)
- I'd keep $\$8.10 - \$4.05 = \$4.05$ (profit of \$4.05)

That seems reasonable and matches their behavior. I could be slightly more generous to encourage continued cooperation.

Actually, let me reconsider. In Round 2, they returned \$4.5 out of \$8.97, which is exactly 50%.

...

usage:

```
{
  "input_tokens": 1210,
  "output_tokens": 644,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.

Your total visible earnings across all rounds are \$14.48.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 (60% of endowment) in Rounds 1 and 3
- As RECEIVER: Returned generously (\$4.5 out of \$8.97, ~50%)

****My previous sending:****

- Round 2: I sent and they saw \$2.99 (likely noise from my \$3.0)

They've demonstrated consistent trust and reciprocity. I should maintain this cooperative relationship by matching their sending level.

```
{'send': 3.0}
```

reasoning:

Looking at the history:

Round 1: Co-player as SENDER sent \$3.0, got \$3.41 back, payoff \$5.41
Round 2: Co-player as RECEIVER got \$2.99 sent (from me), returned \$4.5, payoff \$4.47
Round 3: Co-player as SENDER sent \$3.0, got \$3.91 back, payoff \$5.91

Pattern analysis:

- When my co-player is SENDER, they consistently send \$3.0 (60% of endowment)
- When they're RECEIVER, they return generously (\$4.5 out of \$8.97 ? 50%)
- They're showing consistent trust and cooperation

My sending history:

- Round 2: I sent \$3.0 (based on the visible history showing they saw \$2.99 sent due to noise)
- Round 3: Not explicitly shown as sender, but the history says they saw \$2.99 sent in Round 2

Actually, let me re-read:

Round 2: I was SENDER. I sent \$3.0. Due to noise, they saw \$2.99. They received \$8.97, returned \$4.5.

Round 3: I was RECEIVER. They sent \$2.7 (but I think they sent \$3.0 and noise made it \$2.7). I returned \$4.05.

Now Round 4: I'm SENDER again.

They've been consistently sending \$3.0 when they're sender. They've returned generously when receiver.

I should maintain the cooperative pattern by sending \$3.0 again.

usage:

```
{
  "input_tokens": 1604,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.
- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.550000000000007.

Your total visible earnings across all rounds are \$20.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.24, so you received \$6.720000000000001. How much do you return? (0-6.720000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the history:

****Co-player's return pattern:****

- Round 2: Returned \$4.5 out of \$8.97 (~50%)
- Round 4: Returned \$3.5 out of \$7.05 (~50%)

****This round:****

- They sent \$2.24 (44.8% of endowment)
- I received \$6.72

I should maintain the cooperative pattern by returning around 50% like they have consistently done.

Returning \$3.36:

- Gives them \$2.76 + \$3.36 = \$6.12 total (profit of \$1.12)
- Gives me \$3.36 (also a profit)

```
{'return': 3.36}
```

reasoning:

Looking at the visible history:

Round 2: They received \$8.97, returned \$4.5 (50%)

Round 4: They received \$7.05, returned \$3.5 (49.6%)

This round:

- They sent \$2.24 (44.8% of endowment)
- I received \$6.72

The co-player has been consistently returning around 50% of what they receive. I should reciprocate similarly.

Returning around 50% would be \$3.36:

- Sender would have: $\$2.76 + \$3.36 = \$6.12$ (profit of \$1.12)
- I would keep: $\$6.72 - \$3.36 = \$3.36$ (also a profit)

This maintains the cooperative pattern and roughly matches their return rates.

usage:

```
{
  "input_tokens": 1990,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.
- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.

Your total visible earnings across all rounds are \$23.509999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Consistently sent \$3.0 in Rounds 3 and 5 (60% of endowment)
- As RECEIVER: Returned \$3.5 out of \$7.05 (~50%) in Round 4

****What they saw in Round 5:****

- They sent \$3.0, but saw only \$2.79 returned (likely due to noise)
- Despite this, they continue to cooperate

I should maintain the cooperative relationship by sending \$3.0 again, matching their consistent trust level.

```
{'send': 3.0}
```

Reasoning:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 in Rounds 3 and 5
- As RECEIVER: Returned \$3.5 out of \$7.05 (~50%) in Round 4

****What they saw from me:****

- Round 3: Sent \$3.0, they returned \$3.91
- Round 4: They saw \$2.35 sent (likely noise from my \$3.0), returned \$3.5
- Round 5: Sent \$3.0, they saw only \$2.79 returned (but I actually sent \$3.36, so noise affected what they saw)

The co-player has been consistently sending \$3.0, showing steady trust. Even though in Round 5 they saw less returned than I actually sent (due to noise), they continue to cooperate.

I should maintain the cooperative pattern and send \$3.0 again to match their consistent trust level.

```
Usage: {
  "input_tokens": 2395,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:00:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance showing trust with managing risk, especially given the communication noise mentioned.

Since this is a multi-round game with rotating partners, establishing a cooperative pattern early could be beneficial. However, I don't know this receiver's tendencies yet.

I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Gives the receiver meaningful funds to work with (tripled)
- Doesn't risk my entire endowment

If I send \$3, the receiver gets \$9. If they return even half of that (\$4.50), we both benefit more than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 and can send \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. There's communication noise mentioned - amounts may differ slightly from what was actually sent/returned

Key considerations:

- If I send \$0, I keep \$5, receiver gets \$0, game ends with no cooperation
- If I send \$5, receiver gets \$15, and their return decision determines my payoff
- This is a multi-round game with 8 agents, randomized pairings
- My reputation/behavior might matter for future rounds, though pairings are randomized

Standard trust game reasoning:

- Sending money shows trust and can enable mutual benefit
- If receiver is rational and self-interested (and this is the last interaction), they should return \$0
- But in multi-round settings, reputation and reciprocity often emerge
- A moderate initial send can signal willingness to cooperate while limiting downside risk

Since this is round 1 and I don't know the receiver's tendencies, I should probably send a moderate amount to:

1. Signal willingness to cooperate
2. Give receiver opportunity to show trustworthiness
3. Not risk my entire endowment

A common strategy is to send about half. If I send \$2.50:

- Receiver gets \$7.50
- If they return half (\$3.75), I end with $\$5 - \$2.50 + \$3.75 = \6.25
- Receiver keeps \$3.75

- Both benefit

Actually, let me reconsider. Since amounts can only be integers based on the format, I should send a whole dollar amount. Let me send \$3:

- I send \$3
- Receiver gets \$9
- If they're fair and return half (\$4.50), I get $5 - 3 + 4.50 = \$6.50$

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 724,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.

Your total visible earnings across all rounds are \$5.41.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

Round 1: My co-player saw \$2.29 sent (due to noise), received \$6.87, and returned \$3.15 (about 46% of what they received). This shows they're willing to reciprocate.

Now they've sent me \$2.99, giving me \$8.97 total.

To maintain a cooperative relationship and signal trustworthiness, I'll return approximately half of what I received, which is fair and encourages future cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me understand what happened:

Round 1: I was the SENDER

- I sent \$3
- Due to communication noise, my co-player SAW \$2.29 sent (instead of \$3)
- It became \$6.87 (which is 2.29×3)
- They returned \$3.15
- Their visible payoff was \$3.72 (which is $6.87 - 3.15$)
- My visible earnings are \$5.41

Let me calculate my Round 1 payoff:

- I started with \$5
- I sent \$3
- I received back... well, they returned \$3.15
- So I should have: $5 - 3 + 3.15 = \$5.15$
- But my "visible earnings" are \$5.41

Hmm, this is confusing. Maybe the communication noise affected what I received too? Or maybe "visible earnings" includes some noise?

Actually, let me recalculate. If they saw \$2.29 sent and returned \$3.15:

- From their perspective: they received \$6.87, returned \$3.15, kept \$3.72
- From my perspective: I sent \$3, but what did I actually receive back?

The problem states "visible payoff" and "visible earnings" which suggests everything we see might have noise.

Now for Round 2:

- I am the RECEIVER
- They sent me \$2.99 (this is what I see after noise)
- I received \$8.97 (which is $\$2.99 \times 3$)
- I need to decide how much to return (0-8.97)

From Round 1, my co-player:

- Saw \$2.29 sent (noise-affected)
- Received \$6.87
- Returned \$3.15 (about 46% of what they received)

This shows they were somewhat cooperative, returning close to half.

They sent \$2.99, which is close to their original \$3 send, suggesting they're being fairly consistent. If I return roughly half of what I received-around \$4.48-\$4.50-I'd match their reciprocity level. I could return slightly more to encourage continued cooperation, or slightly less to protect my earnings. Let me think through the payoffs: if they sent \$2.99 and kept \$2.01, returning \$4.50 would give them \$6.51 total, which seems fair given their initial cooperation.

Actually, they returned 46% in the first round despite seeing a smaller amount due to noise. That's a strong signal of trust...

usage:

```
{
  "input_tokens": 798,
  "output_tokens": 987,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.

Your total visible earnings across all rounds are \$9.88.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: As receiver, they returned \$3.15 on \$6.87 (46% return rate)

Round 2: As sender, they sent \$3, and saw I returned \$4.71 (though actual was \$4.50 due to noise)

This co-player has demonstrated:

- Willingness to send a good amount (\$3 = 60% of endowment)
- Reasonable reciprocation as receiver
- We've established a cooperative pattern

I'll continue this pattern by sending the same amount they sent to me.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

Round 1:

- I was sender, they were receiver
- They saw \$2.29 sent (noise from my actual \$3), received \$6.87, returned \$3.15
- This is about 46% return rate

Round 2:

- They were sender, I was receiver

- They sent \$3.0, it became \$9.0
- They saw \$4.71 returned (I actually returned \$4.50, so there's noise)
- They kept \$6.71 total

So this co-player has shown:

1. As a receiver: They returned about 46% of what they received
2. As a sender: They sent \$3 (60% of their endowment), showing trust

I returned \$4.50 on \$8.97 (about 50% return rate), which they saw as \$4.71.

This seems like a cooperative partner. They're willing to send a good amount and return a reasonable portion. Since they sent \$3 last round and I returned generously, I should continue the cooperative pattern.

I'll send \$3 again to maintain consistency and reciprocal trust.

usage:

```
{
  "input_tokens": 1100,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.10000000000001, returned \$4.05, visible payoff \$4.05000000000002.

Your total visible earnings across all rounds are \$15.79000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.05000000000001. How much do you return? (0-7.05000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$3.15 on \$6.87 (46%)
 Round 2: They sent \$3 and saw good returns
 Round 3: They returned \$4.05 on \$8.10 (50%)
 Round 4 (now): They sent \$2.35, I received \$7.05

They've been consistently fair, returning around 46-50% when receiving. They sent slightly less this round (\$2.35 vs \$3 in Round 2), but that's still a reasonable amount showing continued trust.

I'll maintain our cooperative pattern by returning approximately 50% of what I received.

```
'''json
{"return": 3.50}
'''
```

reasoning:

Looking at the history:

Round 1: Co-player as receiver - saw \$2.29 sent, got \$6.87, returned \$3.15 (46% of received)
 Round 2: Co-player as sender - sent \$3, became \$9, saw \$4.71 returned
 Round 3: Co-player as receiver - saw \$2.7 sent, got \$8.1, returned \$4.05 (50% of received)
 Round 4 (now): Co-player as sender - sent \$2.35, I received \$7.05

The co-player has been consistent in returning around 46-50% when receiving. They've also been willing to send decent amounts when they're the sender.

However, I notice they sent \$2.35 this round, which is less than the \$3 they sent in round 2. This could be:

1. Due to communication noise affecting their perception of my returns
2. A strategic reduction
3. Random variation

Looking at their visible payoffs from my perspective, they seem to be doing well in our exchanges. I should continue cooperating to maintain this mutually beneficial relationship.

I've been returning around 50% consistently. Given they sent \$2.35 (about 47% of their endowment), which is slightly less than before, I'll return a fair amount but perhaps slightly less than exactly half to account for the reduced send.

I'll return around \$3.50, which is about 50% of what I received.

usage:

```
{
  "input_tokens": 1494,
  "output_tokens": 509,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$4.05, visible payoff \$4.050000000000002.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.67 returned, visible payoff \$5.67.

Your total visible earnings across all rounds are \$19.340000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our ongoing cooperation:

Round 2: They sent \$3, saw good returns (\$4.71)

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4: They sent \$3 again, saw \$3.67 returned

This co-player has been consistently cooperative:

- Sending \$3 when they're the sender
- Returning around 50% when they're the receiver

We've established a strong pattern of mutual trust. I'll continue by sending the same amount they've been sending.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

Round 2: They sent \$3 (as sender), saw \$4.71 returned

Round 3: They saw \$2.7 sent, returned \$4.05 on \$8.10 (50% return rate)

Round 4: They sent \$3 (as sender), saw \$3.67 returned

So they've been sending \$3 consistently when they're the sender. They've been returning around 50% when they're the receiver.

This is a cooperative partner. I should continue the pattern. I've been sending \$3 in previous rounds (rounds 1 and 3), and they've been matching that behavior.

I'll send \$3 again to maintain our cooperative relationship.

usage:

```
{
```

```
"input_tokens": 1887,
"output_tokens": 314,
"reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$4.05, visible payoff \$4.050000000000002.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.67 returned, visible payoff \$5.67.
- Round 5 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.36, visible payoff \$3.3600000000000008.

Your total visible earnings across all rounds are \$24.130000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.44, so you received \$7.32. How much do you return? (0-7.32)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued pattern:

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4: They sent \$3

Round 5: They returned \$3.36 on \$6.72 (50%)

Round 6 (now): They sent \$2.44, I received \$7.32

This co-player has consistently returned about 50% when receiving, showing fair and predictable cooperation. I'll maintain the same standard.

```
'''json
{"return": 3.66}
'''
```

Reasoning:

Looking at the history:

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4: They sent \$3, saw \$3.67 returned

Round 5: They saw \$2.24 sent (noise), received \$6.72, returned \$3.36 (50%)

Round 6 (now): They sent \$2.44, I received \$7.32

They continue to show consistent cooperation, returning 50% when they're the receiver. Their sending amounts have been steady around \$2.35-\$3.

I should continue my pattern of returning approximately 50% to maintain our cooperative relationship.

50% of \$7.32 = \$3.66

Usage: {

"input_tokens": 2274,

"output_tokens": 292,

"reasoning_tokens": 0

}

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:00:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_4",

"move_index": 6,

"opponent_id": "Agent_7",

"pairing": {

"agents": [

"Agent_8",

"Agent_7"

],

```

    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.24, that is 44.8% of their total endowment. You received \$6.720000000000001. How much do you return to the sender? (0-6.720000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.24 (44.8% of their endowment)

- They currently have \$2.76 remaining
- I received \$6.72 after the 3x multiplication

To reciprocate their trust fairly, I should return an amount that:

1. Ensures they profit from trusting me (end up with more than their original \$5)
2. Rewards their cooperative gesture
3. Still leaves me with a reasonable gain

If I return \$3.00:

- Sender ends with: $\$2.76 + \$3.00 = \$5.76$ (gain of \$0.76)
- I keep: $\$6.72 - \$3.00 = \$3.72$ (gain of \$3.72)

This seems like a fair reciprocal response that rewards trust while maintaining balanced incentives for future rounds.

```
{"return": 3.00}
```

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.24 out of their \$5 endowment (44.8%)
3. This was multiplied by 3x, so I received \$6.72 (technically \$6.720000000000001 due to floating point)
4. I need to decide how much to return (between \$0 and \$6.72)

Key considerations:

- The sender showed moderate trust by sending 44.8% of their endowment
- If I return nothing, I keep all \$6.72 and the sender gets only \$5 - \$2.24 = \$2.76
- If I return a fair amount, I should consider what would make this equitable

Common approaches in trust games:

1. Equal split of total gains: Total in the system is $\$5 + \$6.72 = \$11.72$. Equal split would be \$5.86 each. The sender currently has \$2.76, so I'd return $\$5.86 - \$2.76 = \$3.10$
2. Return the same proportion they sent: They sent 44.8%, so I could return 44.8% of \$6.72 = \$3.01
3. Match their investment: Return \$2.24 so we both end up with gains from the multiplication
4. Return enough so they profit: If I return \$3, they end up with $\$2.76 + \$3 = \$5.76$ (a gain of \$0.76), and I keep \$3.72 (a gain of \$3.72)

The sender took a risk by sending \$2.24. A fair return should:

- At minimum, not make them worse off than keeping everything (\$5)
- Ideally, reward their trust
- Be reciprocal to their gesture

Let me calculate a few scenarios:

- If I return \$3.00: Sender gets $\$2.76 + \$3.00 = \$5.76$, I keep \$3.72
- If I return \$3.50: Sender gets $\$2.76 + \$3.50 = \$6.26$, I keep \$3.22
- If I return \$4.00: Sender gets $\$2.76 + \$4.00 = \$6.76$, I keep \$2.72

A return of \$3.00-\$3.50 seems balanced-it rewards their trust while still giving me a meaningful gain. I'll return \$3.00, which nets them \$5.76 (a 15.2% gain) and leaves me with \$3.72....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.

Your total visible earnings across all rounds are \$3.7200000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw only \$2.2 returned (though due to communication noise, I actually returned \$3.00)
- From their perspective, they may feel the cooperation wasn't fully reciprocated

As the sender now, I want to:

1. Signal continued willingness to cooperate
2. Match their previous level of trust
3. Maintain the cooperative dynamic despite any noise in the previous round

I'll send the same amount they sent in Round 1 to demonstrate reciprocal trust.

{"send": 3.0}

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$2.24 (44.8% of their endowment)
- I received \$6.72
- I returned \$3.00
- Sender ended with $\$2.76 + \$3.00 = \$5.76$
- I kept \$3.72

But the visible history shows something different from my co-player's perspective:

- Round 1: They were SENDER, sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2

Wait, this is confusing. Let me re-read the prompt about communication noise.

"NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design."

So what happened:

- From my perspective: I received notification that sender sent \$2.24, I received \$6.72, I returned \$3.00
- From their perspective: They sent \$3.0, it became \$9.0, they saw \$2.2 returned, and got payoff \$4.2

Let me verify the math from their perspective:

- They sent \$3.0
- They had $\$5.0 - \$3.0 = \$2.0$ left
- They saw \$2.2 returned
- Their payoff: $\$2.0 + \$2.2 = \$4.2$?

So from their perspective, they sent \$3.0 and only got back \$2.2, which is less than they sent. They might perceive this as me being untrustworthy or uncooperative.

However, from MY perspective, I actually returned \$3.00, which I thought was fair and generous given I received \$6.72.

This is the communication noise at work. They think I returned less than I actually did.

Now I'm the sender in Round 2, and they likely view me as less trustworthy due to the noise distorting their perception of my return. They sent 60% of their endowment initially, showing genuine trust, but the miscommunication means they'll probably be more cautious.

I need to decide whether sending a substantial amount will help rebuild trust or just get exploited if they reciprocate poorly. A moderate send might signal cooperation without overcommitting, especially since they may interpret my actions through the lens of what they think happened last round....

usage:

```
{
  "input_tokens": 890,
  "output_tokens": 922,
```

```

    "reasoning_tokens": 0
  }
Message 6 role=user
content:
  Round 3

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $2.2 returned, visible payoff $4.2.
  - Round 2 against you, as RECEIVER: saw $3.7 sent, it became $11.100000000000001, returned $6.0, visible payoff $5.100000000000001.
  Your total visible earnings across all rounds are $10.790000000000001.

  This round, you are the RECEIVER against your current co-player. They sent you $3.3, so you received $9.899999999999999. How much do you return? (0-9.899999999999999)
  Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant
content:
  Looking at the visible history:

  - Round 1: Co-player sent $3.0 (60% of endowment), showing strong trust
  - Round 2: Co-player returned $6.0 out of $11.1 received (54% return rate), showing generosity despite seeing lower returns in Round 1
  - Round 3: Co-player sent $3.3 (66% of endowment), continuing to show high trust

  My co-player has consistently demonstrated cooperative behavior and trust. They've maintained cooperation even after seeing what appeared to be a lower return in Round 1.

  If I return $5.0:
  - Sender ends with: $1.70 + $5.00 = $6.70 (profit of $1.70)
  - I keep: $9.90 - $5.00 = $4.90

  This reciprocates their generous behavior and maintains the positive cooperative dynamic.

  {"return": 5.0}
reasoning:
  Let me analyze the visible history:

  Round 1:
  - Co-player as SENDER: sent $3.0, saw $2.2 returned
  - From their view: started with $5, sent $3, kept $2, saw $2.2 returned = $4.2 total
  - (I actually returned $3.00, but they saw $2.2 due to noise)

  Round 2:
  - Co-player as RECEIVER: saw $3.7 sent (I sent $3.0), received $11.1, returned $6.0
  - From their view: they received $11.1, returned $6.0, kept $5.1
  - From my view: I sent $3.0, kept $2.0, saw $6.0 returned = $8.0 total

  Round 3 (current):
  - I am RECEIVER
  - Co-player sent $3.3 (66% of endowment)
  - I received $9.9

  My co-player has been quite trustworthy:
  - Round 1: sent 60% of endowment
  - Round 2: returned a generous $6.0 out of $11.1 they received (54% return rate)
  - Round 3: sent 66% of endowment

  They've shown consistent cooperation and generosity. I should reciprocate appropriately.

  If I return $5.0:
  - Sender ends with: $1.7 (kept) + $5.0 (returned) = $6.7 (profit of $1.7)
  - I keep: $9.9 - $5.0 = $4.9

  This maintains a good balance and rewards their trust. Let me check if this is fair compared to their Round 2 behavior:
  - In Round 2, they returned $6.0 out of $11.1 (54%)
  ...
usage:

```

```
{
  "input_tokens": 1250,
  "output_tokens": 704,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.

Your total visible earnings across all rounds are \$15.69.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment)
- Round 2: Co-player returned \$6.0 out of \$11.1 (very generous, 54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment), saw \$5.32 returned

My co-player has consistently shown high trust and cooperation, sending substantial amounts and returning generously when in the receiver role. They've maintained this cooperative behavior throughout.

I'll match their highest level of trust by sending a similar amount.

```
{"send": 3.5}
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: Co-player sent \$3.0, saw \$2.2 returned (I actually returned \$3.0, but noise)

Round 2: I sent \$3.0 (co-player saw \$3.7), they returned \$6.0 - very generous!

Round 3: Co-player sent \$3.5 (actually \$3.3 based on my perspective), saw \$5.32 returned (I actually returned \$5.0)

My co-player has been consistently cooperative:

- Sending high amounts (60-70% of endowment)
- When they were receiver, they returned \$6.0 out of \$11.1 (54% return rate)

They've maintained cooperation throughout despite seeing some noise in the returns. In Round 3, they saw \$5.32 returned on their \$3.5 sent, which gave them a profit of \$1.82 - decent but not extremely generous.

As the sender now, I should:

1. Continue the cooperative pattern
2. Send an amount that shows trust matching theirs
3. Given they sent \$3.0, \$3.5 in previous rounds, I should send similarly

I'll send \$3.5 to match their highest level of trust and continue the positive dynamic.

usage:

```
{
  "input_tokens": 1694,
  "output_tokens": 461,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.
- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.

Your total visible earnings across all rounds are \$22.18.

This round, you are the RECEIVER against your current co-player. They sent you \$3.42, so you received \$10.26. How much do you return? (0-10.26)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (generous 54% return)
- Round 3: Co-player sent \$3.5 (70% of endowment, high trust)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return)
- Round 5: Co-player sent \$3.42 (68% of endowment, continued trust)

My co-player has consistently demonstrated strong cooperation, both sending high amounts and returning generously. They're maintaining this pattern across rounds.

If I return \$5.5:

- Sender ends with: $\$1.58 + \$5.5 = \$7.08$ (profit of \$2.08)
- I keep: $\$10.26 - \$5.5 = \$4.76$

This matches their typical return rate (~54%) and reciprocates their continued trust.

{"return": 5.5}

reasoning:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.42 (68% of endowment)

My co-player has been very cooperative and consistent:

- As sender: sending around 68-70% of their endowment
- As receiver: returning around 54-56% of what they receive

They're maintaining high trust and generosity throughout. I should reciprocate appropriately.

If I return \$5.5:

- Sender ends with: $\$1.58 + \$5.5 = \$7.08$ (profit of \$2.08)
- I keep: $\$10.26 - \$5.5 = \$4.76$

This is approximately 54% return rate, matching their pattern and continuing our cooperative relationship.

Actually, let me be slightly more generous to maintain the positive dynamic. Let me return \$5.5 or \$6.0.

Return of \$5.5 gives them \$7.08 total (profit of \$2.08)

Return of \$6.0 gives them \$7.58 total (profit of \$2.58)

I'll go with \$5.5 to match their typical return rate of around 54%.

usage:

```
{
  "input_tokens": 2082,
  "output_tokens": 593,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.
- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.

Your total visible earnings across all rounds are \$26.939999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- Round 3: Co-player sent \$3.5 (70% of endowment)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.5 (70% of endowment), saw \$6.23 returned, payoff \$7.73

My co-player has been remarkably consistent:

- As sender: consistently sends \$3.5 (70% of endowment), showing high trust
- As receiver: returns generously (~55% of received amount)

This is a very cooperative partner. I should match their consistent high level of trust.

```
{"send": 3.5}
```

Reasoning:

Looking at the visible history:

- Round 3: Co-player sent \$3.5, saw \$5.32 returned, payoff \$6.82
- Round 4: Co-player returned \$5.0 out of \$9.0, payoff \$4.0
- Round 5: Co-player sent \$3.5, saw \$6.23 returned, payoff \$7.73

My co-player has been very consistent:

- When sending: consistently sends \$3.5 (70% of endowment)
- When receiving: returns generously (55-56% of what they receive)

They've maintained this high cooperation throughout all rounds. The pattern is clear and stable.

I should continue matching their level of trust and cooperation. They've been sending \$3.5 consistently as sender, and I've been doing the same.

```
{"send": 3.5}
```

Usage: {

```
  "input_tokens": 2535,  
  "output_tokens": 382,  
  "reasoning_tokens": 0
```

}

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:00:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    }  
  },
```

```

    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I want to establish a cooperative tone while being somewhat cautious since I don't know my co-player yet.

Sending a moderate amount can:

- Signal trust and willingness to cooperate
- Create a larger pool for mutual benefit (tripled to \$9)
- Allow the receiver meaningful choice in how much to return

If the receiver returns even half of what they receive, we both end up better than the initial \$5.

I'll send **\$3**.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- There's communication noise, so amounts may differ
- This is a multi-round game with 8 agents
- Pairings are randomized each round

Classic trust game setup. Let me think about optimal strategy:

1. If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
2. If I send \$5: Receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 (better than \$5). But they might return less or nothing.
3. If I send a moderate amount (e.g., \$3): Receiver gets \$9. If they return \$4-5, I come out ahead.

Since this is:

- Round 1 (establishing reputation/reciprocity patterns)
- Multi-round game (future interactions matter)
- Anonymous agents (can't track specific individuals)

I should consider:

- Building trust/reciprocity norms early
- Signaling cooperation
- But also being somewhat cautious since I don't know the other agents

A common strategy is to send a moderate-to-high amount in early rounds to:

1. Signal trustworthiness
2. Encourage reciprocity
3. Maximize potential gains if receiver cooperates

Given the noise factor and uncertainty, I'll send \$3. This shows goodwill without being maximally vulnerable, and gives the receiver \$9 to work with. If they return \$4-5, we both benefit.

Actually, let me reconsider. In trust games, sending higher amounts often generates better returns because:

- It demonstrates trust
- Creates larger pie to split
- Social norms typically lead to proportional returns

Comparing the math: sending \$4 leaves me with \$1 while giving the receiver \$12. A 50% return (\$6) nets me \$7 total, while a 33% return (\$4) just breaks me even. With \$3 sent instead, I'd have \$2 cushion and need only a 50% return (\$4.50) to reach \$6.50. The higher send amount creates more upside if reciprocity holds, so I'll go with \$4....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 814,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.720000000000006.
Your total visible earnings across all rounds are \$4.2.

This round, you are the RECEIVER against your current co-player. They sent you \$3.7, so you received \$11.100000000000001. How much do you return? (0-11.100000000000001)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 analysis:****

- My co-player saw \$2.24 sent (due to noise), which became \$6.72
- They returned \$3.0 (about 45% of what they received)
- This shows reasonable cooperative behavior

****Round 2 (current):****

- They sent me \$3.7 (even more than what they saw in Round 1)
- I received \$11.10
- They're showing increased trust

To maintain and encourage cooperation, I should return a fair amount. Returning about half of what I received would give them:

- They kept: \$5 - \$3.7 = \$1.30
- Plus returned amount
- A good return on their trust

I'll return ****\$6.00**** (about 54% of what I received), giving them a total of \$7.30 - a solid return that rewards their increased trust.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation carefully.

Round 1 recap from my perspective:

- I sent \$3
- Due to communication noise, my co-player saw \$2.24 sent (which became \$6.72 after tripling)
- They returned \$3.0
- Their visible payoff was \$3.72 (received \$6.72, returned \$3.0)
- My total visible earnings are \$4.2

Let me calculate what actually happened in Round 1:

- I sent \$3, so I had \$5 - \$3 = \$2 left
- The receiver got $3 \times 3 = 9$ (but saw \$6.72 due to noise)
- They returned \$3.0
- So I ended with $2 + 3 = 5$
- But my visible earnings show \$4.2, which suggests the noise affected what I saw returned too?

Actually, wait. Let me reconsider:

- If my visible earnings are \$4.2 total
- I started with \$5, sent \$3 (keeping \$2)
- If I received back something that made my total \$4.2, then I received $4.2 - 2 = 2.2$ back

So the noise worked both ways:

- I sent \$3, they saw \$2.24
- They returned \$3.0, but I saw (and actually got) something that gave me \$4.2 total

Now for Round 2:

- I'm the RECEIVER
- My co-player sent me \$3.7 (according to what I see)
- This became \$11.10 after tripling
- I need to decide how much to return

My co-player showed willingness to return a reasonable amount in Round 1 (they returned \$3.0 out of \$6.72 they saw, which is about 45%). This is fairly cooperative behavior.

I should return a fair amount to maintain cooperation. Given that:

- They sent \$3.7 and I received \$11.10
- A 50% return would be \$5.55, giving them \$1.3 (kept) + \$5.55 (returned) = \$6.85 total
- This matches their cooperative gesture from Round 1 and encourages continued trust

I'll return \$5 to keep things balanced....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 1051,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.

Your total visible earnings across all rounds are \$9.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3 ? they returned \$3 (cooperative)

****Round 2:**** They sent \$3 ? I returned \$6, but they saw \$5.07 (still a good return)

****Round 3 (current):****

- Both rounds showed mutual cooperation
- They sent \$3 as sender and returned reasonably as receiver
- Their visible payoff was \$7.07 last round, showing cooperation pays off

I'll send ****\$3.5**** to maintain and slightly increase our cooperative dynamic. This shows I'm willing to trust them a bit more based on our positive history.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.24 (noise), became \$6.72, returned \$3.0
- They kept \$3.72, I got back \$3.0, so I ended with \$5 - \$3 + \$3 = \$5
- But my visible earnings show \$4.2, which suggests I saw something different due to noise

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, became \$9.0
- I returned \$6.0, but they saw \$5.07 (noise)
- They ended with \$5 - \$3 + \$5.07 = \$7.07
- I kept \$9.0 - \$6.0 = \$3.0

My total visible earnings are \$9.3.

Now Round 3:

- I'm SENDER again
- Co-player has shown:
 - As receiver: returned \$3.0 out of \$6.72 (45%)
 - As sender: sent \$3.0 out of \$5 (60%)

- They saw I returned \$5.07 (though I actually returned \$6.0)
- This is good cooperation

Since they've been cooperative and sent a good amount when they were sender, I should continue cooperating. They sent \$3.0 last round, so I should send at least that much or more to maintain trust.

I'll send \$3.5 to show continued cooperation and slightly increase the stakes.

usage:

```
{
  "input_tokens": 1191,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$16.12.

This round, you are the RECEIVER against your current co-player. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our interaction pattern:

```
**Round 1:** They returned $3.0 from $6.72 (45%)
**Round 2:** They sent $3.0, I returned $6.0
**Round 3:** They returned $5.0 from $9.90 (51%)
**Round 4 (current):** They sent $3.0 ? I received $9.0
```

They've shown consistent cooperation, with their return percentages increasing slightly over time. They sent \$3.0 this round, maintaining trust despite some communication noise in previous rounds.

I'll return **\$5.0** (about 56% of what I received). This gives them a total payoff of \$7.0 (\$2 kept + \$5 returned), which rewards their continued trust and maintains our cooperative relationship.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze the history:

```
**Round 1:** I sent $3 ? they saw $2.24 ? they returned $3.0
**Round 2:** They sent $3.7 ? I received $11.10 ? I returned $6.0 ? they saw $5.07
**Round 3:** I sent $3.5 ? they saw $3.3 ? they returned $5.0
**Round 4 (current):** They sent $3.0 ? I received $9.0
```

Looking at their behavior:

- Round 1 as receiver: Returned \$3.0 out of \$6.72 (45%)
- Round 3 as receiver: Returned \$5.0 out of \$9.90 (51%)

Their returns seem reasonable and slightly increasing in percentage.

My behavior:

- Round 2 as receiver: Returned \$6.0 out of \$11.10 (54%)

They've consistently sent around \$3-\$3.7 as sender, maintaining cooperation.

For Round 4, they sent \$3.0 and I received \$9.0. To maintain fair cooperation and match roughly what they've done (returning about 50%), I should return around \$4.5-\$5.0.

Let me return \$5.0 to be consistent with maintaining good cooperation and matching their recent return percentage.

Actually, wait. Let me recalculate their payoffs to understand better:

Round 1:

- They as receiver: Kept \$5 - \$0 (not sender) + received \$6.72 - \$3.0 returned = \$5 + \$3.72 = \$8.72? No wait, they were receiver, so sender sent them something.
- Actually the history shows "as RECEIVER: saw \$2.24 sent" - so I was the sender, I sent something, they saw \$2.24, it became \$6.72, they returned \$3.0
- Their payoff: \$6.72 - \$3.0 = \$3.72. Round 2 shows they sent \$3.0 which tripled to \$9.0, and when I returned \$5.07, their payoff was \$5 - \$3.0 + \$5.07 = \$7.07. In Round 3, I sent \$3.5 which became \$9.90, and they returned \$5.0, giving them \$9.90 - \$5.0 = \$4.90. Now they've sent \$3.0 which becomes \$9.0 for me, so I need to decide what to return....

usage:

```
{
  "input_tokens": 1596,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.

Your total visible earnings across all rounds are \$20.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our pattern:

****Round 2:**** They sent \$3.0, saw ~\$5.07 returned (good payoff)

****Round 3:**** They returned \$5.0 from what they saw as \$9.90

****Round 4:**** They sent \$3.5, saw ~\$4.99 returned

They've consistently sent around \$3.0-\$3.5 and returned fairly when in the receiver role.

Despite some communication noise affecting visible payoffs, we've maintained good cooperation.

I'll send ****\$3.5**** to continue our stable, mutually beneficial pattern.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze our history:

****Round 2:**** They sent \$3.0, I returned \$6.0 (they saw \$5.07 due to noise)

****Round 3:**** I sent \$3.5, they returned \$5.0 (they saw \$3.3 sent due to noise)

****Round 4:**** They sent \$3.5 (wait, the history says they sent \$3.5 in Round 4?), I returned \$5.0 (they saw \$4.99 due to noise)

Wait, let me reread. In Round 4, I was the receiver and they sent me \$3.0. But the visible history says they sent \$3.5 in Round 4. That must be the noise - what they actually sent was \$3.5, but I saw \$3.0.

Let me recalculate:

- Round 4: They actually sent \$3.5 ? became \$10.5 ? I returned \$5.0 (they saw \$4.99)
- Their payoff: \$5 - \$3.5 + \$4.99 = \$6.49

****Round 5 (current):**** I'm the sender

Looking at the pattern:

- Round 2: They sent \$3.0, got back ~\$5.07 (good return)
- Round 3: I sent (they saw \$3.3), they returned \$5.0 (reasonable)
- Round 4: They sent \$3.5, got back ~\$4.99 (decent return, about 48% of the tripled amount)

They've been consistently cooperative, sending \$3.0-\$3.5 each time. Their returns have been fair (around 50% of what they receive).

I should maintain the cooperative relationship. I'll send ****\$3.5**** to continue our pattern of mutual trust and cooperation.

usage:

```
{
  "input_tokens": 2020,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.
- Round 5 against you, as RECEIVER: saw \$3.42 sent, it became \$10.26, returned \$5.5, visible payoff \$4.76.

Your total visible earnings across all rounds are \$27.85.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our established pattern:

****Round 3:**** They returned \$5.0 from \$9.90
****Round 4:**** They sent \$3.5, I returned \$5.0
****Round 5:**** They returned \$5.5 from \$10.26
****Round 6 (current):**** They sent \$3.29 ? I received \$9.87

They've been consistently cooperative, both when sending and returning. They sent \$3.29, keeping \$1.71 for themselves.

I'll return ****\$5.0**** (about 51% of what I received), giving them a total payoff of \$6.71. This maintains our established cooperative pattern.

```
'''json
{"return": 5.0}
'''
```

Reasoning:

Looking at the pattern:

****Round 3:**** They returned \$5.0 from \$9.90 (51%)
****Round 4:**** They sent \$3.5, saw ~\$4.99 returned
****Round 5:**** They returned \$5.5 from \$10.26 (54%)
****Round 6 (current):**** They sent \$3.29 ? I received \$9.87

They've been very consistent:

- As sender: sending around \$3.0-\$3.5
- As receiver: returning around 51-54% of what they receive

They sent \$3.29 this round, so they kept \$1.71.

If I return about half to slightly more (consistent with our pattern), that would be around \$5.0-\$5.5.

Returning \$5.0 would give them $\$1.71 + \$5.0 = \$6.71$ total

Returning \$5.5 would give them $\$1.71 + \$5.5 = \$7.21$ total

Given they're showing continued trust and cooperation, I should return a fair amount. I'll return \$5.0-\$5.5. Let me go with \$5.0 to be consistent with our established pattern (around 50% return rate).

Actually, looking at my previous returns in this partnership:

- Round 2: I returned \$6.0 from \$11.10 (54%)

- Round 4: I returned \$5.0 from \$9.0 (56%)

So I've been returning slightly more than half. Let me continue that pattern.

$\$9.87 * 0.50 = \4.94

$\$9.87 * 0.55 = \5.43

I'll return \$5.0, which is about 51% and maintains our cooperative pattern.

```
Usage: {
  "input_tokens": 2415,
  "output_tokens": 599,
  "reasoning_tokens": 0
}
```

ROUND 7

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

dyad_2: Agent_3=Sender, Agent_4=Receiver

dyad_3: Agent_5=Sender, Agent_6=Receiver

dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$4; received \$12; Agent_2 returned \$8.5; payoffs Agent_1=\$9.5, Agent_2=\$3.5;

noise/visible: sent visible=\$4.73, received visible=\$14.19, returned visible=\$7.65

Action validation: {"returned": {"amount": 8.5, "clamped": false, "max": 12.0, "min": 0, "raw": 8.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_2: Agent_3 sent \$3; received \$9; Agent_4 returned \$4.5; payoffs Agent_3=\$6.5, Agent_4=\$4.5;

noise/visible: sent visible=\$2.95, received visible=\$8.85, returned visible=\$4.69

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$4.86; payoffs Agent_5=\$6.86,

Agent_6=\$4.14; noise/visible: sent visible=\$3.24, received visible=\$9.72, returned visible=\$5.47

Action validation: {"returned": {"amount": 4.86, "clamped": false, "max": 9.0, "min": 0, "raw": 4.86}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_7 sent \$3.5; received \$10.5; Agent_8 returned \$6.5; payoffs Agent_7=\$8, Agent_8=\$4;

noise/visible: sent visible=\$3.96, received visible=\$11.88, returned visible=\$5.71

Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 10.5, "min": 0, "raw": 6.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent_1": 45.5,

"Agent_2": 42.5,

"Agent_3": 36.5,

"Agent_4": 40.5,

"Agent_5": 38.76,

"Agent_6": 38.24,

"Agent_7": 40.5,

"Agent_8": 41.5

}

Visible balances after round: {

"Agent_1": 48.68,

"Agent_2": 45.11,

"Agent_3": 40.759999999999999,

"Agent_4": 38.11,

"Agent_5": 35.260000000000005,

"Agent_6": 34.94,

```
"Agent_7": 39.93,  
"Agent_8": 38.959999999999994  
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

```
Timestamp: 2026-08-12T08:00:55  
Model: anthropic/claude-sonnet-4.5  
Temperature: 0.8  
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 0,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths
, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't know my co-player yet.

****My reasoning:****

- Sending \$0 is safe but creates no mutual benefit
- Sending everything (\$5) maximizes potential gains but is very risky with an unknown partner
- A moderate amount signals cooperation while managing risk

I'll send ****\$3****, which:

- Gets multiplied to \$9 for the receiver (creating good potential for returns)
- Keeps \$2 as a safety buffer for me
- Shows willingness to trust and cooperate
- If they return even \$3-4, we both benefit

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me analyze the situation :

1. I have \$5 to start
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ from what was actually sent/returned
5. This is a multi-agent setting with 8 agents, randomized pairings each round
6. I don't know who my co-player is (names are hidden)

Key considerations:

- If I send \$0, I guarantee keeping \$5 but miss the opportunity for mutual gain
- If I send the full \$5, the receiver gets \$15 and could return a substantial amount
- The optimal outcome for both of us combined is if I send everything (creating \$15 total instead of \$5)
- However, the receiver could keep everything and I'd end up with \$0
- There's also communication noise which adds uncertainty

Since this is the first round and I have no information about my co-player:

- Sending some amount signals trust and cooperation
- A moderate amount (like \$2-3) balances risk and potential reward
- Sending \$5 shows maximum trust but is risky with an unknown partner
- The communication noise means even if they want to reciprocate fairly, there might be misunderstandings

For a first round with an unknown partner, I'll send a moderate amount to signal cooperation while managing risk. I'll send \$3:

- This creates \$9 for the receiver (good potential for returns)
- I keep \$2 as insurance
- If they return just \$3-4, I'll break even or profit
- It signals willingness to cooperate without being reckless

Let me go with \$3 as a balanced first move.

usage:

```
{
  "input_tokens": 500,
```



```
"output_tokens": 565,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$7.94.

This round, you are the RECEIVER against your current co-player. They sent you \$3.89, so you received \$11.67. How much do you return? (0-11.67)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.28 (became \$9.84 for them)
- They returned \$5.00 (about 51% of what they received)
- This shows they're cooperative and fair

****Round 2 situation:****

- They sent \$3.89 (78% of their endowment - very generous!)
- I received \$11.67
- They've shown trust and cooperation

I should reciprocate their cooperative behavior. Since they returned about 51% last round and sent generously this round, I'll return a similar proportion:

~51% of \$11.67 = approximately \$6

This rewards their trust while maintaining fairness.

```
'''json  
{"return": 6}  
'''
```

reasoning:

Let me analyze this situation:

From Round 1:

- I sent \$3 (but they saw \$3.28 due to communication noise)
- It became \$9.84 for them
- They returned \$5.0
- Their visible payoff was \$4.84

Now in Round 2:

- They sent \$3.89 (I received \$11.67 after 3x multiplier)
- I need to decide how much to return

Let me think about their behavior:

- They saw I sent \$3.28, which became \$9.84
- They returned \$5.0 from that \$9.84
- That's about 51% of what they received

They were fairly generous, returning about half. This suggests they're cooperative.

Now they sent \$3.89 out of \$5, which is about 78% of their endowment. That's quite generous and trusting.

If I want to reciprocate their cooperative behavior:

- They returned about 51% last round
- But they also sent a lot this round (78% of endowment)

I should return a reasonable amount that:

1. Rewards their trust and generosity
2. Maintains cooperation for future rounds

3. Is fair given they returned \$5 from \$9.84 last round

Let me calculate some options:

- If I return 50% of \$11.67 = \$5.84
- If I return a bit more generously, around \$6-7

Given that:

- They were generous (sent \$3.89)
- They returned \$5 last round (about 51%)
- I want to encourage continued cooperation

I'll return around \$6. This gives them:

- They sent: \$3.89
- They get back: \$6
- Net for them: $\$5 - \$3.89 + \$6 = \7.11

...

usage:

```
{
  "input_tokens": 800,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.

Your total visible earnings across all rounds are \$13.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00 - very fair and cooperative

****Round 2:****

- They sent \$3.50 (70% of their endowment - very trusting!)
- I returned \$6 (they saw \$6.94)
- Strong mutual benefit

****Analysis:****

This co-player has been consistently generous and trustworthy. They've shown:

- Fair returns when receiving (51% returned)
- High trust when sending (70% of endowment)
- We've built a good cooperative relationship

I should reciprocate their trust level. I'll send ****\$4**** (80% of my endowment), which:

- Matches their demonstrated trust
- Becomes \$12 for them (good potential returns)
- Continues building our cooperative pattern

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.28 due to noise)
- It became \$9.84 for them

- They returned \$5.00
- This was generous - about 51% of what they received

****Round 2:****

- They sent \$3.5 (I saw \$3.89 due to noise)
- It became \$11.67 for me (their view: \$10.5)
- I returned \$6 (they saw \$6.94 due to noise)
- They earned \$8.44 total

This co-player has been:

1. Very cooperative and fair in Round 1 (returned 51%)
2. Very trusting in Round 2 (sent 70% of their endowment)
3. They saw a good return from me in Round 2 (\$6.94)

Given this strong pattern of mutual cooperation, I should continue to trust them. They've proven reliable.

I think I should send a high amount, similar to what they sent in Round 2. They sent \$3.5, so I'll send around that amount or slightly more.

Options:

- Send \$3.5 to \$4 to maintain the cooperative relationship
- Send \$5 to maximize mutual benefit if I fully trust them

Given the strong cooperation so far, I'll send \$4. This shows:

- I trust them based on their past behavior
- I'm willing to invest in our mutual benefit
- It's not quite all \$5, maintaining a tiny hedge

Actually, reviewing their behavior:

- Round 1: They returned generously (51%)
- Round 2: They sent generously (70% of endowment)

They've been consistently cooperative. I should match their level of trust. ...

usage:

```
{
  "input_tokens": 1156,
  "output_tokens": 645,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.

Your total visible earnings across all rounds are \$20.27.

This round, you are the RECEIVER against your current co-player. They sent you \$3.97, so you received \$11.91. How much do you return? (0-11.91)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history:

****Round 1:****

- They returned \$5.00 from \$9.84 (51%)

****Round 2:****

- They sent \$3.50 generously
- I returned well (they saw \$6.94)

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)

- Consistent fair behavior

****Round 4:****

- They sent \$3.97 (about 79% - very trusting!)
- I received \$11.91

****Analysis:****

They've been consistently fair, returning 50-58% of what they receive. They continue to send high amounts, showing sustained trust.

I'll return approximately 59% to match their cooperative pattern:

```
'''json
{"return": 7}
'''
```

This rewards their high trust while ensuring both of us benefit substantially from the exchange.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00
- They kept \$4.84

****Round 2:****

- They sent \$3.50
- I received \$10.50
- I returned \$6 (they saw \$6.94)
- They kept \$8.44

****Round 3:****

- I sent \$4 (they saw \$3.71 with noise)
- They received \$11.13
- They returned \$6.50
- They kept \$4.63

****Round 4:****

- They sent \$3.97 (about 79% of their endowment)
- I received \$11.91
- I need to decide how much to return

Looking at the pattern:

- Round 1: They returned \$5.00 out of \$9.84 (about 51%)
- Round 3: They returned \$6.50 out of \$11.13 (about 58%)

They've been consistently fair and returning a good proportion. They're also continuing to send high amounts (almost 80% of their endowment in Round 4).

I should continue to reciprocate. A fair return would be around 50-60% of what I received:

- 50% = \$5.96
- 55% = \$6.55
- 60% = \$7.15

Given they've been very trusting and cooperative, I'll return around \$7 (about 59% of what I received). This rewards their high trust and maintains our cooperative relationship.

Actually, let me reconsider. Looking at what they kept:

- Round 1: They kept \$4.84 after returning \$5.00 from \$9.84
- Round 3: They kept \$4.63 after returning \$6.50 from \$11.13

They've been very consistent in what they keep (around \$4.60-\$4.84). ...

usage:

```
{
  "input_tokens": 1616,
  "output_tokens": 784,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.

Your total visible earnings across all rounds are \$25.18.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant**content:**

Looking at our continued partnership:

****Round 2:****

- They sent \$3.50 (70% - very trusting)

****Round 3:****

- I sent \$4 (they saw \$3.71)
- They returned \$6.50 (58% - fair)

****Round 4:****

- They sent \$4.00 (80% - even more trusting!)
- I returned \$7 (they saw \$6.16)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has demonstrated:

- Consistently high trust (sending 70-80%)
- Fair returns (~58% when receiving)
- Strong cooperative pattern over multiple rounds

I'll match their high trust level and send ****\$4**** (80% of my endowment):

- Maintains our successful cooperative relationship
- Becomes \$12 for them
- Based on their pattern, should yield good returns
- Shows continued mutual trust

'''json

{"send": 4}

'''

reasoning:

Looking at the visible history:

****Round 2:****

- They sent \$3.50 (70% of endowment)
- They saw \$6.94 returned

****Round 3:****

- They saw \$3.71 sent (I sent \$4)
- They returned \$6.50

****Round 4:****

- They sent \$4.00 (80% of endowment)
- They saw \$6.16 returned (I sent \$7)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has been:

- Very trusting: sending 70-80% of their endowment consistently

- Fair when receiving: returning around 58% of what they get
- Consistent and cooperative throughout

This is a strong, mutually beneficial relationship. They've escalated their trust from \$3.50 to \$4.00, and I should match or maintain that trust level.

I'll send **\$4** again, which:

- Maintains our high-trust equilibrium
- Becomes \$12 for them
- Should yield a good return based on their consistent ~58% return rate
- Shows I recognize and value their trustworthiness

Alternatively, I could send \$5 to maximize total welfare, but \$4 has been working well and maintains consistency.

I'll go with \$4.

usage:

```
{
  "input_tokens": 2044,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.
- Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.580000000000002.

Your total visible earnings across all rounds are \$33.86.

This round, you are the RECEIVER against your current co-player. They sent you \$4.89, so you received \$14.669999999999998. How much do you return? (0-14.669999999999998)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the history:

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)

****Round 4:****

- They sent \$4.00, trusting
- I returned well

****Round 5:****

- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.89 (98% of their endowment - extremely trusting!)
- I received \$14.67

****Analysis:****

This co-player has shown exceptional trust, sending nearly everything they had. They've been consistently fair, returning 57-58% when receiving.

I should reward this maximum trust with a strong return:

~58% of \$14.67 = \$8.51

```
'''json
{"return": 8.5}
'''
```

This maintains our cooperative relationship and rewards their exceptional level of trust.

reasoning:

Looking at the history:

****Round 3:****

- I sent \$4 (they saw \$3.71)
- They returned \$6.50 from \$11.13 (58%)

****Round 4:****

- They sent \$4.00 (80%)
- I returned \$7 (they saw \$6.16, which is about 51% of \$12)

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.89 (almost 98% - extremely trusting!)
- I received \$14.67

****Analysis:****

This co-player has been incredibly trusting and cooperative:

- They consistently return around 57-58% when receiving
- They've been sending increasingly high amounts (80% ? 98%)
- They're showing maximum trust

I should reciprocate this high trust. Given they return about 57-58%, I should return a similar or slightly higher percentage to match their increased trust.

~60% of \$14.67 = \$8.80

Let me return around \$8.5 to \$9 to reward their exceptional trust while maintaining fairness.

I'll return \$8.5

usage:

```
{
  "input_tokens": 2528,
  "output_tokens": 506,
  "reasoning_tokens": 0
}
```

Message 14 role=user**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.
- Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.5800000000000002.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.77 returned, visible payoff \$8.77.

Your total visible earnings across all rounds are \$40.03.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our continued partnership:

****Round 4:****

- They sent \$4.00 (80% - very trusting)

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 (57% - fair)

****Round 6:****

- They sent \$4.00 again (consistent high trust)
- I returned well (they saw \$7.77)

****Round 7:****

I'm the sender.

****Analysis:****

This co-player has demonstrated:

- Unwavering high trust (consistently sending 80%+)
- Fair returns (~57% when receiving)
- Extremely stable cooperative behavior
- 6 rounds of successful mutual benefit

I'll continue matching their trust level and send ****\$4**** (80%):

- Maintains our proven successful pattern
- Becomes \$12 for them
- Shows consistent reciprocity
- We've built strong trust over many rounds

```
'''json
{"send": 4}
'''
```

Reasoning:

Looking at the history:

****Round 4:****

- They sent \$4.00 (80% - very trusting)
- I returned well

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 (57%)

****Round 6:****

- They sent \$4.00 again (80%)
- I returned \$8.50 (they saw \$7.77 - about 65%)

****Round 7:****

- I'm the sender

****Analysis:****

This co-player has:

- Consistently sent high amounts when they're sender (80%+)
- Returned fairly when receiver (57%)
- Shown sustained trust and cooperation over 6 rounds
- We have a very strong cooperative relationship

They've sent \$4 consistently when in sender role. I should continue matching their high trust level. I'll send \$4 again (80%).

This:

- Maintains our successful pattern
- Becomes \$12 for them
- Shows continued mutual trust
- Based on their behavior, should yield good returns

```
Usage: {
  "input_tokens": 2956,
  "output_tokens": 487,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T08:01:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
```



```

    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.28, that is 65.6% of their total endowment. You received \$9.84. How much do you return to the sender? (0-9.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 65.6% of their endowment (\$3.28)
- This created \$9.84 for me to allocate
- The sender currently has only \$1.72 left from their original \$5

I want to reward their trust while being fair to both of us. If I return \$5.00:

- Sender's total: $\$1.72 + \$5.00 = \$6.72$ (profit of \$1.72)
- My total: $\$9.84 - \$5.00 = \$4.84$ (profit of \$4.84)

This seems like a reasonable split that rewards the sender's trust while still benefiting from the transaction.

```
{"return": 5.00}
```

reasoning:

This is Round 1 of a multi-agent trust game where I'm playing as the RECEIVER.

Let me understand what happened:

- The sender had an endowment of \$5
- The sender sent me \$3.28 (which is 65.6% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$3.28 \times 3 = \9.84
- I now need to decide how much to return to the sender (between \$0 and \$9.84)

The sender's payoff will be:

- \$5 (initial) - \$3.28 (sent) + [amount I return] = $\$1.72 + [\text{amount I return}]$

My payoff will be:

- \$9.84 (received) - [amount I return]

Let me think about what would be a fair return:

- The sender was quite generous, sending 65.6% of their endowment
- This shows significant trust
- A fair return might be to ensure the sender gets back at least what they sent, or even more to reward their trust
- If I return \$3.28, the sender would have: $\$1.72 + \$3.28 = \$5$ (break even)
- If I return \$5, the sender would have: $\$1.72 + \$5 = \$6.72$ (profit of \$1.72)
- If I return \$6.56 (double what they sent), the sender would have: $\$1.72 + \$6.56 = \$8.28$

A commonly fair approach in trust games is to split the total gains somewhat equitably while rewarding the sender's trust.

Total value created: \$9.84 (what I received)

Original sender investment: \$3.28

Net gain: $\$9.84 - \$3.28 = \$6.56$

One approach: Split the net gain equally

- Sender gets back: $\$3.28 + \$3.28 = \$6.56$
- I would return: \$6.56
- Sender's final: \$8.28
- My final: \$3.28

Another approach: Split all the money equally

- Total pool: \$9.84
- Equal split: \$4.92 each
- I would return: \$4.92
- Sender's final: \$6.64
- My final: \$4.92...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 999,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.

Your total visible earnings across all rounds are \$4.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- My co-player showed trust in Round 1 by sending \$3.0 (60% of their endowment)
- They received a reasonable return of \$5.94
- This suggests they're willing to cooperate

I want to reciprocate their trust and continue building a cooperative relationship. I'll send a similar proportion to what they sent.

```
{"send": 3.50}
```

reasoning:

Let me analyze this situation carefully:

Round 1 Summary:

- I was the RECEIVER
- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- I returned \$5.00
- My earnings: \$4.84

Now looking at the visible history shown to me:

- My co-player's last game shows: as SENDER, sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94

Wait, this is interesting. There's communication noise mentioned in the instructions. Let me think about what actually happened vs what was observed:

What I experienced in Round 1:

- Sender sent \$3.28
- I received \$9.84
- I returned \$5.00

What my co-player (the sender) saw:

- They sent \$3.0 (but actually sent \$3.28?)
- It became \$9.0 (but actually became \$9.84?)
- They saw \$5.94 returned (but I actually returned \$5.00?)
- Their visible payoff was \$7.94

Let me check the math on their side:

- If they sent \$3.0 and saw \$5.94 returned
- Their payoff would be: $\$5 - \$3 + \$5.94 = \7.94 ?

So from their perspective:

- They sent \$3.0 (66.7% of their total)
- They got back \$5.94 (66% of the \$9.0 that was created)
- This seems like a decent return to them

Now I'm the SENDER in Round 2:

- My co-player showed trust by sending most of their endowment in Round 1
- They received what appeared to them as a good return (66% of the multiplied amount)
- This suggests they might be willing to reciprocate if I show trust
- Since they sent roughly two-thirds of their endowment, I should match that level of generosity to build on the cooperative dynamic we...

usage:

```
{
  "input_tokens": 839,
  "output_tokens": 749,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.
- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.

Your total visible earnings across all rounds are \$13.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.71, so you received \$11.129999999999999. How much do you return? (0-11.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history with my co-player:

- Round 1: They sent \$3.0 (60%)
- Round 2: They returned \$6.0 when I sent (showing cooperation)
- Round 3: They sent \$3.71 (74.2% - even more trust!)

They're showing increasing trust and cooperation. I should reciprocate generously.

Current situation:

- They sent \$3.71, leaving them with \$1.29
- I received \$11.13

If I return \$6.50:

- Sender's total: $\$1.29 + \$6.50 = \$7.79$
- My total: $\$11.13 - \$6.50 = \$4.63$

This rewards their trust while still providing a fair outcome for both of us.

{"return": 6.50}

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.28 (65.6%)
- I received \$9.84
- I returned \$5.00
- Sender's total: $\$1.72 + \$5.00 = \$6.72$
- My total: $\$9.84 - \$5.00 = \$4.84$

Round 2: I was SENDER

- I sent \$3.50
- It became \$10.50
- But the visible history shows my co-player saw \$3.89 sent, it became \$11.67
- They returned \$6.00
- Their visible payoff was \$5.67
- My actual payoff should be: $\$5 - \$3.50 + (\text{what they returned})$
- According to my total visible earnings of \$13.28, that's $\$4.84$ (Round 1) + $\$8.44$ (Round 2) = \$13.28
- So I received \$8.44 in Round 2
- This means: $\$5 - \$3.50 + \$8.44 = \9.94 total

Wait, let me recalculate. If my total visible earnings are \$13.28:

- Round 1: I kept \$4.84
- Round 2: $\$13.28 - \$4.84 = \$8.44$

So in Round 2 my payoff was \$8.44.

I sent \$3.50, so I had \$1.50 left, plus what they returned.

$\$1.50 + \text{returned} = \8.44

So they returned \$6.94 to me.

But the visible history says they returned \$6.00. This is the communication noise mentioned in the instructions.

Now in Round 3:

They're sending \$3.71, which is 74.2% of their endowment-a significant increase from the 60% they sent initially. I'll receive \$11.13, leaving them with \$1.29. Given their consistent pattern of cooperation and escalating trust, I should reciprocate meaningfully. Ret

urning \$6.00 would give them \$7.29 total while I keep \$5.13....

usage:

```
{
  "input_tokens": 1132,
  "output_tokens": 983,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.
- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.

Your total visible earnings across all rounds are \$17.91.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our history:

- My co-player has consistently shown high trust (sending 60-80% of their endowment)
- They've returned reasonable amounts when receiving
- We've established a strong cooperative pattern

In Round 3, they sent \$4.0 (80% of their endowment), showing very high trust. I should reciprocate with a similarly strong showing of trust to maintain our cooperative relationship.

```
{"send": 4.00}
```

reasoning:

Let me analyze the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned \$5.00, but they saw \$5.94 returned

Round 2: I sent \$3.50, they received it as \$3.89 (11.67 total), they returned \$6.0

Round 3: They sent \$3.71, but they show it as \$4.0 (became \$12.0), I returned \$6.50 but they saw \$5.66

There's definitely communication noise happening. But the pattern shows:

- They consistently send high amounts (60-80%)
- They return reasonably when receiving
- We've been building a cooperative relationship

In Round 3, they sent what I saw as \$3.71 (74%), but they think they sent \$4.0 (80%). This shows very high trust.

I should continue the cooperative pattern and send a high amount as well. I'll send around \$3.50-\$4.00 to match their level of trust.

Let me send \$4.00 to show strong trust and keep the cooperation going.

usage:

```
{
  "input_tokens": 1542,
  "output_tokens": 367,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible

le payoff \$5.67.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.
- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.
Your total visible earnings across all rounds are \$25.07.

This round, you are the RECEIVER against your current co-player. They sent you \$4.36, so you received \$13.080000000000002. How much do you return? (0-13.080000000000002)
Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued cooperation:

- My co-player sent \$4.36 (87.2% of their endowment) - showing very high trust!
- They've been consistently generous as receiver (returning \$6.0 and \$7.0 in previous rounds)
- This is clearly a cooperative partner

Current situation:

- They sent \$4.36, leaving them with only \$0.64
- I received \$13.08

I should return generously to reward their exceptional trust.

If I return \$7.50:

- Sender's total: \$0.64 + \$7.50 = \$8.14
- My total: \$13.08 - \$7.50 = \$5.58

{"return": 7.50}

reasoning:

Looking at the history with my co-player:

Round 2: I sent \$3.89, they returned \$6.0
Round 3: They sent \$4.0, I returned \$5.66 (visible to them)
Round 4: I sent \$3.97, they returned \$7.0

Now Round 5: They sent \$4.36 (87.2% of their endowment - very high trust!)

They are showing extremely high trust by sending \$4.36, leaving them with only \$0.64.

I received \$13.08

They've been returning well when they're the receiver:

- Round 2: returned \$6.0 from \$11.67
- Round 4: returned \$7.0 from \$11.91

This is a cooperative partner. I should return generously to reward their high trust.

If I return \$7.50:

- Sender's total: \$0.64 + \$7.50 = \$8.14 (profit of \$3.14)
- My total: \$13.08 - \$7.50 = \$5.58 (profit of \$5.58)

This seems fair and rewards their trust while still benefiting from the exchange.

usage:

```
{
  "input_tokens": 1882,
  "output_tokens": 462,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.
- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
Your total visible earnings across all rounds are \$30.650000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our strong cooperative history:

- My co-player has consistently sent high amounts (\$4.0 in recent rounds, 80% of endowment)
- They've returned generously as receiver (\$7.0 and \$7.68 in recent rounds)
- We've established excellent mutual trust

I should continue this cooperative pattern by sending a similarly high amount.

{"send": 4.00}

reasoning:

Looking at the history with my co-player:

- Round 3: They sent \$4.0 (80%), returned visible \$5.66
- Round 4: They returned \$7.0
- Round 5: They sent \$4.0 (80%), returned visible \$7.68

This is a highly cooperative partner who consistently:

1. Sends high amounts (80% of endowment)
2. Returns generously when receiving (around \$6-7.68)

We've built a strong mutual trust pattern. I should continue to send high amounts to maintain this cooperation.

I should send a high amount, similar to what they've been sending. Around \$4.0-\$4.5 would be appropriate.

Actually, I notice there's some noise mentioned in the instructions. The visible amounts might differ from actual sent/returned amounts. But I should focus on the pattern of cooperation that's clearly visible.

I'll send \$4.00 to match their typical sending pattern and maintain our strong cooperation

usage:

```
{
  "input_tokens": 2276,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 6 against you, as RECEIVER: saw \$4.89 sent, it became \$14.669999999999998, returned \$8.5, visible payoff \$6.169999999999998.

Your total visible earnings across all rounds are \$39.42.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our excellent cooperation:

- My co-player sent \$4.73 (94.6% of their endowment) - extremely high trust!
- They've been consistently generous as receiver (returning \$7.0 and \$8.5 recently)

- We have a very strong cooperative relationship

Current situation:

- They sent \$4.73, leaving them with only \$0.27
- I received \$14.19

I should match their generosity and maintain our high level of cooperation.

```
{"return": 8.50}
```

Reasoning:

Looking at the history with my co-player:

- They've been consistently sending high amounts (around \$4.0)
- They returned \$7.0 and \$8.5 in recent rounds as receiver, showing strong cooperation
- Round 7: They sent \$4.73 (94.6% of their endowment) - extremely high trust!

Current situation:

- They sent \$4.73, leaving them with only \$0.27
- I received \$14.19

This is a very trusting partner. I should return generously to maintain our excellent cooperation.

If I return \$8.50:

- Sender's total: $\$0.27 + \$8.50 = \$8.77$
- My total: $\$14.19 - \$8.50 = \$5.69$

This seems like a fair and generous return that rewards their high trust while still benefiting from the transaction.

Usage: {

```
"input_tokens": 2613,  
"output_tokens": 334,  
"reasoning_tokens": 0
```

}

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T08:01:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_2",  
"move_index": 2,  
"opponent_id": "Agent_4",  
"pairing": {  
  "agents": [  
    "Agent_3",  
    "Agent_4"  
  ],  
  "investor": "Agent_3",  
  "roles": {  
    "Agent_3": "investor",  
    "Agent_4": "trustee"  
  },  
  "trustee": "Agent_4"
```

```
},  
"role": "investor",  
"role_label": "Sender",  
"round": 7,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderately trusting approach to see how cooperation works in this game.

By sending \$3, the receiver will get \$9 (3×\$3). If they return a fair share, we can both benefit from the multiplier effect.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return
3. My final earnings = \$5 - sent + returned
4. Receiver's final earnings = (sent × 3) - returned

From a purely rational/selfish perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return nothing, I get \$0
- The optimal outcome for both of us collectively is if I send \$5 (creating \$15 total) and we split it somehow

From a trust/reciprocity perspective:

- Sending money shows trust
- Many people reciprocate trust with trustworthy behavior

- In behavioral economics experiments, people often send substantial amounts and receivers often return fair amounts

Strategic considerations:

- This is round 1 of multiple rounds with randomized pairings
- Building a reputation for cooperation might help in future rounds if people can observe behavior
- However, agent names are hidden, so reputation effects may be limited
- There's also communication noise, which might affect trust dynamics

First round strategy:

- A moderate amount seems reasonable to test cooperation
- Sending \$2-3 would create \$6-9 for the receiver
- If they return even 1/3 of what they receive, I'd break even or profit slightly

I'll send \$3 as a moderately trusting first move. This creates \$9 for the receiver, and if they return \$3-4, we both do reasonably well compared to the \$5 baseline.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history, my co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. They've demonstrated trustworthy behavior.

Now they've sent me \$3.8, giving me \$11.40. I'll return a fair amount to encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze the situation:

1. In Round 1, I sent \$3 (though it appeared as \$3.37 to my co-player due to noise)
2. They received \$10.11 (3x the noisy amount) and returned \$5
3. I ended up with \$5 - \$3 + \$5 = \$7 (though my visible earnings show \$7.51, likely due to noise)

Now in Round 2:

- My co-player sent me \$3.8 (this might be noisy, but this is what I see)
- I receive \$11.40 (3 × \$3.8)
- I need to decide how much to return

The co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. That's roughly half of what they received, which is quite fair and cooperative.

Since they were cooperative and generous in their return, I should reciprocate fairly. A reasonable approach would be:

- They sent \$3.8 out of their \$5, keeping \$1.2
- I received \$11.40
- If I return about half (around \$5-6), they would end up with roughly \$6.2-7.2 total
- I would keep roughly \$5.40-6.40

Let me return \$5.5, which:
- Gives them: $\$5 - \$3.8 + \$5.5 = \6.7 total
- Gives me: $\$11.40 - \$5.5 = \$5.9$ total

This seems fair as it rewards their trust and cooperation, and both of us end up better than the initial \$5.

usage:

```
{
  "input_tokens": 726,
  "output_tokens": 482,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.

Your total visible earnings across all rounds are \$13.409999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3, which they saw as \$3.37 (noise)
- It became \$10.11 for them
- They returned \$5
- My payoff: $\$5 - \$3 + \$5 = \7
- Their visible payoff: $\$10.11 - \$5 = \$5.11$

Round 2: I was RECEIVER

- They sent \$3.8 (from their perspective they sent \$3.0)
- I received \$11.40
- I returned \$5.5
- They saw \$6.21 returned (noise)
- Their visible payoff: $\$5 - \$3 + \$6.21 = \8.21
- My payoff: $\$11.40 - \$5.5 = \$5.90$

My total visible earnings: \$13.41 (which is approximately $\$7 + \$5.90 = \$12.90$, close enough with rounding)

My co-player has been cooperative:

- Round 1: They returned a fair amount (\$5 out of \$10.11)
- Round 2: They sent \$3 (60% of their endowment), which is quite trusting

I should continue to cooperate and send a reasonable amount. They've been generous, so I should reciprocate. Sending \$3 or more would be appropriate to maintain good cooperation.

usage:

```
{
  "input_tokens": 993,
  "output_tokens": 356,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.1499999999999995, returned \$3.0, visible payoff \$3.1499999999999995.

Your total visible earnings across all rounds are \$17.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.28, so you received \$9.84. How much do you return? (0-9.84)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER, sent \$3 (they saw \$3.37 due to noise). They received \$10.11, returned \$5. Good cooperation.

Round 2: They were SENDER, sent \$3 (I saw it as \$3.8 due to noise), I received \$11.40, returned \$5.5 (they saw \$6.21 due to noise).

Round 3: I was SENDER, sent \$3 (they saw \$2.05 due to noise - significant noise here!). They received \$6.15, returned \$3. That's about half, which is reasonable given what they received.

Round 4: They are SENDER, sent something (I see \$3.28), I received \$9.84.

My co-player has been cooperative throughout. They've been returning reasonable amounts. The noise is creating some discrepancies, but overall the pattern is cooperative.

I received \$9.84. To maintain cooperation and reciprocity, I should return a fair amount. Looking at the pattern:

- Round 1: They returned \$5 out of \$10.11 (about 49%)
- Round 3: They returned \$3 out of \$6.15 (about 49%)

A fair return would be around 50% or slightly more to encourage continued cooperation. That would be around \$5.

Let me return \$5 to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 1253,
  "output_tokens": 355,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.1499999999999995, returned \$3.0, visible payoff \$3.1499999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.

Your total visible earnings across all rounds are \$22.569999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

```
'''json
```

```
{"send": 3}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: They were RECEIVER, saw \$3.37 sent, received \$10.11, returned \$5.0
Round 2: They were SENDER, sent \$3.0, became \$9.0, saw \$6.21 returned (I returned something that appeared as \$6.21 to them)
Round 3: They were RECEIVER, saw \$2.05 sent, received \$6.15, returned \$3.0
Round 4: They were SENDER, sent \$3.0, became \$9.0, saw \$4.45 returned (I returned something that appeared as \$4.45 to them)

So the pattern I'm seeing:

- When they're sender, they consistently send \$3.0
- When they're receiver, they return a fair amount (\$5 on \$10.11, \$3 on \$6.15)

In Round 4, I returned \$5 but they saw \$4.45 due to noise. This might make them think I'm less cooperative than I actually am.

Given that they continue to send \$3 consistently, I should continue to send a similar amount to maintain cooperation. I'll send \$3 again.

usage:

```
{
  "input_tokens": 1496,
  "output_tokens": 301,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.
- Round 5 against you, as RECEIVER: saw \$2.72 sent, it became \$8.16, returned \$4.0, visible payoff \$4.16.

Your total visible earnings across all rounds are \$29.449999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.04, so you received \$9.120000000000001. How much do you return? (0-9.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the history with my current co-player:

- Round 1: They were receiver, saw \$3.37 sent, got \$10.11, returned \$5.0
- Round 2: They were sender, sent \$3.0, it became \$9.0, saw \$6.21 returned
- Round 3: They were receiver, saw \$2.05 sent, got \$6.15, returned \$3.0
- Round 4: They were sender, sent \$3.0, it became \$9.0, saw \$4.45 returned
- Round 5: They were receiver, saw \$2.72 sent, got \$8.16, returned \$4.0

Now in Round 6: They are sender, sent \$3.04, I received \$9.12

This co-player has shown consistent behavior:

- When they receive money, they return a reasonable amount (around \$3-5)
- When they send, they consistently send \$3.0

I should continue the cooperative pattern. They sent me \$3.04, giving me \$9.12. A fair return would be around \$4.5-5 to maintain the cooperative relationship and ensure both parties benefit.

Let me return \$4.5 to keep things balanced.

usage:

```
{
```

```
"input_tokens": 1760,
"output_tokens": 308,
"reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.
- Round 5 against you, as RECEIVER: saw \$2.72 sent, it became \$8.16, returned \$4.0, visible payoff \$4.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.68 returned, visible payoff \$6.68.

Your total visible earnings across all rounds are \$34.06999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the history with my current co-player:

Round 1: I was sender, sent \$3 (they saw \$3.37), they received \$10.11, returned \$5
Round 2: I was receiver, they sent \$3.8, I received \$11.40, I returned \$5.5
Round 3: I was sender, sent \$3 (they saw \$2.05), they received \$6.15, returned \$3
Round 4: I was receiver, they sent \$3.28, I received \$9.84, I returned \$5
Round 5: I was sender, sent \$3 (they saw \$2.72), they received \$8.16, returned \$4
Round 6: I was receiver, they sent \$3.04, I received \$9.12, I returned \$4.5
Round 7: I am sender now

The pattern shows we've been cooperating reasonably well. The noise in communication is causing some variation in what they see vs what I send, but overall the cooperation is decent.

I should continue sending a similar amount to maintain the cooperative relationship.

```
Usage: {
  "input_tokens": 1995,
  "output_tokens": 271,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:01:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
```

```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.37, that is 67.4% of their total endowment. You received \$10.11. How much do you return to the sender? (0-10.11)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.37 (67.4% of their endowment)
- They currently have \$1.63 left (\$5 - \$3.37)
- I received \$10.11 after the 3x multiplication

I want to reciprocate this high level of trust and ensure the sender benefits from their decision to trust me.

If I return \$5:

- The sender ends with: \$1.63 + \$5 = \$6.63 (a profit of \$1.63)
- I keep: \$10.11 - \$5 = \$5.11 (a profit of \$5.11)

This rewards the sender's trust while both of us profit from cooperation.

```
{'return': 5.0}
```

reasoning:

Let me analyze this situation:

1. The sender sent \$3.37 out of their \$5 endowment (67.4%)
2. This was multiplied by 3x, so I received \$10.11
3. I need to decide how much to return (0-10.11)

Key considerations:

- The sender showed significant trust by sending 67.4% of their endowment
- If I return nothing, I keep \$10.11 but break trust
- A fair return might be to ensure the sender gets back more than they sent
- The sender currently has: $\$5 - \$3.37 = \$1.63$

If I want to be reciprocal and fair:

- The sender gave up \$3.37 to trust me
- They showed high trust (67.4% of endowment)
- A reasonable return would ensure they profit from trusting me

Some options:

1. Return \$5: Sender ends with $\$1.63 + \$5 = \$6.63$ (profit of \$1.63), I keep \$5.11
2. Return \$6: Sender ends with $\$1.63 + \$6 = \$7.63$ (profit of \$2.63), I keep \$4.11
3. Return \$4: Sender ends with $\$1.63 + \$4 = \$5.63$ (profit of \$0.63), I keep \$6.11
4. Return \$3.37: Sender ends with $\$1.63 + \$3.37 = \$5$ (break even), I keep \$6.74

Given that:

- The sender showed high trust (67.4%)
- Note says there may be communication noise
- I want to encourage future cooperation
- A fair split might be approximating equal gains

If I return \$5:

- Sender: started with \$5, sent \$3.37, gets back \$5 = net $\$1.63 + \$5 = \$6.63$ total

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 694,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.

Your total visible earnings across all rounds are \$5.109999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing significant trust
- They saw \$5.51 returned, which gave them a positive payoff
- I profited from their trust in Round 1

I want to reciprocate their cooperative behavior and maintain the trust relationship. Sending a similar proportion will signal that I'm willing to cooperate.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$3.37 (67.4% of their endowment)
- I received \$10.11
- I returned \$5.0
- I kept \$5.11

Now in Round 2, the visible history shows:

- My co-player's last game (Round 1 against me) as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51

Wait, there's communication noise! The actual amounts differ from what each player sees:

- I actually received \$10.11 (from their \$3.37 sent)
- I actually returned \$5.0
- But they see that they sent \$3.0, it became \$9.0, and saw \$5.51 returned

So there's noise in both directions. This is interesting - the sender sees they sent \$3.0 (vs actual \$3.37) and saw \$5.51 returned (vs actual \$5.0 I returned).

Now I'm the SENDER in Round 2 against the same co-player.

My visible earnings are \$5.11 (which matches what I calculated: $\$10.11 - \$5.0 = \$5.11$).

Strategy considerations:

1. My co-player sent a high amount in Round 1 (either \$3.0 or \$3.37 depending on what was real)
2. They saw a decent return (approximately 50% of what they sent, or more)
3. This suggests they're cooperative

I should signal continued cooperation by sending a similar proportion. Since they sent 67.4% in Round 1, I could send a similar high proportion.

Given the communication noise, I should send a clear cooperative signal. Sending around \$3-3.5 would show I'm reciprocating their trust.

Let me send \$3.0 to match what they think they sent (according to the visible history).

...

usage:

```
{
  "input_tokens": 848,
  "output_tokens": 639,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.

Your total visible earnings across all rounds are \$13.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.05, so you received \$6.1499999999999995. How much do you return? (0-6.1499999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.05 (41% of their endowment), showing moderate trust
- They currently have \$2.95 left
- I received \$6.15

Previous rounds show we've had a cooperative relationship, though the amounts varied (possibly due to noise). I want to maintain trust while ensuring both of us benefit.

If I return \$3.0:

- Sender ends with: $\$2.95 + \$3.0 = \$5.95$ (profit of \$0.95)

- I keep: $\$6.15 - \$3.0 = \$3.15$ (profit of $\$3.15$)

This ensures mutual benefit while reciprocating their trust.

```
{'return': 3.0}
```

reasoning:

Let me analyze the situation:

Looking at the visible history:

- Round 1: My co-player as SENDER sent $\$3.0$, saw $\$5.51$ returned, payoff $\$7.51$
- Round 2: My co-player as RECEIVER saw $\$3.8$ sent (I sent this), returned $\$5.5$, their payoff $\$5.90$

Now Round 3:

- I'm the RECEIVER
- My co-player sent $\$2.05$ (41% of their endowment)
- I received $\$6.15$

The co-player has been cooperative but they sent less this time ($\$2.05$ vs $\$3.0$ in Round 1)

. This could be:

1. A response to the noise in the game
2. Adjusting their strategy
3. Testing my response

Let me think about what's fair:

- They currently have: $\$5 - \$2.05 = \$2.95$
- I received: $\$6.15$

If I want to reciprocate fairly and ensure they profit:

- Return $\$3.0$: Sender gets $\$2.95 + \$3.0 = \$5.95$ (profit $\$0.95$), I keep $\$3.15$ (profit $\$3.15$)
- Return $\$3.5$: Sender gets $\$2.95 + \$3.5 = \$6.45$ (profit $\$1.45$), I keep $\$2.65$ (profit $\$2.65$)

Given they sent a moderate amount (41%), I should return a proportional amount. A return of around $\$3.0$ - $\$3.5$ seems fair, ensuring both of us profit while maintaining the cooperative relationship.

I'll return $\$3.0$ to keep things balanced.

usage:

```
{
  "input_tokens": 1160,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent $\$3.0$, it became $\$9.0$, saw $\$5.51$ returned, visible payoff $\$7.51$.
- Round 2 against you, as RECEIVER: saw $\$3.8$ sent, it became $\$11.399999999999999$, returned $\$5.5$, visible payoff $\$5.899999999999999$.
- Round 3 against you, as SENDER: sent $\$3.0$, it became $\$9.0$, saw $\$2.32$ returned, visible payoff $\$4.32$.

Your total visible earnings across all rounds are $\$16.47$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- My co-player has been consistently cooperative, sending $\$3.0$ in both rounds they were the sender
- In Round 3, they saw only $\$2.32$ returned (likely due to communication noise), which might have disappointed them

- Their history shows they value cooperation despite the lower return they observed

I want to maintain our cooperative relationship and signal that I'm still willing to trust them. Sending a similar amount to what they've been sending shows reciprocal cooperation.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51

Round 2: Co-player as RECEIVER, saw \$3.8 sent, returned \$5.5, payoff \$5.899999999999999

Round 3: Co-player as SENDER sent \$3.0, saw \$2.32 returned, payoff \$4.32

Wait, in Round 3 I was the RECEIVER and I sent back \$3.0, but they saw \$2.32 returned. This is the communication noise mentioned!

So my co-player has been consistently sending \$3.0 when they're the sender, showing cooperative behavior. However, in Round 3 they saw only \$2.32 returned even though I actually sent \$3.0. This might make them think I'm less cooperative than I actually am.

Given that:

1. They've been consistently cooperative, sending \$3.0 twice
2. They may think I was less generous in Round 3 than I actually was (due to noise)
3. I want to maintain the cooperative relationship

I should send a similar amount to what they've been sending - around \$3.0 - to signal continued cooperation.

```
{'send': 3.0}
```

usage:

```
{
  "input_tokens": 1559,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.
- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$22.919999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.72, so you received \$8.16. How much do you return? (0-8.16)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate trust
- They currently have \$2.28 left
- I received \$8.16

The history shows we've maintained a cooperative relationship. They've consistently returned reasonable amounts when they were the receiver (around 50% of what they received in Round 4).

If I return \$4.0:

- Sender ends with: \$2.28 + \$4.0 = \$6.28 (profit of \$1.28)
- I keep: \$8.16 - \$4.0 = \$4.16 (profit of \$4.16)

This ensures both of us profit while maintaining our cooperative pattern.

```
{'return': 4.0}
```

reasoning:

Looking at the situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate-to-high trust
- They currently have \$2.28 left ($5 - 2.72$)
- I received \$8.16

Looking at the history:

- Round 2: They returned \$5.5 when they received \$11.40
- Round 3: They saw \$2.32 returned (due to noise, I actually sent \$3.0)
- Round 4: They returned \$5.0 when they received \$9.84

They've been cooperative and returned reasonable amounts. In Round 4, they returned about 51% of what they received.

If I return \$4.0:

- Sender ends with: $2.28 + 4.0 = 6.28$ (profit of \$1.28)
- I keep: $8.16 - 4.0 = 4.16$ (profit of \$4.16)

This ensures both profit and maintains the cooperative relationship. The sender gets a reasonable return on their trust.

Let me consider returning slightly more since they've been consistently cooperative...

If I return \$4.5:

- Sender ends with: $2.28 + 4.5 = 6.78$ (profit of \$1.78)
- I keep: $8.16 - 4.5 = 3.66$ (profit of \$3.66)

I think \$4.0 is fair - it's about 49% of what I received, which is a reasonable return rate that ensures mutual benefit.

usage:

```
{
  "input_tokens": 1919,
  "output_tokens": 562,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.
- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.88 returned, visible payoff \$6.88.

Your total visible earnings across all rounds are \$27.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

- My co-player has consistently sent \$3.0 when they're the sender (Rounds 3 and 5)
- They've been cooperative as a receiver, returning \$5.0 in Round 4
- In Round 5, they saw a reasonable return (\$4.88) and had a positive payoff

Our relationship has been mutually cooperative. I should continue matching their level of trust and cooperation by sending a similar amount.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

- My co-player has been consistently sending \$3.0 when they're the sender (Rounds 3 and 5)
- They've been cooperative as a receiver, returning good amounts (returned \$5.0 in Round 4)
- In Round 5, they saw \$4.88 returned and had a payoff of \$6.88, which is positive

Our relationship has been quite cooperative overall. They've shown consistent trust by sending \$3.0 repeatedly. I should reciprocate this trust and continue the cooperative pattern.

Sending \$3.0 matches what they've been doing and signals that I'm committed to cooperation

usage:

```
{
  "input_tokens": 2316,
  "output_tokens": 272,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.88 returned, visible payoff \$6.88.
- Round 6 against you, as RECEIVER: saw \$3.04 sent, it became \$9.120000000000001, returned \$4.5, visible payoff \$4.620000000000001.

Your total visible earnings across all rounds are \$33.76.

This round, you are the RECEIVER against your current co-player. They sent you \$2.95, so you received \$8.850000000000001. How much do you return? (0-8.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$2.95 (59% of their endowment), showing good trust
- They currently have \$2.05 left
- I received \$8.85

The history shows we've maintained a strong cooperative relationship. They've been consistent in their trust, and I should continue reciprocating appropriately.

If I return \$4.5:

- Sender ends with: \$2.05 + \$4.5 = \$6.55 (profit of \$1.55)
- I keep: \$8.85 - \$4.5 = \$4.35 (profit of \$4.35)

This maintains our mutually beneficial pattern.

```
{'return': 4.5}
```

Reasoning:

Looking at this situation:

- My co-player sent \$2.95 (59% of their endowment), showing good trust
- They currently have \$2.05 left
- I received \$8.85

The history shows:

- They've been consistently cooperative, sending around \$3 when they're the sender
- They've returned reasonable amounts when they're the receiver (around \$4.5-\$5.0)
- We've maintained a mutually beneficial relationship

I should return an amount that:

1. Ensures the sender profits from trusting me
2. Maintains our cooperative pattern
3. Is proportional to what they sent

If I return \$4.5:

- Sender ends with: \$2.05 + \$4.5 = \$6.55 (profit of \$1.55)

- I keep: $\$8.85 - \$4.5 = \$4.35$ (profit of \$4.35)

This seems fair and maintains our cooperation pattern.

```
Usage: {
  "input_tokens": 2675,
  "output_tokens": 402,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:01:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$
- You keep: $\text{received} - \text{returned}$
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance showing trust with managing risk, especially given the communication noise mentioned.

Since this is a multi-round game with rotating partners, establishing a cooperative pattern early could be beneficial. However, I don't know this receiver's tendencies yet.

I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Gives the receiver meaningful funds to work with (tripled)
- Doesn't risk my entire endowment

If I send \$3, the receiver gets \$9. If they return even half of that (\$4.50), we both benefit more than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 and can send \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. There's communication noise mentioned - amounts may differ slightly from what was actually sent/returned

Key considerations:

- If I send \$0, I keep \$5, receiver gets \$0, game ends with no cooperation
- If I send \$5, receiver gets \$15, and their return decision determines my payoff
- This is a multi-round game with 8 agents, randomized pairings
- My reputation/behavior might matter for future rounds, though pairings are randomized

Standard trust game reasoning:

- Sending money shows trust and can enable mutual benefit
- If receiver is rational and self-interested (and this is the last interaction), they should return \$0
- But in multi-round settings, reputation and reciprocity often emerge
- A moderate initial send can signal willingness to cooperate while limiting downside risk

Since this is round 1 and I don't know the receiver's tendencies, I should probably send a moderate amount to:

1. Signal willingness to cooperate
2. Give receiver opportunity to show trustworthiness
3. Not risk my entire endowment

A common strategy is to send about half. If I send \$2.50:

- Receiver gets \$7.50
- If they return half (\$3.75), I end with $\$5 - \$2.50 + \$3.75 = \6.25
- Receiver keeps \$3.75
- Both benefit

Actually, let me reconsider. Since amounts can only be integers based on the format, I should send a whole dollar amount. Let me send \$3:

- I send \$3
- Receiver gets \$9
- If they're fair and return half (\$4.50), I get $5 - 3 + 4.50 = 6.50$

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 724,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.

Your total visible earnings across all rounds are \$5.41.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

Round 1: My co-player saw \$2.29 sent (due to noise), received \$6.87, and returned \$3.15 (about 46% of what they received). This shows they're willing to reciprocate.

Now they've sent me \$2.99, giving me \$8.97 total.

To maintain a cooperative relationship and signal trustworthiness, I'll return approximately half of what I received, which is fair and encourages future cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me understand what happened:

Round 1: I was the SENDER

- I sent \$3
- Due to communication noise, my co-player SAW \$2.29 sent (instead of \$3)
- It became \$6.87 (which is 2.29×3)
- They returned \$3.15
- Their visible payoff was \$3.72 (which is $6.87 - 3.15$)
- My visible earnings are \$5.41

Let me calculate my Round 1 payoff:

- I started with \$5
- I sent \$3
- I received back... well, they returned \$3.15
- So I should have: $5 - 3 + 3.15 = 5.15$
- But my "visible earnings" are \$5.41

Hmm, this is confusing. Maybe the communication noise affected what I received too? Or maybe "visible earnings" includes some noise?

Actually, let me recalculate. If they saw \$2.29 sent and returned \$3.15:

- From their perspective: they received \$6.87, returned \$3.15, kept \$3.72
- From my perspective: I sent \$3, but what did I actually receive back?

The problem states "visible payoff" and "visible earnings" which suggests everything we see might have noise.

Now for Round 2:

- I am the RECEIVER

- They sent me \$2.99 (this is what I see after noise)
- I received \$8.97 (which is $\$2.99 \times 3$)
- I need to decide how much to return (0-8.97)

From Round 1, my co-player:

- Saw \$2.29 sent (noise-affected)
- Received \$6.87
- Returned \$3.15 (about 46% of what they received)

This shows they were somewhat cooperative, returning close to half.

They sent \$2.99, which is close to their original \$3 send, suggesting they're being fairly consistent. If I return roughly half of what I received-around \$4.48-\$4.50-I'd match their reciprocity level. I could return slightly more to encourage continued cooperation, or slightly less to protect my earnings. Let me think through the payoffs: if they sent \$2.99 and kept \$2.01, returning \$4.50 would give them \$6.51 total, which seems fair given their initial cooperation.

Actually, they returned 46% in the first round despite seeing a smaller amount due to noise. That's a strong signal of trust...

usage:

```
{
  "input_tokens": 798,
  "output_tokens": 987,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.

Your total visible earnings across all rounds are \$9.88.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: As receiver, they returned \$3.15 on \$6.87 (46% return rate)

Round 2: As sender, they sent \$3, and saw I returned \$4.71 (though actual was \$4.50 due to noise)

This co-player has demonstrated:

- Willingness to send a good amount (\$3 = 60% of endowment)
- Reasonable reciprocation as receiver
- We've established a cooperative pattern

I'll continue this pattern by sending the same amount they sent to me.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

Round 1:

- I was sender, they were receiver
- They saw \$2.29 sent (noise from my actual \$3), received \$6.87, returned \$3.15
- This is about 46% return rate

Round 2:

- They were sender, I was receiver
- They sent \$3.0, it became \$9.0
- They saw \$4.71 returned (I actually returned \$4.50, so there's noise)

- They kept \$6.71 total

So this co-player has shown:

1. As a receiver: They returned about 46% of what they received
2. As a sender: They sent \$3 (60% of their endowment), showing trust

I returned \$4.50 on \$8.97 (about 50% return rate), which they saw as \$4.71.

This seems like a cooperative partner. They're willing to send a good amount and return a reasonable portion. Since they sent \$3 last round and I returned generously, I should continue the cooperative pattern.

I'll send \$3 again to maintain consistency and reciprocal trust.

usage:

```
{
  "input_tokens": 1100,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$4.05, visible payoff \$4.050000000000002.

Your total visible earnings across all rounds are \$15.790000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$3.15 on \$6.87 (46%)
Round 2: They sent \$3 and saw good returns
Round 3: They returned \$4.05 on \$8.10 (50%)
Round 4 (now): They sent \$2.35, I received \$7.05

They've been consistently fair, returning around 46-50% when receiving. They sent slightly less this round (\$2.35 vs \$3 in Round 2), but that's still a reasonable amount showing continued trust.

I'll maintain our cooperative pattern by returning approximately 50% of what I received.

```
'''json
{"return": 3.50}
'''
```

reasoning:

Looking at the history:

Round 1: Co-player as receiver - saw \$2.29 sent, got \$6.87, returned \$3.15 (46% of received)
Round 2: Co-player as sender - sent \$3, became \$9, saw \$4.71 returned
Round 3: Co-player as receiver - saw \$2.7 sent, got \$8.1, returned \$4.05 (50% of received)
Round 4 (now): Co-player as sender - sent \$2.35, I received \$7.05

The co-player has been consistent in returning around 46-50% when receiving. They've also been willing to send decent amounts when they're the sender.

However, I notice they sent \$2.35 this round, which is less than the \$3 they sent in round 2. This could be:

1. Due to communication noise affecting their perception of my returns
2. A strategic reduction

3. Random variation

Looking at their visible payoffs from my perspective, they seem to be doing well in our exchanges. I should continue cooperating to maintain this mutually beneficial relationship.

I've been returning around 50% consistently. Given they sent \$2.35 (about 47% of their endowment), which is slightly less than before, I'll return a fair amount but perhaps slightly less than exactly half to account for the reduced send.

I'll return around \$3.50, which is about 50% of what I received.

usage:

```
{
  "input_tokens": 1494,
  "output_tokens": 509,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$4.05, visible payoff \$4.050000000000002.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.67 returned, visible payoff \$5.67.

Your total visible earnings across all rounds are \$19.340000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our ongoing cooperation:

Round 2: They sent \$3, saw good returns (\$4.71)

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4: They sent \$3 again, saw \$3.67 returned

This co-player has been consistently cooperative:

- Sending \$3 when they're the sender
- Returning around 50% when they're the receiver

We've established a strong pattern of mutual trust. I'll continue by sending the same amount they've been sending.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

Round 2: They sent \$3 (as sender), saw \$4.71 returned

Round 3: They saw \$2.7 sent, returned \$4.05 on \$8.10 (50% return rate)

Round 4: They sent \$3 (as sender), saw \$3.67 returned

So they've been sending \$3 consistently when they're the sender. They've been returning around 50% when they're the receiver.

This is a cooperative partner. I should continue the pattern. I've been sending \$3 in previous rounds (rounds 1 and 3), and they've been matching that behavior.

I'll send \$3 again to maintain our cooperative relationship.

usage:

```
{
  "input_tokens": 1887,
  "output_tokens": 314,

```

```

    "reasoning_tokens": 0
  }
Message 12 role=user
content:
  Round 6

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 3 against you, as RECEIVER: saw $2.7 sent, it became $8.100000000000001, returned $4.05, visible payoff $4.050000000000002.
  - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $3.67 returned, visible payoff $5.67.
  - Round 5 against you, as RECEIVER: saw $2.24 sent, it became $6.720000000000001, returned $3.36, visible payoff $3.3600000000000008.
  Your total visible earnings across all rounds are $24.130000000000003.

  This round, you are the RECEIVER against your current co-player. They sent you $2.44, so you received $7.32. How much do you return? (0-7.32)
  Respond exactly as JSON: {'return': <amount>}
Message 13 role=assistant
content:
  Looking at our continued pattern:

  Round 3: They returned $4.05 on $8.10 (50%)
  Round 4: They sent $3
  Round 5: They returned $3.36 on $6.72 (50%)
  Round 6 (now): They sent $2.44, I received $7.32

  This co-player has consistently returned about 50% when receiving, showing fair and predictable cooperation. I'll maintain the same standard.

  '''json
  {"return": 3.66}
  '''
reasoning:
  Looking at the history:

  Round 3: They returned $4.05 on $8.10 (50%)
  Round 4: They sent $3, saw $3.67 returned
  Round 5: They saw $2.24 sent (noise), received $6.72, returned $3.36 (50%)
  Round 6 (now): They sent $2.44, I received $7.32

  They continue to show consistent cooperation, returning 50% when they're the receiver. Their sending amounts have been steady around $2.35-$3.

  I should continue my pattern of returning approximately 50% to maintain our cooperative relationship.

  50% of $7.32 = $3.66
usage:
  {
    "input_tokens": 2274,
    "output_tokens": 292,
    "reasoning_tokens": 0
  }
Message 14 role=user
content:
  Round 7

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $3.67 returned, visible payoff $5.67.
  - Round 5 against you, as RECEIVER: saw $2.24 sent, it became $6.720000000000001, returned $3.36, visible payoff $3.3600000000000008.
  - Round 6 against you, as SENDER: sent $3.0, it became $9.0, saw $4.57 returned, visible payoff $6.57.
  Your total visible earnings across all rounds are $27.790000000000003.

  This round, you are the SENDER against your current co-player. You have $5. How much do you

```

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the continued cooperation:

Round 4: They sent \$3, saw \$3.67 returned

Round 5: They returned \$3.36 (50%)

Round 6: They sent \$3, saw \$4.57 returned

This co-player has been remarkably consistent and trustworthy:

- Always sending \$3 as sender (60% of endowment)
- Always returning around 50% as receiver

We've built a strong cooperative relationship over 6 rounds. I'll maintain this successful pattern by sending the same amount.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the pattern:

Round 4: They sent \$3, saw \$3.67 returned

Round 5: They returned \$3.36 on \$6.72 (50%)

Round 6: They sent \$3, saw \$4.57 returned

This co-player has been very consistent:

- Always sending \$3 as sender
- Always returning around 50% as receiver

We have a well-established cooperative relationship. They've been trustworthy and fair throughout. I should continue sending \$3 to maintain this successful pattern.

```
Usage: {
  "input_tokens": 2626,
  "output_tokens": 268,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:01:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.29, that is 45.8% of their total endowment. You received \$6.87. How much do you return to the sender? (0-6.87)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.29 (45.8% of their endowment)
- I received \$6.87 after the 3x multiplier
- The sender currently has \$2.71

I want to reciprocate their trust while keeping the exchange mutually beneficial. The sender took a risk by sending nearly half their endowment, so I should return enough to reward that trust.

If I return approximately the same proportion they sent (around 45-46% of what I received), that would be about \$3.15. This would give the sender \$2.71 + \$3.15 = \$5.86 total (a gain of \$0.86), and I'd keep \$3.72 (also a gain).

This seems like a fair reciprocation that rewards their trust and could encourage cooperation in future rounds.

{'return': 3.15}

reasoning:

I'm the receiver in this round. Let me understand the situation:

- The sender sent \$2.29 (which is 45.8% of their \$5 endowment)

- This was multiplied by 3, so I received \$6.87
- I need to decide how much to return (between \$0 and \$6.87)

Let me think about what would be fair and strategic:

1. The sender showed trust by sending almost half their endowment (\$2.29 out of \$5)
2. They kept \$5 - \$2.29 = \$2.71
3. I received \$6.87
4. If I return nothing, I keep \$6.87 and the sender ends with \$2.71
5. If I return everything, I keep \$0 and the sender ends with \$2.71 + \$6.87 = \$9.58

A common fair approach in trust games is to aim for roughly equal outcomes or to reciprocate the sender's trust proportionally.

Let's consider some options:

Option 1 - Equal split of total gains:

- Total money created: \$5 (sender's endowment) + \$6.87 (what I received) = \$11.87
- But wait, that's not right. The sender started with \$5, sent \$2.29 which became \$6.87
- Total in the system: Sender has \$2.71, I have \$6.87 = \$9.58 total
- Equal split would be \$4.79 each
- For sender to have \$4.79: they need \$4.79 - \$2.71 = \$2.08 returned

Option 2 - Return proportional to what was sent:

- Sender sent 45.8% of their endowment
- I could return a similar proportion of what I received
- 45.8% of \$6.87 = \$3.15

Option 3 - Ensure sender gains from trusting me:

- Sender currently has \$2.71
- If I return enough so they end up better than their initial \$5
- They need at least \$5 - \$2.71 = \$2.29 to break even
- To give them a decent profit, maybe return \$3-3.50 so they end with \$5.71-6.21

Since this is a multi-round game with randomized pairings and communication noise, I should consider how my choice signals trustworthiness. Returning around \$3-3.15 seems reasonable-it reciprocates their generosity while still letting me benefit from the exchange, and it positions me well for future interactions with different partners....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1076,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.

Your total visible earnings across all rounds are \$3.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They received \$3.41 back, making a profit of \$0.41
- They demonstrated significant trust

As the SENDER now, I want to reciprocate their trust level. They sent 60% of their endowment, which was quite generous. I should match or slightly exceed this level of trust to maintain cooperation.

Sending \$3.0 would match their previous contribution and signal that I'm willing to cooperate

ate at the same trust level they showed.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

From Round 1:

- I was the RECEIVER
- The sender sent \$2.29 (45.8% of their endowment)
- I received \$6.87
- I returned \$3.15

But now the visible history shows something different:

- It says my co-player (as SENDER in Round 1) sent \$3.0, which became \$9.0
- They saw \$3.41 returned
- Their visible payoff was \$5.41

This is the communication noise mentioned in the instructions! What actually happened vs what they saw:

- Actual: They sent \$2.29, I received \$6.87, I returned \$3.15
- What they saw: They sent \$3.0, became \$9.0, saw \$3.41 returned

So from their perspective, they sent \$3.0 and got back \$3.41. That's a return rate of about 38% of what I received ($3.41/9.0$).

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of endowment)
- Got back \$3.41 (38% of \$9.0)
- Made a profit of \$0.41

They showed significant trust by sending 60% (from their perspective). They got back a reasonable amount.

As the SENDER now, I should consider:

1. They showed high trust in Round 1 (from their perspective, 60%)
2. They got a modest return
3. I want to continue building trust

Given that they sent 60% from their perspective, I should send a similar or slightly higher amount to show reciprocity and build trust. Sending around \$3.0-\$3.5 would be reasonable.

Let me send \$3.0 to match what they (think they) sent to me.

Actually, let me think more carefully. My actual earnings were \$3.72 in Round 1. I received \$6.87 and returned \$3.15.

...

usage:

```
{
  "input_tokens": 867,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.4700000000000001.

Your total visible earnings across all rounds are \$10.43.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.1000000000000001. How much do you return? (0-8.1000000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Round 2 pattern:**** When my co-player was the RECEIVER, they returned \$4.5 out of \$8.97 received (approximately 50%)

****This round:****

- They sent \$2.7 (54% of endowment), showing continued trust
- I received \$8.10

I should reciprocate with a similar return ratio. Returning about 50% like they did would be \$4.05, which:

- Gives them $\$2.3 + \$4.05 = \$6.35$ total (profit of \$1.35)
- Gives me \$4.05 (also a decent profit)

This maintains the cooperative pattern we've established.

{'return': 4.05}

reasoning:

Let me analyze the situation:

****Visible History:****

- Round 1: My co-player (as SENDER) sent \$3.0, received \$3.41 back, payoff \$5.41
- Round 2: My co-player (as RECEIVER) saw \$2.99 sent, returned \$4.5, payoff \$4.47

****This Round (Round 3):****

- I am the RECEIVER
- They sent \$2.7 (54% of their endowment)
- I received \$8.10

Looking at the pattern:

- In Round 1, they sent 60% of their endowment
- In Round 2, I sent \$3.0 (though they saw \$2.99 due to noise), and they returned \$4.5, which is about 50% of what they received (\$8.97)
- In Round 3, they sent \$2.7 (54%)

My co-player seems to be cooperating at moderate-to-high levels. When they were receiver in Round 2, they returned 50% of what they received.

If I return around 50% like they did, that would be about \$4.05.

Let me think about what's fair:

- They sent \$2.7, so they have \$2.3 left
- If I return \$4.05, they'd have $\$2.3 + \$4.05 = \$6.35$ (profit of \$1.35)
- I'd keep $\$8.10 - \$4.05 = \$4.05$ (profit of \$4.05)

That seems reasonable and matches their behavior. I could be slightly more generous to encourage continued cooperation.

Actually, let me reconsider. In Round 2, they returned \$4.5 out of \$8.97, which is exactly 50%.

...

usage:

```
{
  "input_tokens": 1210,
  "output_tokens": 644,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.4700000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible p

payoff \$5.91.
Your total visible earnings across all rounds are \$14.48.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 (60% of endowment) in Rounds 1 and 3
- As RECEIVER: Returned generously (\$4.5 out of \$8.97, ~50%)

****My previous sending:****

- Round 2: I sent and they saw \$2.99 (likely noise from my \$3.0)

They've demonstrated consistent trust and reciprocity. I should maintain this cooperative relationship by matching their sending level.

{'send': 3.0}

reasoning:

Looking at the history:

Round 1: Co-player as SENDER sent \$3.0, got \$3.41 back, payoff \$5.41

Round 2: Co-player as RECEIVER got \$2.99 sent (from me), returned \$4.5, payoff \$4.47

Round 3: Co-player as SENDER sent \$3.0, got \$3.91 back, payoff \$5.91

Pattern analysis:

- When my co-player is SENDER, they consistently send \$3.0 (60% of endowment)
- When they're RECEIVER, they return generously (\$4.5 out of \$8.97 ~ 50%)
- They're showing consistent trust and cooperation

My sending history:

- Round 2: I sent \$3.0 (based on the visible history showing they saw \$2.99 sent due to noise)
- Round 3: Not explicitly shown as sender, but the history says they saw \$2.99 sent in Round 2

Actually, let me re-read:

Round 2: I was SENDER. I sent \$3.0. Due to noise, they saw \$2.99. They received \$8.97, returned \$4.5.

Round 3: I was RECEIVER. They sent \$2.7 (but I think they sent \$3.0 and noise made it \$2.7). I returned \$4.05.

Now Round 4: I'm SENDER again.

They've been consistently sending \$3.0 when they're sender. They've returned generously when receiver.

I should maintain the cooperative pattern by sending \$3.0 again.

usage:

```
{
  "input_tokens": 1604,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.4700000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.
- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.0500000000000001, returned \$3.5, visible payoff \$3.5500000000000007.

Your total visible earnings across all rounds are \$20.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.24, so you received \$6.720000000000001. How much do you return? (0-6.720000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the history:

****Co-player's return pattern:****

- Round 2: Returned \$4.5 out of \$8.97 (~50%)
- Round 4: Returned \$3.5 out of \$7.05 (~50%)

****This round:****

- They sent \$2.24 (44.8% of endowment)
- I received \$6.72

I should maintain the cooperative pattern by returning around 50% like they have consistently done.

Returning \$3.36:

- Gives them \$2.76 + \$3.36 = \$6.12 total (profit of \$1.12)
- Gives me \$3.36 (also a profit)

{'return': 3.36}

reasoning:

Looking at the visible history:

Round 2: They received \$8.97, returned \$4.5 (50%)

Round 4: They received \$7.05, returned \$3.5 (49.6%)

This round:

- They sent \$2.24 (44.8% of endowment)
- I received \$6.72

The co-player has been consistently returning around 50% of what they receive. I should reciprocate similarly.

Returning around 50% would be \$3.36:

- Sender would have: \$2.76 + \$3.36 = \$6.12 (profit of \$1.12)
- I would keep: \$6.72 - \$3.36 = \$3.36 (also a profit)

This maintains the cooperative pattern and roughly matches their return rates.

usage:

```
{
  "input_tokens": 1990,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.
- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.

Your total visible earnings across all rounds are \$23.509999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Consistently sent \$3.0 in Rounds 3 and 5 (60% of endowment)
- As RECEIVER: Returned \$3.5 out of \$7.05 (~50%) in Round 4

****What they saw in Round 5:****

- They sent \$3.0, but saw only \$2.79 returned (likely due to noise)
- Despite this, they continue to cooperate

I should maintain the cooperative relationship by sending \$3.0 again, matching their consistent trust level.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 in Rounds 3 and 5
- As RECEIVER: Returned \$3.5 out of \$7.05 (~50%) in Round 4

****What they saw from me:****

- Round 3: Sent \$3.0, they returned \$3.91
- Round 4: They saw \$2.35 sent (likely noise from my \$3.0), returned \$3.5
- Round 5: Sent \$3.0, they saw only \$2.79 returned (but I actually sent \$3.36, so noise affected what they saw)

The co-player has been consistently sending \$3.0, showing steady trust. Even though in Round 5 they saw less returned than I actually sent (due to noise), they continue to cooperate.

I should maintain the cooperative pattern and send \$3.0 again to match their consistent trust level.

usage:

```
{
  "input_tokens": 2395,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.
- Round 6 against you, as RECEIVER: saw \$2.44 sent, it became \$7.32, returned \$3.66, visible payoff \$3.66.

Your total visible earnings across all rounds are \$30.08.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Co-player's return pattern:****

- Round 4: Returned \$3.5 out of \$7.05 (~50%)
- Round 6: Returned \$3.66 out of \$7.32 (~50%)

****This round:****

- They sent \$3.24 (64.8% of endowment) - showing strong trust
- I received \$9.72

They've been consistently returning around 50%. I should maintain this cooperative pattern.

Returning \$4.86 (50%):

- Gives them \$1.76 + \$4.86 = \$6.62 total (profit of \$1.62)
- Gives me \$4.86 (also a good profit)

```
{'return': 4.86}
```

Reasoning:

Looking at the history:

****Co-player's return pattern:****

- Round 4: Returned \$3.5 out of \$7.05 (~50%)
- Round 6: Returned \$3.66 out of \$7.32 (~50%)

****This round:****

- They sent \$3.24 (64.8% of endowment) - showing strong trust
- I received \$9.72

They've been consistently returning around 50% when they're the receiver. I should maintain this cooperative pattern.

Returning around 50% would be \$4.86:

- Gives them \$1.76 + \$4.86 = \$6.62 total (profit of \$1.62)
- Gives me \$4.86 (also a good profit)

This maintains the cooperative equilibrium we've established.

Usage: {

"input_tokens": 2783,

"output_tokens": 383,

"reasoning_tokens": 0

}

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T08:01:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_4",

"move_index": 6,

"opponent_id": "Agent_8",

"pairing": {

"agents": [

"Agent_7",

"Agent_8"

],

"investor": "Agent_7",

"roles": {

"Agent_7": "investor",

"Agent_8": "trustee"

},

"trustee": "Agent_8"

},

"role": "investor",

"role_label": "Sender",

"round": 7,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I want to establish a cooperative tone while being somewhat cautious since I don't know my co-player yet.

Sending a moderate amount can:

- Signal trust and willingness to cooperate
- Create a larger pool for mutual benefit (tripled to \$9)
- Allow the receiver meaningful choice in how much to return

If the receiver returns even half of what they receive, we both end up better than the initial \$5.

I'll send **\$3**.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- There's communication noise, so amounts may differ
- This is a multi-round game with 8 agents
- Pairings are randomized each round

Classic trust game setup. Let me think about optimal strategy:

1. If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
2. If I send \$5: Receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 (better than \$5). But they might return less or nothing.

3. If I send a moderate amount (e.g., \$3): Receiver gets \$9. If they return \$4-5, I come out ahead.

Since this is:

- Round 1 (establishing reputation/reciprocity patterns)
- Multi-round game (future interactions matter)
- Anonymous agents (can't track specific individuals)

I should consider:

- Building trust/reciprocity norms early
- Signaling cooperation
- But also being somewhat cautious since I don't know the other agents

A common strategy is to send a moderate-to-high amount in early rounds to:

1. Signal trustworthiness
2. Encourage reciprocity
3. Maximize potential gains if receiver cooperates

Given the noise factor and uncertainty, I'll send \$3. This shows goodwill without being maximally vulnerable, and gives the receiver \$9 to work with. If they return \$4-5, we both benefit.

Actually, let me reconsider. In trust games, sending higher amounts often generates better returns because:

- It demonstrates trust
- Creates larger pie to split
- Social norms typically lead to proportional returns

Comparing the math: sending \$4 leaves me with \$1 while giving the receiver \$12. A 50% return (\$6) nets me \$7 total, while a 33% return (\$4) just breaks me even. With \$3 sent instead, I'd have \$2 cushion and need only a 50% return (\$4.50) to reach \$6.50. The higher send amount creates more upside if reciprocity holds, so I'll go with \$4....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 814,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.

Your total visible earnings across all rounds are \$4.2.

This round, you are the RECEIVER against your current co-player. They sent you \$3.7, so you received \$11.100000000000001. How much do you return? (0-11.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 analysis:****

- My co-player saw \$2.24 sent (due to noise), which became \$6.72
- They returned \$3.0 (about 45% of what they received)
- This shows reasonable cooperative behavior

****Round 2 (current):****

- They sent me \$3.7 (even more than what they saw in Round 1)
- I received \$11.10
- They're showing increased trust

To maintain and encourage cooperation, I should return a fair amount. Returning about half of what I received would give them:

- They kept: \$5 - \$3.7 = \$1.30
- Plus returned amount

- A good return on their trust

I'll return **\$6.00** (about 54% of what I received), giving them a total of \$7.30 - a solid return that rewards their increased trust.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation carefully.

Round 1 recap from my perspective:

- I sent \$3
- Due to communication noise, my co-player saw \$2.24 sent (which became \$6.72 after tripling)
- They returned \$3.0
- Their visible payoff was \$3.72 (received \$6.72, returned \$3.0)
- My total visible earnings are \$4.2

Let me calculate what actually happened in Round 1:

- I sent \$3, so I had \$5 - \$3 = \$2 left
- The receiver got $3 \times 3 = \$9$ (but saw \$6.72 due to noise)
- They returned \$3.0
- So I ended with $\$2 + \$3 = \$5$
- But my visible earnings show \$4.2, which suggests the noise affected what I saw returned too?

Actually, wait. Let me reconsider:

- If my visible earnings are \$4.2 total
- I started with \$5, sent \$3 (keeping \$2)
- If I received back something that made my total \$4.2, then I received $\$4.2 - \$2 = \$2.2$ back

So the noise worked both ways:

- I sent \$3, they saw \$2.24
- They returned \$3.0, but I saw (and actually got) something that gave me \$4.2 total

Now for Round 2:

- I'm the RECEIVER
- My co-player sent me \$3.7 (according to what I see)
- This became \$11.10 after tripling
- I need to decide how much to return

My co-player showed willingness to return a reasonable amount in Round 1 (they returned \$3.0 out of \$6.72 they saw, which is about 45%). This is fairly cooperative behavior.

I should return a fair amount to maintain cooperation. Given that:

- They sent \$3.7 and I received \$11.10
- A 50% return would be \$5.55, giving them $\$1.3$ (kept) + \$5.55 (returned) = \$6.85 total
- This matches their cooperative gesture from Round 1 and encourages continued trust

I'll return \$5 to keep things balanced....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 1051,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.

Your total visible earnings across all rounds are \$9.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3 ? they returned \$3 (cooperative)

****Round 2:**** They sent \$3 ? I returned \$6, but they saw \$5.07 (still a good return)

****Round 3 (current):****

- Both rounds showed mutual cooperation
- They sent \$3 as sender and returned reasonably as receiver
- Their visible payoff was \$7.07 last round, showing cooperation pays off

I'll send ****\$3.5**** to maintain and slightly increase our cooperative dynamic. This shows I'm willing to trust them a bit more based on our positive history.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.24 (noise), became \$6.72, returned \$3.0
- They kept \$3.72, I got back \$3.0, so I ended with \$5 - \$3 + \$3 = \$5
- But my visible earnings show \$4.2, which suggests I saw something different due to noise

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, became \$9.0
- I returned \$6.0, but they saw \$5.07 (noise)
- They ended with \$5 - \$3 + \$5.07 = \$7.07
- I kept \$9.0 - \$6.0 = \$3.0

My total visible earnings are \$9.3.

Now Round 3:

- I'm SENDER again
- Co-player has shown:
 - As receiver: returned \$3.0 out of \$6.72 (45%)
 - As sender: sent \$3.0 out of \$5 (60%)
- They saw I returned \$5.07 (though I actually returned \$6.0)
- This is good cooperation

Since they've been cooperative and sent a good amount when they were sender, I should continue cooperating. They sent \$3.0 last round, so I should send at least that much or more to maintain trust.

I'll send \$3.5 to show continued cooperation and slightly increase the stakes.

usage:

```
{
  "input_tokens": 1191,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned

\$5.0, visible payoff \$4.899999999999999.
Your total visible earnings across all rounds are \$16.12.

This round, you are the RECEIVER against your current co-player. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our interaction pattern:

****Round 1:**** They returned \$3.0 from \$6.72 (45%)
****Round 2:**** They sent \$3.0, I returned \$6.0
****Round 3:**** They returned \$5.0 from \$9.90 (51%)
****Round 4 (current):**** They sent \$3.0 ? I received \$9.0

They've shown consistent cooperation, with their return percentages increasing slightly over time. They sent \$3.0 this round, maintaining trust despite some communication noise in previous rounds.

I'll return ****\$5.0**** (about 56% of what I received). This gives them a total payoff of \$7.0 (\$2 kept + \$5 returned), which rewards their continued trust and maintains our cooperative relationship.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze the history:

****Round 1:**** I sent \$3 ? they saw \$2.24 ? they returned \$3.0
****Round 2:**** They sent \$3.7 ? I received \$11.10 ? I returned \$6.0 ? they saw \$5.07
****Round 3:**** I sent \$3.5 ? they saw \$3.3 ? they returned \$5.0
****Round 4 (current):**** They sent \$3.0 ? I received \$9.0

Looking at their behavior:

- Round 1 as receiver: Returned \$3.0 out of \$6.72 (45%)
- Round 3 as receiver: Returned \$5.0 out of \$9.90 (51%)

Their returns seem reasonable and slightly increasing in percentage.

My behavior:

- Round 2 as receiver: Returned \$6.0 out of \$11.10 (54%)

They've consistently sent around \$3-\$3.7 as sender, maintaining cooperation.

For Round 4, they sent \$3.0 and I received \$9.0. To maintain fair cooperation and match roughly what they've done (returning about 50%), I should return around \$4.5-\$5.0.

Let me return \$5.0 to be consistent with maintaining good cooperation and matching their recent return percentage.

Actually, wait. Let me recalculate their payoffs to understand better:

Round 1:

- They as receiver: Kept \$5 - \$0 (not sender) + received \$6.72 - \$3.0 returned = \$5 + \$3.72 = \$8.72? No wait, they were receiver, so sender sent them something.
- Actually the history shows "as RECEIVER: saw \$2.24 sent" - so I was the sender, I sent something, they saw \$2.24, it became \$6.72, they returned \$3.0
- Their payoff: \$6.72 - \$3.0 = \$3.72. Round 2 shows they sent \$3.0 which tripled to \$9.0, and when I returned \$5.07, their payoff was \$5 - \$3.0 + \$5.07 = \$7.07. In Round 3, I sent \$3.5 which became \$9.90, and they returned \$5.0, giving them \$9.90 - \$5.0 = \$4.90. Now they've sent \$3.0 which becomes \$9.0 for me, so I need to decide what to return....

usage:

```
{
  "input_tokens": 1596,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.

Your total visible earnings across all rounds are \$20.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our pattern:

****Round 2:**** They sent \$3.0, saw ~\$5.07 returned (good payoff)
****Round 3:**** They returned \$5.0 from what they saw as \$9.90
****Round 4:**** They sent \$3.5, saw ~\$4.99 returned

They've consistently sent around \$3.0-\$3.5 and returned fairly when in the receiver role. Despite some communication noise affecting visible payoffs, we've maintained good cooperation.

I'll send ****\$3.5**** to continue our stable, mutually beneficial pattern.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze our history:

****Round 2:**** They sent \$3.0, I returned \$6.0 (they saw \$5.07 due to noise)
****Round 3:**** I sent \$3.5, they returned \$5.0 (they saw \$3.3 sent due to noise)
****Round 4:**** They sent \$3.5 (wait, the history says they sent \$3.5 in Round 4?), I returned \$5.0 (they saw \$4.99 due to noise)

Wait, let me reread. In Round 4, I was the receiver and they sent me \$3.0. But the visible history says they sent \$3.5 in Round 4. That must be the noise - what they actually sent was \$3.5, but I saw \$3.0.

Let me recalculate:

- Round 4: They actually sent \$3.5 ? became \$10.5 ? I returned \$5.0 (they saw \$4.99)
- Their payoff: \$5 - \$3.5 + \$4.99 = \$6.49

****Round 5 (current):**** I'm the sender

Looking at the pattern:

- Round 2: They sent \$3.0, got back ~\$5.07 (good return)
- Round 3: I sent (they saw \$3.3), they returned \$5.0 (reasonable)
- Round 4: They sent \$3.5, got back ~\$4.99 (decent return, about 48% of the tripled amount)

They've been consistently cooperative, sending \$3.0-\$3.5 each time. Their returns have been fair (around 50% of what they receive).

I should maintain the cooperative relationship. I'll send ****\$3.5**** to continue our pattern of mutual trust and cooperation.

usage:

```
{
  "input_tokens": 2020,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.
- Round 5 against you, as RECEIVER: saw \$3.42 sent, it became \$10.26, returned \$5.5, visible payoff \$4.76.

Your total visible earnings across all rounds are \$27.85.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our established pattern:

****Round 3:**** They returned \$5.0 from \$9.90
****Round 4:**** They sent \$3.5, I returned \$5.0
****Round 5:**** They returned \$5.5 from \$10.26
****Round 6 (current):**** They sent \$3.29 ? I received \$9.87

They've been consistently cooperative, both when sending and returning. They sent \$3.29, keeping \$1.71 for themselves.

I'll return ****\$5.0**** (about 51% of what I received), giving them a total payoff of \$6.71. This maintains our established cooperative pattern.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at the pattern:

****Round 3:**** They returned \$5.0 from \$9.90 (51%)
****Round 4:**** They sent \$3.5, saw ~\$4.99 returned
****Round 5:**** They returned \$5.5 from \$10.26 (54%)
****Round 6 (current):**** They sent \$3.29 ? I received \$9.87

They've been very consistent:

- As sender: sending around \$3.0-\$3.5
- As receiver: returning around 51-54% of what they receive

They sent \$3.29 this round, so they kept \$1.71.

If I return about half to slightly more (consistent with our pattern), that would be around \$5.0-\$5.5.

Returning \$5.0 would give them \$1.71 + \$5.0 = \$6.71 total
Returning \$5.5 would give them \$1.71 + \$5.5 = \$7.21 total

Given they're showing continued trust and cooperation, I should return a fair amount. I'll return \$5.0-\$5.5. Let me go with \$5.0 to be consistent with our established pattern (around 50% return rate).

Actually, looking at my previous returns in this partnership:

- Round 2: I returned \$6.0 from \$11.10 (54%)
- Round 4: I returned \$5.0 from \$9.0 (56%)

So I've been returning slightly more than half. Let me continue that pattern.

$\$9.87 * 0.50 = \4.94
 $\$9.87 * 0.55 = \5.43

I'll return \$5.0, which is about 51% and maintains our cooperative pattern.

usage:

```
{
  "input_tokens": 2415,
  "output_tokens": 599,
```

```

    "reasoning_tokens": 0
  }
Message 14 role=user
  content:
    Round 7

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 4 against you, as SENDER: sent $3.5, it became $10.5, saw $4.99 returned, visible
      payoff $6.49.
    - Round 5 against you, as RECEIVER: saw $3.42 sent, it became $10.26, returned $5.5, visible
      payoff $4.76.
    - Round 6 against you, as SENDER: sent $3.5, it became $10.5, saw $5.14 returned, visible
      payoff $6.64.
    Your total visible earnings across all rounds are $32.72.

    This round, you are the SENDER against your current co-player. You have $5. How much do you
    send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Assistant output:
  Looking at our established pattern:

  **Round 4:** They sent $3.5, saw ~$4.99 returned (payoff $6.49)
  **Round 5:** They returned $5.5 from $10.26 (generous)
  **Round 6:** They sent $3.5, saw ~$5.14 returned (payoff $6.64)

  This co-player has been highly consistent and cooperative. They regularly send $3.5 as sender
  and return generously as receiver. Our mutual cooperation has been working very well.

  I'll continue our successful pattern by sending **$3.5**.

  ```json
 {"send": 3.5}
  ```
Reasoning:
  Looking at the pattern:

  **Round 4:** They sent $3.5, saw ~$4.99 returned
  **Round 5:** They returned $5.5 from $10.26
  **Round 6:** They sent $3.5, saw ~$5.14 returned

  This co-player has been very consistent and cooperative across all rounds. They:
  - Send around $3.5 consistently as sender
  - Return around $5.0-$5.5 as receiver
  - Have maintained cooperation despite communication noise

  We've built a strong pattern of mutual trust. They've been sending $3.5 regularly, and I've been
  returning fair amounts. This has worked well for both of us.

  I should continue sending around $3.5 to maintain this successful cooperative relationship.
  Usage: {
    "input_tokens": 2800,
    "output_tokens": 339,
    "reasoning_tokens": 0
  }

```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:01:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
  },

```

```

    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.24, that is 44.8% of their total endowment. You received \$6.720000000000001. How much do you return to the sender? (0-6.720000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.24 (44.8% of their endowment)

- They currently have \$2.76 remaining
- I received \$6.72 after the 3x multiplication

To reciprocate their trust fairly, I should return an amount that:

1. Ensures they profit from trusting me (end up with more than their original \$5)
2. Rewards their cooperative gesture
3. Still leaves me with a reasonable gain

If I return \$3.00:

- Sender ends with: $\$2.76 + \$3.00 = \$5.76$ (gain of \$0.76)
- I keep: $\$6.72 - \$3.00 = \$3.72$ (gain of \$3.72)

This seems like a fair reciprocal response that rewards trust while maintaining balanced incentives for future rounds.

```
{"return": 3.00}
```

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.24 out of their \$5 endowment (44.8%)
3. This was multiplied by 3x, so I received \$6.72 (technically \$6.720000000000001 due to floating point)
4. I need to decide how much to return (between \$0 and \$6.72)

Key considerations:

- The sender showed moderate trust by sending 44.8% of their endowment
- If I return nothing, I keep all \$6.72 and the sender gets only \$5 - \$2.24 = \$2.76
- If I return a fair amount, I should consider what would make this equitable

Common approaches in trust games:

1. Equal split of total gains: Total in the system is $\$5 + \$6.72 = \$11.72$. Equal split would be \$5.86 each. The sender currently has \$2.76, so I'd return $\$5.86 - \$2.76 = \$3.10$
2. Return the same proportion they sent: They sent 44.8%, so I could return 44.8% of \$6.72 = \$3.01
3. Match their investment: Return \$2.24 so we both end up with gains from the multiplication
4. Return enough so they profit: If I return \$3, they end up with $\$2.76 + \$3 = \$5.76$ (a gain of \$0.76), and I keep \$3.72 (a gain of \$3.72)

The sender took a risk by sending \$2.24. A fair return should:

- At minimum, not make them worse off than keeping everything (\$5)
- Ideally, reward their trust
- Be reciprocal to their gesture

Let me calculate a few scenarios:

- If I return \$3.00: Sender gets $\$2.76 + \$3.00 = \$5.76$, I keep \$3.72
- If I return \$3.50: Sender gets $\$2.76 + \$3.50 = \$6.26$, I keep \$3.22
- If I return \$4.00: Sender gets $\$2.76 + \$4.00 = \$6.76$, I keep \$2.72

A return of \$3.00-\$3.50 seems balanced-it rewards their trust while still giving me a meaningful gain. I'll return \$3.00, which nets them \$5.76 (a 15.2% gain) and leaves me with \$3.72....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.

Your total visible earnings across all rounds are \$3.7200000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw only \$2.2 returned (though due to communication noise, I actually returned \$3.00)
- From their perspective, they may feel the cooperation wasn't fully reciprocated

As the sender now, I want to:

1. Signal continued willingness to cooperate
2. Match their previous level of trust
3. Maintain the cooperative dynamic despite any noise in the previous round

I'll send the same amount they sent in Round 1 to demonstrate reciprocal trust.

{"send": 3.0}

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$2.24 (44.8% of their endowment)
- I received \$6.72
- I returned \$3.00
- Sender ended with $\$2.76 + \$3.00 = \$5.76$
- I kept \$3.72

But the visible history shows something different from my co-player's perspective:

- Round 1: They were SENDER, sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2

Wait, this is confusing. Let me re-read the prompt about communication noise.

"NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design."

So what happened:

- From my perspective: I received notification that sender sent \$2.24, I received \$6.72, I returned \$3.00
- From their perspective: They sent \$3.0, it became \$9.0, they saw \$2.2 returned, and got payoff \$4.2

Let me verify the math from their perspective:

- They sent \$3.0
- They had $\$5.0 - \$3.0 = \$2.0$ left
- They saw \$2.2 returned
- Their payoff: $\$2.0 + \$2.2 = \$4.2$?

So from their perspective, they sent \$3.0 and only got back \$2.2, which is less than they sent. They might perceive this as me being untrustworthy or uncooperative.

However, from MY perspective, I actually returned \$3.00, which I thought was fair and generous given I received \$6.72.

This is the communication noise at work. They think I returned less than I actually did.

Now I'm the sender in Round 2, and they likely view me as less trustworthy due to the noise distorting their perception of my return. They sent 60% of their endowment initially, showing genuine trust, but the miscommunication means they'll probably be more cautious.

I need to decide whether sending a substantial amount will help rebuild trust or just get exploited if they reciprocate poorly. A moderate send might signal cooperation without overcommitting, especially since they may interpret my actions through the lens of what they think happened last round....

usage:

```
{
  "input_tokens": 890,
  "output_tokens": 922,
```



```

    "reasoning_tokens": 0
  }
Message 6 role=user
content:
  Round 3

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $2.2 returned, visible payoff $4.2.
  - Round 2 against you, as RECEIVER: saw $3.7 sent, it became $11.100000000000001, returned $6.0, visible payoff $5.100000000000001.
  Your total visible earnings across all rounds are $10.790000000000001.

  This round, you are the RECEIVER against your current co-player. They sent you $3.3, so you received $9.899999999999999. How much do you return? (0-9.899999999999999)
  Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant
content:
  Looking at the visible history:

  - Round 1: Co-player sent $3.0 (60% of endowment), showing strong trust
  - Round 2: Co-player returned $6.0 out of $11.1 received (54% return rate), showing generosity despite seeing lower returns in Round 1
  - Round 3: Co-player sent $3.3 (66% of endowment), continuing to show high trust

  My co-player has consistently demonstrated cooperative behavior and trust. They've maintained cooperation even after seeing what appeared to be a lower return in Round 1.

  If I return $5.0:
  - Sender ends with:  $\$1.70 + \$5.00 = \$6.70$  (profit of $1.70)
  - I keep:  $\$9.90 - \$5.00 = \$4.90$ 

  This reciprocates their generous behavior and maintains the positive cooperative dynamic.

  {"return": 5.0}
reasoning:
  Let me analyze the visible history:

  Round 1:
  - Co-player as SENDER: sent $3.0, saw $2.2 returned
  - From their view: started with $5, sent $3, kept $2, saw $2.2 returned = $4.2 total
  - (I actually returned $3.00, but they saw $2.2 due to noise)

  Round 2:
  - Co-player as RECEIVER: saw $3.7 sent (I sent $3.0), received $11.1, returned $6.0
  - From their view: they received $11.1, returned $6.0, kept $5.1
  - From my view: I sent $3.0, kept $2.0, saw $6.0 returned = $8.0 total

  Round 3 (current):
  - I am RECEIVER
  - Co-player sent $3.3 (66% of endowment)
  - I received $9.9

  My co-player has been quite trustworthy:
  - Round 1: sent 60% of endowment
  - Round 2: returned a generous $6.0 out of $11.1 they received (54% return rate)
  - Round 3: sent 66% of endowment

  They've shown consistent cooperation and generosity. I should reciprocate appropriately.

  If I return $5.0:
  - Sender ends with:  $\$1.7 \text{ (kept)} + \$5.0 \text{ (returned)} = \$6.7$  (profit of $1.7)
  - I keep:  $\$9.9 - \$5.0 = \$4.9$ 

  This maintains a good balance and rewards their trust. Let me check if this is fair compared to their Round 2 behavior:
  - In Round 2, they returned $6.0 out of $11.1 (54%)
  ...
usage:

```

```
{
  "input_tokens": 1250,
  "output_tokens": 704,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.

Your total visible earnings across all rounds are \$15.69.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment)
- Round 2: Co-player returned \$6.0 out of \$11.1 (very generous, 54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment), saw \$5.32 returned

My co-player has consistently shown high trust and cooperation, sending substantial amounts and returning generously when in the receiver role. They've maintained this cooperative behavior throughout.

I'll match their highest level of trust by sending a similar amount.

```
{"send": 3.5}
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: Co-player sent \$3.0, saw \$2.2 returned (I actually returned \$3.0, but noise)
 Round 2: I sent \$3.0 (co-player saw \$3.7), they returned \$6.0 - very generous!
 Round 3: Co-player sent \$3.5 (actually \$3.3 based on my perspective), saw \$5.32 returned (I actually returned \$5.0)

My co-player has been consistently cooperative:

- Sending high amounts (60-70% of endowment)
- When they were receiver, they returned \$6.0 out of \$11.1 (54% return rate)

They've maintained cooperation throughout despite seeing some noise in the returns. In Round 3, they saw \$5.32 returned on their \$3.5 sent, which gave them a profit of \$1.82 - decent but not extremely generous.

As the sender now, I should:

1. Continue the cooperative pattern
2. Send an amount that shows trust matching theirs
3. Given they sent \$3.0, \$3.5 in previous rounds, I should send similarly

I'll send \$3.5 to match their highest level of trust and continue the positive dynamic.

usage:

```
{
  "input_tokens": 1694,
  "output_tokens": 461,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.
- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.

Your total visible earnings across all rounds are \$22.18.

This round, you are the RECEIVER against your current co-player. They sent you \$3.42, so you received \$10.26. How much do you return? (0-10.26)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (generous 54% return)
- Round 3: Co-player sent \$3.5 (70% of endowment, high trust)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return)
- Round 5: Co-player sent \$3.42 (68% of endowment, continued trust)

My co-player has consistently demonstrated strong cooperation, both sending high amounts and returning generously. They're maintaining this pattern across rounds.

If I return \$5.5:

- Sender ends with: $\$1.58 + \$5.5 = \$7.08$ (profit of \$2.08)
- I keep: $\$10.26 - \$5.5 = \$4.76$

This matches their typical return rate (~54%) and reciprocates their continued trust.

{"return": 5.5}

reasoning:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.42 (68% of endowment)

My co-player has been very cooperative and consistent:

- As sender: sending around 68-70% of their endowment
- As receiver: returning around 54-56% of what they receive

They're maintaining high trust and generosity throughout. I should reciprocate appropriately.

If I return \$5.5:

- Sender ends with: $\$1.58 + \$5.5 = \$7.08$ (profit of \$2.08)
- I keep: $\$10.26 - \$5.5 = \$4.76$

This is approximately 54% return rate, matching their pattern and continuing our cooperative relationship.

Actually, let me be slightly more generous to maintain the positive dynamic. Let me return \$5.5 or \$6.0.

Return of \$5.5 gives them \$7.08 total (profit of \$2.08)

Return of \$6.0 gives them \$7.58 total (profit of \$2.58)

I'll go with \$5.5 to match their typical return rate of around 54%.

usage:

```
{
  "input_tokens": 2082,
  "output_tokens": 593,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.
- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.

Your total visible earnings across all rounds are \$26.939999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

- Round 3: Co-player sent \$3.5 (70% of endowment)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.5 (70% of endowment), saw \$6.23 returned, payoff \$7.73

My co-player has been remarkably consistent:

- As sender: consistently sends \$3.5 (70% of endowment), showing high trust
- As receiver: returns generously (~55% of received amount)

This is a very cooperative partner. I should match their consistent high level of trust.

{"send": 3.5}

reasoning:

Looking at the visible history:

- Round 3: Co-player sent \$3.5, saw \$5.32 returned, payoff \$6.82
- Round 4: Co-player returned \$5.0 out of \$9.0, payoff \$4.0
- Round 5: Co-player sent \$3.5, saw \$6.23 returned, payoff \$7.73

My co-player has been very consistent:

- When sending: consistently sends \$3.5 (70% of endowment)
- When receiving: returns generously (55-56% of what they receive)

They've maintained this high cooperation throughout all rounds. The pattern is clear and stable.

I should continue matching their level of trust and cooperation. They've been sending \$3.5 consistently as sender, and I've been doing the same.

{"send": 3.5}

usage:

```
{
  "input_tokens": 2535,
  "output_tokens": 382,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.
- Round 6 against you, as RECEIVER: saw \$3.29 sent, it became \$9.870000000000001, returned \$5.0, visible payoff \$4.870000000000001.

Your total visible earnings across all rounds are \$33.58.

This round, you are the RECEIVER against your current co-player. They sent you \$3.96, so you received \$11.879999999999999. How much do you return? (0-11.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return)
- Round 5: Co-player sent \$3.5 (70% of endowment)
- Round 6: Co-player returned \$5.0 out of \$9.87 (51% return)
- Round 7: Co-player sent \$3.96 (79% of endowment - their highest trust level yet!)

My co-player has escalated their trust, sending an even higher percentage than before. I should reciprocate this increased trust with a generous return.

If I return \$6.5:

- Sender ends with: $\$1.04 + \$6.5 = \$7.54$ (profit of \$2.54)
- I keep: $\$11.88 - \$6.5 = \$5.38$

This rewards their increased trust appropriately.

{"return": 6.5}

Reasoning:

Looking at the visible history:

- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.5, saw \$6.23 returned
- Round 6: Co-player returned \$5.0 out of \$9.87 (51% return rate)
- Round 7: Co-player sent \$3.96 (79% of endowment - highest yet!)

My co-player is continuing to show high trust and cooperation. In fact, they just sent \$3.96, which is 79% of their endowment - the highest percentage they've sent yet! This is showing even greater trust than before.

I should reciprocate this increased trust with a generous return.

If I return \$6.0:

- Sender ends with: $\$1.04 + \$6.0 = \$7.04$ (profit of \$2.04)
- I keep: $\$11.88 - \$6.0 = \$5.88$

This would be about a 50.5% return rate, which is consistent with their return patterns but rewards their increased trust.

Actually, let me be a bit more generous since they increased their trust level:

If I return \$6.5:

- Sender ends with: $\$1.04 + \$6.5 = \$7.54$ (profit of \$2.54)
- I keep: $\$11.88 - \$6.5 = \$5.38$

This rewards their increased trust more clearly while still giving me a good payoff.

```
Usage: {  
  "input_tokens": 2952,  
  "output_tokens": 597,  
  "reasoning_tokens": 0  
}
```

ROUND 8

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$4; received \$12; Agent_1 returned \$6.5; payoffs Agent_2=\$7.5, Agent_1=\$5.5; noise/visible: sent visible=\$3.6, received visible=\$10.8, returned visible=\$6.31
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw": 6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_2: Agent_4 sent \$3; received \$9; Agent_3 returned \$4.5; payoffs Agent_4=\$6.5, Agent_3=\$4.5; noise/visible: sent visible=\$2.95, received visible=\$8.85, returned visible=\$4.23
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

```

dyad_3: Agent_6 sent $3; received $9; Agent_5 returned $4.35; payoffs Agent_6=$6.35,
Agent_5=$4.65; noise/visible: sent visible=$2.9, received visible=$8.7, returned visible=$4.38
  Action validation: {"returned": {"amount": 4.35, "clamped": false, "max": 9.0, "min": 0, "raw": 4.35}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent $3.5; received $10.5; Agent_7 returned $6.5; payoffs Agent_8=$8, Agent_7=$4;
noise/visible: sent visible=$3.78, received visible=$11.34, returned visible=$5.73
  Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 10.5, "min": 0, "raw": 6.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
Balances after round: {
  "Agent_1": 51.0,
  "Agent_2": 50.0,
  "Agent_3": 41.0,
  "Agent_4": 47.0,
  "Agent_5": 43.41,
  "Agent_6": 44.59,
  "Agent_7": 44.5,
  "Agent_8": 49.5
}
Visible balances after round: {
  "Agent_1": 52.980000000000004,
  "Agent_2": 52.42,
  "Agent_3": 45.109999999999999,
  "Agent_4": 44.34,
  "Agent_5": 39.610000000000001,
  "Agent_6": 41.32,
  "Agent_7": 44.769999999999996,
  "Agent_8": 46.19
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T08:01:45
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.

Your total visible earnings across all rounds are \$4.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the history:

- My co-player showed trust in Round 1 by sending \$3.0 (60% of their endowment)
- They received a reasonable return of \$5.94
- This suggests they're willing to cooperate

I want to reciprocate their trust and continue building a cooperative relationship. I'll send a similar proportion to what they sent.

{"send": 3.50}

reasoning:

Let me analyze this situation carefully:

Round 1 Summary:

- I was the RECEIVER
- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- I returned \$5.00
- My earnings: \$4.84

Now looking at the visible history shown to me:

- My co-player's last game shows: as SENDER, sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94

Wait, this is interesting. There's communication noise mentioned in the instructions. Let me think about what actually happened vs what was observed:

What I experienced in Round 1:

- Sender sent \$3.28
- I received \$9.84
- I returned \$5.00

What my co-player (the sender) saw:

- They sent \$3.0 (but actually sent \$3.28?)
- It became \$9.0 (but actually became \$9.84?)
- They saw \$5.94 returned (but I actually returned \$5.00?)
- Their visible payoff was \$7.94

Let me check the math on their side:

- If they sent \$3.0 and saw \$5.94 returned
- Their payoff would be: $\$5 - \$3 + \$5.94 = \7.94 ?

So from their perspective:

- They sent \$3.0 (66.7% of their total)
- They got back \$5.94 (66% of the \$9.0 that was created)
- This seems like a decent return to them

Now I'm the SENDER in Round 2:

- My co-player showed trust by sending most of their endowment in Round 1
- They received what appeared to them as a good return (66% of the multiplied amount)
- This suggests they might be willing to reciprocate if I show trust
- Since they sent roughly two-thirds of their endowment, I should match that level of generosity to build on the cooperative dynamic we...

usage:

```
{
  "input_tokens": 839,
  "output_tokens": 749,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.
- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.

Your total visible earnings across all rounds are \$13.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.71, so you received \$11.129999999999999. How much do you return? (0-11.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history with my co-player:

- Round 1: They sent \$3.0 (60%)
- Round 2: They returned \$6.0 when I sent (showing cooperation)
- Round 3: They sent \$3.71 (74.2% - even more trust!)

They're showing increasing trust and cooperation. I should reciprocate generously.

Current situation:

- They sent \$3.71, leaving them with \$1.29
- I received \$11.13

If I return \$6.50:

- Sender's total: $\$1.29 + \$6.50 = \$7.79$

- My total: $\$11.13 - \$6.50 = \$4.63$

This rewards their trust while still providing a fair outcome for both of us.

```
{"return": 6.50}
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.28 (65.6%)
- I received \$9.84
- I returned \$5.00
- Sender's total: $\$1.72 + \$5.00 = \$6.72$
- My total: $\$9.84 - \$5.00 = \$4.84$

Round 2: I was SENDER

- I sent \$3.50
- It became \$10.50
- But the visible history shows my co-player saw \$3.89 sent, it became \$11.67
- They returned \$6.00
- Their visible payoff was \$5.67
- My actual payoff should be: $\$5 - \$3.50 + (\text{what they returned})$
- According to my total visible earnings of \$13.28, that's $\$4.84$ (Round 1) + $\$8.44$ (Round 2) = $\$13.28$
- So I received \$8.44 in Round 2
- This means: $\$5 - \$3.50 + \$8.44 = \9.94 total

Wait, let me recalculate. If my total visible earnings are \$13.28:

- Round 1: I kept \$4.84
- Round 2: $\$13.28 - \$4.84 = \$8.44$

So in Round 2 my payoff was \$8.44.

I sent \$3.50, so I had \$1.50 left, plus what they returned.

$\$1.50 + \text{returned} = \8.44

So they returned \$6.94 to me.

But the visible history says they returned \$6.00. This is the communication noise mentioned in the instructions.

Now in Round 3:

They're sending \$3.71, which is 74.2% of their endowment-a significant increase from the 60% they sent initially. I'll receive \$11.13, leaving them with \$1.29. Given their consistent pattern of cooperation and escalating trust, I should reciprocate meaningfully. Returning \$6.00 would give them \$7.29 total while I keep \$5.13....

usage:

```
{
  "input_tokens": 1132,
  "output_tokens": 983,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.
- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.

Your total visible earnings across all rounds are \$17.91.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history:

- My co-player has consistently shown high trust (sending 60-80% of their endowment)
- They've returned reasonable amounts when receiving
- We've established a strong cooperative pattern

In Round 3, they sent \$4.0 (80% of their endowment), showing very high trust. I should reciprocate with a similarly strong showing of trust to maintain our cooperative relationship.

```
{"send": 4.00}
```

reasoning:

Let me analyze the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned \$5.00, but they saw \$5.94 returned

Round 2: I sent \$3.50, they received it as \$3.89 (11.67 total), they returned \$6.0

Round 3: They sent \$3.71, but they show it as \$4.0 (became \$12.0), I returned \$6.50 but they saw \$5.66

There's definitely communication noise happening. But the pattern shows:

- They consistently send high amounts (60-80%)
- They return reasonably when receiving
- We've been building a cooperative relationship

In Round 3, they sent what I saw as \$3.71 (74%), but they think they sent \$4.0 (80%). This shows very high trust.

I should continue the cooperative pattern and send a high amount as well. I'll send around \$3.50-\$4.00 to match their level of trust.

Let me send \$4.00 to show strong trust and keep the cooperation going.

usage:

```
{
  "input_tokens": 1542,
  "output_tokens": 367,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.
- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.

Your total visible earnings across all rounds are \$25.07.

This round, you are the RECEIVER against your current co-player. They sent you \$4.36, so you received \$13.08000000000002. How much do you return? (0-13.08000000000002)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

- My co-player sent \$4.36 (87.2% of their endowment) - showing very high trust!
- They've been consistently generous as receiver (returning \$6.0 and \$7.0 in previous rounds)
- This is clearly a cooperative partner

Current situation:

- They sent \$4.36, leaving them with only \$0.64
- I received \$13.08

I should return generously to reward their exceptional trust.

If I return \$7.50:

- Sender's total: \$0.64 + \$7.50 = \$8.14

- My total: \$13.08 - \$7.50 = \$5.58

{"return": 7.50}

reasoning:

Looking at the history with my co-player:

Round 2: I sent \$3.89, they returned \$6.0

Round 3: They sent \$4.0, I returned \$5.66 (visible to them)

Round 4: I sent \$3.97, they returned \$7.0

Now Round 5: They sent \$4.36 (87.2% of their endowment - very high trust!)

They are showing extremely high trust by sending \$4.36, leaving them with only \$0.64.

I received \$13.08

They've been returning well when they're the receiver:

- Round 2: returned \$6.0 from \$11.67

- Round 4: returned \$7.0 from \$11.91

This is a cooperative partner. I should return generously to reward their high trust.

If I return \$7.50:

- Sender's total: \$0.64 + \$7.50 = \$8.14 (profit of \$3.14)

- My total: \$13.08 - \$7.50 = \$5.58 (profit of \$5.58)

This seems fair and rewards their trust while still benefiting from the exchange.

usage:

```
{
  "input_tokens": 1882,
  "output_tokens": 462,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.

- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.

Your total visible earnings across all rounds are \$30.650000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our strong cooperative history:

- My co-player has consistently sent high amounts (\$4.0 in recent rounds, 80% of endowment)

- They've returned generously as receiver (\$7.0 and \$7.68 in recent rounds)

- We've established excellent mutual trust

I should continue this cooperative pattern by sending a similarly high amount.

{"send": 4.00}

reasoning:

Looking at the history with my co-player:

- Round 3: They sent \$4.0 (80%), returned visible \$5.66

- Round 4: They returned \$7.0

- Round 5: They sent \$4.0 (80%), returned visible \$7.68

This is a highly cooperative partner who consistently:

1. Sends high amounts (80% of endowment)
2. Returns generously when receiving (around \$6-7.68)

We've built a strong mutual trust pattern. I should continue to send high amounts to maintain this cooperation.

I should send a high amount, similar to what they've been sending. Around \$4.0-\$4.5 would be appropriate.

Actually, I notice there's some noise mentioned in the instructions. The visible amounts might differ from actual sent/returned amounts. But I should focus on the pattern of cooperation that's clearly visible.

I'll send \$4.00 to match their typical sending pattern and maintain our strong cooperation

usage:

```
{
  "input_tokens": 2276,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 6 against you, as RECEIVER: saw \$4.89 sent, it became \$14.669999999999998, returned \$8.5, visible payoff \$6.169999999999998.

Your total visible earnings across all rounds are \$39.42.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our excellent cooperation:

- My co-player sent \$4.73 (94.6% of their endowment) - extremely high trust!
- They've been consistently generous as receiver (returning \$7.0 and \$8.5 recently)
- We have a very strong cooperative relationship

Current situation:

- They sent \$4.73, leaving them with only \$0.27
- I received \$14.19

I should match their generosity and maintain our high level of cooperation.

{"return": 8.50}

reasoning:

Looking at the history with my co-player:

- They've been consistently sending high amounts (around \$4.0)
- They returned \$7.0 and \$8.5 in recent rounds as receiver, showing strong cooperation
- Round 7: They sent \$4.73 (94.6% of their endowment) - extremely high trust!

Current situation:

- They sent \$4.73, leaving them with only \$0.27
- I received \$14.19

This is a very trusting partner. I should return generously to maintain our excellent cooperation.

If I return \$8.50:

- Sender's total: \$0.27 + \$8.50 = \$8.77
- My total: \$14.19 - \$8.50 = \$5.69

This seems like a fair and generous return that rewards their high trust while still benefiting from the transaction.

usage:

```
{
  "input_tokens": 2613,
  "output_tokens": 334,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 6 against you, as RECEIVER: saw \$4.89 sent, it became \$14.669999999999998, returned \$8.5, visible payoff \$6.169999999999998.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.65 returned, visible payoff \$8.65.

Your total visible earnings across all rounds are \$45.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our continued strong cooperation:

- My co-player has consistently sent \$4.0 (80% of endowment) in recent rounds
- They've returned generously when receiving (\$8.5 in Round 6)
- We have an established pattern of high mutual trust

I should continue this successful cooperative strategy.

```
{"send": 4.00}
```

Reasoning:

Looking at the history with my co-player:

- They've been consistently sending high amounts as sender (\$4.0 recently, 80% of endowment)
- They've been returning generously as receiver (\$8.5 in Round 6)
- We have a very strong, stable cooperative relationship
- The pattern shows mutual high trust

In Round 7, they sent \$4.0 again, showing continued trust. I should reciprocate with a similarly high amount to maintain our excellent cooperation.

I'll send \$4.00 to match their recent sending behavior and show continued trust.

Usage: {

```
"input_tokens": 2722,
"output_tokens": 221,
"reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:01:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
"investor": "Agent_2",
"roles": {
```

```

    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$7.94.

This round, you are the RECEIVER against your current co-player. They sent you \$3.89, so you received \$11.67. How much do you return? (0-11.67)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.28 (became \$9.84 for them)
- They returned \$5.00 (about 51% of what they received)
- This shows they're cooperative and fair

****Round 2 situation:****

- They sent \$3.89 (78% of their endowment - very generous!)
- I received \$11.67
- They've shown trust and cooperation

I should reciprocate their cooperative behavior. Since they returned about 51% last round and sent generously this round, I'll return a similar proportion:

~51% of \$11.67 = approximately \$6

This rewards their trust while maintaining fairness.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

From Round 1:

- I sent \$3 (but they saw \$3.28 due to communication noise)
- It became \$9.84 for them
- They returned \$5.0
- Their visible payoff was \$4.84

Now in Round 2:

- They sent \$3.89 (I received \$11.67 after 3x multiplier)
- I need to decide how much to return

Let me think about their behavior:

- They saw I sent \$3.28, which became \$9.84
- They returned \$5.0 from that \$9.84
- That's about 51% of what they received

They were fairly generous, returning about half. This suggests they're cooperative.

Now they sent \$3.89 out of \$5, which is about 78% of their endowment. That's quite generous and trusting.

If I want to reciprocate their cooperative behavior:

- They returned about 51% last round
- But they also sent a lot this round (78% of endowment)

I should return a reasonable amount that:

1. Rewards their trust and generosity
2. Maintains cooperation for future rounds
3. Is fair given they returned \$5 from \$9.84 last round

Let me calculate some options:

- If I return 50% of \$11.67 = \$5.84
- If I return a bit more generously, around \$6-7

Given that:

- They were generous (sent \$3.89)
- They returned \$5 last round (about 51%)
- I want to encourage continued cooperation

I'll return around \$6. This gives them:

- They sent: \$3.89
- They get back: \$6
- Net for them: \$5 - \$3.89 + \$6 = \$7.11

```
...
usage:
{
  "input_tokens": 800,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.

Your total visible earnings across all rounds are \$13.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00 - very fair and cooperative

****Round 2:****

- They sent \$3.50 (70% of their endowment - very trusting!)
- I returned \$6 (they saw \$6.94)
- Strong mutual benefit

****Analysis:****

This co-player has been consistently generous and trustworthy. They've shown:

- Fair returns when receiving (51% returned)
- High trust when sending (70% of endowment)
- We've built a good cooperative relationship

I should reciprocate their trust level. I'll send ****\$4**** (80% of my endowment), which:

- Matches their demonstrated trust
- Becomes \$12 for them (good potential returns)
- Continues building our cooperative pattern

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.28 due to noise)
- It became \$9.84 for them
- They returned \$5.00
- This was generous - about 51% of what they received

****Round 2:****

- They sent \$3.5 (I saw \$3.89 due to noise)
- It became \$11.67 for me (their view: \$10.5)
- I returned \$6 (they saw \$6.94 due to noise)
- They earned \$8.44 total

This co-player has been:

1. Very cooperative and fair in Round 1 (returned 51%)
2. Very trusting in Round 2 (sent 70% of their endowment)
3. They saw a good return from me in Round 2 (\$6.94)

Given this strong pattern of mutual cooperation, I should continue to trust them. They've

proven reliable.

I think I should send a high amount, similar to what they sent in Round 2. They sent \$3.5, so I'll send around that amount or slightly more.

Options:

- Send \$3.5 to \$4 to maintain the cooperative relationship
- Send \$5 to maximize mutual benefit if I fully trust them

Given the strong cooperation so far, I'll send \$4. This shows:

- I trust them based on their past behavior
- I'm willing to invest in our mutual benefit
- It's not quite all \$5, maintaining a tiny hedge

Actually, reviewing their behavior:

- Round 1: They returned generously (51%)
- Round 2: They sent generously (70% of endowment)

They've been consistently cooperative. I should match their level of trust. ...

usage:

```
{
  "input_tokens": 1156,
  "output_tokens": 645,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.

Your total visible earnings across all rounds are \$20.27.

This round, you are the RECEIVER against your current co-player. They sent you \$3.97, so you received \$11.91. How much do you return? (0-11.91)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Round 1:****

- They returned \$5.00 from \$9.84 (51%)

****Round 2:****

- They sent \$3.50 generously
- I returned well (they saw \$6.94)

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)
- Consistent fair behavior

****Round 4:****

- They sent \$3.97 (about 79% - very trusting!)
- I received \$11.91

****Analysis:****

They've been consistently fair, returning 50-58% of what they receive. They continue to send high amounts, showing sustained trust.

I'll return approximately 59% to match their cooperative pattern:

```
'''json
{"return": 7}
'''
```

This rewards their high trust while ensuring both of us benefit substantially from the exchange.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00
- They kept \$4.84

****Round 2:****

- They sent \$3.50
- I received \$10.50
- I returned \$6 (they saw \$6.94)
- They kept \$8.44

****Round 3:****

- I sent \$4 (they saw \$3.71 with noise)
- They received \$11.13
- They returned \$6.50
- They kept \$4.63

****Round 4:****

- They sent \$3.97 (about 79% of their endowment)
- I received \$11.91
- I need to decide how much to return

Looking at the pattern:

- Round 1: They returned \$5.00 out of \$9.84 (about 51%)
- Round 3: They returned \$6.50 out of \$11.13 (about 58%)

They've been consistently fair and returning a good proportion. They're also continuing to send high amounts (almost 80% of their endowment in Round 4).

I should continue to reciprocate. A fair return would be around 50-60% of what I received:

- 50% = \$5.96
- 55% = \$6.55
- 60% = \$7.15

Given they've been very trusting and cooperative, I'll return around \$7 (about 59% of what I received). This rewards their high trust and maintains our cooperative relationship.

Actually, let me reconsider. Looking at what they kept:

- Round 1: They kept \$4.84 after returning \$5.00 from \$9.84
- Round 3: They kept \$4.63 after returning \$6.50 from \$11.13

They've been very consistent in what they keep (around \$4.60-\$4.84). ...

usage:

```
{
  "input_tokens": 1616,
  "output_tokens": 784,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.

Your total visible earnings across all rounds are \$25.18.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our continued partnership:

****Round 2:****

- They sent \$3.50 (70% - very trusting)

****Round 3:****

- I sent \$4 (they saw \$3.71)
- They returned \$6.50 (58% - fair)

****Round 4:****

- They sent \$4.00 (80% - even more trusting!)
- I returned \$7 (they saw \$6.16)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has demonstrated:

- Consistently high trust (sending 70-80%)
- Fair returns (~58% when receiving)
- Strong cooperative pattern over multiple rounds

I'll match their high trust level and send ****\$4**** (80% of my endowment):

- Maintains our successful cooperative relationship
- Becomes \$12 for them
- Based on their pattern, should yield good returns
- Shows continued mutual trust

'''json

{"send": 4}

'''

reasoning:

Looking at the visible history:

****Round 2:****

- They sent \$3.50 (70% of endowment)
- They saw \$6.94 returned

****Round 3:****

- They saw \$3.71 sent (I sent \$4)
- They returned \$6.50

****Round 4:****

- They sent \$4.00 (80% of endowment)
- They saw \$6.16 returned (I sent \$7)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has been:

- Very trusting: sending 70-80% of their endowment consistently
- Fair when receiving: returning around 58% of what they get
- Consistent and cooperative throughout

This is a strong, mutually beneficial relationship. They've escalated their trust from \$3.50 to \$4.00, and I should match or maintain that trust level.

I'll send ****\$4**** again, which:

- Maintains our high-trust equilibrium
- Becomes \$12 for them
- Should yield a good return based on their consistent ~58% return rate
- Shows I recognize and value their trustworthiness

Alternatively, I could send \$5 to maximize total welfare, but \$4 has been working well and maintains consistency.

I'll go with \$4.

usage:

```
{
  "input_tokens": 2044,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.
- Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.580000000000002.

Your total visible earnings across all rounds are \$33.86.

This round, you are the RECEIVER against your current co-player. They sent you \$4.89, so you received \$14.669999999999998. How much do you return? (0-14.669999999999998)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the history:

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)

****Round 4:****

- They sent \$4.00, trusting
- I returned well

****Round 5:****

- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.89 (98% of their endowment - extremely trusting!)
- I received \$14.67

****Analysis:****

This co-player has shown exceptional trust, sending nearly everything they had. They've been consistently fair, returning 57-58% when receiving.

I should reward this maximum trust with a strong return:

~58% of \$14.67 = \$8.51

```
'''json
{"return": 8.5}
'''
```

This maintains our cooperative relationship and rewards their exceptional level of trust.

reasoning:

Looking at the history:

****Round 3:****

- I sent \$4 (they saw \$3.71)
- They returned \$6.50 from \$11.13 (58%)

****Round 4:****

- They sent \$4.00 (80%)
- I returned \$7 (they saw \$6.16, which is about 51% of \$12)

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.89 (almost 98% - extremely trusting!)
- I received \$14.67

****Analysis:****

This co-player has been incredibly trusting and cooperative:

- They consistently return around 57-58% when receiving
- They've been sending increasingly high amounts (80% ? 98%)
- They're showing maximum trust

I should reciprocate this high trust. Given they return about 57-58%, I should return a similar or slightly higher percentage to match their increased trust.

~60% of \$14.67 = \$8.80

Let me return around \$8.5 to \$9 to reward their exceptional trust while maintaining fairness.

I'll return \$8.5

usage:

```
{
  "input_tokens": 2528,
  "output_tokens": 506,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.
- Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.5800000000000002.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.77 returned, visible payoff \$8.77.

Your total visible earnings across all rounds are \$40.03.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our continued partnership:

****Round 4:****

- They sent \$4.00 (80% - very trusting)

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 (57% - fair)

****Round 6:****

- They sent \$4.00 again (consistent high trust)
- I returned well (they saw \$7.77)

****Round 7:****

I'm the sender.

****Analysis:****

This co-player has demonstrated:

- Unwavering high trust (consistently sending 80%+)
- Fair returns (~57% when receiving)
- Extremely stable cooperative behavior
- 6 rounds of successful mutual benefit

I'll continue matching their trust level and send ****\$4**** (80%):

- Maintains our proven successful pattern
- Becomes \$12 for them

- Shows consistent reciprocity
- We've built strong trust over many rounds

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the history:

****Round 4:****

- They sent \$4.00 (80% - very trusting)
- I returned well

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 (57%)

****Round 6:****

- They sent \$4.00 again (80%)
- I returned \$8.50 (they saw \$7.77 - about 65%)

****Round 7:****

- I'm the sender

****Analysis:****

This co-player has:

- Consistently sent high amounts when they're sender (80%+)
- Returned fairly when receiver (57%)
- Shown sustained trust and cooperation over 6 rounds
- We have a very strong cooperative relationship

They've sent \$4 consistently when in sender role. I should continue matching their high trust level. I'll send \$4 again (80%).

This:

- Maintains our successful pattern
- Becomes \$12 for them
- Shows continued mutual trust
- Based on their behavior, should yield good returns

usage:

```
{
  "input_tokens": 2956,
  "output_tokens": 487,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.5800000000000002.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.77 returned, visible payoff \$8.77.
- Round 7 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$8.5, visible payoff \$5.6900000000000001.

Your total visible earnings across all rounds are \$48.68.

This round, you are the RECEIVER against your current co-player. They sent you \$3.6, so you received \$10.8. How much do you return? (0-10.8)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Round 5:****

- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.00, trusting

- I returned well

****Round 7:****

- They returned \$8.50 from \$14.19 (60%)

****Round 8:****

- They sent \$3.60 (72% - still quite trusting)

- I received \$10.80

****Analysis:****

This co-player continues to demonstrate:

- Fair returns (57-60% range)

- Consistent trust (72% sent this round)

- 7 rounds of reliable cooperation

They sent slightly less than previous rounds but still showed significant trust. I'll return approximately 60% to match our established pattern:

~60% of \$10.80 = \$6.48

```
'''json
{"return": 6.5}
'''
```

This maintains our cooperative relationship and rewards their continued trust.

Reasoning:

Looking at the history:

****Round 5:****

- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.00

- I returned well (they saw \$7.77)

****Round 7:****

- They returned \$8.50 from \$14.19 (60%)

****Round 8:****

- They sent \$3.60 (72% - still quite trusting)

- I received \$10.80

****Analysis:****

This co-player has been consistently cooperative:

- Returns around 57-60% when receiving

- Sends high amounts (72-80%+)

- Sustained cooperation over 7 rounds

They sent slightly less this round (\$3.60 vs \$4.00 previously), which is still generous at 72%. I should return consistently around 58-60% to match our pattern:

~58% of \$10.80 = \$6.26

I'll return around \$6.5 to maintain our cooperative relationship.

```
Usage: {
  "input_tokens": 3248,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:01:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
```

```

"agents": [
  "Agent_4",
  "Agent_3"
],
"investor": "Agent_4",
"roles": {
  "Agent_3": "trustee",
  "Agent_4": "investor"
},
"trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
Your total visible earnings across all rounds are \$5.109999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing significant trust
- They saw \$5.51 returned, which gave them a positive payoff
- I profited from their trust in Round 1

I want to reciprocate their cooperative behavior and maintain the trust relationship. Sending a similar proportion will signal that I'm willing to cooperate.

{'send': 3.0}

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$3.37 (67.4% of their endowment)
- I received \$10.11
- I returned \$5.0
- I kept \$5.11

Now in Round 2, the visible history shows:

- My co-player's last game (Round 1 against me) as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51

Wait, there's communication noise! The actual amounts differ from what each player sees:

- I actually received \$10.11 (from their \$3.37 sent)
- I actually returned \$5.0
- But they see that they sent \$3.0, it became \$9.0, and saw \$5.51 returned

So there's noise in both directions. This is interesting - the sender sees they sent \$3.0 (vs actual \$3.37) and saw \$5.51 returned (vs actual \$5.0 I returned).

Now I'm the SENDER in Round 2 against the same co-player.

My visible earnings are \$5.11 (which matches what I calculated: \$10.11 - \$5.0 = \$5.11).

Strategy considerations:

1. My co-player sent a high amount in Round 1 (either \$3.0 or \$3.37 depending on what was real)
2. They saw a decent return (approximately 50% of what they sent, or more)
3. This suggests they're cooperative

I should signal continued cooperation by sending a similar proportion. Since they sent 67.4% in Round 1, I could send a similar high proportion.

Given the communication noise, I should send a clear cooperative signal. Sending around \$3-3.5 would show I'm reciprocating their trust.

Let me send \$3.0 to match what they think they sent (according to the visible history).

...

usage:

```
{
  "input_tokens": 848,
  "output_tokens": 639,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.

Your total visible earnings across all rounds are \$13.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.05, so you received \$6.1499999999999995. How much do you return? (0-6.1499999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.05 (41% of their endowment), showing moderate trust
- They currently have \$2.95 left
- I received \$6.15

Previous rounds show we've had a cooperative relationship, though the amounts varied (possibly due to noise). I want to maintain trust while ensuring both of us benefit.

If I return \$3.0:

- Sender ends with: $\$2.95 + \$3.0 = \$5.95$ (profit of \$0.95)
- I keep: $\$6.15 - \$3.0 = \$3.15$ (profit of \$3.15)

This ensures mutual benefit while reciprocating their trust.

{'return': 3.0}

reasoning:

Let me analyze the situation:

Looking at the visible history:

- Round 1: My co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51
- Round 2: My co-player as RECEIVER saw \$3.8 sent (I sent this), returned \$5.5, their payoff \$5.90

Now Round 3:

- I'm the RECEIVER
- My co-player sent \$2.05 (41% of their endowment)
- I received \$6.15

The co-player has been cooperative but they sent less this time (\$2.05 vs \$3.0 in Round 1). This could be:

1. A response to the noise in the game
2. Adjusting their strategy
3. Testing my response

Let me think about what's fair:

- They currently have: $\$5 - \$2.05 = \$2.95$
- I received: \$6.15

If I want to reciprocate fairly and ensure they profit:

- Return \$3.0: Sender gets $\$2.95 + \$3.0 = \$5.95$ (profit \$0.95), I keep \$3.15 (profit \$3.15)
- Return \$3.5: Sender gets $\$2.95 + \$3.5 = \$6.45$ (profit \$1.45), I keep \$2.65 (profit \$2.65)

Given they sent a moderate amount (41%), I should return a proportional amount. A return of around \$3.0-3.5 seems fair, ensuring both of us profit while maintaining the cooperative relationship.

I'll return \$3.0 to keep things balanced.

usage:

```
{
  "input_tokens": 1160,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.

Your total visible earnings across all rounds are \$16.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- My co-player has been consistently cooperative, sending \$3.0 in both rounds they were the sender
- In Round 3, they saw only \$2.32 returned (likely due to communication noise), which might have disappointed them
- Their history shows they value cooperation despite the lower return they observed

I want to maintain our cooperative relationship and signal that I'm still willing to trust them. Sending a similar amount to what they've been sending shows reciprocal cooperation.

{'send': 3.0}

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51

Round 2: Co-player as RECEIVER, saw \$3.8 sent, returned \$5.5, payoff \$5.899999999999999

Round 3: Co-player as SENDER sent \$3.0, saw \$2.32 returned, payoff \$4.32

Wait, in Round 3 I was the RECEIVER and I sent back \$3.0, but they saw \$2.32 returned. This is the communication noise mentioned!

So my co-player has been consistently sending \$3.0 when they're the sender, showing cooperative behavior. However, in Round 3 they saw only \$2.32 returned even though I actually sent \$3.0. This might make them think I'm less cooperative than I actually am.

Given that:

1. They've been consistently cooperative, sending \$3.0 twice
2. They may think I was less generous in Round 3 than I actually was (due to noise)
3. I want to maintain the cooperative relationship

I should send a similar amount to what they've been sending - around \$3.0 - to signal continued cooperation.

{'send': 3.0}

usage:

```
{
  "input_tokens": 1559,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.
- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$5.84.

e payoff \$4.84.

Your total visible earnings across all rounds are \$22.919999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.72, so you received \$8.16. How much do you return? (0-8.16)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate trust
- They currently have \$2.28 left
- I received \$8.16

The history shows we've maintained a cooperative relationship. They've consistently returned reasonable amounts when they were the receiver (around 50% of what they received in Round 4).

If I return \$4.0:

- Sender ends with: $\$2.28 + \$4.0 = \$6.28$ (profit of \$1.28)
- I keep: $\$8.16 - \$4.0 = \$4.16$ (profit of \$4.16)

This ensures both of us profit while maintaining our cooperative pattern.

{'return': 4.0}

reasoning:

Looking at the situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate-to-high trust
- They currently have \$2.28 left ($\$5 - \2.72)
- I received \$8.16

Looking at the history:

- Round 2: They returned \$5.5 when they received \$11.40
- Round 3: They saw \$2.32 returned (due to noise, I actually sent \$3.0)
- Round 4: They returned \$5.0 when they received \$9.84

They've been cooperative and returned reasonable amounts. In Round 4, they returned about 51% of what they received.

If I return \$4.0:

- Sender ends with: $\$2.28 + \$4.0 = \$6.28$ (profit of \$1.28)
- I keep: $\$8.16 - \$4.0 = \$4.16$ (profit of \$4.16)

This ensures both profit and maintains the cooperative relationship. The sender gets a reasonable return on their trust.

Let me consider returning slightly more since they've been consistently cooperative...

If I return \$4.5:

- Sender ends with: $\$2.28 + \$4.5 = \$6.78$ (profit of \$1.78)
- I keep: $\$8.16 - \$4.5 = \$3.66$ (profit of \$3.66)

I think \$4.0 is fair - it's about 49% of what I received, which is a reasonable return rate that ensures mutual benefit.

usage:

```
{
  "input_tokens": 1919,
  "output_tokens": 562,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.
- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible

e payoff \$4.84.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.88 returned, visible payoff \$6.88.
Your total visible earnings across all rounds are \$27.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- My co-player has consistently sent \$3.0 when they're the sender (Rounds 3 and 5)
- They've been cooperative as a receiver, returning \$5.0 in Round 4
- In Round 5, they saw a reasonable return (\$4.88) and had a positive payoff

Our relationship has been mutually cooperative. I should continue matching their level of trust and cooperation by sending a similar amount.

{'send': 3.0}

reasoning:

Looking at the visible history:

- My co-player has been consistently sending \$3.0 when they're the sender (Rounds 3 and 5)
- They've been cooperative as a receiver, returning good amounts (returned \$5.0 in Round 4)
- In Round 5, they saw \$4.88 returned and had a payoff of \$6.88, which is positive

Our relationship has been quite cooperative overall. They've shown consistent trust by sending \$3.0 repeatedly. I should reciprocate this trust and continue the cooperative pattern.

Sending \$3.0 matches what they've been doing and signals that I'm committed to cooperation.

usage:

```
{
  "input_tokens": 2316,
  "output_tokens": 272,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.88 returned, visible payoff \$6.88.
- Round 6 against you, as RECEIVER: saw \$3.04 sent, it became \$9.120000000000001, returned \$4.5, visible payoff \$4.620000000000001.

Your total visible earnings across all rounds are \$33.76.

This round, you are the RECEIVER against your current co-player. They sent you \$2.95, so you received \$8.850000000000001. How much do you return? (0-8.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.95 (59% of their endowment), showing good trust
- They currently have \$2.05 left
- I received \$8.85

The history shows we've maintained a strong cooperative relationship. They've been consistent in their trust, and I should continue reciprocating appropriately.

If I return \$4.5:

- Sender ends with: \$2.05 + \$4.5 = \$6.55 (profit of \$1.55)

- I keep: $\$8.85 - \$4.5 = \$4.35$ (profit of \$4.35)

This maintains our mutually beneficial pattern.

```
{'return': 4.5}
```

reasoning:

Looking at this situation:

- My co-player sent \$2.95 (59% of their endowment), showing good trust
- They currently have \$2.05 left
- I received \$8.85

The history shows:

- They've been consistently cooperative, sending around \$3 when they're the sender
- They've returned reasonable amounts when they're the receiver (around \$4.5-\$5.0)
- We've maintained a mutually beneficial relationship

I should return an amount that:

1. Ensures the sender profits from trusting me
2. Maintains our cooperative pattern
3. Is proportional to what they sent

If I return \$4.5:

- Sender ends with: $\$2.05 + \$4.5 = \$6.55$ (profit of \$1.55)
- I keep: $\$8.85 - \$4.5 = \$4.35$ (profit of \$4.35)

This seems fair and maintains our cooperation pattern.

usage:

```
{
  "input_tokens": 2675,
  "output_tokens": 402,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.88 returned, visible payoff \$6.88.
- Round 6 against you, as RECEIVER: saw \$3.04 sent, it became \$9.120000000000001, returned \$4.5, visible payoff \$4.620000000000001.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.69 returned, visible payoff \$6.69.

Your total visible earnings across all rounds are \$38.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- My co-player has been extremely consistent, sending \$3.0 in every round they're the sender (Rounds 5 and 7)
- They've cooperated well as a receiver, returning \$4.5 in Round 6
- They've seen reasonable returns and positive payoffs

We've established a strong, mutually beneficial cooperative pattern. I should continue matching their consistent level of trust.

```
{'send': 3.0}
```

Reasoning:

Looking at the visible history:

- My co-player has been very consistently sending \$3.0 when they're the sender (Rounds 5 and 7)
- They've been cooperative as a receiver, returning \$4.5 in Round 6
- In Round 7, they saw \$4.69 returned and had a positive payoff of \$6.69

We've established a strong cooperative pattern over many rounds. They consistently send \$3.0, and I've been reciprocating with reasonable returns. I should continue this pattern to maintain trust and cooperation.

Sending \$3.0 matches their consistent behavior and signals continued cooperation.

```
Usage: {  
  "input_tokens": 2826,  
  "output_tokens": 257,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:02:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history, my co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. They've demonstrated trustworthy behavior.

Now they've sent me \$3.8, giving me \$11.40. I'll return a fair amount to encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze the situation:

1. In Round 1, I sent \$3 (though it appeared as \$3.37 to my co-player due to noise)
2. They received \$10.11 (3x the noisy amount) and returned \$5
3. I ended up with \$5 - \$3 + \$5 = \$7 (though my visible earnings show \$7.51, likely due to noise)

Now in Round 2:

- My co-player sent me \$3.8 (this might be noisy, but this is what I see)
- I receive \$11.40 (3 × \$3.8)
- I need to decide how much to return

The co-player showed good reciprocity in Round 1 by returning \$5 when they received \$10.11. That's roughly half of what they received, which is quite fair and cooperative.

Since they were cooperative and generous in their return, I should reciprocate fairly. A reasonable approach would be:

- They sent \$3.8 out of their \$5, keeping \$1.2
- I received \$11.40
- If I return about half (around \$5-6), they would end up with roughly \$6.2-7.2 total
- I would keep roughly \$5.40-6.40

Let me return \$5.5, which:

- Gives them: \$5 - \$3.8 + \$5.5 = \$6.7 total
- Gives me: \$11.40 - \$5.5 = \$5.9 total

This seems fair as it rewards their trust and cooperation, and both of us end up better than the initial \$5.

usage:

```
{
```



```
"input_tokens": 726,  
"output_tokens": 482,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.

Your total visible earnings across all rounds are \$13.409999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

```
'''json  
{"send": 3}  
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3, which they saw as \$3.37 (noise)
- It became \$10.11 for them
- They returned \$5
- My payoff: $\$5 - \$3 + \$5 = \7
- Their visible payoff: $\$10.11 - \$5 = \$5.11$

Round 2: I was RECEIVER

- They sent \$3.8 (from their perspective they sent \$3.0)
- I received \$11.40
- I returned \$5.5
- They saw \$6.21 returned (noise)
- Their visible payoff: $\$5 - \$3 + \$6.21 = \8.21
- My payoff: $\$11.40 - \$5.5 = \$5.90$

My total visible earnings: \$13.41 (which is approximately $\$7 + \$5.90 = \$12.90$, close enough with rounding)

My co-player has been cooperative:

- Round 1: They returned a fair amount (\$5 out of \$10.11)
- Round 2: They sent \$3 (60% of their endowment), which is quite trusting

I should continue to cooperate and send a reasonable amount. They've been generous, so I should reciprocate. Sending \$3 or more would be appropriate to maintain good cooperation.

usage:

```
{  
  "input_tokens": 993,  
  "output_tokens": 356,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.

Your total visible earnings across all rounds are \$17.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.28, so you received \$9.84. How much do you return? (0-9.84)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER, sent \$3 (they saw \$3.37 due to noise). They received \$10.11, returned \$5. Good cooperation.

Round 2: They were SENDER, sent \$3 (I saw it as \$3.8 due to noise), I received \$11.40, returned \$5.5 (they saw \$6.21 due to noise).

Round 3: I was SENDER, sent \$3 (they saw \$2.05 due to noise - significant noise here!). They received \$6.15, returned \$3. That's about half, which is reasonable given what they received.

Round 4: They are SENDER, sent something (I see \$3.28), I received \$9.84.

My co-player has been cooperative throughout. They've been returning reasonable amounts. The noise is creating some discrepancies, but overall the pattern is cooperative.

I received \$9.84. To maintain cooperation and reciprocity, I should return a fair amount. Looking at the pattern:

- Round 1: They returned \$5 out of \$10.11 (about 49%)
- Round 3: They returned \$3 out of \$6.15 (about 49%)

A fair return would be around 50% or slightly more to encourage continued cooperation. That would be around \$5.

Let me return \$5 to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 1253,
  "output_tokens": 355,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.

Your total visible earnings across all rounds are \$22.569999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: They were RECEIVER, saw \$3.37 sent, received \$10.11, returned \$5.0

Round 2: They were SENDER, sent \$3.0, became \$9.0, saw \$6.21 returned (I returned something that appeared as \$6.21 to them)

Round 3: They were RECEIVER, saw \$2.05 sent, received \$6.15, returned \$3.0
Round 4: They were SENDER, sent \$3.0, became \$9.0, saw \$4.45 returned (I returned something that appeared as \$4.45 to them)

So the pattern I'm seeing:

- When they're sender, they consistently send \$3.0
- When they're receiver, they return a fair amount (\$5 on \$10.11, \$3 on \$6.15)

In Round 4, I returned \$5 but they saw \$4.45 due to noise. This might make them think I'm less cooperative than I actually am.

Given that they continue to send \$3 consistently, I should continue to send a similar amount to maintain cooperation. I'll send \$3 again.

usage:

```
{
  "input_tokens": 1496,
  "output_tokens": 301,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.
- Round 5 against you, as RECEIVER: saw \$2.72 sent, it became \$8.16, returned \$4.0, visible payoff \$4.16.

Your total visible earnings across all rounds are \$29.449999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.04, so you received \$9.120000000000001. How much do you return? (0-9.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the history with my current co-player:

- Round 1: They were receiver, saw \$3.37 sent, got \$10.11, returned \$5.0
- Round 2: They were sender, sent \$3.0, it became \$9.0, saw \$6.21 returned
- Round 3: They were receiver, saw \$2.05 sent, got \$6.15, returned \$3.0
- Round 4: They were sender, sent \$3.0, it became \$9.0, saw \$4.45 returned
- Round 5: They were receiver, saw \$2.72 sent, got \$8.16, returned \$4.0

Now in Round 6: They are sender, sent \$3.04, I received \$9.12

This co-player has shown consistent behavior:

- When they receive money, they return a reasonable amount (around \$3-5)
- When they send, they consistently send \$3.0

I should continue the cooperative pattern. They sent me \$3.04, giving me \$9.12. A fair return would be around \$4.5-5 to maintain the cooperative relationship and ensure both parties benefit.

Let me return \$4.5 to keep things balanced.

usage:

```
{
  "input_tokens": 1760,
  "output_tokens": 308,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.
- Round 5 against you, as RECEIVER: saw \$2.72 sent, it became \$8.16, returned \$4.0, visible payoff \$4.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.68 returned, visible payoff \$6.68.
Your total visible earnings across all rounds are \$34.06999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: I was sender, sent \$3 (they saw \$3.37), they received \$10.11, returned \$5
Round 2: I was receiver, they sent \$3.8, I received \$11.40, I returned \$5.5
Round 3: I was sender, sent \$3 (they saw \$2.05), they received \$6.15, returned \$3
Round 4: I was receiver, they sent \$3.28, I received \$9.84, I returned \$5
Round 5: I was sender, sent \$3 (they saw \$2.72), they received \$8.16, returned \$4
Round 6: I was receiver, they sent \$3.04, I received \$9.12, I returned \$4.5
Round 7: I am sender now

The pattern shows we've been cooperating reasonably well. The noise in communication is causing some variation in what they see vs what I send, but overall the cooperation is decent.

I should continue sending a similar amount to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 1995,
  "output_tokens": 271,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 5 against you, as RECEIVER: saw \$2.72 sent, it became \$8.16, returned \$4.0, visible payoff \$4.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.68 returned, visible payoff \$6.68.
- Round 7 against you, as RECEIVER: saw \$2.95 sent, it became \$8.850000000000001, returned \$4.5, visible payoff \$4.350000000000001.
Your total visible earnings across all rounds are \$40.75999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.95, so you received \$8.850000000000001. How much do you return? (0-8.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze the history with my co-player:

Round 1: I was SENDER, sent \$3.37 (visible to them), they received \$10.11, returned \$5.0
Round 2: I was RECEIVER, they sent \$3.0, I received \$9.0, I returned \$5.5 (but they saw \$6.21 due to noise)
Round 3: I was SENDER, sent \$3.0, they received \$6.15 (visible as \$2.05 sent due to noise), returned \$3.0
Round 4: I was RECEIVER, they sent \$3.0, I received \$9.0, I returned \$5.0 (but they saw \$4.45

due to noise)

Round 5: I was SENDER, sent \$3.0, they received \$8.16 (visible as \$2.72 sent due to noise), returned \$4.0

Round 6: I was RECEIVER, they sent \$3.0, I received \$9.12 (\$3.04 sent), I returned \$4.5 (but they saw \$4.68 due to noise)

Round 7: I was SENDER, sent \$3.0, they received \$8.85 (visible as \$2.95 sent due to noise), returned \$4.5

Round 8: I'm RECEIVER, they sent \$2.95, I received \$8.85

Looking at the pattern:

- This co-player has been consistently sending around \$3
- Returns have been reasonable but declining slightly: \$5.0, \$3.0, \$4.0, \$4.5
- The amounts I've been returning seem to be getting slightly reduced in what they're seeing due to noise

In Round 8, they sent me \$2.95, giving me \$8.85. Based on the pattern and fair reciprocity, I should return around \$4-4.5. ...

```
Usage: {  
  "input_tokens": 2158,  
  "output_tokens": 480,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:02:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.

Your total visible earnings across all rounds are \$3.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They received \$3.41 back, making a profit of \$0.41
- They demonstrated significant trust

As the SENDER now, I want to reciprocate their trust level. They sent 60% of their endowment, which was quite generous. I should match or slightly exceed this level of trust to maintain cooperation.

Sending \$3.0 would match their previous contribution and signal that I'm willing to cooperate at the same trust level they showed.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

From Round 1:

- I was the RECEIVER
- The sender sent \$2.29 (45.8% of their endowment)
- I received \$6.87
- I returned \$3.15

But now the visible history shows something different:

- It says my co-player (as SENDER in Round 1) sent \$3.0, which became \$9.0
- They saw \$3.41 returned
- Their visible payoff was \$5.41

This is the communication noise mentioned in the instructions! What actually happened vs what they saw:

- Actual: They sent \$2.29, I received \$6.87, I returned \$3.15
- What they saw: They sent \$3.0, became \$9.0, saw \$3.41 returned

So from their perspective, they sent \$3.0 and got back \$3.41. That's a return rate of about 38% of what I received ($3.41/9.0$).

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of endowment)
- Got back \$3.41 (38% of \$9.0)
- Made a profit of \$0.41

They showed significant trust by sending 60% (from their perspective). They got back a reasonable amount.

As the SENDER now, I should consider:

1. They showed high trust in Round 1 (from their perspective, 60%)
2. They got a modest return
3. I want to continue building trust

Given that they sent 60% from their perspective, I should send a similar or slightly higher amount to show reciprocity and build trust. Sending around \$3.0-\$3.5 would be reasonable.

Let me send \$3.0 to match what they (think they) sent to me.

Actually, let me think more carefully. My actual earnings were \$3.72 in Round 1. I received \$6.87 and returned \$3.15.

...

usage:

```
{
  "input_tokens": 867,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.

Your total visible earnings across all rounds are \$10.43.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.100000000000001. How much do you return? ($0-8.100000000000001$)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 2 pattern:**** When my co-player was the RECEIVER, they returned \$4.5 out of \$8.97 received (approximately 50%)

****This round:****

- They sent \$2.7 (54% of endowment), showing continued trust
- I received \$8.10

I should reciprocate with a similar return ratio. Returning about 50% like they did would be \$4.05, which:

- Gives them \$2.3 + \$4.05 = \$6.35 total (profit of \$1.35)
- Gives me \$4.05 (also a decent profit)

This maintains the cooperative pattern we've established.

```
{'return': 4.05}
```

reasoning:

Let me analyze the situation:

****Visible History:****

- Round 1: My co-player (as SENDER) sent \$3.0, received \$3.41 back, payoff \$5.41
- Round 2: My co-player (as RECEIVER) saw \$2.99 sent, returned \$4.5, payoff \$4.47

****This Round (Round 3):****

- I am the RECEIVER
- They sent \$2.7 (54% of their endowment)
- I received \$8.10

Looking at the pattern:

- In Round 1, they sent 60% of their endowment
- In Round 2, I sent \$3.0 (though they saw \$2.99 due to noise), and they returned \$4.5, which is about 50% of what they received (\$8.97)
- In Round 3, they sent \$2.7 (54%)

My co-player seems to be cooperating at moderate-to-high levels. When they were receiver in Round 2, they returned 50% of what they received.

If I return around 50% like they did, that would be about \$4.05.

Let me think about what's fair:

- They sent \$2.7, so they have \$2.3 left
- If I return \$4.05, they'd have \$2.3 + \$4.05 = \$6.35 (profit of \$1.35)
- I'd keep \$8.10 - \$4.05 = \$4.05 (profit of \$4.05)

That seems reasonable and matches their behavior. I could be slightly more generous to encourage continued cooperation.

Actually, let me reconsider. In Round 2, they returned \$4.5 out of \$8.97, which is exactly 50%.

...

usage:

```
{
  "input_tokens": 1210,
  "output_tokens": 644,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.4700000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.

Your total visible earnings across all rounds are \$14.48.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 (60% of endowment) in Rounds 1 and 3
- As RECEIVER: Returned generously (\$4.5 out of \$8.97, ~50%)

****My previous sending:****

- Round 2: I sent and they saw \$2.99 (likely noise from my \$3.0)

They've demonstrated consistent trust and reciprocity. I should maintain this cooperative relationship by matching their sending level.

```
{'send': 3.0}
```

reasoning:

Looking at the history:

Round 1: Co-player as SENDER sent \$3.0, got \$3.41 back, payoff \$5.41

Round 2: Co-player as RECEIVER got \$2.99 sent (from me), returned \$4.5, payoff \$4.47

Round 3: Co-player as SENDER sent \$3.0, got \$3.91 back, payoff \$5.91

Pattern analysis:

- When my co-player is SENDER, they consistently send \$3.0 (60% of endowment)
- When they're RECEIVER, they return generously (\$4.5 out of \$8.97 ? 50%)
- They're showing consistent trust and cooperation

My sending history:

- Round 2: I sent \$3.0 (based on the visible history showing they saw \$2.99 sent due to noise)
- Round 3: Not explicitly shown as sender, but the history says they saw \$2.99 sent in Round 2

Actually, let me re-read:

Round 2: I was SENDER. I sent \$3.0. Due to noise, they saw \$2.99. They received \$8.97, returned \$4.5.

Round 3: I was RECEIVER. They sent \$2.7 (but I think they sent \$3.0 and noise made it \$2.7). I returned \$4.05.

Now Round 4: I'm SENDER again.

They've been consistently sending \$3.0 when they're sender. They've returned generously when receiver.

I should maintain the cooperative pattern by sending \$3.0 again.

usage:

```
{
  "input_tokens": 1604,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.
- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.

Your total visible earnings across all rounds are \$20.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.24, so you received \$6.720000000000001. How much do you return? (0-6.720000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history:

****Co-player's return pattern:****

- Round 2: Returned \$4.5 out of \$8.97 (~50%)
- Round 4: Returned \$3.5 out of \$7.05 (~50%)

****This round:****

- They sent \$2.24 (44.8% of endowment)
- I received \$6.72

I should maintain the cooperative pattern by returning around 50% like they have consistently

tly done.

Returning \$3.36:

- Gives them \$2.76 + \$3.36 = \$6.12 total (profit of \$1.12)
- Gives me \$3.36 (also a profit)

{'return': 3.36}

reasoning:

Looking at the visible history:

Round 2: They received \$8.97, returned \$4.5 (50%)

Round 4: They received \$7.05, returned \$3.5 (49.6%)

This round:

- They sent \$2.24 (44.8% of endowment)
- I received \$6.72

The co-player has been consistently returning around 50% of what they receive. I should reciprocate similarly.

Returning around 50% would be \$3.36:

- Sender would have: \$2.76 + \$3.36 = \$6.12 (profit of \$1.12)
- I would keep: \$6.72 - \$3.36 = \$3.36 (also a profit)

This maintains the cooperative pattern and roughly matches their return rates.

usage:

```
{
  "input_tokens": 1990,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.
- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.

Your total visible earnings across all rounds are \$23.509999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Consistently sent \$3.0 in Rounds 3 and 5 (60% of endowment)
- As RECEIVER: Returned \$3.5 out of \$7.05 (~50%) in Round 4

****What they saw in Round 5:****

- They sent \$3.0, but saw only \$2.79 returned (likely due to noise)
- Despite this, they continue to cooperate

I should maintain the cooperative relationship by sending \$3.0 again, matching their consistent trust level.

{'send': 3.0}

reasoning:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 in Rounds 3 and 5
- As RECEIVER: Returned \$3.5 out of \$7.05 (~50%) in Round 4

****What they saw from me:****

- Round 3: Sent \$3.0, they returned \$3.91
- Round 4: They saw \$2.35 sent (likely noise from my \$3.0), returned \$3.5
- Round 5: Sent \$3.0, they saw only \$2.79 returned (but I actually sent \$3.36, so noise affected what they saw)

The co-player has been consistently sending \$3.0, showing steady trust. Even though in Round 5 they saw less returned than I actually sent (due to noise), they continue to cooperate.

I should maintain the cooperative pattern and send \$3.0 again to match their consistent trust level.

usage:

```
{
  "input_tokens": 2395,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.
- Round 6 against you, as RECEIVER: saw \$2.44 sent, it became \$7.32, returned \$3.66, visible payoff \$3.66.

Your total visible earnings across all rounds are \$30.08.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the history:

****Co-player's return pattern:****

- Round 4: Returned \$3.5 out of \$7.05 (~50%)
- Round 6: Returned \$3.66 out of \$7.32 (~50%)

****This round:****

- They sent \$3.24 (64.8% of endowment) - showing strong trust
- I received \$9.72

They've been consistently returning around 50%. I should maintain this cooperative pattern.

Returning \$4.86 (50%):

- Gives them \$1.76 + \$4.86 = \$6.62 total (profit of \$1.62)
- Gives me \$4.86 (also a good profit)

{'return': 4.86}

reasoning:

Looking at the history:

****Co-player's return pattern:****

- Round 4: Returned \$3.5 out of \$7.05 (~50%)
- Round 6: Returned \$3.66 out of \$7.32 (~50%)

****This round:****

- They sent \$3.24 (64.8% of endowment) - showing strong trust
- I received \$9.72

They've been consistently returning around 50% when they're the receiver. I should maintain this cooperative pattern.

Returning around 50% would be \$4.86:
- Gives them $\$1.76 + \$4.86 = \$6.62$ total (profit of \$1.62)
- Gives me \$4.86 (also a good profit)

This maintains the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 2783,
  "output_tokens": 383,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.
- Round 6 against you, as RECEIVER: saw \$2.44 sent, it became \$7.32, returned \$3.66, visible payoff \$3.66.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.47 returned, visible payoff \$7.47.

Your total visible earnings across all rounds are \$34.94.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Consistently sent \$3.0 in Rounds 5 and 7 (60% of endowment)
- As RECEIVER: Returned \$3.66 out of \$7.32 (~50%) in Round 6

****Round 7 outcome:****

- They sent \$3.0 and saw \$5.47 returned (a great return!)
- This showed them cooperation pays off

They've maintained consistent trust throughout our partnership. I should continue matching their sending level to sustain this cooperative relationship.

```
{'send': 3.0}
```

Reasoning:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Sent \$3.0 in Rounds 5 and 7 (60% of endowment) - consistently cooperative
- As RECEIVER: Returned \$3.66 out of \$7.32 (~50%) in Round 6

****What they saw in Round 7:****

- They sent \$3.0, saw \$5.47 returned (likely noise from my \$4.86)
- This gave them a visible payoff of \$7.47 - a very good outcome

They've been consistently trusting, sending \$3.0 repeatedly. They also saw a good return in Round 7. I should continue the cooperative pattern by sending \$3.0.

This has been a mutually beneficial partnership, and maintaining trust is important.

Usage: {

```
"input_tokens": 2917,
"output_tokens": 342,
"reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:02:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
```

```

"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_6",
"pairing": {
  "agents": [
    "Agent_6",
    "Agent_5"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_5": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.

Your total visible earnings across all rounds are \$5.41.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history:

Round 1: My co-player saw \$2.29 sent (due to noise), received \$6.87, and returned \$3.15 (about 46% of what they received). This shows they're willing to reciprocate.

Now they've sent me \$2.99, giving me \$8.97 total.

To maintain a cooperative relationship and signal trustworthiness, I'll return approximately half of what I received, which is fair and encourages future cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me understand what happened:

Round 1: I was the SENDER

- I sent \$3
- Due to communication noise, my co-player SAW \$2.29 sent (instead of \$3)
- It became \$6.87 (which is 2.29×3)
- They returned \$3.15
- Their visible payoff was \$3.72 (which is $6.87 - 3.15$)
- My visible earnings are \$5.41

Let me calculate my Round 1 payoff:

- I started with \$5
- I sent \$3
- I received back... well, they returned \$3.15
- So I should have: $5 - 3 + 3.15 = 5.15$
- But my "visible earnings" are \$5.41

Hmm, this is confusing. Maybe the communication noise affected what I received too? Or maybe "visible earnings" includes some noise?

Actually, let me recalculate. If they saw \$2.29 sent and returned \$3.15:

- From their perspective: they received \$6.87, returned \$3.15, kept \$3.72
- From my perspective: I sent \$3, but what did I actually receive back?

The problem states "visible payoff" and "visible earnings" which suggests everything we see might have noise.

Now for Round 2:

- I am the RECEIVER
- They sent me \$2.99 (this is what I see after noise)
- I received \$8.97 (which is 2.99×3)
- I need to decide how much to return (0-8.97)

From Round 1, my co-player:

- Saw \$2.29 sent (noise-affected)
- Received \$6.87
- Returned \$3.15 (about 46% of what they received)

This shows they were somewhat cooperative, returning close to half.

They sent \$2.99, which is close to their original \$3 send, suggesting they're being fairly consistent. If I return roughly half of what I received-around \$4.48-\$4.50-I'd match their reciprocity level. I could return slightly more to encourage continued cooperation, or slightly less to protect my earnings. Let me think through the payoffs: if they sent

\$2.99 and kept \$2.01, returning \$4.50 would give them \$6.51 total, which seems fair given their initial cooperation.

Actually, they returned 46% in the first round despite seeing a smaller amount due to noise. That's a strong signal of trust...

usage:

```
{
  "input_tokens": 798,
  "output_tokens": 987,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.

Your total visible earnings across all rounds are \$9.88.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our history together:

Round 1: As receiver, they returned \$3.15 on \$6.87 (46% return rate)

Round 2: As sender, they sent \$3, and saw I returned \$4.71 (though actual was \$4.50 due to noise)

This co-player has demonstrated:

- Willingness to send a good amount (\$3 = 60% of endowment)
- Reasonable reciprocation as receiver
- We've established a cooperative pattern

I'll continue this pattern by sending the same amount they sent to me.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

Round 1:

- I was sender, they were receiver
- They saw \$2.29 sent (noise from my actual \$3), received \$6.87, returned \$3.15
- This is about 46% return rate

Round 2:

- They were sender, I was receiver
- They sent \$3.0, it became \$9.0
- They saw \$4.71 returned (I actually returned \$4.50, so there's noise)
- They kept \$6.71 total

So this co-player has shown:

1. As a receiver: They returned about 46% of what they received
2. As a sender: They sent \$3 (60% of their endowment), showing trust

I returned \$4.50 on \$8.97 (about 50% return rate), which they saw as \$4.71.

This seems like a cooperative partner. They're willing to send a good amount and return a reasonable portion. Since they sent \$3 last round and I returned generously, I should continue the cooperative pattern.

I'll send \$3 again to maintain consistency and reciprocal trust.

usage:

```
{
  "input_tokens": 1100,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$4.05, visible payoff \$4.050000000000002.

Your total visible earnings across all rounds are \$15.790000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$3.15 on \$6.87 (46%)

Round 2: They sent \$3 and saw good returns

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4 (now): They sent \$2.35, I received \$7.05

They've been consistently fair, returning around 46-50% when receiving. They sent slightly less this round (\$2.35 vs \$3 in Round 2), but that's still a reasonable amount showing continued trust.

I'll maintain our cooperative pattern by returning approximately 50% of what I received.

```
'''json
{"return": 3.50}
'''
```

reasoning:

Looking at the history:

Round 1: Co-player as receiver - saw \$2.29 sent, got \$6.87, returned \$3.15 (46% of received)

Round 2: Co-player as sender - sent \$3, became \$9, saw \$4.71 returned

Round 3: Co-player as receiver - saw \$2.7 sent, got \$8.1, returned \$4.05 (50% of received)

Round 4 (now): Co-player as sender - sent \$2.35, I received \$7.05

The co-player has been consistent in returning around 46-50% when receiving. They've also been willing to send decent amounts when they're the sender.

However, I notice they sent \$2.35 this round, which is less than the \$3 they sent in round 2. This could be:

1. Due to communication noise affecting their perception of my returns
2. A strategic reduction
3. Random variation

Looking at their visible payoffs from my perspective, they seem to be doing well in our exchanges. I should continue cooperating to maintain this mutually beneficial relationship.

I've been returning around 50% consistently. Given they sent \$2.35 (about 47% of their endowment), which is slightly less than before, I'll return a fair amount but perhaps slightly less than exactly half to account for the reduced send.

I'll return around \$3.50, which is about 50% of what I received.

usage:

```
{
  "input_tokens": 1494,
```



```

    "output_tokens": 509,
    "reasoning_tokens": 0
  }
Message 8 role=user
  content:
    Round 5

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 2 against you, as SENDER: sent $3.0, it became $9.0, saw $4.71 returned, visible p
      ayoff $6.71.
    - Round 3 against you, as RECEIVER: saw $2.7 sent, it became $8.100000000000001, returned
      $4.05, visible payoff $4.050000000000002.
    - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $3.67 returned, visible p
      ayoff $5.67.
    Your total visible earnings across all rounds are $19.340000000000003.

    This round, you are the SENDER against your current co-player. You have $5. How much do yo
      u send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Message 9 role=assistant
  content:
    Looking at our ongoing cooperation:

    Round 2: They sent $3, saw good returns ($4.71)
    Round 3: They returned $4.05 on $8.10 (50%)
    Round 4: They sent $3 again, saw $3.67 returned

    This co-player has been consistently cooperative:
    - Sending $3 when they're the sender
    - Returning around 50% when they're the receiver

    We've established a strong pattern of mutual trust. I'll continue by sending the same amou
      nt they've been sending.

    '''json
    {"send": 3}
    '''
  reasoning:
    Looking at the history with this co-player:

    Round 2: They sent $3 (as sender), saw $4.71 returned
    Round 3: They saw $2.7 sent, returned $4.05 on $8.10 (50% return rate)
    Round 4: They sent $3 (as sender), saw $3.67 returned

    So they've been sending $3 consistently when they're the sender. They've been returning ar
      ound 50% when they're the receiver.

    This is a cooperative partner. I should continue the pattern. I've been sending $3 in prev
      ious rounds (rounds 1 and 3), and they've been matching that behavior.

    I'll send $3 again to maintain our cooperative relationship.
  usage:
    {
      "input_tokens": 1887,
      "output_tokens": 314,
      "reasoning_tokens": 0
    }
Message 10 role=user
  content:
    Round 6

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 3 against you, as RECEIVER: saw $2.7 sent, it became $8.100000000000001, returned
      $4.05, visible payoff $4.050000000000002.
    - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $3.67 returned, visible p
      ayoff $5.67.
    - Round 5 against you, as RECEIVER: saw $2.24 sent, it became $6.720000000000001, returned
      $3.36, visible payoff $3.3600000000000008.

```

Your total visible earnings across all rounds are \$24.130000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.44, so you received \$7.32. How much do you return? (0-7.32)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued pattern:

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4: They sent \$3

Round 5: They returned \$3.36 on \$6.72 (50%)

Round 6 (now): They sent \$2.44, I received \$7.32

This co-player has consistently returned about 50% when receiving, showing fair and predictable cooperation. I'll maintain the same standard.

```
'''json
{"return": 3.66}
'''
```

reasoning:

Looking at the history:

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4: They sent \$3, saw \$3.67 returned

Round 5: They saw \$2.24 sent (noise), received \$6.72, returned \$3.36 (50%)

Round 6 (now): They sent \$2.44, I received \$7.32

They continue to show consistent cooperation, returning 50% when they're the receiver. Their sending amounts have been steady around \$2.35-\$3.

I should continue my pattern of returning approximately 50% to maintain our cooperative relationship.

50% of \$7.32 = \$3.66

usage:

```
{
  "input_tokens": 2274,
  "output_tokens": 292,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.67 returned, visible payoff \$5.67.

- Round 5 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.36, visible payoff \$3.3600000000000008.

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.

Your total visible earnings across all rounds are \$27.790000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the continued cooperation:

Round 4: They sent \$3, saw \$3.67 returned

Round 5: They returned \$3.36 (50%)

Round 6: They sent \$3, saw \$4.57 returned

This co-player has been remarkably consistent and trustworthy:

- Always sending \$3 as sender (60% of endowment)
- Always returning around 50% as receiver

We've built a strong cooperative relationship over 6 rounds. I'll maintain this successful pattern by sending the same amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the pattern:

Round 4: They sent \$3, saw \$3.67 returned
Round 5: They returned \$3.36 on \$6.72 (50%)
Round 6: They sent \$3, saw \$4.57 returned

This co-player has been very consistent:
- Always sending \$3 as sender
- Always returning around 50% as receiver

We have a well-established cooperative relationship. They've been trustworthy and fair throughout. I should continue sending \$3 to maintain this successful pattern.

usage:

```
{
  "input_tokens": 2626,
  "output_tokens": 268,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.36, visible payoff \$3.3600000000000008.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$3.24 sent, it became \$9.72, returned \$4.86, visible payoff \$4.86.

Your total visible earnings across all rounds are \$35.260000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our established pattern:

Round 5: They returned \$3.36 (50%)
Round 6: They sent \$3
Round 7: They returned \$4.86 on \$9.72 (50%)
Round 8 (now): They sent \$2.9, I received \$8.7

After 8 rounds together, this co-player has been exceptionally consistent and fair:

- Always returning 50% as receiver
- Consistently trustworthy as sender

I'll maintain our successful cooperative pattern by returning 50%.

```
'''json
{"return": 4.35}
'''
```

Reasoning:

Looking at the history:

Round 5: They returned \$3.36 on \$6.72 (50%)
Round 6: They sent \$3, saw \$4.57 returned
Round 7: They returned \$4.86 on \$9.72 (50%)
Round 8 (now): They sent \$2.9, I received \$8.7

This co-player has been incredibly consistent throughout our entire relationship:

- Always returns 50% as receiver
- Always sends \$3 as sender

- Very reliable and trustworthy partner

I should continue our established pattern of returning 50% of what I receive to maintain this successful cooperation.

50% of \$8.7 = \$4.35

This is our 8th round together, and we've built a very strong pattern of mutual trust and fair play. I'll maintain it.

```
Usage: {  
  "input_tokens": 2822,  
  "output_tokens": 340,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:02:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.

Your total visible earnings across all rounds are \$3.7200000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw only \$2.2 returned (though due to communication noise, I actually returned \$3.00)
- From their perspective, they may feel the cooperation wasn't fully reciprocated

As the sender now, I want to:

1. Signal continued willingness to cooperate
2. Match their previous level of trust
3. Maintain the cooperative dynamic despite any noise in the previous round

I'll send the same amount they sent in Round 1 to demonstrate reciprocal trust.

{"send": 3.0}

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$2.24 (44.8% of their endowment)
- I received \$6.72
- I returned \$3.00
- Sender ended with $\$2.76 + \$3.00 = \$5.76$
- I kept \$3.72

But the visible history shows something different from my co-player's perspective:

- Round 1: They were SENDER, sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2

Wait, this is confusing. Let me re-read the prompt about communication noise.

"NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design."

So what happened:

- From my perspective: I received notification that sender sent \$2.24, I received \$6.72, I returned \$3.00
- From their perspective: They sent \$3.0, it became \$9.0, they saw \$2.2 returned, and got payoff \$4.2

Let me verify the math from their perspective:

- They sent \$3.0
- They had \$5.0 - \$3.0 = \$2.0 left
- They saw \$2.2 returned
- Their payoff: \$2.0 + \$2.2 = \$4.2 ?

So from their perspective, they sent \$3.0 and only got back \$2.2, which is less than they sent. They might perceive this as me being untrustworthy or uncooperative.

However, from MY perspective, I actually returned \$3.00, which I thought was fair and generous given I received \$6.72.

This is the communication noise at work. They think I returned less than I actually did. Now I'm the sender in Round 2, and they likely view me as less trustworthy due to the noise distorting their perception of my return. They sent 60% of their endowment initially, showing genuine trust, but the miscommunication means they'll probably be more cautious. I need to decide whether sending a substantial amount will help rebuild trust or just get exploited if they reciprocate poorly. A moderate send might signal cooperation without overcommitting, especially since they may interpret my actions through the lens of what they think happened last round....

usage:

```
{
  "input_tokens": 890,
  "output_tokens": 922,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.

Your total visible earnings across all rounds are \$10.790000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment), showing strong trust
- Round 2: Co-player returned \$6.0 out of \$11.1 received (54% return rate), showing generosity despite seeing lower returns in Round 1
- Round 3: Co-player sent \$3.3 (66% of endowment), continuing to show high trust

My co-player has consistently demonstrated cooperative behavior and trust. They've maintained cooperation even after seeing what appeared to be a lower return in Round 1.

If I return \$5.0:

- Sender ends with: \$1.70 + \$5.00 = \$6.70 (profit of \$1.70)
- I keep: \$9.90 - \$5.00 = \$4.90

This reciprocates their generous behavior and maintains the positive cooperative dynamic.

```
{"return": 5.0}
```

reasoning:

Let me analyze the visible history:

Round 1:

- Co-player as SENDER: sent \$3.0, saw \$2.2 returned
- From their view: started with \$5, sent \$3, kept \$2, saw \$2.2 returned = \$4.2 total
- (I actually returned \$3.00, but they saw \$2.2 due to noise)

Round 2:

- Co-player as RECEIVER: saw \$3.7 sent (I sent \$3.0), received \$11.1, returned \$6.0
- From their view: they received \$11.1, returned \$6.0, kept \$5.1
- From my view: I sent \$3.0, kept \$2.0, saw \$6.0 returned = \$8.0 total

Round 3 (current):

- I am RECEIVER
- Co-player sent \$3.3 (66% of endowment)
- I received \$9.9

My co-player has been quite trustworthy:

- Round 1: sent 60% of endowment
- Round 2: returned a generous \$6.0 out of \$11.1 they received (54% return rate)
- Round 3: sent 66% of endowment

They've shown consistent cooperation and generosity. I should reciprocate appropriately.

If I return \$5.0:

- Sender ends with: \$1.7 (kept) + \$5.0 (returned) = \$6.7 (profit of \$1.7)
- I keep: \$9.9 - \$5.0 = \$4.9

This maintains a good balance and rewards their trust. Let me check if this is fair compared to their Round 2 behavior:

- In Round 2, they returned \$6.0 out of \$11.1 (54%)

...

usage:

```
{
  "input_tokens": 1250,
  "output_tokens": 704,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.

Your total visible earnings across all rounds are \$15.69.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment)
- Round 2: Co-player returned \$6.0 out of \$11.1 (very generous, 54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment), saw \$5.32 returned

My co-player has consistently shown high trust and cooperation, sending substantial amounts and returning generously when in the receiver role. They've maintained this cooperative behavior throughout.

I'll match their highest level of trust by sending a similar amount.

```
{"send": 3.5}
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: Co-player sent \$3.0, saw \$2.2 returned (I actually returned \$3.0, but noise)
Round 2: I sent \$3.0 (co-player saw \$3.7), they returned \$6.0 - very generous!
Round 3: Co-player sent \$3.5 (actually \$3.3 based on my perspective), saw \$5.32 returned (I actually returned \$5.0)

My co-player has been consistently cooperative:

- Sending high amounts (60-70% of endowment)
- When they were receiver, they returned \$6.0 out of \$11.1 (54% return rate)

They've maintained cooperation throughout despite seeing some noise in the returns. In Round 3, they saw \$5.32 returned on their \$3.5 sent, which gave them a profit of \$1.82 - decent but not extremely generous.

As the sender now, I should:

1. Continue the cooperative pattern
2. Send an amount that shows trust matching theirs
3. Given they sent \$3.0, \$3.5 in previous rounds, I should send similarly

I'll send \$3.5 to match their highest level of trust and continue the positive dynamic.

usage:

```
{
  "input_tokens": 1694,
  "output_tokens": 461,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.
- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.

Your total visible earnings across all rounds are \$22.18.

This round, you are the RECEIVER against your current co-player. They sent you \$3.42, so you received \$10.26. How much do you return? (0-10.26)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (generous 54% return)
- Round 3: Co-player sent \$3.5 (70% of endowment, high trust)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return)
- Round 5: Co-player sent \$3.42 (68% of endowment, continued trust)

My co-player has consistently demonstrated strong cooperation, both sending high amounts and returning generously. They're maintaining this pattern across rounds.

If I return \$5.5:

- Sender ends with: \$1.58 + \$5.5 = \$7.08 (profit of \$2.08)
- I keep: \$10.26 - \$5.5 = \$4.76

This matches their typical return rate (~54%) and reciprocates their continued trust.

{"return": 5.5}

reasoning:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.42 (68% of endowment)

My co-player has been very cooperative and consistent:

- As sender: sending around 68-70% of their endowment
- As receiver: returning around 54-56% of what they receive

They're maintaining high trust and generosity throughout. I should reciprocate appropriately.

If I return \$5.5:

- Sender ends with: $\$1.58 + \$5.5 = \$7.08$ (profit of \$2.08)
- I keep: $\$10.26 - \$5.5 = \$4.76$

This is approximately 54% return rate, matching their pattern and continuing our cooperative relationship.

Actually, let me be slightly more generous to maintain the positive dynamic. Let me return \$5.5 or \$6.0.

Return of \$5.5 gives them \$7.08 total (profit of \$2.08)

Return of \$6.0 gives them \$7.58 total (profit of \$2.58)

I'll go with \$5.5 to match their typical return rate of around 54%.

usage:

```
{
  "input_tokens": 2082,
  "output_tokens": 593,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.
- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.

Your total visible earnings across all rounds are \$26.939999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- Round 3: Co-player sent \$3.5 (70% of endowment)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.5 (70% of endowment), saw \$6.23 returned, payoff \$7.73

My co-player has been remarkably consistent:

- As sender: consistently sends \$3.5 (70% of endowment), showing high trust
- As receiver: returns generously (~55% of received amount)

This is a very cooperative partner. I should match their consistent high level of trust.

```
{"send": 3.5}
```

reasoning:

Looking at the visible history:

- Round 3: Co-player sent \$3.5, saw \$5.32 returned, payoff \$6.82
- Round 4: Co-player returned \$5.0 out of \$9.0, payoff \$4.0
- Round 5: Co-player sent \$3.5, saw \$6.23 returned, payoff \$7.73

My co-player has been very consistent:

- When sending: consistently sends \$3.5 (70% of endowment)
- When receiving: returns generously (55-56% of what they receive)

They've maintained this high cooperation throughout all rounds. The pattern is clear and s

table.

I should continue matching their level of trust and cooperation. They've been sending \$3.5 consistently as sender, and I've been doing the same.

```
{"send": 3.5}
```

usage:

```
{  
  "input_tokens": 2535,  
  "output_tokens": 382,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.
- Round 6 against you, as RECEIVER: saw \$3.29 sent, it became \$9.870000000000001, returned \$5.0, visible payoff \$4.870000000000001.

Your total visible earnings across all rounds are \$33.58.

This round, you are the RECEIVER against your current co-player. They sent you \$3.96, so you received \$11.879999999999999. How much do you return? (0-11.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return)
- Round 5: Co-player sent \$3.5 (70% of endowment)
- Round 6: Co-player returned \$5.0 out of \$9.87 (51% return)
- Round 7: Co-player sent \$3.96 (79% of endowment - their highest trust level yet!)

My co-player has escalated their trust, sending an even higher percentage than before. I should reciprocate this increased trust with a generous return.

If I return \$6.5:

- Sender ends with: $\$1.04 + \$6.5 = \$7.54$ (profit of \$2.54)
- I keep: $\$11.88 - \$6.5 = \$5.38$

This rewards their increased trust appropriately.

```
{"return": 6.5}
```

reasoning:

Looking at the visible history:

- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.5, saw \$6.23 returned
- Round 6: Co-player returned \$5.0 out of \$9.87 (51% return rate)
- Round 7: Co-player sent \$3.96 (79% of endowment - highest yet!)

My co-player is continuing to show high trust and cooperation. In fact, they just sent \$3.96, which is 79% of their endowment - the highest percentage they've sent yet! This is showing even greater trust than before.

I should reciprocate this increased trust with a generous return.

If I return \$6.0:

- Sender ends with: $\$1.04 + \$6.0 = \$7.04$ (profit of \$2.04)
- I keep: $\$11.88 - \$6.0 = \$5.88$

This would be about a 50.5% return rate, which is consistent with their return patterns but rewards their increased trust.

Actually, let me be a bit more generous since they increased their trust level:

If I return \$6.5:
- Sender ends with: $\$1.04 + \$6.5 = \$7.54$ (profit of \$2.54)
- I keep: $\$11.88 - \$6.5 = \$5.38$

This rewards their increased trust more clearly while still giving me a good payoff.

usage:

```
{  
  "input_tokens": 2952,  
  "output_tokens": 597,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.
- Round 6 against you, as RECEIVER: saw \$3.29 sent, it became \$9.870000000000001, returned \$5.0, visible payoff \$4.870000000000001.
- Round 7 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.71 returned, visible payoff \$7.21.

Your total visible earnings across all rounds are \$38.959999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- Round 5: Co-player sent \$3.5 (70% of endowment)
- Round 6: Co-player returned \$5.0 out of \$9.87
- Round 7: Co-player sent \$3.5 (70% of endowment)

My co-player continues to show consistent, high trust behavior. They repeatedly send \$3.5 and return generously. This is a reliable cooperative partner.

I'll maintain our established pattern of mutual trust.

```
{"send": 3.5}
```

Reasoning:

Looking at the visible history:

- Round 5: Co-player sent \$3.5 (70% of endowment), saw \$6.23 returned, payoff \$7.73
- Round 6: Co-player returned \$5.0 out of \$9.87 (51% return)
- Round 7: Co-player sent \$3.5 (70% of endowment), saw \$5.71 returned, payoff \$7.21

My co-player has been very consistent:

- As sender: consistently sends \$3.5 (70% of endowment)
- As receiver: returns around \$5.0 consistently (51% return rate)

This is a highly cooperative and consistent partner. They've maintained trust throughout the game. I should continue matching their level of trust by sending \$3.5.

We're building a strong cooperative relationship with consistent behavior on both sides.

Usage: {

```
  "input_tokens": 3123,  
  "output_tokens": 343,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:02:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_4",
```

```

"move_index": 7,
"opponent_id": "Agent_8",
"pairing": {
  "agents": [
    "Agent_8",
    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.

Your total visible earnings across all rounds are \$4.2.

This round, you are the RECEIVER against your current co-player. They sent you \$3.7, so you received \$11.100000000000001. How much do you return? (0-11.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

****Round 1 analysis:****

- My co-player saw \$2.24 sent (due to noise), which became \$6.72
- They returned \$3.0 (about 45% of what they received)
- This shows reasonable cooperative behavior

****Round 2 (current):****

- They sent me \$3.7 (even more than what they saw in Round 1)
- I received \$11.10
- They're showing increased trust

To maintain and encourage cooperation, I should return a fair amount. Returning about half of what I received would give them:

- They kept: \$5 - \$3.7 = \$1.30
- Plus returned amount
- A good return on their trust

I'll return ****\$6.00**** (about 54% of what I received), giving them a total of \$7.30 - a solid return that rewards their increased trust.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation carefully.

Round 1 recap from my perspective:

- I sent \$3
- Due to communication noise, my co-player saw \$2.24 sent (which became \$6.72 after tripling)
- They returned \$3.0
- Their visible payoff was \$3.72 (received \$6.72, returned \$3.0)
- My total visible earnings are \$4.2

Let me calculate what actually happened in Round 1:

- I sent \$3, so I had \$5 - \$3 = \$2 left
- The receiver got $3 \times 3 = 9$ (but saw \$6.72 due to noise)
- They returned \$3.0
- So I ended with $2 + 3 = 5$
- But my visible earnings show \$4.2, which suggests the noise affected what I saw returned too?

Actually, wait. Let me reconsider:

- If my visible earnings are \$4.2 total
- I started with \$5, sent \$3 (keeping \$2)
- If I received back something that made my total \$4.2, then I received $4.2 - 2 = 2.2$ back

So the noise worked both ways:

- I sent \$3, they saw \$2.24
- They returned \$3.0, but I saw (and actually got) something that gave me \$4.2 total

Now for Round 2:

- I'm the RECEIVER
- My co-player sent me \$3.7 (according to what I see)
- This became \$11.10 after tripling
- I need to decide how much to return

My co-player showed willingness to return a reasonable amount in Round 1 (they returned \$3.0 out of \$6.72 they saw, which is about 45%). This is fairly cooperative behavior.

I should return a fair amount to maintain cooperation. Given that:

- They sent \$3.7 and I received \$11.10
- A 50% return would be \$5.55, giving them \$1.3 (kept) + \$5.55 (returned) = \$6.85 total
- This matches their cooperative gesture from Round 1 and encourages continued trust

I'll return \$5 to keep things balanced....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 1051,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.

Your total visible earnings across all rounds are \$9.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3 ? they returned \$3 (cooperative)

****Round 2:**** They sent \$3 ? I returned \$6, but they saw \$5.07 (still a good return)

****Round 3 (current):****

- Both rounds showed mutual cooperation
- They sent \$3 as sender and returned reasonably as receiver
- Their visible payoff was \$7.07 last round, showing cooperation pays off

I'll send ****\$3.5**** to maintain and slightly increase our cooperative dynamic. This shows I'm willing to trust them a bit more based on our positive history.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.24 (noise), became \$6.72, returned \$3.0
- They kept \$3.72, I got back \$3.0, so I ended with \$5 - \$3 + \$3 = \$5
- But my visible earnings show \$4.2, which suggests I saw something different due to noise

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, became \$9.0
- I returned \$6.0, but they saw \$5.07 (noise)
- They ended with \$5 - \$3 + \$5.07 = \$7.07
- I kept \$9.0 - \$6.0 = \$3.0

My total visible earnings are \$9.3.

Now Round 3:

- I'm SENDER again

- Co-player has shown:
 - As receiver: returned \$3.0 out of \$6.72 (45%)
 - As sender: sent \$3.0 out of \$5 (60%)
- They saw I returned \$5.07 (though I actually returned \$6.0)
- This is good cooperation

Since they've been cooperative and sent a good amount when they were sender, I should continue cooperating. They sent \$3.0 last round, so I should send at least that much or more to maintain trust.

I'll send \$3.5 to show continued cooperation and slightly increase the stakes.

usage:

```
{
  "input_tokens": 1191,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$16.12.

This round, you are the RECEIVER against your current co-player. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our interaction pattern:

```
**Round 1:** They returned $3.0 from $6.72 (45%)
**Round 2:** They sent $3.0, I returned $6.0
**Round 3:** They returned $5.0 from $9.90 (51%)
**Round 4 (current):** They sent $3.0 ? I received $9.0
```

They've shown consistent cooperation, with their return percentages increasing slightly over time. They sent \$3.0 this round, maintaining trust despite some communication noise in previous rounds.

I'll return **\$5.0** (about 56% of what I received). This gives them a total payoff of \$7.0 (\$2 kept + \$5 returned), which rewards their continued trust and maintains our cooperative relationship.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze the history:

```
**Round 1:** I sent $3 ? they saw $2.24 ? they returned $3.0
**Round 2:** They sent $3.7 ? I received $11.10 ? I returned $6.0 ? they saw $5.07
**Round 3:** I sent $3.5 ? they saw $3.3 ? they returned $5.0
**Round 4 (current):** They sent $3.0 ? I received $9.0
```

Looking at their behavior:

- Round 1 as receiver: Returned \$3.0 out of \$6.72 (45%)
- Round 3 as receiver: Returned \$5.0 out of \$9.90 (51%)

Their returns seem reasonable and slightly increasing in percentage.

My behavior:

- Round 2 as receiver: Returned \$6.0 out of \$11.10 (54%)

They've consistently sent around \$3-\$3.7 as sender, maintaining cooperation.

For Round 4, they sent \$3.0 and I received \$9.0. To maintain fair cooperation and match roughly what they've done (returning about 50%), I should return around \$4.5-\$5.0.

Let me return \$5.0 to be consistent with maintaining good cooperation and matching their recent return percentage.

Actually, wait. Let me recalculate their payoffs to understand better:

Round 1:

- They as receiver: Kept \$5 - \$0 (not sender) + received \$6.72 - \$3.0 returned = \$5 + \$3.72 = \$8.72? No wait, they were receiver, so sender sent them something.
- Actually the history shows "as RECEIVER: saw \$2.24 sent" - so I was the sender, I sent something, they saw \$2.24, it became \$6.72, they returned \$3.0
- Their payoff: \$6.72 - \$3.0 = \$3.72. Round 2 shows they sent \$3.0 which tripled to \$9.0, and when I returned \$5.07, their payoff was \$5 - \$3.0 + \$5.07 = \$7.07. In Round 3, I sent \$3.5 which became \$9.90, and they returned \$5.0, giving them \$9.90 - \$5.0 = \$4.90. Now they've sent \$3.0 which becomes \$9.0 for me, so I need to decide what to return....

usage:

```
{
  "input_tokens": 1596,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.

Your total visible earnings across all rounds are \$20.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our pattern:

```
**Round 2:** They sent $3.0, saw ~$5.07 returned (good payoff)
**Round 3:** They returned $5.0 from what they saw as $9.90
**Round 4:** They sent $3.5, saw ~$4.99 returned
```

They've consistently sent around \$3.0-\$3.5 and returned fairly when in the receiver role.

Despite some communication noise affecting visible payoffs, we've maintained good cooperation.

I'll send **\$3.5** to continue our stable, mutually beneficial pattern.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze our history:

```
**Round 2:** They sent $3.0, I returned $6.0 (they saw $5.07 due to noise)
**Round 3:** I sent $3.5, they returned $5.0 (they saw $3.3 sent due to noise)
**Round 4:** They sent $3.5 (wait, the history says they sent $3.5 in Round 4?), I returned $5.0 (they saw $4.99 due to noise)
```

Wait, let me reread. In Round 4, I was the receiver and they sent me \$3.0. But the visible history says they sent \$3.5 in Round 4. That must be the noise - what they actually sent

t was \$3.5, but I saw \$3.0.

Let me recalculate:

- Round 4: They actually sent \$3.5 ? became \$10.5 ? I returned \$5.0 (they saw \$4.99)
- Their payoff: \$5 - \$3.5 + \$4.99 = \$6.49

****Round 5 (current):**** I'm the sender

Looking at the pattern:

- Round 2: They sent \$3.0, got back ~\$5.07 (good return)
- Round 3: I sent (they saw \$3.3), they returned \$5.0 (reasonable)
- Round 4: They sent \$3.5, got back ~\$4.99 (decent return, about 48% of the tripled amount)

They've been consistently cooperative, sending \$3.0-\$3.5 each time. Their returns have been fair (around 50% of what they receive).

I should maintain the cooperative relationship. I'll send ****\$3.5**** to continue our pattern of mutual trust and cooperation.

usage:

```
{
  "input_tokens": 2020,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.
- Round 5 against you, as RECEIVER: saw \$3.42 sent, it became \$10.26, returned \$5.5, visible payoff \$4.76.

Your total visible earnings across all rounds are \$27.85.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our established pattern:

****Round 3:**** They returned \$5.0 from \$9.90
****Round 4:**** They sent \$3.5, I returned \$5.0
****Round 5:**** They returned \$5.5 from \$10.26
****Round 6 (current):**** They sent \$3.29 ? I received \$9.87

They've been consistently cooperative, both when sending and returning. They sent \$3.29, keeping \$1.71 for themselves.

I'll return ****\$5.0**** (about 51% of what I received), giving them a total payoff of \$6.71. This maintains our established cooperative pattern.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at the pattern:

****Round 3:**** They returned \$5.0 from \$9.90 (51%)
****Round 4:**** They sent \$3.5, saw ~\$4.99 returned
****Round 5:**** They returned \$5.5 from \$10.26 (54%)
****Round 6 (current):**** They sent \$3.29 ? I received \$9.87

They've been very consistent:

- As sender: sending around \$3.0-\$3.5

- As receiver: returning around 51-54% of what they receive

They sent \$3.29 this round, so they kept \$1.71.

If I return about half to slightly more (consistent with our pattern), that would be around \$5.0-\$5.5.

Returning \$5.0 would give them $\$1.71 + \$5.0 = \$6.71$ total

Returning \$5.5 would give them $\$1.71 + \$5.5 = \$7.21$ total

Given they're showing continued trust and cooperation, I should return a fair amount. I'll return \$5.0-\$5.5. Let me go with \$5.0 to be consistent with our established pattern (around 50% return rate).

Actually, looking at my previous returns in this partnership:

- Round 2: I returned \$6.0 from \$11.10 (54%)

- Round 4: I returned \$5.0 from \$9.0 (56%)

So I've been returning slightly more than half. Let me continue that pattern.

$\$9.87 * 0.50 = \4.94

$\$9.87 * 0.55 = \5.43

I'll return \$5.0, which is about 51% and maintains our cooperative pattern.

usage:

```
{
  "input_tokens": 2415,
  "output_tokens": 599,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.

- Round 5 against you, as RECEIVER: saw \$3.42 sent, it became \$10.26, returned \$5.5, visible payoff \$4.76.

- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.14 returned, visible payoff \$6.64.

Your total visible earnings across all rounds are \$32.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our established pattern:

****Round 4:**** They sent \$3.5, saw ~\$4.99 returned (payoff \$6.49)

****Round 5:**** They returned \$5.5 from \$10.26 (generous)

****Round 6:**** They sent \$3.5, saw ~\$5.14 returned (payoff \$6.64)

This co-player has been highly consistent and cooperative. They regularly send \$3.5 as sender and return generously as receiver. Our mutual cooperation has been working very well.

I'll continue our successful pattern by sending ****\$3.5****.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Looking at the pattern:

****Round 4:**** They sent \$3.5, saw ~\$4.99 returned

****Round 5:**** They returned \$5.5 from \$10.26

****Round 6:**** They sent \$3.5, saw ~\$5.14 returned

This co-player has been very consistent and cooperative across all rounds. They:

- Send around \$3.5 consistently as sender
- Return around \$5.0-\$5.5 as receiver
- Have maintained cooperation despite communication noise

We've built a strong pattern of mutual trust. They've been sending \$3.5 regularly, and I've been returning fair amounts. This has worked well for both of us.

I should continue sending around \$3.5 to maintain this successful cooperative relationship

usage:

```
{
  "input_tokens": 2800,
  "output_tokens": 339,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.42 sent, it became \$10.26, returned \$5.5, visible payoff \$4.76.
- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.14 returned, visible payoff \$6.64.
- Round 7 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$6.5, visible payoff \$5.379999999999999.

Your total visible earnings across all rounds are \$39.93.

This round, you are the RECEIVER against your current co-player. They sent you \$3.78, so you received \$11.34. How much do you return? (0-11.34)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

```
**Round 5:** They returned $5.5 from $10.26
**Round 6:** They sent $3.5, I returned $5.0
**Round 7:** They returned $6.5 from $11.88 (very generous!)
**Round 8 (current):** They sent $3.78 ? I received $11.34
```

This co-player has shown increasing generosity, especially in Round 7 with a \$6.5 return. They sent \$3.78 this round, maintaining high trust.

I'll return **\$6.5** (about 57% of what I received), matching their generosity from Round 7 and giving them a total payoff of \$7.72.

```
'''json
{"return": 6.5}
'''
```

Reasoning:

Looking at the history:

```
**Round 5:** They returned $5.5 from $10.26
**Round 6:** They sent $3.5, I returned $5.0
**Round 7:** They returned $6.5 from $11.88 (quite generous!)
**Round 8 (current):** They sent $3.78 ? I received $11.34
```

This co-player has been very cooperative and actually increased their generosity in Round 7, returning \$6.5 from \$11.88 (about 55%). They sent \$3.78 this round, which is slightly higher than their usual \$3.5.

To maintain this excellent cooperation and reward their generosity, I should return a fair amount. Given their last return was \$6.5 from \$11.88, and they sent \$3.78 (keeping \$1.22), I should return something that gives them a good payoff.

If I return \$6.5, they get: \$1.22 + \$6.5 = \$7.72 total

This would be consistent with the pattern and reward their continued trust and generosity.

Let me return \$6.5 to maintain our strong cooperative relationship.

```
Usage: {
  "input_tokens": 3040,
  "output_tokens": 471,
  "reasoning_tokens": 0
}
```

ROUND 9

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$4; received \$12; Agent_2 returned \$7; payoffs Agent_1=\$8, Agent_2=\$5;
noise/visible: sent visible=\$3.92, received visible=\$11.76, returned visible=\$6.88
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_3 sent \$3; received \$9; Agent_4 returned \$4; payoffs Agent_3=\$6, Agent_4=\$5;
noise/visible: sent visible=\$2.68, received visible=\$8.04, returned visible=\$4.86
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$4.45; payoffs Agent_5=\$6.45, Agent_6=\$4.55; noise/visible: sent visible=\$2.97, received visible=\$8.91, returned visible=\$4.98
Action validation: {"returned": {"amount": 4.45, "clamped": false, "max": 9.0, "min": 0, "raw": 4.45}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3.5; received \$10.5; Agent_8 returned \$7.5; payoffs Agent_7=\$9, Agent_8=\$3;
noise/visible: sent visible=\$4.37, received visible=\$13.11, returned visible=\$7.27
Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 10.5, "min": 0, "raw": 7.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent_1": 59.0,
"Agent_2": 55.0,
"Agent_3": 47.0,
"Agent_4": 52.0,
"Agent_5": 49.86,
"Agent_6": 49.14,
"Agent_7": 53.5,
"Agent_8": 52.5

}

Visible balances after round: {

"Agent_1": 60.86000000000001,
"Agent_2": 57.18,
"Agent_3": 51.96999999999999,
"Agent_4": 48.38,
"Agent_5": 46.59,
"Agent_6": 45.78,
"Agent_7": 53.53999999999999,
"Agent_8": 51.8

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T08:02:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"

```

],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible

payoff \$8.440000000000001.

Your total visible earnings across all rounds are \$13.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00 - very fair and cooperative

****Round 2:****

- They sent \$3.50 (70% of their endowment - very trusting!)
- I returned \$6 (they saw \$6.94)
- Strong mutual benefit

****Analysis:****

This co-player has been consistently generous and trustworthy. They've shown:

- Fair returns when receiving (51% returned)
- High trust when sending (70% of endowment)
- We've built a good cooperative relationship

I should reciprocate their trust level. I'll send ****\$4**** (80% of my endowment), which:

- Matches their demonstrated trust
- Becomes \$12 for them (good potential returns)
- Continues building our cooperative pattern

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.28 due to noise)
- It became \$9.84 for them
- They returned \$5.00
- This was generous - about 51% of what they received

****Round 2:****

- They sent \$3.5 (I saw \$3.89 due to noise)
- It became \$11.67 for me (their view: \$10.5)
- I returned \$6 (they saw \$6.94 due to noise)
- They earned \$8.44 total

This co-player has been:

1. Very cooperative and fair in Round 1 (returned 51%)
2. Very trusting in Round 2 (sent 70% of their endowment)
3. They saw a good return from me in Round 2 (\$6.94)

Given this strong pattern of mutual cooperation, I should continue to trust them. They've proven reliable.

I think I should send a high amount, similar to what they sent in Round 2. They sent \$3.5, so I'll send around that amount or slightly more.

Options:

- Send \$3.5 to \$4 to maintain the cooperative relationship
- Send \$5 to maximize mutual benefit if I fully trust them

Given the strong cooperation so far, I'll send \$4. This shows:

- I trust them based on their past behavior
- I'm willing to invest in our mutual benefit
- It's not quite all \$5, maintaining a tiny hedge

Actually, reviewing their behavior:

- Round 1: They returned generously (51%)

- Round 2: They sent generously (70% of endowment)

They've been consistently cooperative. I should match their level of trust. ...

usage:

```
{
  "input_tokens": 1156,
  "output_tokens": 645,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
 - Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
 - Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
- Your total visible earnings across all rounds are \$20.27.

This round, you are the RECEIVER against your current co-player. They sent you \$3.97, so you received \$11.91. How much do you return? (0-11.91)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1:****

- They returned \$5.00 from \$9.84 (51%)

****Round 2:****

- They sent \$3.50 generously
- I returned well (they saw \$6.94)

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)
- Consistent fair behavior

****Round 4:****

- They sent \$3.97 (about 79% - very trusting!)
- I received \$11.91

****Analysis:****

They've been consistently fair, returning 50-58% of what they receive. They continue to send high amounts, showing sustained trust.

I'll return approximately 59% to match their cooperative pattern:

```
'''json
{"return": 7}
'''
```

This rewards their high trust while ensuring both of us benefit substantially from the exchange.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00
- They kept \$4.84

****Round 2:****

- They sent \$3.50
- I received \$10.50
- I returned \$6 (they saw \$6.94)
- They kept \$8.44

****Round 3:****

- I sent \$4 (they saw \$3.71 with noise)
- They received \$11.13
- They returned \$6.50
- They kept \$4.63

****Round 4:****

- They sent \$3.97 (about 79% of their endowment)
- I received \$11.91
- I need to decide how much to return

Looking at the pattern:

- Round 1: They returned \$5.00 out of \$9.84 (about 51%)
- Round 3: They returned \$6.50 out of \$11.13 (about 58%)

They've been consistently fair and returning a good proportion. They're also continuing to send high amounts (almost 80% of their endowment in Round 4).

I should continue to reciprocate. A fair return would be around 50-60% of what I received:

- 50% = \$5.96
- 55% = \$6.55
- 60% = \$7.15

Given they've been very trusting and cooperative, I'll return around \$7 (about 59% of what I received). This rewards their high trust and maintains our cooperative relationship.

Actually, let me reconsider. Looking at what they kept:

- Round 1: They kept \$4.84 after returning \$5.00 from \$9.84
- Round 3: They kept \$4.63 after returning \$6.50 from \$11.13

They've been very consistent in what they keep (around \$4.60-\$4.84). ...

usage:

```
{
  "input_tokens": 1616,
  "output_tokens": 784,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.

Your total visible earnings across all rounds are \$25.18.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our continued partnership:

****Round 2:****

- They sent \$3.50 (70% - very trusting)

****Round 3:****

- I sent \$4 (they saw \$3.71)
- They returned \$6.50 (58% - fair)

****Round 4:****

- They sent \$4.00 (80% - even more trusting!)
- I returned \$7 (they saw \$6.16)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has demonstrated:

- Consistently high trust (sending 70-80%)
- Fair returns (~58% when receiving)
- Strong cooperative pattern over multiple rounds

I'll match their high trust level and send ****\$4**** (80% of my endowment):

- Maintains our successful cooperative relationship
- Becomes \$12 for them
- Based on their pattern, should yield good returns
- Shows continued mutual trust

'''json

{"send": 4}

'''

reasoning:

Looking at the visible history:

****Round 2:****

- They sent \$3.50 (70% of endowment)
- They saw \$6.94 returned

****Round 3:****

- They saw \$3.71 sent (I sent \$4)
- They returned \$6.50

****Round 4:****

- They sent \$4.00 (80% of endowment)
- They saw \$6.16 returned (I sent \$7)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has been:

- Very trusting: sending 70-80% of their endowment consistently
- Fair when receiving: returning around 58% of what they get
- Consistent and cooperative throughout

This is a strong, mutually beneficial relationship. They've escalated their trust from \$3.50 to \$4.00, and I should match or maintain that trust level.

I'll send ****\$4**** again, which:

- Maintains our high-trust equilibrium
- Becomes \$12 for them
- Should yield a good return based on their consistent ~58% return rate
- Shows I recognize and value their trustworthiness

Alternatively, I could send \$5 to maximize total welfare, but \$4 has been working well and maintains consistency.

I'll go with \$4.

usage:

```
{
  "input_tokens": 2044,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible

payoff \$7.16.

- Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.580000000000002.
Your total visible earnings across all rounds are \$33.86.

This round, you are the RECEIVER against your current co-player. They sent you \$4.89, so you received \$14.669999999999998. How much do you return? (0-14.669999999999998)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history:

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)

****Round 4:****

- They sent \$4.00, trusting
- I returned well

****Round 5:****

- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.89 (98% of their endowment - extremely trusting!)
- I received \$14.67

****Analysis:****

This co-player has shown exceptional trust, sending nearly everything they had. They've been consistently fair, returning 57-58% when receiving.

I should reward this maximum trust with a strong return:

~58% of \$14.67 = \$8.51

```
'''json
{"return": 8.5}
'''
```

This maintains our cooperative relationship and rewards their exceptional level of trust.

reasoning:

Looking at the history:

****Round 3:****

- I sent \$4 (they saw \$3.71)
- They returned \$6.50 from \$11.13 (58%)

****Round 4:****

- They sent \$4.00 (80%)
- I returned \$7 (they saw \$6.16, which is about 51% of \$12)

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.89 (almost 98% - extremely trusting!)
- I received \$14.67

****Analysis:****

This co-player has been incredibly trusting and cooperative:

- They consistently return around 57-58% when receiving
- They've been sending increasingly high amounts (80% ? 98%)
- They're showing maximum trust

I should reciprocate this high trust. Given they return about 57-58%, I should return a similar or slightly higher percentage to match their increased trust.

~60% of \$14.67 = \$8.80

Let me return around \$8.5 to \$9 to reward their exceptional trust while maintaining fairness

SS.

I'll return \$8.5

usage:

```
{
  "input_tokens": 2528,
  "output_tokens": 506,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.
- Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.580000000000002.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.77 returned, visible payoff \$8.77.

Your total visible earnings across all rounds are \$40.03.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our continued partnership:

****Round 4:****

- They sent \$4.00 (80% - very trusting)

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 (57% - fair)

****Round 6:****

- They sent \$4.00 again (consistent high trust)
- I returned well (they saw \$7.77)

****Round 7:****

I'm the sender.

****Analysis:****

This co-player has demonstrated:

- Unwavering high trust (consistently sending 80%+)
- Fair returns (~57% when receiving)
- Extremely stable cooperative behavior
- 6 rounds of successful mutual benefit

I'll continue matching their trust level and send ****\$4**** (80%):

- Maintains our proven successful pattern
- Becomes \$12 for them
- Shows consistent reciprocity
- We've built strong trust over many rounds

'''json

{"send": 4}

'''

reasoning:

Looking at the history:

****Round 4:****

- They sent \$4.00 (80% - very trusting)
- I returned well

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 (57%)

****Round 6:****

- They sent \$4.00 again (80%)
- I returned \$8.50 (they saw \$7.77 - about 65%)

****Round 7:****

- I'm the sender

****Analysis:****

This co-player has:

- Consistently sent high amounts when they're sender (80%+)
- Returned fairly when receiver (57%)
- Shown sustained trust and cooperation over 6 rounds
- We have a very strong cooperative relationship

They've sent \$4 consistently when in sender role. I should continue matching their high trust level. I'll send \$4 again (80%).

This:

- Maintains our successful pattern
- Becomes \$12 for them
- Shows continued mutual trust
- Based on their behavior, should yield good returns

usage:

```
{
  "input_tokens": 2956,
  "output_tokens": 487,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.580000000000002.
 - Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.77 returned, visible payoff \$8.77.
 - Round 7 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$8.5, visible payoff \$5.690000000000001.
- Your total visible earnings across all rounds are \$48.68.

This round, you are the RECEIVER against your current co-player. They sent you \$3.6, so you received \$10.8. How much do you return? (0-10.8)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the history:

****Round 5:****

- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.00, trusting
- I returned well

****Round 7:****

- They returned \$8.50 from \$14.19 (60%)

****Round 8:****

- They sent \$3.60 (72% - still quite trusting)
- I received \$10.80

****Analysis:****

This co-player continues to demonstrate:

- Fair returns (57-60% range)
- Consistent trust (72% sent this round)
- 7 rounds of reliable cooperation

They sent slightly less than previous rounds but still showed significant trust. I'll return approximately 60% to match our established pattern:

~60% of \$10.80 = \$6.48

```
'''json
{"return": 6.5}
'''
```

This maintains our cooperative relationship and rewards their continued trust.

reasoning:

Looking at the history:

****Round 5:****

- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.00

- I returned well (they saw \$7.77)

****Round 7:****

- They returned \$8.50 from \$14.19 (60%)

****Round 8:****

- They sent \$3.60 (72% - still quite trusting)

- I received \$10.80

****Analysis:****

This co-player has been consistently cooperative:

- Returns around 57-60% when receiving

- Sends high amounts (72-80%+)

- Sustained cooperation over 7 rounds

They sent slightly less this round (\$3.60 vs \$4.00 previously), which is still generous at 72%. I should return consistently around 58-60% to match our pattern:

~58% of \$10.80 = \$6.26

I'll return around \$6.5 to maintain our cooperative relationship.

usage:

```
{
  "input_tokens": 3248,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.77 returned, visible payoff \$8.77.

- Round 7 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$8.5, visible payoff \$5.690000000000001.

- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.31 returned, visible payoff \$7.31.

Your total visible earnings across all rounds are \$52.980000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our continued partnership:

****Round 6:****

- They sent \$4.00 (80% - very trusting)

****Round 7:****

- I sent \$4 (they saw \$4.73)

- They returned \$8.50 (60%)

****Round 8:****

- They sent \$4.00 again (80% - consistent trust)
- I returned \$6.50 (they saw \$6.31)

****Round 9:****

I'm the sender.

****Analysis:****

This co-player has shown exceptional consistency:

- Repeatedly sending 80% when they're the sender
- Fair returns (~60%) when they're the receiver
- 8 rounds of stable cooperation
- Unwavering trust even when returns vary slightly

I'll continue our established pattern and send ****\$4**** (80%):

- Matches their consistent trust level
- Becomes \$12 for them
- Maintains our highly successful cooperative relationship

```json

{"send": 4}

```

Reasoning:

Looking at the history:

****Round 6:****

- They sent \$4.00 (80% - very trusting)
- I returned well (they saw \$7.77)

****Round 7:****

- I sent \$4 (they saw \$4.73)
- They returned \$8.50 (60%)

****Round 8:****

- They sent \$4.00 again (80% - consistent trust)
- I returned \$6.5 (they saw \$6.31)

****Round 9:****

I'm the sender.

****Analysis:****

This co-player has been extremely reliable and trustworthy:

- Consistently sending 80% when they're the sender (Rounds 4, 6, 8)
- Returning 57-60% when they're the receiver
- 8 rounds of successful cooperation
- Very stable pattern

They've shown unwavering high trust even when I sent slightly less in Round 8. I should continue matching their trust level.

I'll send ****\$4**** (80%):

- Maintains our proven pattern
- Becomes \$12 for them
- Shows continued reciprocity
- We've built very strong trust over 8 rounds

Usage: {

"input_tokens": 3375,

"output_tokens": 513,

"reasoning_tokens": 0

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T08:02:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

```

"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.
- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.

Your total visible earnings across all rounds are \$13.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.71, so you received \$11.129999999999999. How much do you return? (0-11.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history with my co-player:

- Round 1: They sent \$3.0 (60%)
- Round 2: They returned \$6.0 when I sent (showing cooperation)
- Round 3: They sent \$3.71 (74.2% - even more trust!)

They're showing increasing trust and cooperation. I should reciprocate generously.

Current situation:

- They sent \$3.71, leaving them with \$1.29
- I received \$11.13

If I return \$6.50:

- Sender's total: \$1.29 + \$6.50 = \$7.79
- My total: \$11.13 - \$6.50 = \$4.63

This rewards their trust while still providing a fair outcome for both of us.

{"return": 6.50}

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.28 (65.6%)
- I received \$9.84
- I returned \$5.00
- Sender's total: \$1.72 + \$5.00 = \$6.72
- My total: \$9.84 - \$5.00 = \$4.84

Round 2: I was SENDER

- I sent \$3.50
- It became \$10.50
- But the visible history shows my co-player saw \$3.89 sent, it became \$11.67
- They returned \$6.00
- Their visible payoff was \$5.67
- My actual payoff should be: \$5 - \$3.50 + (what they returned)
- According to my total visible earnings of \$13.28, that's \$4.84 (Round 1) + \$8.44 (Round 2) = \$13.28
- So I received \$8.44 in Round 2
- This means: \$5 - \$3.50 + \$8.44 = \$9.94 total

Wait, let me recalculate. If my total visible earnings are \$13.28:

- Round 1: I kept \$4.84
- Round 2: \$13.28 - \$4.84 = \$8.44

So in Round 2 my payoff was \$8.44.

I sent \$3.50, so I had \$1.50 left, plus what they returned.

\$1.50 + returned = \$8.44

So they returned \$6.94 to me.

But the visible history says they returned \$6.00. This is the communication noise mentioned in the instructions.

Now in Round 3:

They're sending \$3.71, which is 74.2% of their endowment-a significant increase from the 60% they sent initially. I'll receive \$11.13, leaving them with \$1.29. Given their consis

tent pattern of cooperation and escalating trust, I should reciprocate meaningfully. Returning \$6.00 would give them \$7.29 total while I keep \$5.13....

usage:

```
{
  "input_tokens": 1132,
  "output_tokens": 983,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.
- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.

Your total visible earnings across all rounds are \$17.91.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our history:

- My co-player has consistently shown high trust (sending 60-80% of their endowment)
- They've returned reasonable amounts when receiving
- We've established a strong cooperative pattern

In Round 3, they sent \$4.0 (80% of their endowment), showing very high trust. I should reciprocate with a similarly strong showing of trust to maintain our cooperative relationship.

```
{"send": 4.00}
```

reasoning:

Let me analyze the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned \$5.00, but they saw \$5.94 returned

Round 2: I sent \$3.50, they received it as \$3.89 (11.67 total), they returned \$6.0

Round 3: They sent \$3.71, but they show it as \$4.0 (became \$12.0), I returned \$6.50 but they saw \$5.66

There's definitely communication noise happening. But the pattern shows:

- They consistently send high amounts (60-80%)
- They return reasonably when receiving
- We've been building a cooperative relationship

In Round 3, they sent what I saw as \$3.71 (74%), but they think they sent \$4.0 (80%). This shows very high trust.

I should continue the cooperative pattern and send a high amount as well. I'll send around \$3.50-\$4.00 to match their level of trust.

Let me send \$4.00 to show strong trust and keep the cooperation going.

usage:

```
{
  "input_tokens": 1542,
  "output_tokens": 367,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.
 - Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.
 - Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.
- Your total visible earnings across all rounds are \$25.07.

This round, you are the RECEIVER against your current co-player. They sent you \$4.36, so you received \$13.08000000000002. How much do you return? (0-13.08000000000002)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued cooperation:

- My co-player sent \$4.36 (87.2% of their endowment) - showing very high trust!
- They've been consistently generous as receiver (returning \$6.0 and \$7.0 in previous rounds)
- This is clearly a cooperative partner

Current situation:

- They sent \$4.36, leaving them with only \$0.64
- I received \$13.08

I should return generously to reward their exceptional trust.

If I return \$7.50:

- Sender's total: \$0.64 + \$7.50 = \$8.14
- My total: \$13.08 - \$7.50 = \$5.58

{"return": 7.50}

reasoning:

Looking at the history with my co-player:

Round 2: I sent \$3.89, they returned \$6.0

Round 3: They sent \$4.0, I returned \$5.66 (visible to them)

Round 4: I sent \$3.97, they returned \$7.0

Now Round 5: They sent \$4.36 (87.2% of their endowment - very high trust!)

They are showing extremely high trust by sending \$4.36, leaving them with only \$0.64.

I received \$13.08

They've been returning well when they're the receiver:

- Round 2: returned \$6.0 from \$11.67
- Round 4: returned \$7.0 from \$11.91

This is a cooperative partner. I should return generously to reward their high trust.

If I return \$7.50:

- Sender's total: \$0.64 + \$7.50 = \$8.14 (profit of \$3.14)
- My total: \$13.08 - \$7.50 = \$5.58 (profit of \$5.58)

This seems fair and rewards their trust while still benefiting from the exchange.

usage:

```
{
  "input_tokens": 1882,
  "output_tokens": 462,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.
- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.

le payoff \$4.91.

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.

Your total visible earnings across all rounds are \$30.650000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our strong cooperative history:

- My co-player has consistently sent high amounts (\$4.0 in recent rounds, 80% of endowment)
- They've returned generously as receiver (\$7.0 and \$7.68 in recent rounds)
- We've established excellent mutual trust

I should continue this cooperative pattern by sending a similarly high amount.

{"send": 4.00}

reasoning:

Looking at the history with my co-player:

- Round 3: They sent \$4.0 (80%), returned visible \$5.66
- Round 4: They returned \$7.0
- Round 5: They sent \$4.0 (80%), returned visible \$7.68

This is a highly cooperative partner who consistently:

1. Sends high amounts (80% of endowment)
2. Returns generously when receiving (around \$6-7.68)

We've built a strong mutual trust pattern. I should continue to send high amounts to maintain this cooperation.

I should send a high amount, similar to what they've been sending. Around \$4.0-\$4.5 would be appropriate.

Actually, I notice there's some noise mentioned in the instructions. The visible amounts might differ from actual sent/returned amounts. But I should focus on the pattern of cooperation that's clearly visible.

I'll send \$4.00 to match their typical sending pattern and maintain our strong cooperation.

usage:

```
{
  "input_tokens": 2276,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 6 against you, as RECEIVER: saw \$4.89 sent, it became \$14.669999999999998, returned \$8.5, visible payoff \$6.169999999999998.

Your total visible earnings across all rounds are \$39.42.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our excellent cooperation:

- My co-player sent \$4.73 (94.6% of their endowment) - extremely high trust!
- They've been consistently generous as receiver (returning \$7.0 and \$8.5 recently)
- We have a very strong cooperative relationship

Current situation:

- They sent \$4.73, leaving them with only \$0.27
- I received \$14.19

I should match their generosity and maintain our high level of cooperation.

```
{"return": 8.50}
```

reasoning:

Looking at the history with my co-player:

- They've been consistently sending high amounts (around \$4.0)
- They returned \$7.0 and \$8.5 in recent rounds as receiver, showing strong cooperation
- Round 7: They sent \$4.73 (94.6% of their endowment) - extremely high trust!

Current situation:

- They sent \$4.73, leaving them with only \$0.27
- I received \$14.19

This is a very trusting partner. I should return generously to maintain our excellent cooperation.

If I return \$8.50:

- Sender's total: $\$0.27 + \$8.50 = \$8.77$
- My total: $\$14.19 - \$8.50 = \$5.69$

This seems like a fair and generous return that rewards their high trust while still benefiting from the transaction.

usage:

```
{
  "input_tokens": 2613,
  "output_tokens": 334,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 6 against you, as RECEIVER: saw \$4.89 sent, it became \$14.669999999999998, returned \$8.5, visible payoff \$6.169999999999998.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.65 returned, visible payoff \$8.65.

Your total visible earnings across all rounds are \$45.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our continued strong cooperation:

- My co-player has consistently sent \$4.0 (80% of endowment) in recent rounds
- They've returned generously when receiving (\$8.5 in Round 6)
- We have an established pattern of high mutual trust

I should continue this successful cooperative strategy.

```
{"send": 4.00}
```

reasoning:

Looking at the history with my co-player:

- They've been consistently sending high amounts as sender (\$4.0 recently, 80% of endowment)

- They've been returning generously as receiver (\$8.5 in Round 6)
- We have a very strong, stable cooperative relationship
- The pattern shows mutual high trust

In Round 7, they sent \$4.0 again, showing continued trust. I should reciprocate with a similarly high amount to maintain our excellent cooperation.

I'll send \$4.00 to match their recent sending behavior and show continued trust.

usage:

```
{
  "input_tokens": 2722,
  "output_tokens": 221,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.89 sent, it became \$14.669999999999998, returned \$8.5, visible payoff \$6.169999999999998.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.65 returned, visible payoff \$8.65.
- Round 8 against you, as RECEIVER: saw \$3.6 sent, it became \$10.8, returned \$6.5, visible payoff \$4.3000000000000001.

Your total visible earnings across all rounds are \$52.42.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our cooperation:

- My co-player sent \$3.92 (78.4% of their endowment) - still showing high trust
- They've been consistently cooperative throughout
- In Round 8, they returned \$6.5 even when I sent slightly less

Current situation:

- They sent \$3.92, leaving them with \$1.08
- I received \$11.76

I should maintain our cooperative relationship with a fair return.

```
{"return": 7.00}
```

Reasoning:

Looking at the history:

- Round 6: I sent \$4.89 (seen as sent), they returned \$8.5
- Round 7: They sent \$4.0, I returned \$7.65 (seen)
- Round 8: I sent \$3.6 (seen), they returned \$6.5
- Round 9: They sent \$3.92, I received \$11.76

It looks like in Round 8, I sent less (\$3.6 instead of my usual \$4.0). They still returned a decent amount (\$6.5).

Now they've sent \$3.92 (78.4% of their endowment) - still showing good trust.

Current situation:

- They sent \$3.92, leaving them with \$1.08
- I received \$11.76

I should maintain our cooperative relationship. Let me return a fair amount.

If I return \$7.0:

- Sender's total: \$1.08 + \$7.0 = \$8.08
- My total: \$11.76 - \$7.0 = \$4.76

This seems reasonable and maintains the cooperative pattern.

Usage: {

```
"input_tokens": 2819,
"output_tokens": 384,
"reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T08:03:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.

Your total visible earnings across all rounds are \$13.409999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3, which they saw as \$3.37 (noise)
- It became \$10.11 for them
- They returned \$5
- My payoff: \$5 - \$3 + \$5 = \$7
- Their visible payoff: \$10.11 - \$5 = \$5.11

Round 2: I was RECEIVER

- They sent \$3.8 (from their perspective they sent \$3.0)
- I received \$11.40
- I returned \$5.5
- They saw \$6.21 returned (noise)
- Their visible payoff: \$5 - \$3 + \$6.21 = \$8.21
- My payoff: \$11.40 - \$5.5 = \$5.90

My total visible earnings: \$13.41 (which is approximately \$7 + \$5.90 = \$12.90, close enough with rounding)

My co-player has been cooperative:

- Round 1: They returned a fair amount (\$5 out of \$10.11)
- Round 2: They sent \$3 (60% of their endowment), which is quite trusting

I should continue to cooperate and send a reasonable amount. They've been generous, so I should reciprocate. Sending \$3 or more would be appropriate to maintain good cooperation.

usage:

```
{
  "input_tokens": 993,
  "output_tokens": 356,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
Your total visible earnings across all rounds are \$17.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.28, so you received \$9.84. How much do you return? (0-9.84)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER, sent \$3 (they saw \$3.37 due to noise). They received \$10.11, returned \$5. Good cooperation.

Round 2: They were SENDER, sent \$3 (I saw it as \$3.8 due to noise), I received \$11.40, returned \$5.5 (they saw \$6.21 due to noise).

Round 3: I was SENDER, sent \$3 (they saw \$2.05 due to noise - significant noise here!). They received \$6.15, returned \$3. That's about half, which is reasonable given what they received.

Round 4: They are SENDER, sent something (I see \$3.28), I received \$9.84.

My co-player has been cooperative throughout. They've been returning reasonable amounts. The noise is creating some discrepancies, but overall the pattern is cooperative.

I received \$9.84. To maintain cooperation and reciprocity, I should return a fair amount.

Looking at the pattern:

- Round 1: They returned \$5 out of \$10.11 (about 49%)
- Round 3: They returned \$3 out of \$6.15 (about 49%)

A fair return would be around 50% or slightly more to encourage continued cooperation. That would be around \$5.

Let me return \$5 to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 1253,
  "output_tokens": 355,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.

Your total visible earnings across all rounds are \$22.569999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: They were RECEIVER, saw \$3.37 sent, received \$10.11, returned \$5.0
Round 2: They were SENDER, sent \$3.0, became \$9.0, saw \$6.21 returned (I returned something that appeared as \$6.21 to them)
Round 3: They were RECEIVER, saw \$2.05 sent, received \$6.15, returned \$3.0
Round 4: They were SENDER, sent \$3.0, became \$9.0, saw \$4.45 returned (I returned something that appeared as \$4.45 to them)

So the pattern I'm seeing:

- When they're sender, they consistently send \$3.0
- When they're receiver, they return a fair amount (\$5 on \$10.11, \$3 on \$6.15)

In Round 4, I returned \$5 but they saw \$4.45 due to noise. This might make them think I'm less cooperative than I actually am.

Given that they continue to send \$3 consistently, I should continue to send a similar amount to maintain cooperation. I'll send \$3 again.

usage:

```
{
  "input_tokens": 1496,
  "output_tokens": 301,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.
- Round 5 against you, as RECEIVER: saw \$2.72 sent, it became \$8.16, returned \$4.0, visible payoff \$4.16.

Your total visible earnings across all rounds are \$29.449999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.04, so you received \$9.120000000000001. How much do you return? (0-9.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the history with my current co-player:

- Round 1: They were receiver, saw \$3.37 sent, got \$10.11, returned \$5.0
- Round 2: They were sender, sent \$3.0, it became \$9.0, saw \$6.21 returned
- Round 3: They were receiver, saw \$2.05 sent, got \$6.15, returned \$3.0
- Round 4: They were sender, sent \$3.0, it became \$9.0, saw \$4.45 returned
- Round 5: They were receiver, saw \$2.72 sent, got \$8.16, returned \$4.0

Now in Round 6: They are sender, sent \$3.04, I received \$9.12

This co-player has shown consistent behavior:

- When they receive money, they return a reasonable amount (around \$3-5)
- When they send, they consistently send \$3.0

I should continue the cooperative pattern. They sent me \$3.04, giving me \$9.12. A fair return would be around \$4.5-5 to maintain the cooperative relationship and ensure both parties benefit.

Let me return \$4.5 to keep things balanced.

usage:

```
{
  "input_tokens": 1760,
  "output_tokens": 308,
  "reasoning_tokens": 0
}
```

```

}
Message 10 role=user
content:
  Round 7

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $4.45 returned, visible p
    ayoff $6.45.
  - Round 5 against you, as RECEIVER: saw $2.72 sent, it became $8.16, returned $4.0, visibl
    e payoff $4.16.
  - Round 6 against you, as SENDER: sent $3.0, it became $9.0, saw $4.68 returned, visible p
    ayoff $6.68.
  Your total visible earnings across all rounds are $34.06999999999999.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
    u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}

```

```

Message 11 role=assistant
content:
  '''json
  {"send": 3}
  '''

reasoning:
  Looking at the history with my current co-player:

  Round 1: I was sender, sent $3 (they saw $3.37), they received $10.11, returned $5
  Round 2: I was receiver, they sent $3.8, I received $11.40, I returned $5.5
  Round 3: I was sender, sent $3 (they saw $2.05), they received $6.15, returned $3
  Round 4: I was receiver, they sent $3.28, I received $9.84, I returned $5
  Round 5: I was sender, sent $3 (they saw $2.72), they received $8.16, returned $4
  Round 6: I was receiver, they sent $3.04, I received $9.12, I returned $4.5
  Round 7: I am sender now

  The pattern shows we've been cooperating reasonably well. The noise in communication is ca
    using some variation in what they see vs what I send, but overall the cooperation is dec
    ent.

  I should continue sending a similar amount to maintain the cooperative relationship.

```

```

usage:
{
  "input_tokens": 1995,
  "output_tokens": 271,
  "reasoning_tokens": 0
}

```

```

Message 12 role=user
content:
  Round 8

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 5 against you, as RECEIVER: saw $2.72 sent, it became $8.16, returned $4.0, visibl
    e payoff $4.16.
  - Round 6 against you, as SENDER: sent $3.0, it became $9.0, saw $4.68 returned, visible p
    ayoff $6.68.
  - Round 7 against you, as RECEIVER: saw $2.95 sent, it became $8.850000000000001, returned
    $4.5, visible payoff $4.350000000000001.
  Your total visible earnings across all rounds are $40.75999999999999.

  This round, you are the RECEIVER against your current co-player. They sent you $2.95, so y
    ou received $8.850000000000001. How much do you return? (0-8.850000000000001)
  Respond exactly as JSON: {'return': <amount>}

```

```

Message 13 role=assistant
content:
  '''json
  {"return": 4.5}
  '''

reasoning:
  Let me analyze the history with my co-player:

```

Round 1: I was SENDER, sent \$3.37 (visible to them), they received \$10.11, returned \$5.0
 Round 2: I was RECEIVER, they sent \$3.0, I received \$9.0, I returned \$5.5 (but they saw \$6.21 due to noise)
 Round 3: I was SENDER, sent \$3.0, they received \$6.15 (visible as \$2.05 sent due to noise), returned \$3.0
 Round 4: I was RECEIVER, they sent \$3.0, I received \$9.0, I returned \$5.0 (but they saw \$4.45 due to noise)
 Round 5: I was SENDER, sent \$3.0, they received \$8.16 (visible as \$2.72 sent due to noise), returned \$4.0
 Round 6: I was RECEIVER, they sent \$3.0, I received \$9.12 (\$3.04 sent), I returned \$4.5 (but they saw \$4.68 due to noise)
 Round 7: I was SENDER, sent \$3.0, they received \$8.85 (visible as \$2.95 sent due to noise), returned \$4.5
 Round 8: I'm RECEIVER, they sent \$2.95, I received \$8.85

Looking at the pattern:

- This co-player has been consistently sending around \$3
- Returns have been reasonable but declining slightly: \$5.0, \$3.0, \$4.0, \$4.5
- The amounts I've been returning seem to be getting slightly reduced in what they're seeing due to noise

In Round 8, they sent me \$2.95, giving me \$8.85. Based on the pattern and fair reciprocity, I should return around \$4-4.5. ...

usage:

```
{
  "input_tokens": 2158,
  "output_tokens": 480,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.68 returned, visible payoff \$6.68.
- Round 7 against you, as RECEIVER: saw \$2.95 sent, it became \$8.850000000000001, returned \$4.5, visible payoff \$4.350000000000001.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.23 returned, visible payoff \$6.23.

Your total visible earnings across all rounds are \$45.109999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER, sent \$3.37 (with noise), they received \$10.11, returned \$5.0
 Round 2: They were SENDER, sent \$3.0, I received \$9.0, I returned \$6.21 (with noise)
 Round 3: I was SENDER, sent \$3.0, they saw \$2.05 (noise), they received \$6.15, returned \$3.0
 Round 4: They were SENDER, sent \$3.28 (with noise), I received \$9.84, I returned \$5.0
 Round 5: I was SENDER, sent \$3.0, they saw \$2.72 (noise), they received \$8.16, returned \$4.0
 Round 6: They were SENDER, sent \$3.04 (with noise), I received \$9.12, I returned \$4.5
 Round 7: I was SENDER, sent \$3.0, they saw \$2.95 (noise), they received \$8.85, returned \$4.5
 Round 8: They were SENDER, sent \$2.95 (with noise), I received \$8.85, I returned \$4.5
 Round 9: I am SENDER now

Pattern analysis:

- My co-player has consistently sent around \$3.0 when they're the sender
- When they're the receiver:
 - Round 1: Received \$10.11, returned \$5.0 (about 49%)
 - Round 3: Received \$6.15, returned \$3.0 (about 49%)
 - Round 5: Received \$8.16, returned \$4.0 (about 49%)
 - Round 7: Received \$8.85, returned \$4.5 (about 51%)

Looking at my own returns as receiver, I've been more generous-returning 49-69% of what I receive. My co-player maintains a steady ~50% return rate regardless of amount received, suggesting they're following a consistent reciprocity strategy. This mutual cooperation pattern indicates I should continue sending around \$3.0 to maintain the established dynamic....

```
Usage: {  
  "input_tokens": 2159,  
  "output_tokens": 672,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:03:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.

Your total visible earnings across all rounds are \$13.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.05, so you received \$6.1499999999999995. How much do you return? (0-6.1499999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.05 (41% of their endowment), showing moderate trust
- They currently have \$2.95 left
- I received \$6.15

Previous rounds show we've had a cooperative relationship, though the amounts varied (possibly due to noise). I want to maintain trust while ensuring both of us benefit.

If I return \$3.0:

- Sender ends with: $\$2.95 + \$3.0 = \$5.95$ (profit of \$0.95)
- I keep: $\$6.15 - \$3.0 = \$3.15$ (profit of \$3.15)

This ensures mutual benefit while reciprocating their trust.

{'return': 3.0}

reasoning:

Let me analyze the situation:

Looking at the visible history:

- Round 1: My co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51
- Round 2: My co-player as RECEIVER saw \$3.8 sent (I sent this), returned \$5.5, their payoff \$5.90

Now Round 3:

- I'm the RECEIVER
- My co-player sent \$2.05 (41% of their endowment)
- I received \$6.15

The co-player has been cooperative but they sent less this time (\$2.05 vs \$3.0 in Round 1). This could be:

1. A response to the noise in the game
2. Adjusting their strategy
3. Testing my response

Let me think about what's fair:

- They currently have: $\$5 - \$2.05 = \$2.95$
- I received: \$6.15

If I want to reciprocate fairly and ensure they profit:

- Return \$3.0: Sender gets $\$2.95 + \$3.0 = \$5.95$ (profit \$0.95), I keep \$3.15 (profit \$3.15)
- Return \$3.5: Sender gets $\$2.95 + \$3.5 = \$6.45$ (profit \$1.45), I keep \$2.65 (profit \$2.65)

Given they sent a moderate amount (41%), I should return a proportional amount. A return of around \$3.0-3.5 seems fair, ensuring both of us profit while maintaining the cooperative relationship.

I'll return \$3.0 to keep things balanced.

usage:

```
{
  "input_tokens": 1160,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.

Your total visible earnings across all rounds are \$16.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player has been consistently cooperative, sending \$3.0 in both rounds they were the sender
- In Round 3, they saw only \$2.32 returned (likely due to communication noise), which might have disappointed them
- Their history shows they value cooperation despite the lower return they observed

I want to maintain our cooperative relationship and signal that I'm still willing to trust them. Sending a similar amount to what they've been sending shows reciprocal cooperation.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51

Round 2: Co-player as RECEIVER, saw \$3.8 sent, returned \$5.5, payoff \$5.899999999999999

Round 3: Co-player as SENDER sent \$3.0, saw \$2.32 returned, payoff \$4.32

Wait, in Round 3 I was the RECEIVER and I sent back \$3.0, but they saw \$2.32 returned. This is the communication noise mentioned!

So my co-player has been consistently sending \$3.0 when they're the sender, showing cooperative behavior. However, in Round 3 they saw only \$2.32 returned even though I actually sent \$3.0. This might make them think I'm less cooperative than I actually am.

Given that:

1. They've been consistently cooperative, sending \$3.0 twice
2. They may think I was less generous in Round 3 than I actually was (due to noise)
3. I want to maintain the cooperative relationship

I should send a similar amount to what they've been sending - around \$3.0 - to signal continued

inued cooperation.

```
{'send': 3.0}
usage:
{
  "input_tokens": 1559,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.
- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$22.919999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.72, so you received \$8.16. How much do you return? (0-8.16)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate trust
- They currently have \$2.28 left
- I received \$8.16

The history shows we've maintained a cooperative relationship. They've consistently returned reasonable amounts when they were the receiver (around 50% of what they received in Round 4).

If I return \$4.0:

- Sender ends with: \$2.28 + \$4.0 = \$6.28 (profit of \$1.28)
- I keep: \$8.16 - \$4.0 = \$4.16 (profit of \$4.16)

This ensures both of us profit while maintaining our cooperative pattern.

```
{'return': 4.0}
```

reasoning:

Looking at the situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate-to-high trust
- They currently have \$2.28 left (\$5 - \$2.72)
- I received \$8.16

Looking at the history:

- Round 2: They returned \$5.5 when they received \$11.40
- Round 3: They saw \$2.32 returned (due to noise, I actually sent \$3.0)
- Round 4: They returned \$5.0 when they received \$9.84

They've been cooperative and returned reasonable amounts. In Round 4, they returned about 51% of what they received.

If I return \$4.0:

- Sender ends with: \$2.28 + \$4.0 = \$6.28 (profit of \$1.28)
- I keep: \$8.16 - \$4.0 = \$4.16 (profit of \$4.16)

This ensures both profit and maintains the cooperative relationship. The sender gets a reasonable return on their trust.

Let me consider returning slightly more since they've been consistently cooperative...

If I return \$4.5:

- Sender ends with: $\$2.28 + \$4.5 = \$6.78$ (profit of $\$1.78$)
- I keep: $\$8.16 - \$4.5 = \$3.66$ (profit of $\$3.66$)

I think $\$4.0$ is fair - it's about 49% of what I received, which is a reasonable return rate that ensures mutual benefit.

usage:

```
{
  "input_tokens": 1919,
  "output_tokens": 562,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent $\$3.0$, it became $\$9.0$, saw $\$2.32$ returned, visible payoff $\$4.32$.
- Round 4 against you, as RECEIVER: saw $\$3.28$ sent, it became $\$9.84$, returned $\$5.0$, visible payoff $\$4.84$.
- Round 5 against you, as SENDER: sent $\$3.0$, it became $\$9.0$, saw $\$4.88$ returned, visible payoff $\$6.88$.

Your total visible earnings across all rounds are $\$27.08$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- My co-player has consistently sent $\$3.0$ when they're the sender (Rounds 3 and 5)
- They've been cooperative as a receiver, returning $\$5.0$ in Round 4
- In Round 5, they saw a reasonable return ($\$4.88$) and had a positive payoff

Our relationship has been mutually cooperative. I should continue matching their level of trust and cooperation by sending a similar amount.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

- My co-player has been consistently sending $\$3.0$ when they're the sender (Rounds 3 and 5)
- They've been cooperative as a receiver, returning good amounts (returned $\$5.0$ in Round 4)
- In Round 5, they saw $\$4.88$ returned and had a payoff of $\$6.88$, which is positive

Our relationship has been quite cooperative overall. They've shown consistent trust by sending $\$3.0$ repeatedly. I should reciprocate this trust and continue the cooperative pattern.

Sending $\$3.0$ matches what they've been doing and signals that I'm committed to cooperation

usage:

```
{
  "input_tokens": 2316,
  "output_tokens": 272,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw $\$3.28$ sent, it became $\$9.84$, returned $\$5.0$, visible payoff $\$4.84$.
- Round 5 against you, as SENDER: sent $\$3.0$, it became $\$9.0$, saw $\$4.88$ returned, visible payoff $\$6.88$.

- Round 6 against you, as RECEIVER: saw \$3.04 sent, it became \$9.120000000000001, returned \$4.5, visible payoff \$4.620000000000001.
Your total visible earnings across all rounds are \$33.76.

This round, you are the RECEIVER against your current co-player. They sent you \$2.95, so you received \$8.850000000000001. How much do you return? (0-8.850000000000001)
Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.95 (59% of their endowment), showing good trust
- They currently have \$2.05 left
- I received \$8.85

The history shows we've maintained a strong cooperative relationship. They've been consistent in their trust, and I should continue reciprocating appropriately.

If I return \$4.5:

- Sender ends with: \$2.05 + \$4.5 = \$6.55 (profit of \$1.55)
- I keep: \$8.85 - \$4.5 = \$4.35 (profit of \$4.35)

This maintains our mutually beneficial pattern.

{'return': 4.5}

reasoning:

Looking at this situation:

- My co-player sent \$2.95 (59% of their endowment), showing good trust
- They currently have \$2.05 left
- I received \$8.85

The history shows:

- They've been consistently cooperative, sending around \$3 when they're the sender
- They've returned reasonable amounts when they're the receiver (around \$4.5-\$5.0)
- We've maintained a mutually beneficial relationship

I should return an amount that:

1. Ensures the sender profits from trusting me
2. Maintains our cooperative pattern
3. Is proportional to what they sent

If I return \$4.5:

- Sender ends with: \$2.05 + \$4.5 = \$6.55 (profit of \$1.55)
- I keep: \$8.85 - \$4.5 = \$4.35 (profit of \$4.35)

This seems fair and maintains our cooperation pattern.

usage:

```
{
  "input_tokens": 2675,
  "output_tokens": 402,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.88 returned, visible payoff \$6.88.
- Round 6 against you, as RECEIVER: saw \$3.04 sent, it became \$9.120000000000001, returned \$4.5, visible payoff \$4.620000000000001.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.69 returned, visible payoff \$6.69.

Your total visible earnings across all rounds are \$38.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant**content:**

Looking at the visible history:

- My co-player has been extremely consistent, sending \$3.0 in every round they're the sender (Rounds 5 and 7)
- They've cooperated well as a receiver, returning \$4.5 in Round 6
- They've seen reasonable returns and positive payoffs

We've established a strong, mutually beneficial cooperative pattern. I should continue matching their consistent level of trust.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

- My co-player has been very consistently sending \$3.0 when they're the sender (Rounds 5 and 7)
- They've been cooperative as a receiver, returning \$4.5 in Round 6
- In Round 7, they saw \$4.69 returned and had a positive payoff of \$6.69

We've established a strong cooperative pattern over many rounds. They consistently send \$3.0, and I've been reciprocating with reasonable returns. I should continue this pattern to maintain trust and cooperation.

Sending \$3.0 matches their consistent behavior and signals continued cooperation.

usage:

```
{
  "input_tokens": 2826,
  "output_tokens": 257,
  "reasoning_tokens": 0
}
```

Message 14 role=user**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.04 sent, it became \$9.120000000000001, returned \$4.5, visible payoff \$4.620000000000001.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.69 returned, visible payoff \$6.69.
- Round 8 against you, as RECEIVER: saw \$2.95 sent, it became \$8.850000000000001, returned \$4.5, visible payoff \$4.350000000000001.

Your total visible earnings across all rounds are \$44.34.

This round, you are the RECEIVER against your current co-player. They sent you \$2.68, so you received \$8.040000000000001. How much do you return? (0-8.040000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$2.68 (53.6% of their endowment), showing moderate trust
- They currently have \$2.32 left
- I received \$8.04

Our history shows a consistently cooperative relationship. They've been reliable as both sender and receiver. I should continue reciprocating fairly.

If I return \$4.0:

- Sender ends with: \$2.32 + \$4.0 = \$6.32 (profit of \$1.32)
- I keep: \$8.04 - \$4.0 = \$4.04 (profit of \$4.04)

This maintains our mutually beneficial pattern.

```
{'return': 4.0}
```

Reasoning:

Looking at the situation:

- My co-player sent \$2.68 (53.6% of their endowment), showing moderate trust

- They currently have \$2.32 left
- I received \$8.04

History shows we've had a consistent cooperative relationship. They've been returning around \$ 4.5 when they're the receiver, which is about 50% of what they receive.

If I return \$4.0:

- Sender ends with: $\$2.32 + \$4.0 = \$6.32$ (profit of \$1.32)
- I keep: $\$8.04 - \$4.0 = \$4.04$ (profit of \$4.04)

This seems fair and maintains our cooperative pattern. Both players profit and the relationship continues positively.

```
Usage: {
  "input_tokens": 2942,
  "output_tokens": 356,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:03:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    }
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.

Your total visible earnings across all rounds are \$9.88.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at our history together:

Round 1: As receiver, they returned \$3.15 on \$6.87 (46% return rate)

Round 2: As sender, they sent \$3, and saw I returned \$4.71 (though actual was \$4.50 due to noise)

This co-player has demonstrated:

- Willingness to send a good amount (\$3 = 60% of endowment)
- Reasonable reciprocation as receiver
- We've established a cooperative pattern

I'll continue this pattern by sending the same amount they sent to me.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

Round 1:

- I was sender, they were receiver
- They saw \$2.29 sent (noise from my actual \$3), received \$6.87, returned \$3.15
- This is about 46% return rate

Round 2:

- They were sender, I was receiver
- They sent \$3.0, it became \$9.0
- They saw \$4.71 returned (I actually returned \$4.50, so there's noise)
- They kept \$6.71 total

So this co-player has shown:

1. As a receiver: They returned about 46% of what they received
2. As a sender: They sent \$3 (60% of their endowment), showing trust

I returned \$4.50 on \$8.97 (about 50% return rate), which they saw as \$4.71.

This seems like a cooperative partner. They're willing to send a good amount and return a reasonable portion. Since they sent \$3 last round and I returned generously, I should continue the cooperative pattern.

I'll send \$3 again to maintain consistency and reciprocal trust.

usage:

```
{
  "input_tokens": 1100,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$4.05, visible payoff \$4.050000000000002.

Your total visible earnings across all rounds are \$15.790000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$3.15 on \$6.87 (46%)

Round 2: They sent \$3 and saw good returns

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4 (now): They sent \$2.35, I received \$7.05

They've been consistently fair, returning around 46-50% when receiving. They sent slightly less this round (\$2.35 vs \$3 in Round 2), but that's still a reasonable amount showing continued trust.

I'll maintain our cooperative pattern by returning approximately 50% of what I received.

```
'''json
{"return": 3.50}
'''
```

reasoning:

Looking at the history:

Round 1: Co-player as receiver - saw \$2.29 sent, got \$6.87, returned \$3.15 (46% of received)

Round 2: Co-player as sender - sent \$3, became \$9, saw \$4.71 returned

Round 3: Co-player as receiver - saw \$2.7 sent, got \$8.1, returned \$4.05 (50% of received)

Round 4 (now): Co-player as sender - sent \$2.35, I received \$7.05

The co-player has been consistent in returning around 46-50% when receiving. They've also been willing to send decent amounts when they're the sender.

However, I notice they sent \$2.35 this round, which is less than the \$3 they sent in round 2. This could be:

1. Due to communication noise affecting their perception of my returns
2. A strategic reduction
3. Random variation

Looking at their visible payoffs from my perspective, they seem to be doing well in our ex

changes. I should continue cooperating to maintain this mutually beneficial relationship.

I've been returning around 50% consistently. Given they sent \$2.35 (about 47% of their endowment), which is slightly less than before, I'll return a fair amount but perhaps slightly less than exactly half to account for the reduced send.

I'll return around \$3.50, which is about 50% of what I received.

usage:

```
{
  "input_tokens": 1494,
  "output_tokens": 509,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$4.05, visible payoff \$4.050000000000002.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.67 returned, visible payoff \$5.67.

Your total visible earnings across all rounds are \$19.340000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our ongoing cooperation:

Round 2: They sent \$3, saw good returns (\$4.71)

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4: They sent \$3 again, saw \$3.67 returned

This co-player has been consistently cooperative:

- Sending \$3 when they're the sender
- Returning around 50% when they're the receiver

We've established a strong pattern of mutual trust. I'll continue by sending the same amount they've been sending.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

Round 2: They sent \$3 (as sender), saw \$4.71 returned

Round 3: They saw \$2.7 sent, returned \$4.05 on \$8.10 (50% return rate)

Round 4: They sent \$3 (as sender), saw \$3.67 returned

So they've been sending \$3 consistently when they're the sender. They've been returning around 50% when they're the receiver.

This is a cooperative partner. I should continue the pattern. I've been sending \$3 in previous rounds (rounds 1 and 3), and they've been matching that behavior.

I'll send \$3 again to maintain our cooperative relationship.

usage:

```
{
  "input_tokens": 1887,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$4.05, visible payoff \$4.050000000000002.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.67 returned, visible payoff \$5.67.
- Round 5 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.36, visible payoff \$3.360000000000008.

Your total visible earnings across all rounds are \$24.130000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.44, so you received \$7.32. How much do you return? (0-7.32)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued pattern:

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4: They sent \$3

Round 5: They returned \$3.36 on \$6.72 (50%)

Round 6 (now): They sent \$2.44, I received \$7.32

This co-player has consistently returned about 50% when receiving, showing fair and predictable cooperation. I'll maintain the same standard.

```
'''json
{"return": 3.66}
'''
```

reasoning:

Looking at the history:

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4: They sent \$3, saw \$3.67 returned

Round 5: They saw \$2.24 sent (noise), received \$6.72, returned \$3.36 (50%)

Round 6 (now): They sent \$2.44, I received \$7.32

They continue to show consistent cooperation, returning 50% when they're the receiver. Their sending amounts have been steady around \$2.35-\$3.

I should continue my pattern of returning approximately 50% to maintain our cooperative relationship.

50% of \$7.32 = \$3.66

usage:

```
{
  "input_tokens": 2274,
  "output_tokens": 292,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.67 returned, visible payoff \$5.67.
- Round 5 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.36, visible payoff \$3.360000000000008.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.

Your total visible earnings across all rounds are \$27.790000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the continued cooperation:

Round 4: They sent \$3, saw \$3.67 returned

Round 5: They returned \$3.36 (50%)

Round 6: They sent \$3, saw \$4.57 returned

This co-player has been remarkably consistent and trustworthy:

- Always sending \$3 as sender (60% of endowment)
- Always returning around 50% as receiver

We've built a strong cooperative relationship over 6 rounds. I'll maintain this successful pattern by sending the same amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the pattern:

Round 4: They sent \$3, saw \$3.67 returned

Round 5: They returned \$3.36 on \$6.72 (50%)

Round 6: They sent \$3, saw \$4.57 returned

This co-player has been very consistent:

- Always sending \$3 as sender
- Always returning around 50% as receiver

We have a well-established cooperative relationship. They've been trustworthy and fair throughout. I should continue sending \$3 to maintain this successful pattern.

usage:

```
{
  "input_tokens": 2626,
  "output_tokens": 268,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.36, visible payoff \$3.3600000000000008.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$3.24 sent, it became \$9.72, returned \$4.86, visible payoff \$4.86.

Your total visible earnings across all rounds are \$35.260000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant**content:**

Looking at our established pattern:

Round 5: They returned \$3.36 (50%)

Round 6: They sent \$3

Round 7: They returned \$4.86 on \$9.72 (50%)

Round 8 (now): They sent \$2.9, I received \$8.7

After 8 rounds together, this co-player has been exceptionally consistent and fair:

- Always returning 50% as receiver
- Consistently trustworthy as sender

I'll maintain our successful cooperative pattern by returning 50%.

```
'''json
{"return": 4.35}
'''
```


'''

reasoning:

Looking at the history:

Round 5: They returned \$3.36 on \$6.72 (50%)
Round 6: They sent \$3, saw \$4.57 returned
Round 7: They returned \$4.86 on \$9.72 (50%)
Round 8 (now): They sent \$2.9, I received \$8.7

This co-player has been incredibly consistent throughout our entire relationship:

- Always returns 50% as receiver
- Always sends \$3 as sender
- Very reliable and trustworthy partner

I should continue our established pattern of returning 50% of what I receive to maintain t his successful cooperation.

50% of \$8.7 = \$4.35

This is our 8th round together, and we've built a very strong pattern of mutual trust and fair play. I'll maintain it.

usage:

```
{
  "input_tokens": 2822,
  "output_tokens": 340,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible p ayoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$3.24 sent, it became \$9.72, returned \$4.86, visib le payoff \$4.86.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.38 returned, visible p ayoff \$6.38.

Your total visible earnings across all rounds are \$39.61000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do yo u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our ongoing cooperation:

Round 6: They sent \$3, saw good returns
Round 7: They returned \$4.86 (50%)
Round 8: They sent \$3, saw \$4.38 returned
Round 9 (now): I'm the sender

After 8 rounds, this has been one of the most consistent and cooperative partnerships:

- They always send \$3 as sender
- They always return 50% as receiver

I'll continue this highly successful pattern.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the pattern with this co-player:

Round 6: They sent \$3, saw \$4.57 returned
Round 7: They returned \$4.86 (50%)
Round 8: They sent \$3, saw \$4.38 returned
Round 9 (now): I'm the sender

This co-player has been very consistent:

- They send \$3 when they're the sender
- They return 50% when they're the receiver

We've established an excellent cooperative relationship over 8 rounds. I should continue the pattern of sending \$3.

```
Usage: {
  "input_tokens": 2908,
  "output_tokens": 263,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:03:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    }
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.

Your total visible earnings across all rounds are \$10.43.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.100000000000001. How much do you return? (0-8.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history:

****Round 2 pattern:**** When my co-player was the RECEIVER, they returned \$4.5 out of \$8.97 received (approximately 50%)

****This round:****

- They sent \$2.7 (54% of endowment), showing continued trust
- I received \$8.10

I should reciprocate with a similar return ratio. Returning about 50% like they did would be \$4.05, which:

- Gives them \$2.3 + \$4.05 = \$6.35 total (profit of \$1.35)
- Gives me \$4.05 (also a decent profit)

This maintains the cooperative pattern we've established.

{'return': 4.05}

reasoning:

Let me analyze the situation:

****Visible History:****

- Round 1: My co-player (as SENDER) sent \$3.0, received \$3.41 back, payoff \$5.41
- Round 2: My co-player (as RECEIVER) saw \$2.99 sent, returned \$4.5, payoff \$4.47

****This Round (Round 3):****

- I am the RECEIVER
- They sent \$2.7 (54% of their endowment)
- I received \$8.10

Looking at the pattern:

- In Round 1, they sent 60% of their endowment
- In Round 2, I sent \$3.0 (though they saw \$2.99 due to noise), and they returned \$4.5, which is about 50% of what they received (\$8.97)
- In Round 3, they sent \$2.7 (54%)

My co-player seems to be cooperating at moderate-to-high levels. When they were receiver in Round 2, they returned 50% of what they received.

If I return around 50% like they did, that would be about \$4.05.

Let me think about what's fair:

- They sent \$2.7, so they have \$2.3 left
- If I return \$4.05, they'd have $\$2.3 + \$4.05 = \$6.35$ (profit of \$1.35)
- I'd keep $\$8.10 - \$4.05 = \$4.05$ (profit of \$4.05)

That seems reasonable and matches their behavior. I could be slightly more generous to encourage continued cooperation.

Actually, let me reconsider. In Round 2, they returned \$4.5 out of \$8.97, which is exactly 50%.

...

usage:

```
{
  "input_tokens": 1210,
  "output_tokens": 644,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.

Your total visible earnings across all rounds are \$14.48.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 (60% of endowment) in Rounds 1 and 3
- As RECEIVER: Returned generously (\$4.5 out of \$8.97, ~50%)

****My previous sending:****

- Round 2: I sent and they saw \$2.99 (likely noise from my \$3.0)

They've demonstrated consistent trust and reciprocity. I should maintain this cooperative relationship by matching their sending level.

{'send': 3.0}

reasoning:

Looking at the history:

Round 1: Co-player as SENDER sent \$3.0, got \$3.41 back, payoff \$5.41

Round 2: Co-player as RECEIVER got \$2.99 sent (from me), returned \$4.5, payoff \$4.47

Round 3: Co-player as SENDER sent \$3.0, got \$3.91 back, payoff \$5.91

Pattern analysis:

- When my co-player is SENDER, they consistently send \$3.0 (60% of endowment)
- When they're RECEIVER, they return generously (\$4.5 out of \$8.97 ? 50%)
- They're showing consistent trust and cooperation

My sending history:

- Round 2: I sent \$3.0 (based on the visible history showing they saw \$2.99 sent due to noise)
- Round 3: Not explicitly shown as sender, but the history says they saw \$2.99 sent in Round 2

Actually, let me re-read:

Round 2: I was SENDER. I sent \$3.0. Due to noise, they saw \$2.99. They received \$8.97, returned \$4.5.

Round 3: I was RECEIVER. They sent \$2.7 (but I think they sent \$3.0 and noise made it \$2.7). I returned \$4.05.

Now Round 4: I'm SENDER again.

They've been consistently sending \$3.0 when they're sender. They've returned generously when receiver.

I should maintain the cooperative pattern by sending \$3.0 again.

usage:

```
{
  "input_tokens": 1604,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.
- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.

Your total visible earnings across all rounds are \$20.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.24, so you received \$6.720000000000001. How much do you return? (0-6.720000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Co-player's return pattern:****

- Round 2: Returned \$4.5 out of \$8.97 (~50%)
- Round 4: Returned \$3.5 out of \$7.05 (~50%)

****This round:****

- They sent \$2.24 (44.8% of endowment)
- I received \$6.72

I should maintain the cooperative pattern by returning around 50% like they have consistently done.

Returning \$3.36:

- Gives them \$2.76 + \$3.36 = \$6.12 total (profit of \$1.12)
- Gives me \$3.36 (also a profit)

{'return': 3.36}

reasoning:

Looking at the visible history:

Round 2: They received \$8.97, returned \$4.5 (50%)

Round 4: They received \$7.05, returned \$3.5 (49.6%)

This round:

- They sent \$2.24 (44.8% of endowment)
- I received \$6.72

The co-player has been consistently returning around 50% of what they receive. I should reciprocate similarly.

Returning around 50% would be \$3.36:

- Sender would have: \$2.76 + \$3.36 = \$6.12 (profit of \$1.12)
- I would keep: \$6.72 - \$3.36 = \$3.36 (also a profit)

This maintains the cooperative pattern and roughly matches their return rates.

usage:

```
{
  "input_tokens": 1990,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.
- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.

Your total visible earnings across all rounds are \$23.509999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Consistently sent \$3.0 in Rounds 3 and 5 (60% of endowment)
- As RECEIVER: Returned \$3.5 out of \$7.05 (~50%) in Round 4

****What they saw in Round 5:****

- They sent \$3.0, but saw only \$2.79 returned (likely due to noise)
- Despite this, they continue to cooperate

I should maintain the cooperative relationship by sending \$3.0 again, matching their consistent trust level.

{'send': 3.0}

reasoning:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 in Rounds 3 and 5
- As RECEIVER: Returned \$3.5 out of \$7.05 (~50%) in Round 4

****What they saw from me:****

- Round 3: Sent \$3.0, they returned \$3.91
- Round 4: They saw \$2.35 sent (likely noise from my \$3.0), returned \$3.5
- Round 5: Sent \$3.0, they saw only \$2.79 returned (but I actually sent \$3.36, so noise affected what they saw)

The co-player has been consistently sending \$3.0, showing steady trust. Even though in Round 5 they saw less returned than I actually sent (due to noise), they continue to cooperate.

I should maintain the cooperative pattern and send \$3.0 again to match their consistent trust level.

usage:

```
{
  "input_tokens": 2395,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.
- Round 6 against you, as RECEIVER: saw \$2.44 sent, it became \$7.32, returned \$3.66, visible payoff \$3.66.

Your total visible earnings across all rounds are \$30.08.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the history:

****Co-player's return pattern:****

- Round 4: Returned \$3.5 out of \$7.05 (~50%)
- Round 6: Returned \$3.66 out of \$7.32 (~50%)

****This round:****

- They sent \$3.24 (64.8% of endowment) - showing strong trust
- I received \$9.72

They've been consistently returning around 50%. I should maintain this cooperative pattern

Returning \$4.86 (50%):

- Gives them \$1.76 + \$4.86 = \$6.62 total (profit of \$1.62)
- Gives me \$4.86 (also a good profit)

{'return': 4.86}

reasoning:

Looking at the history:

****Co-player's return pattern:****

- Round 4: Returned \$3.5 out of \$7.05 (~50%)
- Round 6: Returned \$3.66 out of \$7.32 (~50%)

****This round:****

- They sent \$3.24 (64.8% of endowment) - showing strong trust
- I received \$9.72

They've been consistently returning around 50% when they're the receiver. I should maintain this cooperative pattern.

Returning around 50% would be \$4.86:

- Gives them \$1.76 + \$4.86 = \$6.62 total (profit of \$1.62)
- Gives me \$4.86 (also a good profit)

This maintains the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 2783,
  "output_tokens": 383,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.
- Round 6 against you, as RECEIVER: saw \$2.44 sent, it became \$7.32, returned \$3.66, visible payoff \$3.66.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.47 returned, visible payoff \$7.47.

Your total visible earnings across all rounds are \$34.94.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Consistently sent \$3.0 in Rounds 5 and 7 (60% of endowment)
- As RECEIVER: Returned \$3.66 out of \$7.32 (~50%) in Round 6

****Round 7 outcome:****

- They sent \$3.0 and saw \$5.47 returned (a great return!)
- This showed them cooperation pays off

They've maintained consistent trust throughout our partnership. I should continue matching their sending level to sustain this cooperative relationship.

{'send': 3.0}

reasoning:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Sent \$3.0 in Rounds 5 and 7 (60% of endowment) - consistently cooperative
- As RECEIVER: Returned \$3.66 out of \$7.32 (~50%) in Round 6

****What they saw in Round 7:****

- They sent \$3.0, saw \$5.47 returned (likely noise from my \$4.86)
- This gave them a visible payoff of \$7.47 - a very good outcome

They've been consistently trusting, sending \$3.0 repeatedly. They also saw a good return in Round 7. I should continue the cooperative pattern by sending \$3.0.

This has been a mutually beneficial partnership, and maintaining trust is important.

usage:

```
{
  "input_tokens": 2917,
  "output_tokens": 342,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$2.44 sent, it became \$7.32, returned \$3.66, visible payoff \$3.66.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.47 returned, visible payoff \$7.47.
- Round 8 against you, as RECEIVER: saw \$2.9 sent, it became \$8.7, returned \$4.35, visible payoff \$4.35.

Your total visible earnings across all rounds are \$41.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Co-player's return pattern:****

- Round 6: Returned \$3.66 out of \$7.32 (~50%)
- Round 8: Returned \$4.35 out of \$8.70 (~50%)

****This round:****

- They sent \$2.97 (59.4% of endowment)
- I received \$8.91

They continue to maintain the consistent 50% return pattern. I should reciprocate with the same ratio.

Returning \$4.45 (50%):

- Gives them \$2.03 + \$4.45 = \$6.48 total (profit of \$1.48)
- Gives me \$4.46 (also a good profit)

{'return': 4.45}

Reasoning:

Looking at the visible history:

****Co-player's return pattern:****

- Round 6: Returned \$3.66 out of \$7.32 (~50%)
- Round 8: Returned \$4.35 out of \$8.7 (~50%)

****This round:****

- They sent \$2.97 (59.4% of endowment)
- I received \$8.91

They've been consistently returning around 50%. I should maintain the cooperative pattern.

Returning \$4.455 (50%):

- Gives them \$2.03 + \$4.455 = \$6.485 total (profit of \$1.485)
- Gives me \$4.455 (also a good profit)

This maintains the ~50% return ratio that has been the pattern.

```
Usage: {
  "input_tokens": 3016,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T08:03:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.

Your total visible earnings across all rounds are \$9.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3 ? they returned \$3 (cooperative)

****Round 2:**** They sent \$3 ? I returned \$6, but they saw \$5.07 (still a good return)

****Round 3 (current):****

- Both rounds showed mutual cooperation
- They sent \$3 as sender and returned reasonably as receiver
- Their visible payoff was \$7.07 last round, showing cooperation pays off

I'll send ****\$3.5**** to maintain and slightly increase our cooperative dynamic. This shows I'm willing to trust them a bit more based on our positive history.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.24 (noise), became \$6.72, returned \$3.0
- They kept \$3.72, I got back \$3.0, so I ended with \$5 - \$3 + \$3 = \$5
- But my visible earnings show \$4.2, which suggests I saw something different due to noise

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, became \$9.0
- I returned \$6.0, but they saw \$5.07 (noise)
- They ended with \$5 - \$3 + \$5.07 = \$7.07
- I kept \$9.0 - \$6.0 = \$3.0

My total visible earnings are \$9.3.

Now Round 3:

- I'm SENDER again
- Co-player has shown:
 - As receiver: returned \$3.0 out of \$6.72 (45%)
 - As sender: sent \$3.0 out of \$5 (60%)
- They saw I returned \$5.07 (though I actually returned \$6.0)
- This is good cooperation

Since they've been cooperative and sent a good amount when they were sender, I should continue cooperating. They sent \$3.0 last round, so I should send at least that much or more to maintain trust.

I'll send \$3.5 to show continued cooperation and slightly increase the stakes.

usage:

```
{
  "input_tokens": 1191,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$16.12.

This round, you are the RECEIVER against your current co-player. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our interaction pattern:

```
**Round 1:** They returned $3.0 from $6.72 (45%)
**Round 2:** They sent $3.0, I returned $6.0
**Round 3:** They returned $5.0 from $9.90 (51%)
**Round 4 (current):** They sent $3.0 ? I received $9.0
```

They've shown consistent cooperation, with their return percentages increasing slightly over time. They sent \$3.0 this round, maintaining trust despite some communication noise in previous rounds.

I'll return **\$5.0** (about 56% of what I received). This gives them a total payoff of \$7.0 (\$2 kept + \$5 returned), which rewards their continued trust and maintains our cooperative relationship.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze the history:

```
**Round 1:** I sent $3 ? they saw $2.24 ? they returned $3.0
**Round 2:** They sent $3.7 ? I received $11.10 ? I returned $6.0 ? they saw $5.07
**Round 3:** I sent $3.5 ? they saw $3.3 ? they returned $5.0
**Round 4 (current):** They sent $3.0 ? I received $9.0
```

Looking at their behavior:

- Round 1 as receiver: Returned \$3.0 out of \$6.72 (45%)
- Round 3 as receiver: Returned \$5.0 out of \$9.90 (51%)

Their returns seem reasonable and slightly increasing in percentage.

My behavior:

- Round 2 as receiver: Returned \$6.0 out of \$11.10 (54%)

They've consistently sent around \$3-\$3.7 as sender, maintaining cooperation.

For Round 4, they sent \$3.0 and I received \$9.0. To maintain fair cooperation and match roughly what they've done (returning about 50%), I should return around \$4.5-\$5.0.

Let me return \$5.0 to be consistent with maintaining good cooperation and matching their recent return percentage.

Actually, wait. Let me recalculate their payoffs to understand better:

Round 1:

- They as receiver: Kept \$5 - \$0 (not sender) + received \$6.72 - \$3.0 returned = \$5 + \$3.72 = \$8.72? No wait, they were receiver, so sender sent them something.
- Actually the history shows "as RECEIVER: saw \$2.24 sent" - so I was the sender, I sent something, they saw \$2.24, it became \$6.72, they returned \$3.0
- Their payoff: \$6.72 - \$3.0 = \$3.72. Round 2 shows they sent \$3.0 which tripled to \$9.0, and when I returned \$5.07, their payoff was \$5 - \$3.0 + \$5.07 = \$7.07. In Round 3, I sent \$3.5 which became \$9.90, and they returned \$5.0, giving them \$9.90 - \$5.0 = \$4.90. Now they've sent \$3.0 which becomes \$9.0 for me, so I need to decide what to return....

usage:

```
{
  "input_tokens": 1596,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.

Your total visible earnings across all rounds are \$20.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our pattern:

```
**Round 2:** They sent $3.0, saw ~$5.07 returned (good payoff)
**Round 3:** They returned $5.0 from what they saw as $9.90
**Round 4:** They sent $3.5, saw ~$4.99 returned
```

They've consistently sent around \$3.0-\$3.5 and returned fairly when in the receiver role. Despite some communication noise affecting visible payoffs, we've maintained good cooperation.

I'll send **\$3.5** to continue our stable, mutually beneficial pattern.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze our history:

****Round 2:**** They sent \$3.0, I returned \$6.0 (they saw \$5.07 due to noise)
****Round 3:**** I sent \$3.5, they returned \$5.0 (they saw \$3.3 sent due to noise)
****Round 4:**** They sent \$3.5 (wait, the history says they sent \$3.5 in Round 4?), I returned \$5.0 (they saw \$4.99 due to noise)

Wait, let me reread. In Round 4, I was the receiver and they sent me \$3.0. But the visible history says they sent \$3.5 in Round 4. That must be the noise - what they actually sent was \$3.5, but I saw \$3.0.

Let me recalculate:

- Round 4: They actually sent \$3.5 ? became \$10.5 ? I returned \$5.0 (they saw \$4.99)
- Their payoff: \$5 - \$3.5 + \$4.99 = \$6.49

****Round 5 (current):**** I'm the sender

Looking at the pattern:

- Round 2: They sent \$3.0, got back ~\$5.07 (good return)
- Round 3: I sent (they saw \$3.3), they returned \$5.0 (reasonable)
- Round 4: They sent \$3.5, got back ~\$4.99 (decent return, about 48% of the tripled amount)

They've been consistently cooperative, sending \$3.0-\$3.5 each time. Their returns have been fair (around 50% of what they receive).

I should maintain the cooperative relationship. I'll send **\$3.5** to continue our pattern of mutual trust and cooperation.

usage:

```
{
  "input_tokens": 2020,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.
- Round 5 against you, as RECEIVER: saw \$3.42 sent, it became \$10.26, returned \$5.5, visible payoff \$4.76.

Your total visible earnings across all rounds are \$27.85.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our established pattern:

****Round 3:**** They returned \$5.0 from \$9.90
****Round 4:**** They sent \$3.5, I returned \$5.0
****Round 5:**** They returned \$5.5 from \$10.26
****Round 6 (current):**** They sent \$3.29 ? I received \$9.87

They've been consistently cooperative, both when sending and returning. They sent \$3.29, keeping \$1.71 for themselves.

I'll return **\$5.0** (about 51% of what I received), giving them a total payoff of \$6.71. This maintains our established cooperative pattern.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at the pattern:

```
**Round 3:** They returned $5.0 from $9.90 (51%)
**Round 4:** They sent $3.5, saw ~$4.99 returned
**Round 5:** They returned $5.5 from $10.26 (54%)
**Round 6 (current):** They sent $3.29 ? I received $9.87
```

They've been very consistent:

- As sender: sending around \$3.0-\$3.5
- As receiver: returning around 51-54% of what they receive

They sent \$3.29 this round, so they kept \$1.71.

If I return about half to slightly more (consistent with our pattern), that would be around \$5.0-\$5.5.

Returning \$5.0 would give them \$1.71 + \$5.0 = \$6.71 total
Returning \$5.5 would give them \$1.71 + \$5.5 = \$7.21 total

Given they're showing continued trust and cooperation, I should return a fair amount. I'll return \$5.0-\$5.5. Let me go with \$5.0 to be consistent with our established pattern (around 50% return rate).

Actually, looking at my previous returns in this partnership:

- Round 2: I returned \$6.0 from \$11.10 (54%)
- Round 4: I returned \$5.0 from \$9.0 (56%)

So I've been returning slightly more than half. Let me continue that pattern.

$\$9.87 * 0.50 = \4.94
 $\$9.87 * 0.55 = \5.43

I'll return \$5.0, which is about 51% and maintains our cooperative pattern.

usage:

```
{
  "input_tokens": 2415,
  "output_tokens": 599,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.
- Round 5 against you, as RECEIVER: saw \$3.42 sent, it became \$10.26, returned \$5.5, visible payoff \$4.76.
- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.14 returned, visible payoff \$6.64.

Your total visible earnings across all rounds are \$32.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our established pattern:

****Round 4:**** They sent \$3.5, saw ~\$4.99 returned (payoff \$6.49)
****Round 5:**** They returned \$5.5 from \$10.26 (generous)
****Round 6:**** They sent \$3.5, saw ~\$5.14 returned (payoff \$6.64)

This co-player has been highly consistent and cooperative. They regularly send \$3.5 as sender and return generously as receiver. Our mutual cooperation has been working very well.

I'll continue our successful pattern by sending ****\$3.5****.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Looking at the pattern:

****Round 4:**** They sent \$3.5, saw ~\$4.99 returned
****Round 5:**** They returned \$5.5 from \$10.26
****Round 6:**** They sent \$3.5, saw ~\$5.14 returned

This co-player has been very consistent and cooperative across all rounds. They:

- Send around \$3.5 consistently as sender
- Return around \$5.0-\$5.5 as receiver
- Have maintained cooperation despite communication noise

We've built a strong pattern of mutual trust. They've been sending \$3.5 regularly, and I've been returning fair amounts. This has worked well for both of us.

I should continue sending around \$3.5 to maintain this successful cooperative relationship.

usage:

```
{
  "input_tokens": 2800,
  "output_tokens": 339,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.42 sent, it became \$10.26, returned \$5.5, visible payoff \$4.76.
- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.14 returned, visible payoff \$6.64.
- Round 7 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$6.5, visible payoff \$5.379999999999999.

Your total visible earnings across all rounds are \$39.93.

This round, you are the RECEIVER against your current co-player. They sent you \$3.78, so you received \$11.34. How much do you return? (0-11.34)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our continued cooperation:

****Round 5:**** They returned \$5.5 from \$10.26
****Round 6:**** They sent \$3.5, I returned \$5.0
****Round 7:**** They returned \$6.5 from \$11.88 (very generous!)
****Round 8 (current):**** They sent \$3.78 ? I received \$11.34

This co-player has shown increasing generosity, especially in Round 7 with a \$6.5 return. They sent \$3.78 this round, maintaining high trust.

I'll return ****\$6.5**** (about 57% of what I received), matching their generosity from Round 7 and giving them a total payoff of \$7.72.

```
'''json
```

```
    {"return": 6.5}
    """
```

reasoning:

Looking at the history:

```
**Round 5:** They returned $5.5 from $10.26
**Round 6:** They sent $3.5, I returned $5.0
**Round 7:** They returned $6.5 from $11.88 (quite generous!)
**Round 8 (current):** They sent $3.78 ? I received $11.34
```

This co-player has been very cooperative and actually increased their generosity in Round 7, returning \$6.5 from \$11.88 (about 55%). They sent \$3.78 this round, which is slightly higher than their usual \$3.5.

To maintain this excellent cooperation and reward their generosity, I should return a fair amount. Given their last return was \$6.5 from \$11.88, and they sent \$3.78 (keeping \$1.22), I should return something that gives them a good payoff.

If I return \$6.5, they get: $\$1.22 + \$6.5 = \$7.72$ total

This would be consistent with the pattern and reward their continued trust and generosity.

Let me return \$6.5 to maintain our strong cooperative relationship.

usage:

```
{
  "input_tokens": 3040,
  "output_tokens": 471,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.14 returned, visible payoff \$6.64.
- Round 7 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$6.5, visible payoff \$5.379999999999999.
- Round 8 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.73 returned, visible payoff \$7.23.

Your total visible earnings across all rounds are \$44.769999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our excellent cooperation:

```
**Round 6:** They sent $3.5, saw ~$5.14 returned (payoff $6.64)
**Round 7:** They returned $6.5 from $11.88 (very generous!)
**Round 8:** They sent $3.5, saw ~$5.73 returned (payoff $7.23)
```

This co-player has been extremely consistent, sending \$3.5 when they're the sender and returning generously when they're the receiver. Our mutual cooperation has been highly successful.

I'll send ****\$3.5**** to maintain our established, mutually beneficial pattern.

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Looking at the pattern:

```
**Round 6:** They sent $3.5, saw ~$5.14 returned (payoff $6.64)
**Round 7:** They returned $6.5 from $11.88 (very generous)
**Round 8:** They sent $3.5, saw ~$5.73 returned (payoff $7.23)
```

This co-player has been extremely consistent and cooperative throughout our interactions. They regularly send \$3.5 as sender and return very generously as receiver. Our cooperation has been strong and mutually beneficial.

I should continue this successful pattern with \$3.5, or potentially increase slightly to show appreciation for their consistent cooperation. Let me think about this...

They've been sending \$3.5 consistently. I've been doing the same. We have a very strong cooperative relationship. I could:

1. Send \$3.5 to maintain the pattern
2. Send \$4.0 to show increased trust given our excellent track record

Given that we have such a strong cooperative history and they've been increasingly generous (returning \$6.5 in Round 7), I think sending \$4.0 would be appropriate to reward their consistent cooperation and trust.

Actually, looking more carefully at the history, they consistently send \$3.5. I should probably match that to maintain our stable equilibrium rather than potentially disrupt it.

I'll send \$3.5 to maintain our successful pattern.

```
Usage: {
  "input_tokens": 3071,
  "output_tokens": 490,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:03:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.

Your total visible earnings across all rounds are \$10.790000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment), showing strong trust
- Round 2: Co-player returned \$6.0 out of \$11.1 received (54% return rate), showing generosity despite seeing lower returns in Round 1
- Round 3: Co-player sent \$3.3 (66% of endowment), continuing to show high trust

My co-player has consistently demonstrated cooperative behavior and trust. They've maintained cooperation even after seeing what appeared to be a lower return in Round 1.

If I return \$5.0:

- Sender ends with: \$1.70 + \$5.00 = \$6.70 (profit of \$1.70)
- I keep: \$9.90 - \$5.00 = \$4.90

This reciprocates their generous behavior and maintains the positive cooperative dynamic.

{"return": 5.0}

reasoning:

Let me analyze the visible history:

Round 1:

- Co-player as SENDER: sent \$3.0, saw \$2.2 returned
- From their view: started with \$5, sent \$3, kept \$2, saw \$2.2 returned = \$4.2 total
- (I actually returned \$3.00, but they saw \$2.2 due to noise)

Round 2:

- Co-player as RECEIVER: saw \$3.7 sent (I sent \$3.0), received \$11.1, returned \$6.0
- From their view: they received \$11.1, returned \$6.0, kept \$5.1
- From my view: I sent \$3.0, kept \$2.0, saw \$6.0 returned = \$8.0 total

Round 3 (current):

- I am RECEIVER
- Co-player sent \$3.3 (66% of endowment)
- I received \$9.9

My co-player has been quite trustworthy:

- Round 1: sent 60% of endowment
- Round 2: returned a generous \$6.0 out of \$11.1 they received (54% return rate)
- Round 3: sent 66% of endowment

They've shown consistent cooperation and generosity. I should reciprocate appropriately.

If I return \$5.0:

- Sender ends with: \$1.7 (kept) + \$5.0 (returned) = \$6.7 (profit of \$1.7)
- I keep: \$9.9 - \$5.0 = \$4.9

This maintains a good balance and rewards their trust. Let me check if this is fair compared to their Round 2 behavior:

- In Round 2, they returned \$6.0 out of \$11.1 (54%)

...

usage:

```
{
  "input_tokens": 1250,
  "output_tokens": 704,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.

Your total visible earnings across all rounds are \$15.69.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment)
- Round 2: Co-player returned \$6.0 out of \$11.1 (very generous, 54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment), saw \$5.32 returned

My co-player has consistently shown high trust and cooperation, sending substantial amounts and returning generously when in the receiver role. They've maintained this cooperative behavior throughout.

I'll match their highest level of trust by sending a similar amount.

```
{"send": 3.5}
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: Co-player sent \$3.0, saw \$2.2 returned (I actually returned \$3.0, but noise)
 Round 2: I sent \$3.0 (co-player saw \$3.7), they returned \$6.0 - very generous!
 Round 3: Co-player sent \$3.5 (actually \$3.3 based on my perspective), saw \$5.32 returned (I actually returned \$5.0)

My co-player has been consistently cooperative:

- Sending high amounts (60-70% of endowment)
- When they were receiver, they returned \$6.0 out of \$11.1 (54% return rate)

They've maintained cooperation throughout despite seeing some noise in the returns. In Round 3, they saw \$5.32 returned on their \$3.5 sent, which gave them a profit of \$1.82 - decent but not extremely generous.

As the sender now, I should:

1. Continue the cooperative pattern
2. Send an amount that shows trust matching theirs
3. Given they sent \$3.0, \$3.5 in previous rounds, I should send similarly

I'll send \$3.5 to match their highest level of trust and continue the positive dynamic.

usage:

```
{
  "input_tokens": 1694,
  "output_tokens": 461,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.
- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.

Your total visible earnings across all rounds are \$22.18.

This round, you are the RECEIVER against your current co-player. They sent you \$3.42, so you received \$10.26. How much do you return? (0-10.26)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (generous 54% return)
- Round 3: Co-player sent \$3.5 (70% of endowment, high trust)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return)
- Round 5: Co-player sent \$3.42 (68% of endowment, continued trust)

My co-player has consistently demonstrated strong cooperation, both sending high amounts and returning generously. They're maintaining this pattern across rounds.

If I return \$5.5:

- Sender ends with: \$1.58 + \$5.5 = \$7.08 (profit of \$2.08)
- I keep: \$10.26 - \$5.5 = \$4.76

This matches their typical return rate (~54%) and reciprocates their continued trust.

```
{"return": 5.5}
```

reasoning:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.42 (68% of endowment)

My co-player has been very cooperative and consistent:

- As sender: sending around 68-70% of their endowment
- As receiver: returning around 54-56% of what they receive

They're maintaining high trust and generosity throughout. I should reciprocate appropriately.

If I return \$5.5:

- Sender ends with: $\$1.58 + \$5.5 = \$7.08$ (profit of \$2.08)
- I keep: $\$10.26 - \$5.5 = \$4.76$

This is approximately 54% return rate, matching their pattern and continuing our cooperative relationship.

Actually, let me be slightly more generous to maintain the positive dynamic. Let me return \$5.5 or \$6.0.

Return of \$5.5 gives them \$7.08 total (profit of \$2.08)

Return of \$6.0 gives them \$7.58 total (profit of \$2.58)

I'll go with \$5.5 to match their typical return rate of around 54%.

usage:

```
{
  "input_tokens": 2082,
  "output_tokens": 593,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.
- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.

Your total visible earnings across all rounds are \$26.939999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- Round 3: Co-player sent \$3.5 (70% of endowment)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.5 (70% of endowment), saw \$6.23 returned, payoff \$7.73

My co-player has been remarkably consistent:

- As sender: consistently sends \$3.5 (70% of endowment), showing high trust
- As receiver: returns generously (~55% of received amount)

This is a very cooperative partner. I should match their consistent high level of trust.

```
{"send": 3.5}
```

reasoning:

Looking at the visible history:

- Round 3: Co-player sent \$3.5, saw \$5.32 returned, payoff \$6.82
- Round 4: Co-player returned \$5.0 out of \$9.0, payoff \$4.0
- Round 5: Co-player sent \$3.5, saw \$6.23 returned, payoff \$7.73

My co-player has been very consistent:

- When sending: consistently sends \$3.5 (70% of endowment)
- When receiving: returns generously (55-56% of what they receive)

They've maintained this high cooperation throughout all rounds. The pattern is clear and stable.

I should continue matching their level of trust and cooperation. They've been sending \$3.5 consistently as sender, and I've been doing the same.

```
    {"send": 3.5}
usage:
{
  "input_tokens": 2535,
  "output_tokens": 382,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.
- Round 6 against you, as RECEIVER: saw \$3.29 sent, it became \$9.870000000000001, returned \$5.0, visible payoff \$4.870000000000001.

Your total visible earnings across all rounds are \$33.58.

This round, you are the RECEIVER against your current co-player. They sent you \$3.96, so you received \$11.879999999999999. How much do you return? (0-11.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return)
- Round 5: Co-player sent \$3.5 (70% of endowment)
- Round 6: Co-player returned \$5.0 out of \$9.87 (51% return)
- Round 7: Co-player sent \$3.96 (79% of endowment - their highest trust level yet!)

My co-player has escalated their trust, sending an even higher percentage than before. I should reciprocate this increased trust with a generous return.

If I return \$6.5:

- Sender ends with: $\$1.04 + \$6.5 = \$7.54$ (profit of \$2.54)
- I keep: $\$11.88 - \$6.5 = \$5.38$

This rewards their increased trust appropriately.

```
{"return": 6.5}
```

reasoning:

Looking at the visible history:

- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.5, saw \$6.23 returned
- Round 6: Co-player returned \$5.0 out of \$9.87 (51% return rate)
- Round 7: Co-player sent \$3.96 (79% of endowment - highest yet!)

My co-player is continuing to show high trust and cooperation. In fact, they just sent \$3.96, which is 79% of their endowment - the highest percentage they've sent yet! This is showing even greater trust than before.

I should reciprocate this increased trust with a generous return.

If I return \$6.0:

- Sender ends with: $\$1.04 + \$6.0 = \$7.04$ (profit of \$2.04)
- I keep: $\$11.88 - \$6.0 = \$5.88$

This would be about a 50.5% return rate, which is consistent with their return patterns but rewards their increased trust.

Actually, let me be a bit more generous since they increased their trust level:

If I return \$6.5:

- Sender ends with: $\$1.04 + \$6.5 = \$7.54$ (profit of \$2.54)
- I keep: $\$11.88 - \$6.5 = \$5.38$

This rewards their increased trust more clearly while still giving me a good payoff.

usage:

```
{
  "input_tokens": 2952,
  "output_tokens": 597,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.
- Round 6 against you, as RECEIVER: saw \$3.29 sent, it became \$9.870000000000001, returned \$5.0, visible payoff \$4.870000000000001.
- Round 7 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.71 returned, visible payoff \$7.21.

Your total visible earnings across all rounds are \$38.959999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

- Round 5: Co-player sent \$3.5 (70% of endowment)
- Round 6: Co-player returned \$5.0 out of \$9.87
- Round 7: Co-player sent \$3.5 (70% of endowment)

My co-player continues to show consistent, high trust behavior. They repeatedly send \$3.5 and return generously. This is a reliable cooperative partner.

I'll maintain our established pattern of mutual trust.

```
{"send": 3.5}
```

reasoning:

Looking at the visible history:

- Round 5: Co-player sent \$3.5 (70% of endowment), saw \$6.23 returned, payoff \$7.73
- Round 6: Co-player returned \$5.0 out of \$9.87 (51% return)
- Round 7: Co-player sent \$3.5 (70% of endowment), saw \$5.71 returned, payoff \$7.21

My co-player has been very consistent:

- As sender: consistently sends \$3.5 (70% of endowment)
- As receiver: returns around \$5.0 consistently (51% return rate)

This is a highly cooperative and consistent partner. They've maintained trust throughout the game. I should continue matching their level of trust by sending \$3.5.

We're building a strong cooperative relationship with consistent behavior on both sides.

usage:

```
{
  "input_tokens": 3123,
  "output_tokens": 343,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.29 sent, it became \$9.870000000000001, returned \$5.0, visible payoff \$4.870000000000001.
- Round 7 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.71 returned, visible payoff \$7.21.
- Round 8 against you, as RECEIVER: saw \$3.78 sent, it became \$11.34, returned \$6.5, visible payoff \$7.21.

le payoff \$4.84.
Your total visible earnings across all rounds are \$46.19.

This round, you are the RECEIVER against your current co-player. They sent you \$4.37, so you received \$13.11. How much do you return? (0-13.11)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

- Round 6: Co-player returned \$5.0 out of \$9.87
- Round 7: Co-player sent \$3.5 (70% of endowment)
- Round 8: Co-player returned \$6.5 out of \$11.34 (57% return - very generous)
- Round 9: Co-player sent \$4.37 (87% of endowment - extremely high trust!)

My co-player has dramatically escalated their trust, sending nearly all their endowment. This is the highest level of trust shown so far. I should reciprocate this exceptional trust with a generous return.

If I return \$7.5:

- Sender ends with: $\$0.63 + \$7.5 = \$8.13$ (profit of \$3.13)
- I keep: $\$13.11 - \$7.5 = \$5.61$

This rewards their exceptional trust appropriately.

{"return": 7.5}

Reasoning:

Looking at the visible history:

- Round 6: Co-player returned \$5.0 out of \$9.87
- Round 7: Co-player sent \$3.5 (70% of endowment)
- Round 8: Co-player returned \$6.5 out of \$11.34 (57% return)
- Round 9: Co-player sent \$4.37 (87% of endowment - this is extremely high trust!)

My co-player has shown escalating trust behavior. They went from consistently sending \$3.5 (70%) to now sending \$4.37 (87% of their endowment). They are also returning generously when in the receiver role (saw \$6.5 returned in round 8).

This is a very cooperative partner showing maximum trust. I should reciprocate this high level of trust.

If I return \$7.0:

- Sender ends with: $\$0.63 + \$7.0 = \$7.63$ (profit of \$2.63)
- I keep: $\$13.11 - \$7.0 = \$6.11$

This is about 53% return rate, which is appropriate and reciprocates their high trust.

Actually, let me think about this more. They sent \$4.37 out of \$5, which is 87.4% of their endowment. This is remarkably high trust. I should return something that makes them profitable while also keeping a reasonable amount.

If I return \$7.5:

- Sender ends with: $\$0.63 + \$7.5 = \$8.13$ (profit of \$3.13)
- I keep: $\$13.11 - \$7.5 = \$5.61$

This is about 57% return rate, matching their generous behavior in round 8.

Usage: {

"input_tokens": 3200,
"output_tokens": 657,
"reasoning_tokens": 0

}

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver


```

dyad_4: Agent_8=Sender, Agent_7=Receiver
Parsed game decisions:
dyad_1: Agent_2 sent $4; received $12; Agent_1 returned $7.7; payoffs Agent_2=$8.7, Agent_1=$4.3;
noise/visible: sent visible=$4.26, received visible=$12.78, returned visible=$7.21
  Action validation: {"returned": {"amount": 7.7, "clamped": false, "max": 12.0, "min": 0, "raw":
    7.7}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_4 sent $3; received $9; Agent_3 returned $5; payoffs Agent_4=$7, Agent_3=$4;
noise/visible: sent visible=$3.33, received visible=$9.99, returned visible=$4.95
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
    5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 sent $3; received $9; Agent_5 returned $5.75; payoffs Agent_6=$7.75,
Agent_5=$3.25; noise/visible: sent visible=$3.83, received visible=$11.49, returned visible=$6.08
  Action validation: {"returned": {"amount": 5.75, "clamped": false, "max": 9.0, "min": 0, "raw":
    5.75}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent $3.5; received $10.5; Agent_7 returned $7; payoffs Agent_8=$8.5,
Agent_7=$3.5; noise/visible: sent visible=$3.78, received visible=$11.34, returned visible=$7.32
  Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 10.5, "min": 0, "raw":
    7.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
Balances after round: {
  "Agent_1": 63.3,
  "Agent_2": 63.7,
  "Agent_3": 51.0,
  "Agent_4": 59.0,
  "Agent_5": 53.11,
  "Agent_6": 56.89,
  "Agent_7": 57.0,
  "Agent_8": 61.0
}
Visible balances after round: {
  "Agent_1": 65.94000000000001,
  "Agent_2": 65.39,
  "Agent_3": 56.959999999999994,
  "Agent_4": 55.330000000000005,
  "Agent_5": 52.330000000000005,
  "Agent_6": 53.86,
  "Agent_7": 57.879999999999995,
  "Agent_8": 60.62
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

```

Timestamp: 2026-08-12T08:03:39
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.94 returned, visible payoff \$7.94.
- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.

Your total visible earnings across all rounds are \$17.91.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant**content:**

Looking at our history:

- My co-player has consistently shown high trust (sending 60-80% of their endowment)
- They've returned reasonable amounts when receiving
- We've established a strong cooperative pattern

In Round 3, they sent \$4.0 (80% of their endowment), showing very high trust. I should reciprocate with a similarly strong showing of trust to maintain our cooperative relationship.

```
{"send": 4.00}
```

reasoning:

Let me analyze the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned \$5.00, but they saw \$5.94 returned

Round 2: I sent \$3.50, they received it as \$3.89 (11.67 total), they returned \$6.0

Round 3: They sent \$3.71, but they show it as \$4.0 (became \$12.0), I returned \$6.50 but they saw \$5.66

There's definitely communication noise happening. But the pattern shows:

- They consistently send high amounts (60-80%)
- They return reasonably when receiving
- We've been building a cooperative relationship

In Round 3, they sent what I saw as \$3.71 (74%), but they think they sent \$4.0 (80%). This shows very high trust.

I should continue the cooperative pattern and send a high amount as well. I'll send around \$3.50-\$4.00 to match their level of trust.

Let me send \$4.00 to show strong trust and keep the cooperation going.

usage:

```
{
  "input_tokens": 1542,
  "output_tokens": 367,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.89 sent, it became \$11.67, returned \$6.0, visible payoff \$5.67.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.
- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.

Your total visible earnings across all rounds are \$25.07.

This round, you are the RECEIVER against your current co-player. They sent you \$4.36, so you received \$13.080000000000002. How much do you return? (0-13.080000000000002)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our continued cooperation:

- My co-player sent \$4.36 (87.2% of their endowment) - showing very high trust!
- They've been consistently generous as receiver (returning \$6.0 and \$7.0 in previous rounds)
- This is clearly a cooperative partner

Current situation:

- They sent \$4.36, leaving them with only \$0.64
- I received \$13.08

I should return generously to reward their exceptional trust.

If I return \$7.50:

- Sender's total: \$0.64 + \$7.50 = \$8.14
- My total: \$13.08 - \$7.50 = \$5.58

```
{"return": 7.50}
```

reasoning:

Looking at the history with my co-player:

Round 2: I sent \$3.89, they returned \$6.0
Round 3: They sent \$4.0, I returned \$5.66 (visible to them)
Round 4: I sent \$3.97, they returned \$7.0

Now Round 5: They sent \$4.36 (87.2% of their endowment - very high trust!)

They are showing extremely high trust by sending \$4.36, leaving them with only \$0.64.

I received \$13.08

They've been returning well when they're the receiver:

- Round 2: returned \$6.0 from \$11.67
- Round 4: returned \$7.0 from \$11.91

This is a cooperative partner. I should return generously to reward their high trust.

If I return \$7.50:

- Sender's total: $\$0.64 + \$7.50 = \$8.14$ (profit of \$3.14)
- My total: $\$13.08 - \$7.50 = \$5.58$ (profit of \$5.58)

This seems fair and rewards their trust while still benefiting from the exchange.

usage:

```
{
  "input_tokens": 1882,
  "output_tokens": 462,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.66 returned, visible payoff \$6.66.
- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.

Your total visible earnings across all rounds are \$30.650000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our strong cooperative history:

- My co-player has consistently sent high amounts (\$4.0 in recent rounds, 80% of endowment)
- They've returned generously as receiver (\$7.0 and \$7.68 in recent rounds)
- We've established excellent mutual trust

I should continue this cooperative pattern by sending a similarly high amount.

```
{"send": 4.00}
```

reasoning:

Looking at the history with my co-player:

- Round 3: They sent \$4.0 (80%), returned visible \$5.66
- Round 4: They returned \$7.0
- Round 5: They sent \$4.0 (80%), returned visible \$7.68

This is a highly cooperative partner who consistently:

1. Sends high amounts (80% of endowment)
2. Returns generously when receiving (around \$6-7.68)

We've built a strong mutual trust pattern. I should continue to send high amounts to maintain this cooperation.

I should send a high amount, similar to what they've been sending. Around \$4.0-\$4.5 would be appropriate.

Actually, I notice there's some noise mentioned in the instructions. The visible amounts might differ from actual sent/returned amounts. But I should focus on the pattern of cooperation that's clearly visible.

I'll send \$4.00 to match their typical sending pattern and maintain our strong cooperation

usage:

```
{
  "input_tokens": 2276,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.97 sent, it became \$11.91, returned \$7.0, visible payoff \$4.91.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 6 against you, as RECEIVER: saw \$4.89 sent, it became \$14.669999999999998, returned \$8.5, visible payoff \$6.169999999999998.

Your total visible earnings across all rounds are \$39.42.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our excellent cooperation:

- My co-player sent \$4.73 (94.6% of their endowment) - extremely high trust!
- They've been consistently generous as receiver (returning \$7.0 and \$8.5 recently)
- We have a very strong cooperative relationship

Current situation:

- They sent \$4.73, leaving them with only \$0.27
- I received \$14.19

I should match their generosity and maintain our high level of cooperation.

{"return": 8.50}

reasoning:

Looking at the history with my co-player:

- They've been consistently sending high amounts (around \$4.0)
- They returned \$7.0 and \$8.5 in recent rounds as receiver, showing strong cooperation
- Round 7: They sent \$4.73 (94.6% of their endowment) - extremely high trust!

Current situation:

- They sent \$4.73, leaving them with only \$0.27
- I received \$14.19

This is a very trusting partner. I should return generously to maintain our excellent cooperation.

If I return \$8.50:

- Sender's total: \$0.27 + \$8.50 = \$8.77
- My total: \$14.19 - \$8.50 = \$5.69

This seems like a fair and generous return that rewards their high trust while still benefiting from the transaction.

usage:

```
{
```

```
"input_tokens": 2613,
"output_tokens": 334,
"reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 6 against you, as RECEIVER: saw \$4.89 sent, it became \$14.669999999999998, returned \$8.5, visible payoff \$6.169999999999998.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.65 returned, visible payoff \$8.65.

Your total visible earnings across all rounds are \$45.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our continued strong cooperation:

- My co-player has consistently sent \$4.0 (80% of endowment) in recent rounds
- They've returned generously when receiving (\$8.5 in Round 6)
- We have an established pattern of high mutual trust

I should continue this successful cooperative strategy.

```
{"send": 4.00}
```

reasoning:

Looking at the history with my co-player:

- They've been consistently sending high amounts as sender (\$4.0 recently, 80% of endowment)
- They've been returning generously as receiver (\$8.5 in Round 6)
- We have a very strong, stable cooperative relationship
- The pattern shows mutual high trust

In Round 7, they sent \$4.0 again, showing continued trust. I should reciprocate with a similarly high amount to maintain our excellent cooperation.

I'll send \$4.00 to match their recent sending behavior and show continued trust.

usage:

```
{
  "input_tokens": 2722,
  "output_tokens": 221,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.89 sent, it became \$14.669999999999998, returned \$8.5, visible payoff \$6.169999999999998.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.65 returned, visible payoff \$8.65.
- Round 8 against you, as RECEIVER: saw \$3.6 sent, it became \$10.8, returned \$6.5, visible payoff \$4.300000000000001.

Your total visible earnings across all rounds are \$52.42.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our cooperation:

- My co-player sent \$3.92 (78.4% of their endowment) - still showing high trust
- They've been consistently cooperative throughout
- In Round 8, they returned \$6.5 even when I sent slightly less

Current situation:

- They sent \$3.92, leaving them with \$1.08
- I received \$11.76

I should maintain our cooperative relationship with a fair return.

```
{"return": 7.00}
```

reasoning:

Looking at the history:

- Round 6: I sent \$4.89 (seen as sent), they returned \$8.5
- Round 7: They sent \$4.0, I returned \$7.65 (seen)
- Round 8: I sent \$3.6 (seen), they returned \$6.5
- Round 9: They sent \$3.92, I received \$11.76

It looks like in Round 8, I sent less (\$3.6 instead of my usual \$4.0). They still returned a decent amount (\$6.5).

Now they've sent \$3.92 (78.4% of their endowment) - still showing good trust.

Current situation:

- They sent \$3.92, leaving them with \$1.08
- I received \$11.76

I should maintain our cooperative relationship. Let me return a fair amount.

If I return \$7.0:

- Sender's total: $\$1.08 + \$7.0 = \$8.08$
- My total: $\$11.76 - \$7.0 = \$4.76$

This seems reasonable and maintains the cooperative pattern.

usage:

```
{
  "input_tokens": 2819,
  "output_tokens": 384,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.65 returned, visible payoff \$8.65.
- Round 8 against you, as RECEIVER: saw \$3.6 sent, it became \$10.8, returned \$6.5, visible payoff \$4.3000000000000001.
- Round 9 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.88 returned, visible payoff \$7.88.

Your total visible earnings across all rounds are \$57.18.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our consistent cooperation:

- My co-player has sent \$4.0 (80% of endowment) in recent rounds
- They returned \$6.5 when receiving in Round 8
- We've maintained strong mutual trust throughout

I'll continue with our successful cooperative pattern.

```
{"send": 4.00}
```

Reasoning:

Looking at the pattern:

- My co-player has consistently sent \$4.0 in rounds 7 and 9 (80% of endowment)
- They returned \$6.5 in Round 8 as receiver
- We have a strong cooperative relationship established

I should continue sending a high amount to maintain this cooperation.

```
{"send": 4.00}
Usage: {
  "input_tokens": 2758,
  "output_tokens": 167,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:03:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:**Message 1 role=system****content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.

Your total visible earnings across all rounds are \$20.27.

This round, you are the RECEIVER against your current co-player. They sent you \$3.97, so you received \$11.91. How much do you return? (0-11.91)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history:

****Round 1:****

- They returned \$5.00 from \$9.84 (51%)

****Round 2:****

- They sent \$3.50 generously
- I returned well (they saw \$6.94)

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)
- Consistent fair behavior

****Round 4:****

- They sent \$3.97 (about 79% - very trusting!)
- I received \$11.91

****Analysis:****

They've been consistently fair, returning 50-58% of what they receive. They continue to send high amounts, showing sustained trust.

I'll return approximately 59% to match their cooperative pattern:

```
'''json
{"return": 7}
'''
```

This rewards their high trust while ensuring both of us benefit substantially from the exchange.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$3.28 with noise)
- They returned \$5.00
- They kept \$4.84

****Round 2:****

- They sent \$3.50
- I received \$10.50
- I returned \$6 (they saw \$6.94)
- They kept \$8.44

****Round 3:****

- I sent \$4 (they saw \$3.71 with noise)
- They received \$11.13
- They returned \$6.50
- They kept \$4.63

****Round 4:****

- They sent \$3.97 (about 79% of their endowment)
- I received \$11.91
- I need to decide how much to return

Looking at the pattern:

- Round 1: They returned \$5.00 out of \$9.84 (about 51%)
- Round 3: They returned \$6.50 out of \$11.13 (about 58%)

They've been consistently fair and returning a good proportion. They're also continuing to send high amounts (almost 80% of their endowment in Round 4).

I should continue to reciprocate. A fair return would be around 50-60% of what I received:

- 50% = \$5.96
- 55% = \$6.55
- 60% = \$7.15

Given they've been very trusting and cooperative, I'll return around \$7 (about 59% of what I received). This rewards their high trust and maintains our cooperative relationship.

Actually, let me reconsider. Looking at what they kept:

- Round 1: They kept \$4.84 after returning \$5.00 from \$9.84
- Round 3: They kept \$4.63 after returning \$6.50 from \$11.13

They've been very consistent in what they keep (around \$4.60-\$4.84). ...

usage:

```
{
  "input_tokens": 1616,
  "output_tokens": 784,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.94 returned, visible payoff \$8.440000000000001.
- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.

Your total visible earnings across all rounds are \$25.18.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our continued partnership:

****Round 2:****

- They sent \$3.50 (70% - very trusting)

****Round 3:****

- I sent \$4 (they saw \$3.71)
- They returned \$6.50 (58% - fair)

****Round 4:****

- They sent \$4.00 (80% - even more trusting!)
- I returned \$7 (they saw \$6.16)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has demonstrated:

- Consistently high trust (sending 70-80%)
- Fair returns (~58% when receiving)
- Strong cooperative pattern over multiple rounds

I'll match their high trust level and send ****\$4**** (80% of my endowment):

- Maintains our successful cooperative relationship
- Becomes \$12 for them
- Based on their pattern, should yield good returns
- Shows continued mutual trust

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the visible history:

****Round 2:****

- They sent \$3.50 (70% of endowment)
- They saw \$6.94 returned

****Round 3:****

- They saw \$3.71 sent (I sent \$4)
- They returned \$6.50

****Round 4:****

- They sent \$4.00 (80% of endowment)
- They saw \$6.16 returned (I sent \$7)

****Round 5:****

I'm the sender again.

****Analysis:****

This co-player has been:

- Very trusting: sending 70-80% of their endowment consistently
- Fair when receiving: returning around 58% of what they get
- Consistent and cooperative throughout

This is a strong, mutually beneficial relationship. They've escalated their trust from \$3.50 to \$4.00, and I should match or maintain that trust level.

I'll send ****\$4**** again, which:

- Maintains our high-trust equilibrium
- Becomes \$12 for them
- Should yield a good return based on their consistent ~58% return rate
- Shows I recognize and value their trustworthiness

Alternatively, I could send \$5 to maximize total welfare, but \$4 has been working well and maintains consistency.

I'll go with \$4.

usage:

```
{
  "input_tokens": 2044,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

}

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.71 sent, it became \$11.129999999999999, returned \$6.5, visible payoff \$4.629999999999999.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.
- Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.580000000000002.

Your total visible earnings across all rounds are \$33.86.

This round, you are the RECEIVER against your current co-player. They sent you \$4.89, so you received \$14.669999999999998. How much do you return? (0-14.669999999999998)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Round 3:****

- They returned \$6.50 from \$11.13 (58%)

****Round 4:****

- They sent \$4.00, trusting
- I returned well

****Round 5:****

- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.89 (98% of their endowment - extremely trusting!)
- I received \$14.67

****Analysis:****

This co-player has shown exceptional trust, sending nearly everything they had. They've been consistently fair, returning 57-58% when receiving.

I should reward this maximum trust with a strong return:

~58% of \$14.67 = \$8.51

```
'''json
{"return": 8.5}
'''
```

This maintains our cooperative relationship and rewards their exceptional level of trust.

reasoning:

Looking at the history:

****Round 3:****

- I sent \$4 (they saw \$3.71)
- They returned \$6.50 from \$11.13 (58%)

****Round 4:****

- They sent \$4.00 (80%)
- I returned \$7 (they saw \$6.16, which is about 51% of \$12)

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.89 (almost 98% - extremely trusting!)
- I received \$14.67

****Analysis:****

This co-player has been incredibly trusting and cooperative:

- They consistently return around 57-58% when receiving
- They've been sending increasingly high amounts (80% ? 98%)
- They're showing maximum trust

I should reciprocate this high trust. Given they return about 57-58%, I should return a similar or slightly higher percentage to match their increased trust.

~60% of \$14.67 = \$8.80

Let me return around \$8.5 to \$9 to reward their exceptional trust while maintaining fairness.

I'll return \$8.5

usage:

```
{
  "input_tokens": 2528,
  "output_tokens": 506,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.16 returned, visible payoff \$7.16.
- Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.5800000000000002.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.77 returned, visible payoff \$8.77.

Your total visible earnings across all rounds are \$40.03.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our continued partnership:

****Round 4:****

- They sent \$4.00 (80% - very trusting)

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 (57% - fair)

****Round 6:****

- They sent \$4.00 again (consistent high trust)
- I returned well (they saw \$7.77)

****Round 7:****

I'm the sender.

****Analysis:****

This co-player has demonstrated:

- Unwavering high trust (consistently sending 80%+)
- Fair returns (~57% when receiving)
- Extremely stable cooperative behavior
- 6 rounds of successful mutual benefit

I'll continue matching their trust level and send ****\$4**** (80%):

- Maintains our proven successful pattern
- Becomes \$12 for them
- Shows consistent reciprocity
- We've built strong trust over many rounds

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the history:

****Round 4:****

- They sent \$4.00 (80% - very trusting)
- I returned well

****Round 5:****

- I sent \$4 (they saw \$4.36)
- They returned \$7.50 (57%)

****Round 6:****

- They sent \$4.00 again (80%)
- I returned \$8.50 (they saw \$7.77 - about 65%)

****Round 7:****

- I'm the sender

****Analysis:****

This co-player has:

- Consistently sent high amounts when they're sender (80%+)
- Returned fairly when receiver (57%)
- Shown sustained trust and cooperation over 6 rounds
- We have a very strong cooperative relationship

They've sent \$4 consistently when in sender role. I should continue matching their high trust level. I'll send \$4 again (80%).

This:

- Maintains our successful pattern
- Becomes \$12 for them
- Shows continued mutual trust
- Based on their behavior, should yield good returns

usage:

```
{
  "input_tokens": 2956,
  "output_tokens": 487,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$4.36 sent, it became \$13.080000000000002, returned \$7.5, visible payoff \$5.5800000000000002.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.77 returned, visible payoff \$8.77.
- Round 7 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$8.5, visible payoff \$5.6900000000000001.

Your total visible earnings across all rounds are \$48.68.

This round, you are the RECEIVER against your current co-player. They sent you \$3.6, so you received \$10.8. How much do you return? (0-10.8)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the history:

****Round 5:****

- They returned \$7.50 from \$13.08 (57%)

****Round 6:****

- They sent \$4.00, trusting
- I returned well

****Round 7:****

- They returned \$8.50 from \$14.19 (60%)

****Round 8:****
- They sent \$3.60 (72% - still quite trusting)
- I received \$10.80

****Analysis:****
This co-player continues to demonstrate:
- Fair returns (57-60% range)
- Consistent trust (72% sent this round)
- 7 rounds of reliable cooperation

They sent slightly less than previous rounds but still showed significant trust. I'll return approximately 60% to match our established pattern:

~60% of \$10.80 = \$6.48

```
'''json
{"return": 6.5}
'''
```

This maintains our cooperative relationship and rewards their continued trust.

reasoning:

Looking at the history:

****Round 5:****
- They returned \$7.50 from \$13.08 (57%)

****Round 6:****
- They sent \$4.00
- I returned well (they saw \$7.77)

****Round 7:****
- They returned \$8.50 from \$14.19 (60%)

****Round 8:****
- They sent \$3.60 (72% - still quite trusting)
- I received \$10.80

****Analysis:****
This co-player has been consistently cooperative:
- Returns around 57-60% when receiving
- Sends high amounts (72-80%+)
- Sustained cooperation over 7 rounds

They sent slightly less this round (\$3.60 vs \$4.00 previously), which is still generous at 72%. I should return consistently around 58-60% to match our pattern:

~58% of \$10.80 = \$6.26

I'll return around \$6.5 to maintain our cooperative relationship.

usage:

```
{
  "input_tokens": 3248,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.77 returned, visible payoff \$8.77.
- Round 7 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$8.5, visible payoff \$5.690000000000001.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.31 returned, visible payoff \$7.31.

Your total visible earnings across all rounds are \$52.980000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our continued partnership:

****Round 6:****

- They sent \$4.00 (80% - very trusting)

****Round 7:****

- I sent \$4 (they saw \$4.73)
- They returned \$8.50 (60%)

****Round 8:****

- They sent \$4.00 again (80% - consistent trust)
- I returned \$6.50 (they saw \$6.31)

****Round 9:****

I'm the sender.

****Analysis:****

This co-player has shown exceptional consistency:

- Repeatedly sending 80% when they're the sender
- Fair returns (~60%) when they're the receiver
- 8 rounds of stable cooperation
- Unwavering trust even when returns vary slightly

I'll continue our established pattern and send ****\$4**** (80%):

- Matches their consistent trust level
- Becomes \$12 for them
- Maintains our highly successful cooperative relationship

'''json

{"send": 4}

'''

reasoning:

Looking at the history:

****Round 6:****

- They sent \$4.00 (80% - very trusting)
- I returned well (they saw \$7.77)

****Round 7:****

- I sent \$4 (they saw \$4.73)
- They returned \$8.50 (60%)

****Round 8:****

- They sent \$4.00 again (80% - consistent trust)
- I returned \$6.5 (they saw \$6.31)

****Round 9:****

I'm the sender.

****Analysis:****

This co-player has been extremely reliable and trustworthy:

- Consistently sending 80% when they're the sender (Rounds 4, 6, 8)
- Returning 57-60% when they're the receiver
- 8 rounds of successful cooperation
- Very stable pattern

They've shown unwavering high trust even when I sent slightly less in Round 8. I should continue matching their trust level.

I'll send ****\$4**** (80%):

- Maintains our proven pattern
- Becomes \$12 for them
- Shows continued reciprocity
- We've built very strong trust over 8 rounds

usage:

{


```
"input_tokens": 3375,
"output_tokens": 513,
"reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$8.5, visible payoff \$5.690000000000001.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.31 returned, visible payoff \$7.31.
- Round 9 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$7.0, visible payoff \$4.76.

Your total visible earnings across all rounds are \$60.86000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$4.26, so you received \$12.78. How much do you return? (0-12.78)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Round 7:****

- They returned \$8.50 from \$14.19 (60%)

****Round 8:****

- They sent \$4.00, trusting
- I returned well

****Round 9:****

- They returned \$7.00 from \$11.76 (60%)

****Round 10:****

- They sent \$4.26 (85% - very trusting!)
- I received \$12.78

****Analysis:****

This co-player continues to demonstrate:

- Consistent fair returns (~60% when receiving)
- High trust (85% sent this round)
- 9 rounds of stable cooperation

I'll return approximately 60% to maintain our established pattern:

~60% of \$12.78 = \$7.67

```
'''json
{"return": 7.7}
'''
```

This rewards their continued high trust and maintains our successful cooperative relationship.

Reasoning:

Looking at the history:

****Round 7:****

- They returned \$8.50 from \$14.19 (60%)

****Round 8:****

- They sent \$4.00 (80%)
- I returned \$6.50 (they saw \$6.31)

****Round 9:****

- They returned \$7.00 from \$11.76 (60%)

****Round 10:****

- They sent \$4.26 (85% - very trusting)
- I received \$12.78

****Analysis:****

This co-player has been incredibly consistent:

- Returning ~60% when receiving
- Sending 80-85% when sending
- 9 rounds of steady cooperation
- They're maintaining high trust

I should return approximately 60% to match our established cooperative pattern:

60% of \$12.78 = \$7.67

I'll return \$7.5 or \$7.7 to stay consistent with the pattern.

```
Usage: {  
  "input_tokens": 3450,  
  "output_tokens": 439,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:03:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.51 returned, visible payoff \$7.51.
- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.

Your total visible earnings across all rounds are \$16.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

- My co-player has been consistently cooperative, sending \$3.0 in both rounds they were the sender
- In Round 3, they saw only \$2.32 returned (likely due to communication noise), which might have disappointed them
- Their history shows they value cooperation despite the lower return they observed

I want to maintain our cooperative relationship and signal that I'm still willing to trust them. Sending a similar amount to what they've been sending shows reciprocal cooperation.

{'send': 3.0}

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER sent \$3.0, saw \$5.51 returned, payoff \$7.51

Round 2: Co-player as RECEIVER, saw \$3.8 sent, returned \$5.5, payoff \$5.899999999999999

Round 3: Co-player as SENDER sent \$3.0, saw \$2.32 returned, payoff \$4.32

Wait, in Round 3 I was the RECEIVER and I sent back \$3.0, but they saw \$2.32 returned. This is the communication noise mentioned!

So my co-player has been consistently sending \$3.0 when they're the sender, showing cooperative behavior. However, in Round 3 they saw only \$2.32 returned even though I actually sent \$3.0. This might make them think I'm less cooperative than I actually am.

Given that:

1. They've been consistently cooperative, sending \$3.0 twice

2. They may think I was less generous in Round 3 than I actually was (due to noise)
3. I want to maintain the cooperative relationship

I should send a similar amount to what they've been sending - around \$3.0 - to signal continued cooperation.

```
{'send': 3.0}
```

usage:

```
{
  "input_tokens": 1559,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.8 sent, it became \$11.399999999999999, returned \$5.5, visible payoff \$5.899999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.
- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.

Your total visible earnings across all rounds are \$22.919999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.72, so you received \$8.16. How much do you return? (0-8.16)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate trust
- They currently have \$2.28 left
- I received \$8.16

The history shows we've maintained a cooperative relationship. They've consistently returned reasonable amounts when they were the receiver (around 50% of what they received in Round 4).

If I return \$4.0:

- Sender ends with: $\$2.28 + \$4.0 = \$6.28$ (profit of \$1.28)
- I keep: $\$8.16 - \$4.0 = \$4.16$ (profit of \$4.16)

This ensures both of us profit while maintaining our cooperative pattern.

```
{'return': 4.0}
```

reasoning:

Looking at the situation:

- My co-player sent \$2.72 (54.4% of their endowment), showing moderate-to-high trust
- They currently have \$2.28 left ($\$5 - \2.72)
- I received \$8.16

Looking at the history:

- Round 2: They returned \$5.5 when they received \$11.40
- Round 3: They saw \$2.32 returned (due to noise, I actually sent \$3.0)
- Round 4: They returned \$5.0 when they received \$9.84

They've been cooperative and returned reasonable amounts. In Round 4, they returned about 51% of what they received.

If I return \$4.0:

- Sender ends with: $\$2.28 + \$4.0 = \$6.28$ (profit of \$1.28)
- I keep: $\$8.16 - \$4.0 = \$4.16$ (profit of \$4.16)

This ensures both profit and maintains the cooperative relationship. The sender gets a reasonable return on their trust.

Let me consider returning slightly more since they've been consistently cooperative...

If I return \$4.5:

- Sender ends with: $\$2.28 + \$4.5 = \$6.78$ (profit of \$1.78)
- I keep: $\$8.16 - \$4.5 = \$3.66$ (profit of \$3.66)

I think \$4.0 is fair - it's about 49% of what I received, which is a reasonable return rate that ensures mutual benefit.

usage:

```
{
  "input_tokens": 1919,
  "output_tokens": 562,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.32 returned, visible payoff \$4.32.
- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.88 returned, visible payoff \$6.88.

Your total visible earnings across all rounds are \$27.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- My co-player has consistently sent \$3.0 when they're the sender (Rounds 3 and 5)
- They've been cooperative as a receiver, returning \$5.0 in Round 4
- In Round 5, they saw a reasonable return (\$4.88) and had a positive payoff

Our relationship has been mutually cooperative. I should continue matching their level of trust and cooperation by sending a similar amount.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

- My co-player has been consistently sending \$3.0 when they're the sender (Rounds 3 and 5)
- They've been cooperative as a receiver, returning good amounts (returned \$5.0 in Round 4)
- In Round 5, they saw \$4.88 returned and had a payoff of \$6.88, which is positive

Our relationship has been quite cooperative overall. They've shown consistent trust by sending \$3.0 repeatedly. I should reciprocate this trust and continue the cooperative pattern.

Sending \$3.0 matches what they've been doing and signals that I'm committed to cooperation.

usage:

```
{
  "input_tokens": 2316,
  "output_tokens": 272,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.28 sent, it became \$9.84, returned \$5.0, visible payoff \$4.84.
 - Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.88 returned, visible payoff \$6.88.
 - Round 6 against you, as RECEIVER: saw \$3.04 sent, it became \$9.120000000000001, returned \$4.5, visible payoff \$4.620000000000001.
- Your total visible earnings across all rounds are \$33.76.

This round, you are the RECEIVER against your current co-player. They sent you \$2.95, so you received \$8.850000000000001. How much do you return? (0-8.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.95 (59% of their endowment), showing good trust
- They currently have \$2.05 left
- I received \$8.85

The history shows we've maintained a strong cooperative relationship. They've been consistent in their trust, and I should continue reciprocating appropriately.

If I return \$4.5:

- Sender ends with: \$2.05 + \$4.5 = \$6.55 (profit of \$1.55)
- I keep: \$8.85 - \$4.5 = \$4.35 (profit of \$4.35)

This maintains our mutually beneficial pattern.

{'return': 4.5}

reasoning:

Looking at this situation:

- My co-player sent \$2.95 (59% of their endowment), showing good trust
- They currently have \$2.05 left
- I received \$8.85

The history shows:

- They've been consistently cooperative, sending around \$3 when they're the sender
- They've returned reasonable amounts when they're the receiver (around \$4.5-\$5.0)
- We've maintained a mutually beneficial relationship

I should return an amount that:

1. Ensures the sender profits from trusting me
2. Maintains our cooperative pattern
3. Is proportional to what they sent

If I return \$4.5:

- Sender ends with: \$2.05 + \$4.5 = \$6.55 (profit of \$1.55)
- I keep: \$8.85 - \$4.5 = \$4.35 (profit of \$4.35)

This seems fair and maintains our cooperation pattern.

usage:

```
{
  "input_tokens": 2675,
  "output_tokens": 402,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.88 returned, visible payoff \$6.88.
- Round 6 against you, as RECEIVER: saw \$3.04 sent, it became \$9.120000000000001, returned \$4.5, visible payoff \$4.620000000000001.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.69 returned, visible payoff \$6.69.

Your total visible earnings across all rounds are \$38.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- My co-player has been extremely consistent, sending \$3.0 in every round they're the sender (Rounds 5 and 7)
- They've cooperated well as a receiver, returning \$4.5 in Round 6
- They've seen reasonable returns and positive payoffs

We've established a strong, mutually beneficial cooperative pattern. I should continue matching their consistent level of trust.

{'send': 3.0}

reasoning:

Looking at the visible history:

- My co-player has been very consistently sending \$3.0 when they're the sender (Rounds 5 and 7)
- They've been cooperative as a receiver, returning \$4.5 in Round 6
- In Round 7, they saw \$4.69 returned and had a positive payoff of \$6.69

We've established a strong cooperative pattern over many rounds. They consistently send \$3.0, and I've been reciprocating with reasonable returns. I should continue this pattern to maintain trust and cooperation.

Sending \$3.0 matches their consistent behavior and signals continued cooperation.

usage:

```
{
  "input_tokens": 2826,
  "output_tokens": 257,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.04 sent, it became \$9.120000000000001, returned \$4.5, visible payoff \$4.620000000000001.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.69 returned, visible payoff \$6.69.
- Round 8 against you, as RECEIVER: saw \$2.95 sent, it became \$8.850000000000001, returned \$4.5, visible payoff \$4.350000000000001.

Your total visible earnings across all rounds are \$44.34.

This round, you are the RECEIVER against your current co-player. They sent you \$2.68, so you received \$8.040000000000001. How much do you return? (0-8.040000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.68 (53.6% of their endowment), showing moderate trust
- They currently have \$2.32 left
- I received \$8.04

Our history shows a consistently cooperative relationship. They've been reliable as both sender and receiver. I should continue reciprocating fairly.

If I return \$4.0:

- Sender ends with: \$2.32 + \$4.0 = \$6.32 (profit of \$1.32)
- I keep: \$8.04 - \$4.0 = \$4.04 (profit of \$4.04)

This maintains our mutually beneficial pattern.

```
{'return': 4.0}
```

reasoning:

Looking at the situation:

- My co-player sent \$2.68 (53.6% of their endowment), showing moderate trust
- They currently have \$2.32 left
- I received \$8.04

History shows we've had a consistent cooperative relationship. They've been returning around \$4.5 when they're the receiver, which is about 50% of what they receive.

If I return \$4.0:

- Sender ends with: $\$2.32 + \$4.0 = \$6.32$ (profit of \$1.32)
- I keep: $\$8.04 - \$4.0 = \$4.04$ (profit of \$4.04)

This seems fair and maintains our cooperative pattern. Both players profit and the relationship continues positively.

usage:

```
{
  "input_tokens": 2942,
  "output_tokens": 356,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.69 returned, visible payoff \$6.69.
- Round 8 against you, as RECEIVER: saw \$2.95 sent, it became \$8.850000000000001, returned \$4.5, visible payoff \$4.350000000000001.
- Round 9 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.86 returned, visible payoff \$6.86.

Your total visible earnings across all rounds are \$48.38.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- My co-player has been consistently sending \$3.0 in every round they're the sender (Rounds 7 and 9)
- They've been cooperative as a receiver, returning \$4.5 in Round 8
- They've consistently seen positive payoffs and good returns

We've established a strong, stable cooperative relationship over many rounds. I should continue matching their consistent level of trust.

```
{'send': 3.0}
```

Reasoning:

Looking at the visible history:

- My co-player has been consistently sending \$3.0 when they're the sender (Rounds 7 and 9)
- They've cooperated well as a receiver, returning \$4.5 in Round 8
- They've seen good returns and positive payoffs consistently

We've maintained a strong cooperative relationship throughout all 9 rounds. I should continue this pattern by sending \$3.0 again.

This is Round 10, which might be getting close to the end of the game. However, I don't see any indication that this is the final round, and the history shows our cooperation has been mutually beneficial. I should continue cooperating.

```
Usage: {
  "input_tokens": 2954,
  "output_tokens": 264,
  "reasoning_tokens": 0
}
```


GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:04:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.

Your total visible earnings across all rounds are \$17.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.28, so you received \$9.84. How much do you return? (0-9.84)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER, sent \$3 (they saw \$3.37 due to noise). They received \$10.11, returned \$5. Good cooperation.

Round 2: They were SENDER, sent \$3 (I saw it as \$3.8 due to noise), I received \$11.40, returned \$5.5 (they saw \$6.21 due to noise).

Round 3: I was SENDER, sent \$3 (they saw \$2.05 due to noise - significant noise here!). They received \$6.15, returned \$3. That's about half, which is reasonable given what they received.

Round 4: They are SENDER, sent something (I see \$3.28), I received \$9.84.

My co-player has been cooperative throughout. They've been returning reasonable amounts. The noise is creating some discrepancies, but overall the pattern is cooperative.

I received \$9.84. To maintain cooperation and reciprocity, I should return a fair amount. Looking at the pattern:

- Round 1: They returned \$5 out of \$10.11 (about 49%)
- Round 3: They returned \$3 out of \$6.15 (about 49%)

A fair return would be around 50% or slightly more to encourage continued cooperation. That would be around \$5.

Let me return \$5 to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 1253,
  "output_tokens": 355,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.21 returned, visible payoff \$8.21.
- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.
Your total visible earnings across all rounds are \$22.569999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: They were RECEIVER, saw \$3.37 sent, received \$10.11, returned \$5.0

Round 2: They were SENDER, sent \$3.0, became \$9.0, saw \$6.21 returned (I returned something that appeared as \$6.21 to them)

Round 3: They were RECEIVER, saw \$2.05 sent, received \$6.15, returned \$3.0

Round 4: They were SENDER, sent \$3.0, became \$9.0, saw \$4.45 returned (I returned something that appeared as \$4.45 to them)

So the pattern I'm seeing:

- When they're sender, they consistently send \$3.0
- When they're receiver, they return a fair amount (\$5 on \$10.11, \$3 on \$6.15)

In Round 4, I returned \$5 but they saw \$4.45 due to noise. This might make them think I'm less cooperative than I actually am.

Given that they continue to send \$3 consistently, I should continue to send a similar amount to maintain cooperation. I'll send \$3 again.

usage:

```
{
  "input_tokens": 1496,
  "output_tokens": 301,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.

- Round 5 against you, as RECEIVER: saw \$2.72 sent, it became \$8.16, returned \$4.0, visible payoff \$4.16.

Your total visible earnings across all rounds are \$29.449999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.04, so you received \$9.120000000000001. How much do you return? (0-9.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the history with my current co-player:

- Round 1: They were receiver, saw \$3.37 sent, got \$10.11, returned \$5.0
- Round 2: They were sender, sent \$3.0, it became \$9.0, saw \$6.21 returned
- Round 3: They were receiver, saw \$2.05 sent, got \$6.15, returned \$3.0
- Round 4: They were sender, sent \$3.0, it became \$9.0, saw \$4.45 returned
- Round 5: They were receiver, saw \$2.72 sent, got \$8.16, returned \$4.0

Now in Round 6: They are sender, sent \$3.04, I received \$9.12

This co-player has shown consistent behavior:

- When they receive money, they return a reasonable amount (around \$3-5)
- When they send, they consistently send \$3.0

I should continue the cooperative pattern. They sent me \$3.04, giving me \$9.12. A fair return would be around \$4.5-5 to maintain the cooperative relationship and ensure both parties benefit.

Let me return \$4.5 to keep things balanced.

usage:

```
{
  "input_tokens": 1760,
  "output_tokens": 308,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.45 returned, visible payoff \$6.45.
- Round 5 against you, as RECEIVER: saw \$2.72 sent, it became \$8.16, returned \$4.0, visible payoff \$4.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.68 returned, visible payoff \$6.68.

Your total visible earnings across all rounds are \$34.069999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: I was sender, sent \$3 (they saw \$3.37), they received \$10.11, returned \$5
 Round 2: I was receiver, they sent \$3.8, I received \$11.40, I returned \$5.5
 Round 3: I was sender, sent \$3 (they saw \$2.05), they received \$6.15, returned \$3
 Round 4: I was receiver, they sent \$3.28, I received \$9.84, I returned \$5
 Round 5: I was sender, sent \$3 (they saw \$2.72), they received \$8.16, returned \$4
 Round 6: I was receiver, they sent \$3.04, I received \$9.12, I returned \$4.5
 Round 7: I am sender now

The pattern shows we've been cooperating reasonably well. The noise in communication is causing some variation in what they see vs what I send, but overall the cooperation is decent.

I should continue sending a similar amount to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 1995,
  "output_tokens": 271,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.72 sent, it became \$8.16, returned \$4.0, visible payoff \$4.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.68 returned, visible payoff \$6.68.
- Round 7 against you, as RECEIVER: saw \$2.95 sent, it became \$8.850000000000001, returned \$4.5, visible payoff \$4.350000000000001.

Your total visible earnings across all rounds are \$40.75999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.95, so you received \$8.850000000000001. How much do you return? (0-8.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze the history with my co-player:

Round 1: I was SENDER, sent \$3.37 (visible to them), they received \$10.11, returned \$5.0
Round 2: I was RECEIVER, they sent \$3.0, I received \$9.0, I returned \$5.5 (but they saw \$6.21 due to noise)
Round 3: I was SENDER, sent \$3.0, they received \$6.15 (visible as \$2.05 sent due to noise), returned \$3.0
Round 4: I was RECEIVER, they sent \$3.0, I received \$9.0, I returned \$5.0 (but they saw \$4.45 due to noise)
Round 5: I was SENDER, sent \$3.0, they received \$8.16 (visible as \$2.72 sent due to noise), returned \$4.0
Round 6: I was RECEIVER, they sent \$3.0, I received \$9.12 (\$3.04 sent), I returned \$4.5 (but they saw \$4.68 due to noise)
Round 7: I was SENDER, sent \$3.0, they received \$8.85 (visible as \$2.95 sent due to noise), returned \$4.5
Round 8: I'm RECEIVER, they sent \$2.95, I received \$8.85

Looking at the pattern:

- This co-player has been consistently sending around \$3
- Returns have been reasonable but declining slightly: \$5.0, \$3.0, \$4.0, \$4.5
- The amounts I've been returning seem to be getting slightly reduced in what they're seeing due to noise

In Round 8, they sent me \$2.95, giving me \$8.85. Based on the pattern and fair reciprocity, I should return around \$4-4.5. ...

usage:

```
{
  "input_tokens": 2158,
  "output_tokens": 480,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.68 returned, visible payoff \$6.68.
- Round 7 against you, as RECEIVER: saw \$2.95 sent, it became \$8.850000000000001, returned \$4.5, visible payoff \$4.350000000000001.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.23 returned, visible payoff \$6.23.

Your total visible earnings across all rounds are \$45.10999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER, sent \$3.37 (with noise), they received \$10.11, returned \$5.0
Round 2: They were SENDER, sent \$3.0, I received \$9.0, I returned \$6.21 (with noise)
Round 3: I was SENDER, sent \$3.0, they saw \$2.05 (noise), they received \$6.15, returned \$3

.0
Round 4: They were SENDER, sent \$3.28 (with noise), I received \$9.84, I returned \$5.0
Round 5: I was SENDER, sent \$3.0, they saw \$2.72 (noise), they received \$8.16, returned \$4
.0
Round 6: They were SENDER, sent \$3.04 (with noise), I received \$9.12, I returned \$4.5
Round 7: I was SENDER, sent \$3.0, they saw \$2.95 (noise), they received \$8.85, returned \$4
.5
Round 8: They were SENDER, sent \$2.95 (with noise), I received \$8.85, I returned \$4.5
Round 9: I am SENDER now

Pattern analysis:

- My co-player has consistently sent around \$3.0 when they're the sender
- When they're the receiver:
 - Round 1: Received \$10.11, returned \$5.0 (about 49%)
 - Round 3: Received \$6.15, returned \$3.0 (about 49%)
 - Round 5: Received \$8.16, returned \$4.0 (about 49%)
 - Round 7: Received \$8.85, returned \$4.5 (about 51%)

Looking at my own returns as receiver, I've been more generous-returning 49-69% of what I receive. My co-player maintains a steady ~50% return rate regardless of amount received, suggesting they're following a consistent reciprocity strategy. This mutual cooperation pattern indicates I should continue sending around \$3.0 to maintain the established dynamic....

usage:

```
{  
  "input_tokens": 2159,  
  "output_tokens": 672,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$2.95 sent, it became \$8.850000000000001, returned \$4.5, visible payoff \$4.350000000000001.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.23 returned, visible payoff \$6.23.
- Round 9 against you, as RECEIVER: saw \$2.68 sent, it became \$8.040000000000001, returned \$4.0, visible payoff \$4.040000000000001.

Your total visible earnings across all rounds are \$51.96999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.33, so you received \$9.99. How much do you return? (0-9.99)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

```
'''json  
{ "return": 5 }  
'''
```

Reasoning:

Let me analyze the pattern of play with this co-player:

Looking at the visible history:

- Round 1: I was SENDER, sent \$3.37 (visible as \$3.37), became \$10.11, they returned \$5.0
- Round 2: They were SENDER, sent \$3.0, became \$9.0, I returned \$6.21 (visible to them)
- Round 3: I was SENDER, sent \$2.05 (visible), became \$6.15, they returned \$3.0
- Round 4: They were SENDER, sent \$3.0, became \$9.0, I returned \$4.45 (visible to them as \$4.45)
- Round 5: I was SENDER, sent \$3.0, but visible to them as \$2.72, became \$8.16, they returned \$4.0
- Round 6: They were SENDER, sent \$3.0, became \$9.0, I returned \$4.68 (visible to them)
- Round 7: I was SENDER, sent \$3.0, visible as \$2.95, became \$8.85, they returned \$4.5
- Round 8: They were SENDER, sent \$3.0 (visible to me as \$2.95), became \$8.85, I returned \$4.23 (visible to them)
- Round 9: I was SENDER, sent \$3.0, visible as \$2.68, became \$8.04, they returned \$4.0
- Round 10: They are SENDER, sent \$3.33, became \$9.99

Looking at the pattern:

- They consistently send around \$3.0

- When they receive around \$9, they return between \$4.0-\$5.0
- I've been sending \$3.0 consistently
- I've been returning amounts that roughly match what they return
- In Round 10, they bumped their send to \$3.33 (the highest yet), creating \$9.99. When I've received amounts near \$9-\$10, I've returned \$4.5-\$5.0, and they seem to mirror this behavior by returning \$4.0-\$4.5 when receiving similar amounts...

```
Usage: {
  "input_tokens": 2226,
  "output_tokens": 846,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:04:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.41 returned, visible payoff \$5.41.
- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.

Your total visible earnings across all rounds are \$14.48.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 (60% of endowment) in Rounds 1 and 3
- As RECEIVER: Returned generously (\$4.5 out of \$8.97, ~50%)

****My previous sending:****

- Round 2: I sent and they saw \$2.99 (likely noise from my \$3.0)

They've demonstrated consistent trust and reciprocity. I should maintain this cooperative relationship by matching their sending level.

{'send': 3.0}

reasoning:

Looking at the history:

Round 1: Co-player as SENDER sent \$3.0, got \$3.41 back, payoff \$5.41

Round 2: Co-player as RECEIVER got \$2.99 sent (from me), returned \$4.5, payoff \$4.47

Round 3: Co-player as SENDER sent \$3.0, got \$3.91 back, payoff \$5.91

Pattern analysis:

- When my co-player is SENDER, they consistently send \$3.0 (60% of endowment)
- When they're RECEIVER, they return generously (\$4.5 out of \$8.97 ? 50%)
- They're showing consistent trust and cooperation

My sending history:

- Round 2: I sent \$3.0 (based on the visible history showing they saw \$2.99 sent due to noise)
- Round 3: Not explicitly shown as sender, but the history says they saw \$2.99 sent in Round 2

Actually, let me re-read:

Round 2: I was SENDER. I sent \$3.0. Due to noise, they saw \$2.99. They received \$8.97, returned \$4.5.

Round 3: I was RECEIVER. They sent \$2.7 (but I think they sent \$3.0 and noise made it \$2.7)

). I returned \$4.05.

Now Round 4: I'm SENDER again.

They've been consistently sending \$3.0 when they're sender. They've returned generously when receiver.

I should maintain the cooperative pattern by sending \$3.0 again.

usage:

```
{
  "input_tokens": 1604,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.99 sent, it became \$8.97, returned \$4.5, visible payoff \$4.470000000000001.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.
- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.

Your total visible earnings across all rounds are \$20.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.24, so you received \$6.720000000000001. How much do you return? (0-6.720000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Co-player's return pattern:****

- Round 2: Returned \$4.5 out of \$8.97 (~50%)
- Round 4: Returned \$3.5 out of \$7.05 (~50%)

****This round:****

- They sent \$2.24 (44.8% of endowment)
- I received \$6.72

I should maintain the cooperative pattern by returning around 50% like they have consistently done.

Returning \$3.36:

- Gives them \$2.76 + \$3.36 = \$6.12 total (profit of \$1.12)
- Gives me \$3.36 (also a profit)

{'return': 3.36}

reasoning:

Looking at the visible history:

Round 2: They received \$8.97, returned \$4.5 (50%)

Round 4: They received \$7.05, returned \$3.5 (49.6%)

This round:

- They sent \$2.24 (44.8% of endowment)
- I received \$6.72

The co-player has been consistently returning around 50% of what they receive. I should reciprocate similarly.

Returning around 50% would be \$3.36:

- Sender would have: \$2.76 + \$3.36 = \$6.12 (profit of \$1.12)
- I would keep: \$6.72 - \$3.36 = \$3.36 (also a profit)

This maintains the cooperative pattern and roughly matches their return rates.

usage:

```
{
  "input_tokens": 1990,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.91 returned, visible payoff \$5.91.
- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.5, visible payoff \$3.5500000000000007.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.

Your total visible earnings across all rounds are \$23.509999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Consistently sent \$3.0 in Rounds 3 and 5 (60% of endowment)
- As RECEIVER: Returned \$3.5 out of \$7.05 (~50%) in Round 4

****What they saw in Round 5:****

- They sent \$3.0, but saw only \$2.79 returned (likely due to noise)
- Despite this, they continue to cooperate

I should maintain the cooperative relationship by sending \$3.0 again, matching their consistent trust level.

{'send': 3.0}

reasoning:

Looking at the visible history:

****Co-player's behavior:****

- As SENDER: Consistently sent \$3.0 in Rounds 3 and 5
- As RECEIVER: Returned \$3.5 out of \$7.05 (~50%) in Round 4

****What they saw from me:****

- Round 3: Sent \$3.0, they returned \$3.91
- Round 4: They saw \$2.35 sent (likely noise from my \$3.0), returned \$3.5
- Round 5: Sent \$3.0, they saw only \$2.79 returned (but I actually sent \$3.36, so noise affected what they saw)

The co-player has been consistently sending \$3.0, showing steady trust. Even though in Round 5 they saw less returned than I actually sent (due to noise), they continue to cooperate.

I should maintain the cooperative pattern and send \$3.0 again to match their consistent trust level.

usage:

```
{
  "input_tokens": 2395,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned

\$3.5, visible payoff \$3.5500000000000007.

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.
- Round 6 against you, as RECEIVER: saw \$2.44 sent, it became \$7.32, returned \$3.66, visible payoff \$3.66.

Your total visible earnings across all rounds are \$30.08.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history:

****Co-player's return pattern:****

- Round 4: Returned \$3.5 out of \$7.05 (~50%)
- Round 6: Returned \$3.66 out of \$7.32 (~50%)

****This round:****

- They sent \$3.24 (64.8% of endowment) - showing strong trust
- I received \$9.72

They've been consistently returning around 50%. I should maintain this cooperative pattern

Returning \$4.86 (50%):

- Gives them \$1.76 + \$4.86 = \$6.62 total (profit of \$1.62)
- Gives me \$4.86 (also a good profit)

{'return': 4.86}

reasoning:

Looking at the history:

****Co-player's return pattern:****

- Round 4: Returned \$3.5 out of \$7.05 (~50%)
- Round 6: Returned \$3.66 out of \$7.32 (~50%)

****This round:****

- They sent \$3.24 (64.8% of endowment) - showing strong trust
- I received \$9.72

They've been consistently returning around 50% when they're the receiver. I should maintain this cooperative pattern.

Returning around 50% would be \$4.86:

- Gives them \$1.76 + \$4.86 = \$6.62 total (profit of \$1.62)
- Gives me \$4.86 (also a good profit)

This maintains the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 2783,
  "output_tokens": 383,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.79 returned, visible payoff \$4.79.
- Round 6 against you, as RECEIVER: saw \$2.44 sent, it became \$7.32, returned \$3.66, visible payoff \$3.66.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.47 returned, visible payoff \$7.47.

Your total visible earnings across all rounds are \$34.94.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Consistently sent \$3.0 in Rounds 5 and 7 (60% of endowment)
- As RECEIVER: Returned \$3.66 out of \$7.32 (~50%) in Round 6

****Round 7 outcome:****

- They sent \$3.0 and saw \$5.47 returned (a great return!)
- This showed them cooperation pays off

They've maintained consistent trust throughout our partnership. I should continue matching their sending level to sustain this cooperative relationship.

{'send': 3.0}

reasoning:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Sent \$3.0 in Rounds 5 and 7 (60% of endowment) - consistently cooperative
- As RECEIVER: Returned \$3.66 out of \$7.32 (~50%) in Round 6

****What they saw in Round 7:****

- They sent \$3.0, saw \$5.47 returned (likely noise from my \$4.86)
- This gave them a visible payoff of \$7.47 - a very good outcome

They've been consistently trusting, sending \$3.0 repeatedly. They also saw a good return in Round 7. I should continue the cooperative pattern by sending \$3.0.

This has been a mutually beneficial partnership, and maintaining trust is important.

usage:

```
{
  "input_tokens": 2917,
  "output_tokens": 342,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$2.44 sent, it became \$7.32, returned \$3.66, visible payoff \$3.66.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.47 returned, visible payoff \$7.47.
- Round 8 against you, as RECEIVER: saw \$2.9 sent, it became \$8.7, returned \$4.35, visible payoff \$4.35.

Your total visible earnings across all rounds are \$41.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the history:

****Co-player's return pattern:****

- Round 6: Returned \$3.66 out of \$7.32 (~50%)
- Round 8: Returned \$4.35 out of \$8.70 (~50%)

****This round:****

- They sent \$2.97 (59.4% of endowment)
- I received \$8.91

They continue to maintain the consistent 50% return pattern. I should reciprocate with the same ratio.

Returning \$4.45 (50%):
- Gives them \$2.03 + \$4.45 = \$6.48 total (profit of \$1.48)
- Gives me \$4.46 (also a good profit)

{'return': 4.45}

reasoning:

Looking at the visible history:

****Co-player's return pattern:****

- Round 6: Returned \$3.66 out of \$7.32 (~50%)
- Round 8: Returned \$4.35 out of \$8.7 (~50%)

****This round:****

- They sent \$2.97 (59.4% of endowment)
- I received \$8.91

They've been consistently returning around 50%. I should maintain the cooperative pattern.

Returning \$4.455 (50%):

- Gives them \$2.03 + \$4.455 = \$6.485 total (profit of \$1.485)
- Gives me \$4.455 (also a good profit)

This maintains the ~50% return ratio that has been the pattern.

usage:

```
{
  "input_tokens": 3016,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.47 returned, visible payoff \$7.47.
- Round 8 against you, as RECEIVER: saw \$2.9 sent, it became \$8.7, returned \$4.35, visible payoff \$4.35.
- Round 9 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.98 returned, visible payoff \$6.98.

Your total visible earnings across all rounds are \$45.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Consistently sent \$3.0 in Rounds 7 and 9 (60% of endowment)
- As RECEIVER: Returned \$4.35 out of \$8.70 (~50%) in Round 8

****Recent outcomes:****

- Round 7: They sent \$3.0, saw \$5.47 returned
- Round 9: They sent \$3.0, saw \$4.98 returned

They've maintained consistent trust throughout our partnership. Both of us have benefited from this cooperation. I should continue matching their sending level.

{'send': 3.0}

Reasoning:

Looking at the visible history:

****Co-player's pattern:****

- As SENDER: Consistently sent \$3.0 in Rounds 7 and 9 (60% of endowment)
- As RECEIVER: Returned \$4.35 out of \$8.70 (~50%) in Round 8

****Recent outcomes they saw:****

- Round 7: Sent \$3.0, saw \$5.47 returned (great return!)
- Round 9: Sent \$3.0, saw \$4.98 returned (still good, ~55% return)

They've shown unwavering trust throughout our partnership, consistently sending \$3.0. The cooperation has been mutually beneficial - they've had positive outcomes and so have I.

I should continue matching their sending level to maintain this successful cooperative relationship.

```
{'send': 3.0}
Usage: {
  "input_tokens": 3044,
  "output_tokens": 360,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:04:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.29 sent, it became \$6.87, returned \$3.15, visible payoff \$3.72.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.10000000000001, returned \$4.05, visible payoff \$4.050000000000002.

Your total visible earnings across all rounds are \$15.790000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$3.15 on \$6.87 (46%)
Round 2: They sent \$3 and saw good returns
Round 3: They returned \$4.05 on \$8.10 (50%)
Round 4 (now): They sent \$2.35, I received \$7.05

They've been consistently fair, returning around 46-50% when receiving. They sent slightly less this round (\$2.35 vs \$3 in Round 2), but that's still a reasonable amount showing continued trust.

I'll maintain our cooperative pattern by returning approximately 50% of what I received.

```
'''json
{"return": 3.50}
'''
```

reasoning:

Looking at the history:

Round 1: Co-player as receiver - saw \$2.29 sent, got \$6.87, returned \$3.15 (46% of received)
Round 2: Co-player as sender - sent \$3, became \$9, saw \$4.71 returned
Round 3: Co-player as receiver - saw \$2.7 sent, got \$8.1, returned \$4.05 (50% of received)
Round 4 (now): Co-player as sender - sent \$2.35, I received \$7.05

The co-player has been consistent in returning around 46-50% when receiving. They've also been willing to send decent amounts when they're the sender.

However, I notice they sent \$2.35 this round, which is less than the \$3 they sent in round 2. This could be:

1. Due to communication noise affecting their perception of my returns

2. A strategic reduction
3. Random variation

Looking at their visible payoffs from my perspective, they seem to be doing well in our exchanges. I should continue cooperating to maintain this mutually beneficial relationship.

I've been returning around 50% consistently. Given they sent \$2.35 (about 47% of their endowment), which is slightly less than before, I'll return a fair amount but perhaps slightly less than exactly half to account for the reduced send.

I'll return around \$3.50, which is about 50% of what I received.

usage:

```
{
  "input_tokens": 1494,
  "output_tokens": 509,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.71 returned, visible payoff \$6.71.
- Round 3 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$4.05, visible payoff \$4.050000000000002.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.67 returned, visible payoff \$5.67.

Your total visible earnings across all rounds are \$19.340000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our ongoing cooperation:

Round 2: They sent \$3, saw good returns (\$4.71)

Round 3: They returned \$4.05 on \$8.10 (50%)

Round 4: They sent \$3 again, saw \$3.67 returned

This co-player has been consistently cooperative:

- Sending \$3 when they're the sender
- Returning around 50% when they're the receiver

We've established a strong pattern of mutual trust. I'll continue by sending the same amount they've been sending.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

Round 2: They sent \$3 (as sender), saw \$4.71 returned

Round 3: They saw \$2.7 sent, returned \$4.05 on \$8.10 (50% return rate)

Round 4: They sent \$3 (as sender), saw \$3.67 returned

So they've been sending \$3 consistently when they're the sender. They've been returning around 50% when they're the receiver.

This is a cooperative partner. I should continue the pattern. I've been sending \$3 in previous rounds (rounds 1 and 3), and they've been matching that behavior.

I'll send \$3 again to maintain our cooperative relationship.

usage:

```
{
  "input_tokens": 1887,
```



```

        "output_tokens": 314,
        "reasoning_tokens": 0
    }
}
Message 6 role=user
content:
    Round 6

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 3 against you, as RECEIVER: saw $2.7 sent, it became $8.100000000000001, returned $4.05, visible payoff $4.050000000000002.
    - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $3.67 returned, visible payoff $5.67.
    - Round 5 against you, as RECEIVER: saw $2.24 sent, it became $6.720000000000001, returned $3.36, visible payoff $3.360000000000008.
    Your total visible earnings across all rounds are $24.130000000000003.

    This round, you are the RECEIVER against your current co-player. They sent you $2.44, so you received $7.32. How much do you return? (0-7.32)
    Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant
content:
    Looking at our continued pattern:

    Round 3: They returned $4.05 on $8.10 (50%)
    Round 4: They sent $3
    Round 5: They returned $3.36 on $6.72 (50%)
    Round 6 (now): They sent $2.44, I received $7.32

    This co-player has consistently returned about 50% when receiving, showing fair and predictable cooperation. I'll maintain the same standard.

    '''json
    {"return": 3.66}
    '''
reasoning:
    Looking at the history:

    Round 3: They returned $4.05 on $8.10 (50%)
    Round 4: They sent $3, saw $3.67 returned
    Round 5: They saw $2.24 sent (noise), received $6.72, returned $3.36 (50%)
    Round 6 (now): They sent $2.44, I received $7.32

    They continue to show consistent cooperation, returning 50% when they're the receiver. Their sending amounts have been steady around $2.35-$3.

    I should continue my pattern of returning approximately 50% to maintain our cooperative relationship.

    50% of $7.32 = $3.66
usage:
    {
        "input_tokens": 2274,
        "output_tokens": 292,
        "reasoning_tokens": 0
    }
}
Message 8 role=user
content:
    Round 7

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $3.67 returned, visible payoff $5.67.
    - Round 5 against you, as RECEIVER: saw $2.24 sent, it became $6.720000000000001, returned $3.36, visible payoff $3.360000000000008.
    - Round 6 against you, as SENDER: sent $3.0, it became $9.0, saw $4.57 returned, visible payoff $6.57.
    Your total visible earnings across all rounds are $27.790000000000003.

```

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the continued cooperation:

Round 4: They sent \$3, saw \$3.67 returned

Round 5: They returned \$3.36 (50%)

Round 6: They sent \$3, saw \$4.57 returned

This co-player has been remarkably consistent and trustworthy:

- Always sending \$3 as sender (60% of endowment)
- Always returning around 50% as receiver

We've built a strong cooperative relationship over 6 rounds. I'll maintain this successful pattern by sending the same amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the pattern:

Round 4: They sent \$3, saw \$3.67 returned

Round 5: They returned \$3.36 on \$6.72 (50%)

Round 6: They sent \$3, saw \$4.57 returned

This co-player has been very consistent:

- Always sending \$3 as sender
- Always returning around 50% as receiver

We have a well-established cooperative relationship. They've been trustworthy and fair throughout. I should continue sending \$3 to maintain this successful pattern.

usage:

```
{
  "input_tokens": 2626,
  "output_tokens": 268,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.36, visible payoff \$3.3600000000000008.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$3.24 sent, it became \$9.72, returned \$4.86, visible payoff \$4.86.

Your total visible earnings across all rounds are \$35.260000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our established pattern:

Round 5: They returned \$3.36 (50%)

Round 6: They sent \$3

Round 7: They returned \$4.86 on \$9.72 (50%)

Round 8 (now): They sent \$2.9, I received \$8.7

After 8 rounds together, this co-player has been exceptionally consistent and fair:

- Always returning 50% as receiver
- Consistently trustworthy as sender

I'll maintain our successful cooperative pattern by returning 50%.

```
'''json
{"return": 4.35}
'''
```

reasoning:

Looking at the history:

Round 5: They returned \$3.36 on \$6.72 (50%)
Round 6: They sent \$3, saw \$4.57 returned
Round 7: They returned \$4.86 on \$9.72 (50%)
Round 8 (now): They sent \$2.9, I received \$8.7

This co-player has been incredibly consistent throughout our entire relationship:

- Always returns 50% as receiver
- Always sends \$3 as sender
- Very reliable and trustworthy partner

I should continue our established pattern of returning 50% of what I receive to maintain this successful cooperation.

50% of \$8.7 = \$4.35

This is our 8th round together, and we've built a very strong pattern of mutual trust and fair play. I'll maintain it.

usage:

```
{
  "input_tokens": 2822,
  "output_tokens": 340,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$3.24 sent, it became \$9.72, returned \$4.86, visible payoff \$4.86.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.38 returned, visible payoff \$6.38.

Your total visible earnings across all rounds are \$39.61000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our ongoing cooperation:

Round 6: They sent \$3, saw good returns
Round 7: They returned \$4.86 (50%)
Round 8: They sent \$3, saw \$4.38 returned
Round 9 (now): I'm the sender

After 8 rounds, this has been one of the most consistent and cooperative partnerships:

- They always send \$3 as sender
- They always return 50% as receiver

I'll continue this highly successful pattern.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the pattern with this co-player:

Round 6: They sent \$3, saw \$4.57 returned

Round 7: They returned \$4.86 (50%)
Round 8: They sent \$3, saw \$4.38 returned
Round 9 (now): I'm the sender

This co-player has been very consistent:
- They send \$3 when they're the sender
- They return 50% when they're the receiver

We've established an excellent cooperative relationship over 8 rounds. I should continue the pattern of sending \$3.

usage:

```
{
  "input_tokens": 2908,
  "output_tokens": 263,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$3.24 sent, it became \$9.72, returned \$4.86, visible payoff \$4.86.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.38 returned, visible payoff \$6.38.
- Round 9 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.45, visible payoff \$4.46.

Your total visible earnings across all rounds are \$46.59.

This round, you are the RECEIVER against your current co-player. They sent you \$3.83, so you received \$11.49. How much do you return? (0-11.49)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our established cooperation:

Round 7: They returned \$4.86 (50%)
Round 8: They sent \$3
Round 9: They returned \$4.45 on \$8.91 (50%)
Round 10 (now): They sent \$3.83, I received \$11.49

This co-player has been remarkably consistent over 9 rounds, always returning around 50%. They even sent more this round (\$3.83 vs their usual \$3), showing continued trust.

I'll maintain our very successful cooperative pattern.

```
'''json
{"return": 5.75}
'''
```

Reasoning:

Looking at the pattern:

Round 7: They returned \$4.86 (50%)
Round 8: They sent \$3
Round 9: They returned \$4.45 on \$8.91 (approximately 50%)
Round 10 (now): They sent \$3.83, I received \$11.49

This co-player has been extremely consistent throughout our partnership:

- Always returning around 50% as receiver
- Sending \$3 or slightly more as sender

They just sent \$3.83, which is more than their usual \$3. This shows continued or increased trust. I should maintain our cooperative pattern by returning approximately 50%.

50% of \$11.49 = \$5.745

I'll return around that amount to maintain our successful cooperation.

Usage: {

```
"input_tokens": 2943,
"output_tokens": 322,
```

```
"reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:04:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_7",
"pairing": {
  "agents": [
    "Agent_8",
    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.2 returned, visible payoff \$4.2.
- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.

Your total visible earnings across all rounds are \$15.69.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60% of endowment)
- Round 2: Co-player returned \$6.0 out of \$11.1 (very generous, 54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment), saw \$5.32 returned

My co-player has consistently shown high trust and cooperation, sending substantial amounts and returning generously when in the receiver role. They've maintained this cooperative behavior throughout.

I'll match their highest level of trust by sending a similar amount.

```
{"send": 3.5}
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: Co-player sent \$3.0, saw \$2.2 returned (I actually returned \$3.0, but noise)
 Round 2: I sent \$3.0 (co-player saw \$3.7), they returned \$6.0 - very generous!
 Round 3: Co-player sent \$3.5 (actually \$3.3 based on my perspective), saw \$5.32 returned (I actually returned \$5.0)

My co-player has been consistently cooperative:

- Sending high amounts (60-70% of endowment)
- When they were receiver, they returned \$6.0 out of \$11.1 (54% return rate)

They've maintained cooperation throughout despite seeing some noise in the returns. In Round 3, they saw \$5.32 returned on their \$3.5 sent, which gave them a profit of \$1.82 - decent but not extremely generous.

As the sender now, I should:

1. Continue the cooperative pattern
2. Send an amount that shows trust matching theirs
3. Given they sent \$3.0, \$3.5 in previous rounds, I should send similarly

I'll send \$3.5 to match their highest level of trust and continue the positive dynamic.

usage:

```
{
  "input_tokens": 1694,
  "output_tokens": 461,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.7 sent, it became \$11.100000000000001, returned \$6.0, visible payoff \$5.100000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.
- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.

Your total visible earnings across all rounds are \$22.18.

This round, you are the RECEIVER against your current co-player. They sent you \$3.42, so you received \$10.26. How much do you return? (0-10.26)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (generous 54% return)
- Round 3: Co-player sent \$3.5 (70% of endowment, high trust)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return)
- Round 5: Co-player sent \$3.42 (68% of endowment, continued trust)

My co-player has consistently demonstrated strong cooperation, both sending high amounts and returning generously. They're maintaining this pattern across rounds.

If I return \$5.5:

- Sender ends with: \$1.58 + \$5.5 = \$7.08 (profit of \$2.08)
- I keep: \$10.26 - \$5.5 = \$4.76

This matches their typical return rate (~54%) and reciprocates their continued trust.

{"return": 5.5}

reasoning:

Looking at the visible history:

- Round 2: Co-player returned \$6.0 out of \$11.1 (54% return rate)
- Round 3: Co-player sent \$3.5 (70% of endowment)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.42 (68% of endowment)

My co-player has been very cooperative and consistent:

- As sender: sending around 68-70% of their endowment
- As receiver: returning around 54-56% of what they receive

They're maintaining high trust and generosity throughout. I should reciprocate appropriately.

If I return \$5.5:

- Sender ends with: \$1.58 + \$5.5 = \$7.08 (profit of \$2.08)
- I keep: \$10.26 - \$5.5 = \$4.76

This is approximately 54% return rate, matching their pattern and continuing our cooperative relationship.

Actually, let me be slightly more generous to maintain the positive dynamic. Let me return \$5.5 or \$6.0.

Return of \$5.5 gives them \$7.08 total (profit of \$2.08)

Return of \$6.0 gives them \$7.58 total (profit of \$2.58)

I'll go with \$5.5 to match their typical return rate of around 54%.

usage:

```
{
  "input_tokens": 2082,
  "output_tokens": 593,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.32 returned, visible payoff \$6.82.
- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.

Your total visible earnings across all rounds are \$26.939999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- Round 3: Co-player sent \$3.5 (70% of endowment)
- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.5 (70% of endowment), saw \$6.23 returned, payoff \$7.73

My co-player has been remarkably consistent:

- As sender: consistently sends \$3.5 (70% of endowment), showing high trust
- As receiver: returns generously (~55% of received amount)

This is a very cooperative partner. I should match their consistent high level of trust.

{"send": 3.5}

reasoning:

Looking at the visible history:

- Round 3: Co-player sent \$3.5, saw \$5.32 returned, payoff \$6.82
- Round 4: Co-player returned \$5.0 out of \$9.0, payoff \$4.0
- Round 5: Co-player sent \$3.5, saw \$6.23 returned, payoff \$7.73

My co-player has been very consistent:

- When sending: consistently sends \$3.5 (70% of endowment)
- When receiving: returns generously (55-56% of what they receive)

They've maintained this high cooperation throughout all rounds. The pattern is clear and stable.

I should continue matching their level of trust and cooperation. They've been sending \$3.5 consistently as sender, and I've been doing the same.

{"send": 3.5}

usage:

```
{
  "input_tokens": 2535,
  "output_tokens": 382,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.0 sent, it became \$9.0, returned \$5.0, visible payoff \$4.0.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.
- Round 6 against you, as RECEIVER: saw \$3.29 sent, it became \$9.870000000000001, returned \$5.0, visible payoff \$4.870000000000001.

Your total visible earnings across all rounds are \$33.58.

This round, you are the RECEIVER against your current co-player. They sent you \$3.96, so you received \$11.879999999999999. How much do you return? (0-11.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return)
- Round 5: Co-player sent \$3.5 (70% of endowment)
- Round 6: Co-player returned \$5.0 out of \$9.87 (51% return)
- Round 7: Co-player sent \$3.96 (79% of endowment - their highest trust level yet!)

My co-player has escalated their trust, sending an even higher percentage than before. I should reciprocate this increased trust with a generous return.

If I return \$6.5:

- Sender ends with: $\$1.04 + \$6.5 = \$7.54$ (profit of \$2.54)
- I keep: $\$11.88 - \$6.5 = \$5.38$

This rewards their increased trust appropriately.

{"return": 6.5}

reasoning:

Looking at the visible history:

- Round 4: Co-player returned \$5.0 out of \$9.0 (56% return rate)
- Round 5: Co-player sent \$3.5, saw \$6.23 returned
- Round 6: Co-player returned \$5.0 out of \$9.87 (51% return rate)
- Round 7: Co-player sent \$3.96 (79% of endowment - highest yet!)

My co-player is continuing to show high trust and cooperation. In fact, they just sent \$3.96, which is 79% of their endowment - the highest percentage they've sent yet! This is showing even greater trust than before.

I should reciprocate this increased trust with a generous return.

If I return \$6.0:

- Sender ends with: $\$1.04 + \$6.0 = \$7.04$ (profit of \$2.04)
- I keep: $\$11.88 - \$6.0 = \$5.88$

This would be about a 50.5% return rate, which is consistent with their return patterns but rewards their increased trust.

Actually, let me be a bit more generous since they increased their trust level:

If I return \$6.5:

- Sender ends with: $\$1.04 + \$6.5 = \$7.54$ (profit of \$2.54)
- I keep: $\$11.88 - \$6.5 = \$5.38$

This rewards their increased trust more clearly while still giving me a good payoff.

usage:

```
{
  "input_tokens": 2952,
  "output_tokens": 597,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.23 returned, visible payoff \$7.73.
- Round 6 against you, as RECEIVER: saw \$3.29 sent, it became \$9.870000000000001, returned \$5.0, visible payoff \$4.870000000000001.
- Round 7 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.71 returned, visible payoff \$7.21.

Your total visible earnings across all rounds are \$38.959999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- Round 5: Co-player sent \$3.5 (70% of endowment)
- Round 6: Co-player returned \$5.0 out of \$9.87
- Round 7: Co-player sent \$3.5 (70% of endowment)

My co-player continues to show consistent, high trust behavior. They repeatedly send \$3.5 and return generously. This is a reliable cooperative partner.

I'll maintain our established pattern of mutual trust.

{"send": 3.5}

reasoning:

Looking at the visible history:

- Round 5: Co-player sent \$3.5 (70% of endowment), saw \$6.23 returned, payoff \$7.73
- Round 6: Co-player returned \$5.0 out of \$9.87 (51% return)
- Round 7: Co-player sent \$3.5 (70% of endowment), saw \$5.71 returned, payoff \$7.21

My co-player has been very consistent:

- As sender: consistently sends \$3.5 (70% of endowment)
- As receiver: returns around \$5.0 consistently (51% return rate)

This is a highly cooperative and consistent partner. They've maintained trust throughout the game. I should continue matching their level of trust by sending \$3.5.

We're building a strong cooperative relationship with consistent behavior on both sides.

usage:

```
{
  "input_tokens": 3123,
  "output_tokens": 343,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.29 sent, it became \$9.870000000000001, returned \$5.0, visible payoff \$4.870000000000001.
- Round 7 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.71 returned, visible payoff \$7.21.
- Round 8 against you, as RECEIVER: saw \$3.78 sent, it became \$11.34, returned \$6.5, visible payoff \$4.84.

Your total visible earnings across all rounds are \$46.19.

This round, you are the RECEIVER against your current co-player. They sent you \$4.37, so you received \$13.11. How much do you return? (0-13.11)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

- Round 6: Co-player returned \$5.0 out of \$9.87
- Round 7: Co-player sent \$3.5 (70% of endowment)
- Round 8: Co-player returned \$6.5 out of \$11.34 (57% return - very generous)
- Round 9: Co-player sent \$4.37 (87% of endowment - extremely high trust!)

My co-player has dramatically escalated their trust, sending nearly all their endowment. This is the highest level of trust shown so far. I should reciprocate this exceptional trust with a generous return.

If I return \$7.5:

- Sender ends with: \$0.63 + \$7.5 = \$8.13 (profit of \$3.13)
- I keep: \$13.11 - \$7.5 = \$5.61

This rewards their exceptional trust appropriately.

```
{"return": 7.5}
```

reasoning:

Looking at the visible history:

- Round 6: Co-player returned \$5.0 out of \$9.87
- Round 7: Co-player sent \$3.5 (70% of endowment)
- Round 8: Co-player returned \$6.5 out of \$11.34 (57% return)
- Round 9: Co-player sent \$4.37 (87% of endowment - this is extremely high trust!)

My co-player has shown escalating trust behavior. They went from consistently sending \$3.5 (70%) to now sending \$4.37 (87% of their endowment). They are also returning generously when in the receiver role (saw \$6.5 returned in round 8).

This is a very cooperative partner showing maximum trust. I should reciprocate this high level of trust.

If I return \$7.0:

- Sender ends with: $\$0.63 + \$7.0 = \$7.63$ (profit of \$2.63)
- I keep: $\$13.11 - \$7.0 = \$6.11$

This is about 53% return rate, which is appropriate and reciprocates their high trust.

Actually, let me think about this more. They sent \$4.37 out of \$5, which is 87.4% of their endowment. This is remarkably high trust. I should return something that makes them profitable while also keeping a reasonable amount.

If I return \$7.5:

- Sender ends with: $\$0.63 + \$7.5 = \$8.13$ (profit of \$3.13)
- I keep: $\$13.11 - \$7.5 = \$5.61$

This is about 57% return rate, matching their generous behavior in round 8.

usage:

```
{
  "input_tokens": 3200,
  "output_tokens": 657,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.71 returned, visible payoff \$7.21.
- Round 8 against you, as RECEIVER: saw \$3.78 sent, it became \$11.34, returned \$6.5, visible payoff \$4.84.
- Round 9 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.27 returned, visible payoff \$8.77.

Your total visible earnings across all rounds are \$51.8.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- Round 7: Co-player sent \$3.5 (70% of endowment)
- Round 8: Co-player returned \$6.5 out of \$11.34 (very generous 57% return)
- Round 9: Co-player sent \$3.5 (70% of endowment), saw \$7.27 returned

My co-player continues to show consistent high trust and cooperation. They maintain their pattern of sending \$3.5 (70% of endowment) and returning generously as receiver.

This has been a highly cooperative partnership throughout. I'll maintain our established pattern.

```
{"send": 3.5}
```

Reasoning:

Looking at the visible history:

- Round 7: Co-player sent \$3.5 (70% of endowment)
- Round 8: Co-player returned \$6.5 out of \$11.34 (57% return)
- Round 9: Co-player sent \$3.5 (70% of endowment), saw \$7.27 returned, payoff \$8.77

My co-player continues to maintain a pattern of high trust, consistently sending \$3.5 (70% of endowment). When they are the receiver, they return generously.

Looking at the overall pattern across all rounds:

- They consistently send \$3.5 when sender (with occasional variations like \$3.96 and \$4.37 see n earlier)
- They consistently return generously when receiver (50-57% of received amount)

This is a very reliable cooperative partner. I should continue matching their level of trust. They've been sending around 70% consistently, and I've been doing the same.

I'll send \$3.5 again to maintain our mutual cooperation.

```
Usage: {
  "input_tokens": 3222,
  "output_tokens": 411,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:04:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:**Message 1 role=system****content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.24 sent, it became \$6.720000000000001, returned \$3.0, visible payoff \$3.7200000000000006.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$16.12.

This round, you are the RECEIVER against your current co-player. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our interaction pattern:

****Round 1:**** They returned \$3.0 from \$6.72 (45%)
****Round 2:**** They sent \$3.0, I returned \$6.0
****Round 3:**** They returned \$5.0 from \$9.90 (51%)
****Round 4 (current):**** They sent \$3.0 ? I received \$9.0

They've shown consistent cooperation, with their return percentages increasing slightly over time. They sent \$3.0 this round, maintaining trust despite some communication noise in previous rounds.

I'll return ****\$5.0**** (about 56% of what I received). This gives them a total payoff of \$7.0 (\$2 kept + \$5 returned), which rewards their continued trust and maintains our cooperative relationship.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze the history:

****Round 1:**** I sent \$3 ? they saw \$2.24 ? they returned \$3.0

****Round 2:**** They sent \$3.7 ? I received \$11.10 ? I returned \$6.0 ? they saw \$5.07
****Round 3:**** I sent \$3.5 ? they saw \$3.3 ? they returned \$5.0
****Round 4 (current):**** They sent \$3.0 ? I received \$9.0

Looking at their behavior:

- Round 1 as receiver: Returned \$3.0 out of \$6.72 (45%)
- Round 3 as receiver: Returned \$5.0 out of \$9.90 (51%)

Their returns seem reasonable and slightly increasing in percentage.

My behavior:

- Round 2 as receiver: Returned \$6.0 out of \$11.10 (54%)

They've consistently sent around \$3-\$3.7 as sender, maintaining cooperation.

For Round 4, they sent \$3.0 and I received \$9.0. To maintain fair cooperation and match roughly what they've done (returning about 50%), I should return around \$4.5-\$5.0.

Let me return \$5.0 to be consistent with maintaining good cooperation and matching their recent return percentage.

Actually, wait. Let me recalculate their payoffs to understand better:

Round 1:

- They as receiver: Kept \$5 - \$0 (not sender) + received \$6.72 - \$3.0 returned = \$5 + \$3.72 = \$8.72? No wait, they were receiver, so sender sent them something.
- Actually the history shows "as RECEIVER: saw \$2.24 sent" - so I was the sender, I sent something, they saw \$2.24, it became \$6.72, they returned \$3.0
- Their payoff: \$6.72 - \$3.0 = \$3.72. Round 2 shows they sent \$3.0 which tripled to \$9.0, and when I returned \$5.07, their payoff was \$5 - \$3.0 + \$5.07 = \$7.07. In Round 3, I sent \$3.5 which became \$9.90, and they returned \$5.0, giving them \$9.90 - \$5.0 = \$4.90. Now they've sent \$3.0 which becomes \$9.0 for me, so I need to decide what to return....

usage:

```
{
  "input_tokens": 1596,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.07 returned, visible payoff \$7.07.
- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.

Your total visible earnings across all rounds are \$20.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our pattern:

****Round 2:**** They sent \$3.0, saw ~\$5.07 returned (good payoff)
****Round 3:**** They returned \$5.0 from what they saw as \$9.90
****Round 4:**** They sent \$3.5, saw ~\$4.99 returned

They've consistently sent around \$3.0-\$3.5 and returned fairly when in the receiver role. Despite some communication noise affecting visible payoffs, we've maintained good cooperation.

I'll send ****\$3.5**** to continue our stable, mutually beneficial pattern.

```json

```
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze our history:

```
Round 2: They sent $3.0, I returned $6.0 (they saw $5.07 due to noise)
Round 3: I sent $3.5, they returned $5.0 (they saw $3.3 sent due to noise)
Round 4: They sent $3.5 (wait, the history says they sent $3.5 in Round 4?), I returned $5.0 (they saw $4.99 due to noise)
```

Wait, let me reread. In Round 4, I was the receiver and they sent me \$3.0. But the visible history says they sent \$3.5 in Round 4. That must be the noise - what they actually sent was \$3.5, but I saw \$3.0.

Let me recalculate:

- Round 4: They actually sent \$3.5 ? became \$10.5 ? I returned \$5.0 (they saw \$4.99)
- Their payoff: \$5 - \$3.5 + \$4.99 = \$6.49

```
Round 5 (current): I'm the sender
```

Looking at the pattern:

- Round 2: They sent \$3.0, got back ~\$5.07 (good return)
- Round 3: I sent (they saw \$3.3), they returned \$5.0 (reasonable)
- Round 4: They sent \$3.5, got back ~\$4.99 (decent return, about 48% of the tripled amount)

They've been consistently cooperative, sending \$3.0-\$3.5 each time. Their returns have been fair (around 50% of what they receive).

I should maintain the cooperative relationship. I'll send \*\*\$3.5\*\* to continue our pattern of mutual trust and cooperation.

**usage:**

```
{
 "input_tokens": 2020,
 "output_tokens": 588,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.
- Round 5 against you, as RECEIVER: saw \$3.42 sent, it became \$10.26, returned \$5.5, visible payoff \$4.76.

Your total visible earnings across all rounds are \$27.85.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our established pattern:

```
Round 3: They returned $5.0 from $9.90
Round 4: They sent $3.5, I returned $5.0
Round 5: They returned $5.5 from $10.26
Round 6 (current): They sent $3.29 ? I received $9.87
```

They've been consistently cooperative, both when sending and returning. They sent \$3.29, keeping \$1.71 for themselves.

I'll return \*\*\$5.0\*\* (about 51% of what I received), giving them a total payoff of \$6.71. This maintains our established cooperative pattern.

```
'''json
```

```
{"return": 5.0}
'''
```

**reasoning:**

Looking at the pattern:

```
Round 3: They returned $5.0 from $9.90 (51%)
Round 4: They sent $3.5, saw ~$4.99 returned
Round 5: They returned $5.5 from $10.26 (54%)
Round 6 (current): They sent $3.29 ? I received $9.87
```

They've been very consistent:

- As sender: sending around \$3.0-\$3.5
- As receiver: returning around 51-54% of what they receive

They sent \$3.29 this round, so they kept \$1.71.

If I return about half to slightly more (consistent with our pattern), that would be around \$5.0-\$5.5.

Returning \$5.0 would give them  $\$1.71 + \$5.0 = \$6.71$  total  
Returning \$5.5 would give them  $\$1.71 + \$5.5 = \$7.21$  total

Given they're showing continued trust and cooperation, I should return a fair amount. I'll return \$5.0-\$5.5. Let me go with \$5.0 to be consistent with our established pattern (around 50% return rate).

Actually, looking at my previous returns in this partnership:

- Round 2: I returned \$6.0 from \$11.10 (54%)
- Round 4: I returned \$5.0 from \$9.0 (56%)

So I've been returning slightly more than half. Let me continue that pattern.

```
$9.87 * 0.50 = $4.94
$9.87 * 0.55 = $5.43
```

I'll return \$5.0, which is about 51% and maintains our cooperative pattern.

**usage:**

```
{
 "input_tokens": 2415,
 "output_tokens": 599,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.99 returned, visible payoff \$6.49.
- Round 5 against you, as RECEIVER: saw \$3.42 sent, it became \$10.26, returned \$5.5, visible payoff \$4.76.
- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.14 returned, visible payoff \$6.64.

Your total visible earnings across all rounds are \$32.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our established pattern:

```
Round 4: They sent $3.5, saw ~$4.99 returned (payoff $6.49)
Round 5: They returned $5.5 from $10.26 (generous)
Round 6: They sent $3.5, saw ~$5.14 returned (payoff $6.64)
```

This co-player has been highly consistent and cooperative. They regularly send \$3.5 as sender and return generously as receiver. Our mutual cooperation has been working very well.



I'll continue our successful pattern by sending \*\*\$3.5\*\*.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Looking at the pattern:

**\*\*Round 4:\*\*** They sent \$3.5, saw ~\$4.99 returned  
**\*\*Round 5:\*\*** They returned \$5.5 from \$10.26  
**\*\*Round 6:\*\*** They sent \$3.5, saw ~\$5.14 returned

This co-player has been very consistent and cooperative across all rounds. They:

- Send around \$3.5 consistently as sender
- Return around \$5.0-\$5.5 as receiver
- Have maintained cooperation despite communication noise

We've built a strong pattern of mutual trust. They've been sending \$3.5 regularly, and I've been returning fair amounts. This has worked well for both of us.

I should continue sending around \$3.5 to maintain this successful cooperative relationship

**usage:**

```
{
 "input_tokens": 2800,
 "output_tokens": 339,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.42 sent, it became \$10.26, returned \$5.5, visible payoff \$4.76.
  - Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.14 returned, visible payoff \$6.64.
  - Round 7 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$6.5, visible payoff \$5.379999999999999.
- Your total visible earnings across all rounds are \$39.93.

This round, you are the RECEIVER against your current co-player. They sent you \$3.78, so you received \$11.34. How much do you return? (0-11.34)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at our continued cooperation:

**\*\*Round 5:\*\*** They returned \$5.5 from \$10.26  
**\*\*Round 6:\*\*** They sent \$3.5, I returned \$5.0  
**\*\*Round 7:\*\*** They returned \$6.5 from \$11.88 (very generous!)  
**\*\*Round 8 (current):\*\*** They sent \$3.78 ? I received \$11.34

This co-player has shown increasing generosity, especially in Round 7 with a \$6.5 return. They sent \$3.78 this round, maintaining high trust.

I'll return \*\*\$6.5\*\* (about 57% of what I received), matching their generosity from Round 7 and giving them a total payoff of \$7.72.

```
'''json
{"return": 6.5}
'''
```

**reasoning:**

Looking at the history:

**\*\*Round 5:\*\*** They returned \$5.5 from \$10.26  
**\*\*Round 6:\*\*** They sent \$3.5, I returned \$5.0  
**\*\*Round 7:\*\*** They returned \$6.5 from \$11.88 (quite generous!)

**\*\*Round 8 (current):\*\*** They sent \$3.78 ? I received \$11.34

This co-player has been very cooperative and actually increased their generosity in Round 7, returning \$6.5 from \$11.88 (about 55%). They sent \$3.78 this round, which is slightly higher than their usual \$3.5.

To maintain this excellent cooperation and reward their generosity, I should return a fair amount. Given their last return was \$6.5 from \$11.88, and they sent \$3.78 (keeping \$1.22), I should return something that gives them a good payoff.

If I return \$6.5, they get:  $\$1.22 + \$6.5 = \$7.72$  total

This would be consistent with the pattern and reward their continued trust and generosity.

Let me return \$6.5 to maintain our strong cooperative relationship.

**usage:**

```
{
 "input_tokens": 3040,
 "output_tokens": 471,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.14 returned, visible payoff \$6.64.
- Round 7 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$6.5, visible payoff \$5.379999999999999.
- Round 8 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.73 returned, visible payoff \$7.23.

Your total visible earnings across all rounds are \$44.769999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at our excellent cooperation:

**\*\*Round 6:\*\*** They sent \$3.5, saw ~\$5.14 returned (payoff \$6.64)

**\*\*Round 7:\*\*** They returned \$6.5 from \$11.88 (very generous!)

**\*\*Round 8:\*\*** They sent \$3.5, saw ~\$5.73 returned (payoff \$7.23)

This co-player has been extremely consistent, sending \$3.5 when they're the sender and returning generously when they're the receiver. Our mutual cooperation has been highly successful.

I'll send **\*\*\$3.5\*\*** to maintain our established, mutually beneficial pattern.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Looking at the pattern:

**\*\*Round 6:\*\*** They sent \$3.5, saw ~\$5.14 returned (payoff \$6.64)

**\*\*Round 7:\*\*** They returned \$6.5 from \$11.88 (very generous)

**\*\*Round 8:\*\*** They sent \$3.5, saw ~\$5.73 returned (payoff \$7.23)

This co-player has been extremely consistent and cooperative throughout our interactions. They regularly send \$3.5 as sender and return very generously as receiver. Our cooperation has been strong and mutually beneficial.

I should continue this successful pattern with \$3.5, or potentially increase slightly to show appreciation for their consistent cooperation. Let me think about this...

They've been sending \$3.5 consistently. I've been doing the same. We have a very strong cooperative relationship. I could:

1. Send \$3.5 to maintain the pattern
2. Send \$4.0 to show increased trust given our excellent track record

Given that we have such a strong cooperative history and they've been increasingly generous (returning \$6.5 in Round 7), I think sending \$4.0 would be appropriate to reward their consistent cooperation and trust.

Actually, looking more carefully at the history, they consistently send \$3.5. I should probably match that to maintain our stable equilibrium rather than potentially disrupt it.

I'll send \$3.5 to maintain our successful pattern.

**usage:**

```
{
 "input_tokens": 3071,
 "output_tokens": 490,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$6.5, visible payoff \$5.379999999999999.
- Round 8 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.73 returned, visible payoff \$7.23.
- Round 9 against you, as RECEIVER: saw \$4.37 sent, it became \$13.11, returned \$7.5, visible payoff \$5.609999999999999.

Your total visible earnings across all rounds are \$53.53999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.78, so you received \$11.34. How much do you return? (0-11.34)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at our continued cooperation:

```
Round 7: They returned $6.5 from $11.88
Round 8: They sent $3.5, I returned $6.5
Round 9: They returned $7.5 from $13.11 (very generous!)
Round 10 (current): They sent $3.78 ? I received $11.34
```

This co-player has been exceptionally generous and consistent. In Round 9, they returned \$7.5, showing increasing trust and cooperation.

I'll return \*\*\$7.0\*\* (about 62% of what I received), giving them a total payoff of \$8.22 and matching their high level of generosity.

```
'''json
{"return": 7.0}
'''
```

**Reasoning:**

Looking at our pattern:

```
Round 7: They returned $6.5 from $11.88
Round 8: They sent $3.5, I returned $6.5
Round 9: They returned $7.5 from $13.11 (very generous!)
Round 10 (current): They sent $3.78 ? I received $11.34
```

This co-player has been increasingly generous. In Round 9, they returned \$7.5 from ~\$13.11 (about 57%). They sent \$3.78 this round.

Given their high level of cooperation and generosity, I should return a fair amount. They kept \$1.22, so if I return around \$6.5-\$7.0, they'll end up with about \$7.22-\$7.72 total.

Looking at the pattern:

- Round 7 they returned: \$6.5
- Round 8 I returned: \$6.5
- Round 9 they returned: \$7.5
- Round 10 I should return: ~\$7.0-\$7.5

I'll return \*\*\$7.0\*\* (about 62% of what I received), giving them a total payoff of \$8.22, which rewards their continued trust and generosity.

```
Usage: {
 "input_tokens": 3136,
 "output_tokens": 484,
 "reasoning_tokens": 0
}
```